Organic Chemistry Booklet

Lectures 1 – 12

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Each lecture is split into topics with its study guide questions (multiple choice). Answers are at the end of each lecture.

1 Structural Isomerism

In this lecture

1. Isomers & Structural Isomers 13 questions
2. Stereoisomers & Geometric Isomerism 3 questions
3. Chirality & Chiral Carbon 6 questions
4. Optical Activity & Enantiomers 8 questions
5. Diastereomers & Meso Compounds 15 questions

1.1 Isomers & Structural Isomers

Q1 What are isomers?

A. Compounds with the same molecular formula

B. Compounds with the same structural formula

C. Compounds with the same physical properties

D. Compounds with different molecular weights

Q2 Structural (constitutional) isomers have:

A. The same molecular formula but atoms bonded in a different order

B. The same order of bonding but a different arrangement in space

C. Different molecular formulas

D. Identical physical and chemical properties

Q3 Dimethyl ether (CH_3OCH_3) and ethanol (CH_3CH_2OH) have the same physical properties.

True

False

Q4 Butane ($CH_3CH_2CH_2CH_3$) and methylpropane (isobutane, $(CH_3)_2CHCH_3$) are structural isomers of C_4H_{10} with different boiling points.

True

False

Q5 Which of the following is **NOT** a structural isomer of C_4H_8?

A. 1-Butene ($CH_2{=}CHCH_2CH_3$)

B. Cyclobutane

C. Methylcyclopropane

D. Butane ($CH_3CH_2CH_2CH_3$)

Q6 Which of the following is **NOT** an isomer of $C_4H_{10}O$?

A. 1-Butanol

B. 2-Methyl-2-propanol

C. Diethyl ether ($CH_3CH_2OCH_2CH_3$)

D. 1-Propanol

Q7 Which of the following is **NOT** a structural isomer of C_6H_{14}?

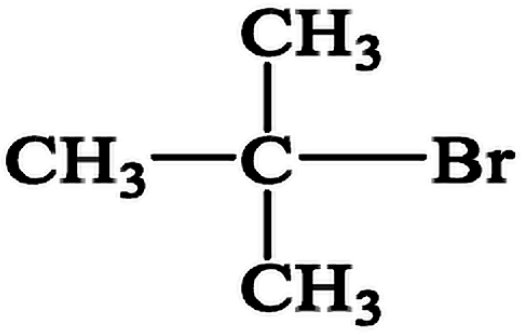
A. Hexane

B. 2-Methylpentane

C. 2,2-Dimethylbutane

D. Cyclohexane

Q8 Which compound is a structural isomer of this compound, $(CH_3)_3CBr$ (2-bromo-2-methylpropane)?



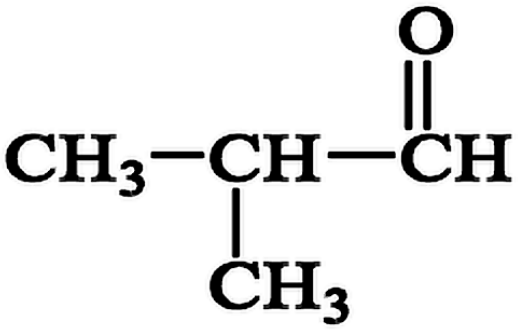
A. $CH_3CH(CH_3)CH_2Br$

B. $(CH_3)_3CCl$

C. CH_3CH_2Br

D. $CH_3C(CH_3)_2Br$

Q9 Which compound is a structural isomer of this compound, 2-methylpropanal (C_4H_8O)?



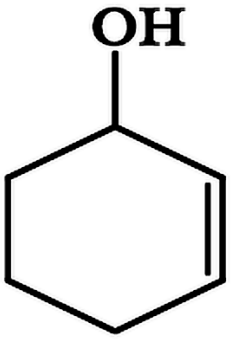
A. Butanone ($CH_3COCH_2CH_3$)

B. Propanal (CH_3CH_2CHO)

C. 2-Methylpropanoic acid ($(CH_3)_2CHCOOH$)

D. 2-Methylpropan-1-ol ($(CH_3)_2CHCH_2OH$)

Q10 Which compound is a structural isomer of this compound, cyclohex-2-en-1-ol ($C_6H_{10}O$)?



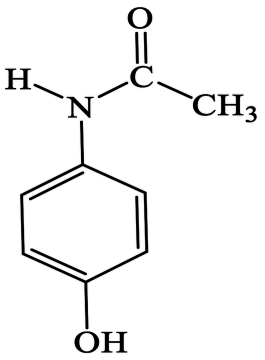
A. Cyclohex-3-en-1-ol

B. Cyclohexanol

C. Phenol

D. Cyclohexane-1,2-diol

Q11 This compound ($C_8H_9NO_2$) has the OH at the para position (opposite the NH group). Which is a structural isomer of it?



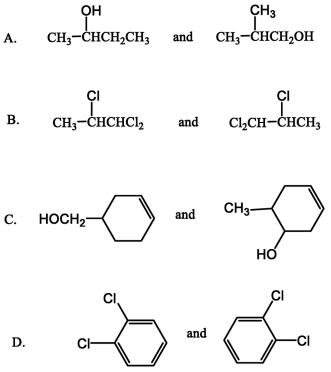
A. The same compound with the OH at the meta position

B. The same compound without the OH group

C. The same compound drawn upside down

D. The same compound with two OH groups

Q12 Which of the following pairs represent structural isomers?



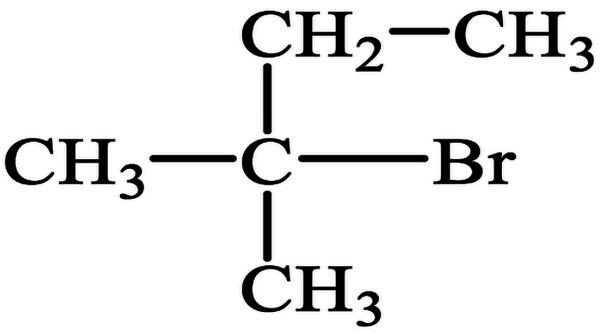
A. A and C

B. A only

C. B and D

D. A, B, C and D

Q13 Which is a structural isomer of 2-bromo-2-methylbutane, shown below?



A. 1-Bromopentane

B. 2-Bromobutane

C. 2-Bromo-2-methylpropane

D. 1-Bromopropane

1.2 Stereoisomers & Geometric Isomerism

Q14 What type of isomerism exists in compounds containing carbon-carbon double bonds and cyclic compounds?

A. Geometric isomerism

B. Structural isomerism

C. Meso isomerism

D. Functional isomerism

Q15 What is the major difference between constitutional isomers and stereoisomers?

A. Constitutional isomers are bonded in different orders; stereoisomers are bonded in the same order but oriented differently in space

B. Stereoisomers are bonded in different orders; constitutional isomers differ only in space

C. Stereoisomers have different molecular formulas

D. There is no difference

Q16 Which of the following is **NOT** a type of stereoisomer?

A. Geometric isomers

B. Enantiomers

C. Diastereomers

D. Constitutional isomers

1.3 Chirality & Chiral Carbon

Q17 Which of the following is a **chiral** object?

A. A hand

B. A ball

C. A cup

D. A conical flask

Q18 An achiral object can be superposed on its mirror image.

True

False

Q19 A chiral carbon is an sp^3-hybridized carbon with:

A. Four different atoms or groups attached

B. Two identical groups attached

C. A double bond

D. Four identical groups attached

Q20 Another name for a chiral carbon is:

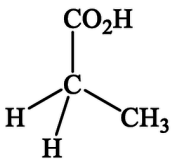
A. Stereocenter

B. Meso center

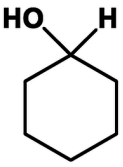
C. Achiral carbon

D. sp^2 carbon

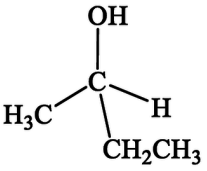
Q21 Which of the following molecules contains a chiral carbon?



a.
Propanoic acid



b.
Cyclohexanol



c.
2-Butanol

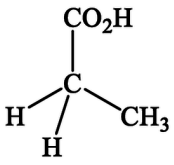
A. a. Propanoic acid

B. b. Cyclohexanol

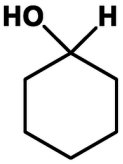
C. c. 2-Butanol

D. a and c

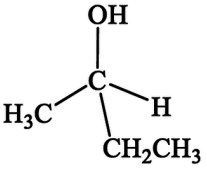
Q22 Which of the following compounds can exist as a pair of optically active compounds (enantiomers or mirror images)?



a.
Propanoic acid



b.
Cyclohexanol



c.
2-Butanol

A. a. Propanoic acid

B. b. Cyclohexanol

C. c. 2-Butanol

D. a and c

1.4 Optical Activity & Enantiomers

Q23 A dextrorotatory enantiomer rotates plane-polarized light:

A. To the right (clockwise)

B. To the left (counterclockwise)

C. It does not rotate light

D. In both directions

Q24 Which instrument measures the rotation of plane-polarized light by a chiral compound?

A. Polarimeter

B. Spectrometer

C. Thermometer

D. Barometer

Q25 A compound that rotates plane-polarized light is called optically active.

True

False

Q26 Which property is **different** for a pair of enantiomers?

A. Interaction with plane-polarized light

B. Boiling point

C. Molecular weight

D. Molecular formula

Q27 Which **chemical** property is different for a pair of enantiomers?

A. Their reactions with achiral reagents such as water

B. Their interactions with other chiral substances or molecules

C. Their molecular formula

D. The order of bonding of their atoms

Q28 Which of the following is **NOT** a property of enantiomers?

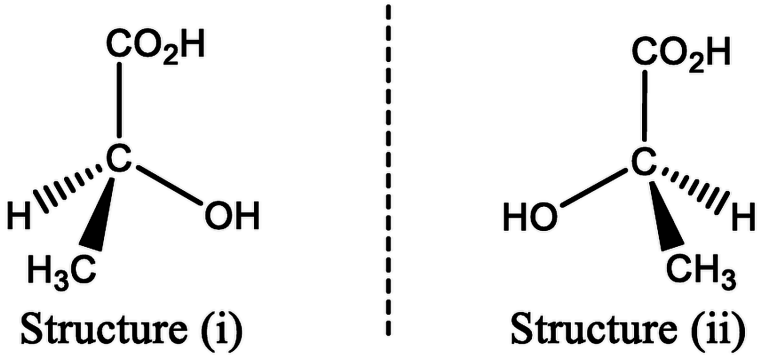
A. They are non-superimposable mirror images

B. They rotate plane-polarized light in opposite directions

C. They have a plane of symmetry

D. They differ in their interactions with other chiral substances

Q29 L(+)-Lactic acid is produced in muscles during exercise. Where is D(−)-lactic acid found? (The two mirror-image forms of lactic acid are shown below.)



A. Sour milk

B. Blood

C. Muscles

D. Sea water

Q30 Compounds having the same order of bonded atoms but differing only in the direction of rotation of plane-polarized light are:

A. Constitutional isomers

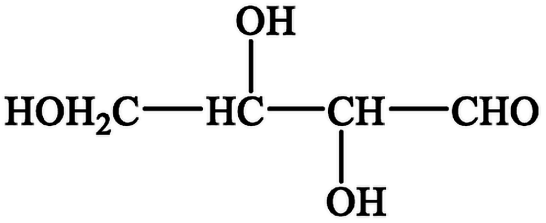
B. Diastereomers

C. Enantiomers

D. Geometric isomers

1.5 Diastereomers & Meso Compounds

Q31 How many stereoisomers are expected for compound (i), 2,3,4-trihydroxybutanal?



(i)

2,3,4-Trihydroxybutanal

A. 2

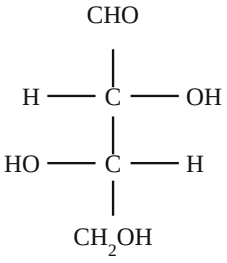
B. 3

C. 4

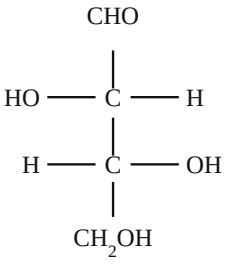
D. 8

Q32 The four stereoisomers of compound (i), 2,3,4-trihydroxybutanal, are shown below as Fischer projections. A and C are:

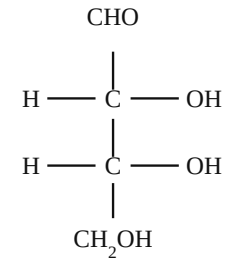
A



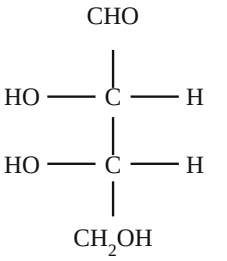
B



C



D



A. Enantiomers

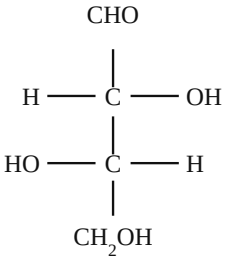
B. Diastereomers

C. Meso compounds

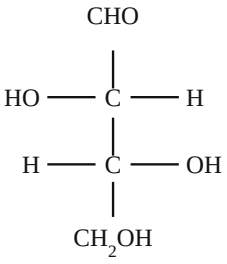
D. The same compound

Q33 The four stereoisomers of compound (i), 2,3,4-trihydroxybutanal, are shown below as Fischer projections. A and B are:

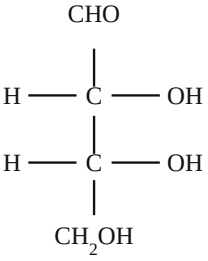
A



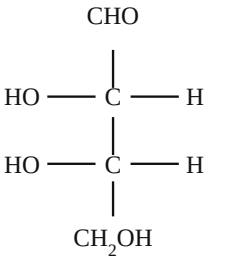
B



C



D



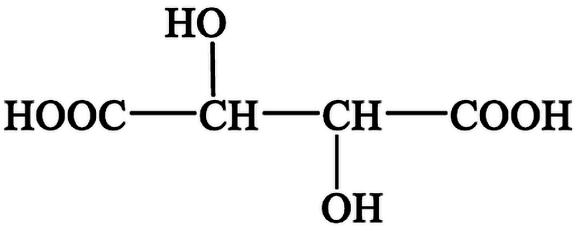
A. Enantiomers

B. Diastereomers

C. Meso compounds

D. The same compound

Q34 How many **different** stereoisomers does compound (ii), 2,3-dihydroxysuccinic acid, have?

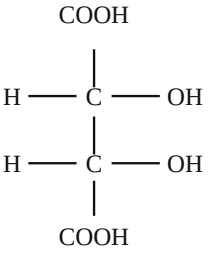


(ii)
2,3-Dihydroxysuccinic acid

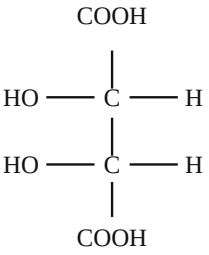
A. 2
B. 3
C. 4
D. 8

Q35 Two of the stereoisomers of compound (ii), 2,3-dihydroxysuccinic acid, are shown below as Fischer projections. G and H are:

G



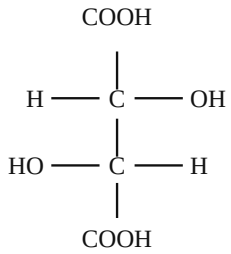
H



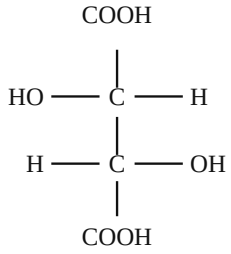
A. Enantiomers
B. Diastereomers
C. The same meso compound
D. Geometric isomers

Q36 Stereoisomers E and F of compound (ii), 2,3-dihydroxysuccinic acid, are shown below. E and F are:

E



F



A. Enantiomers

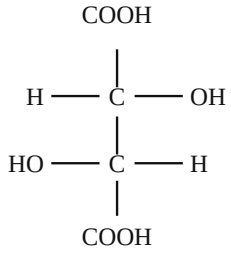
B. Diastereomers

C. The same meso compound

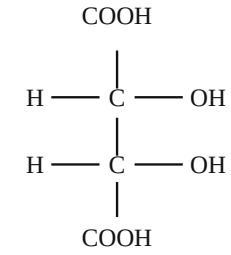
D. Structural isomers

Q37 For compound (ii), 2,3-dihydroxysuccinic acid, what is the relationship between stereoisomer E and the meso form?

E



Meso



A. Enantiomers

B. Diastereomers

C. The same compound

D. Structural isomers

Q38 A meso compound:

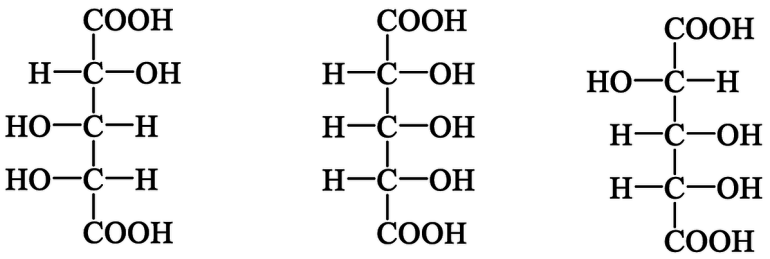
A. Has an internal plane of symmetry and is optically inactive

B. Has only one stereocenter

C. Rotates plane-polarized light

D. Is a pair of enantiomers

Q39 Which Fischer projection is the **meso** form?



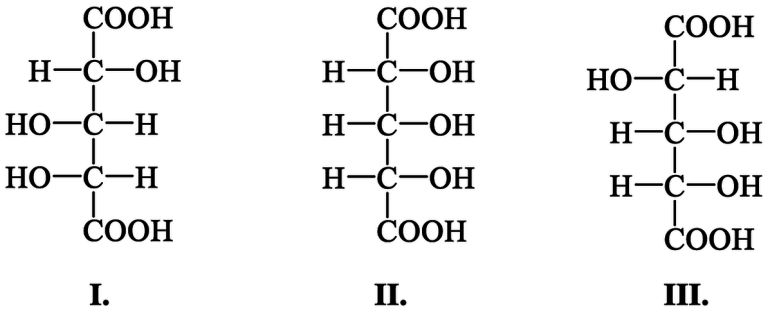
A. I

B. II

C. III

D. None

Q40 In the Fischer projections shown below, which pair are enantiomers?



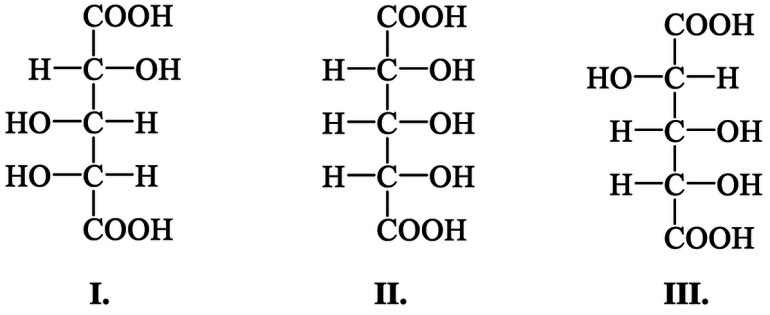
A. I and II

B. I and III

C. II and III

D. None

Q41 In the Fischer projections shown below, which pairs are **diastereomers**?



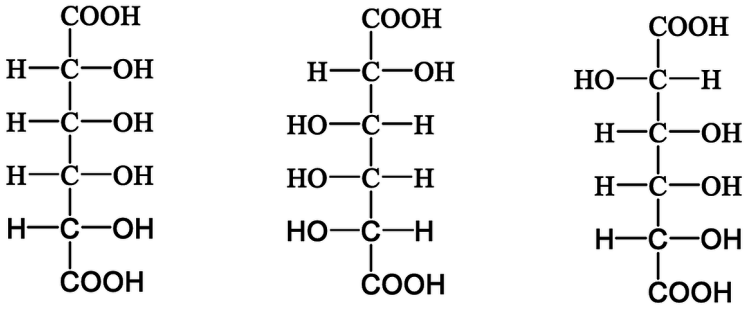
A. I and III only

B. I and II, and I and III

C. I and II, and II and III

D. II and III only

Q42 Which molecule is the **meso** form?



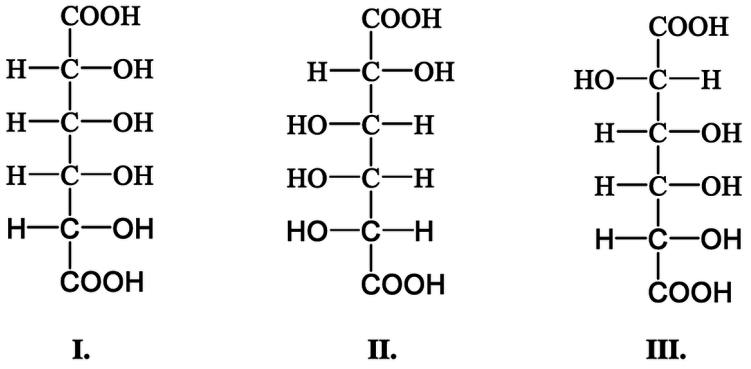
A. I

B. II

C. III

D. None

Q43 In the Fischer projections shown below, II and III are:



A. Enantiomers

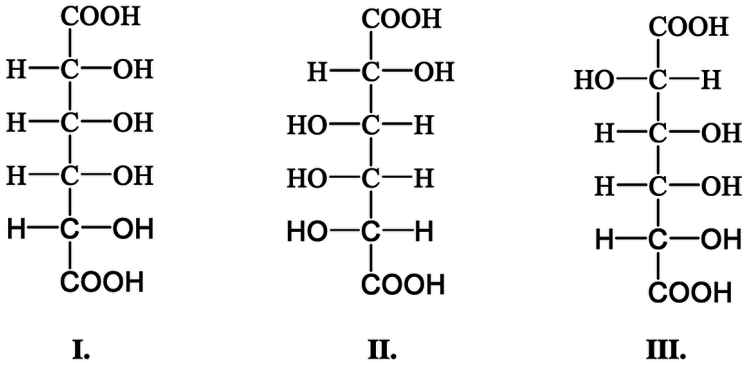
B. Diastereomers

C. Meso forms

D. Geometric isomers

E. The same compound

Q44 In the Fischer projections shown below, I and II are:



A. Enantiomers

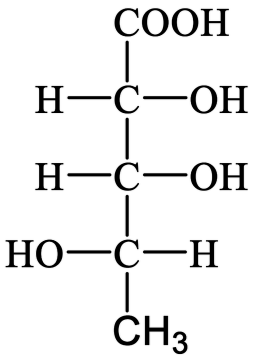
B. Diastereomers

C. Meso forms

D. Geometric isomers

E. The same compound

Q45 How many stereoisomers would you expect for the molecule shown below?



A. 2
B. 4
C. 8
D. 16
E. 32

Answers · Lecture 1

1.1 Isomers & Structural Isomers

Q1 **A. Compounds with the same molecular formula**

Isomers are two or more compounds having the same molecular formula, hence the same molecular weight.

Q2 **A. The same molecular formula but atoms bonded in a different order**

Structural isomers have the same molecular formula, but their atoms are bonded in different orders, so they have different physical and chemical properties.

Q3 **False**

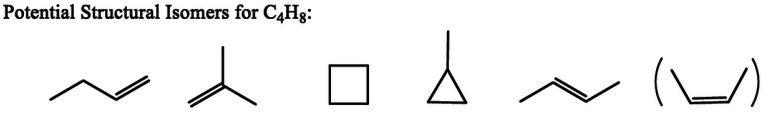
Both are C_2H_6O, but dimethyl ether is a gas with a very low boiling point, while ethanol is a liquid with a higher boiling point.

Q4 **True**

True. Both are C_4H_{10} (same molecular formula) but their atoms are bonded in different orders, so they have different physical properties. The study guide gives butane bp = −5° and methylpropane (isobutane) bp = −12°. (Note: the commonly listed boiling point of butane is about −0.5 °C; either way, the branched isomer boils lower.)

Q5 **D. Butane ($CH_3CH_2CH_2CH_3$)**

Butane is C_4H_{10}, not C_4H_8. The other three are all C_4H_8. The doctor's figure below shows all the C_4H_8 structural isomers (1-butene, 2-methylpropene, cyclobutane, methylcyclopropane, trans-2-butene, and in brackets cis-2-butene); butane is not among them.

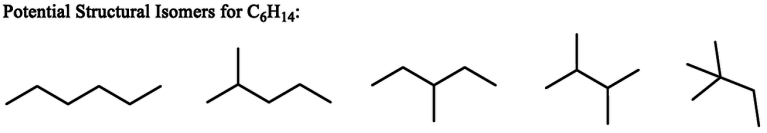


Q6 **D. 1-Propanol**

1-Propanol is C_3H_8O (only 3 carbons). The others are all $C_4H_{10}O$.

Q7 **D. Cyclohexane**

Cyclohexane is C_6H_{12}, not C_6H_{14}. The doctor's five structural isomers of C_6H_{14} are hexane, 2-methylpentane, 3-methylpentane, 2,3-dimethylbutane and 2,2-dimethylbutane (shown below).

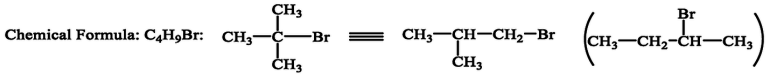


Q8 **A. $CH_3CH(CH_3)CH_2Br$**

$(CH_3)_3CBr$ has the formula C_4H_9Br (4 C, 9 H, 1 Br). $CH_3CH(CH_3)CH_2Br$ also has 4 C, 9 H and 1 Br, but the Br is on an end carbon instead of the middle carbon, so the atoms are bonded in a different order → structural isomer.

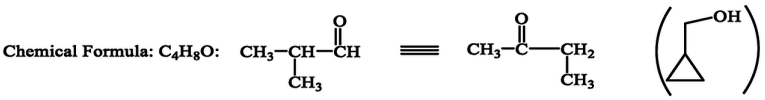
- $(CH_3)_3CCl$ has Cl instead of Br → different formula.
- CH_3CH_2Br has only 2 carbons → different formula.
- $CH_3C(CH_3)_2Br$ is the same molecule, just written in a different way.

In the doctor's answer below, the compound on the left is the question compound; the ones on the right (after the ≡ sign, and in brackets) are its structural isomers. Here the ≡ sign only separates the question compound from its isomers; it does not mean "identical".



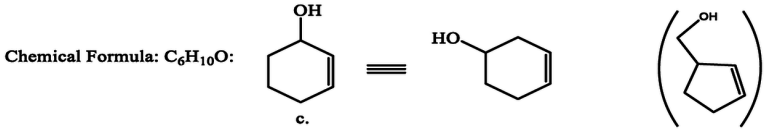
Q9 **A. Butanone ($CH_3COCH_2CH_3$)**

Butanone, $CH_3–CO–CH_2–CH_3$, is also C_4H_8O, but its atoms are bonded in a different order, so it is a structural isomer. The doctor also gives cyclopropylmethanol (in brackets) as another isomer. The other options have different formulas: propanal is C_3H_6O, 2-methylpropanoic acid is $C_4H_8O_2$, and 2-methylpropan-1-ol is $C_4H_{10}O$. (In the doctor's answer, the compound on the left is the question compound; the ones after the ≡ sign, and in brackets, are its structural isomers.)



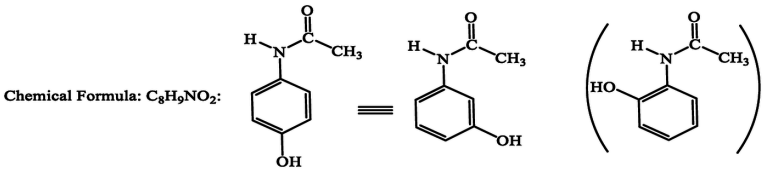
Q10 **A. Cyclohex-3-en-1-ol**

Moving the OH to a different ring carbon (cyclohex-3-en-1-ol) keeps the formula $C_6H_{10}O$ but changes the order of bonding, so it is a structural isomer. The doctor also gives (cyclopent-2-enyl)methanol (in brackets) as another isomer. Cyclohexanol is $C_6H_{12}O$, phenol is C_6H_6O, and cyclohexane-1,2-diol is $C_6H_{12}O_2$. (In the doctor's answer, the compound on the left is the question compound; the ones after the ≡ sign, and in brackets, are its structural isomers.)



Q11 **A. The same compound with the OH at the meta position**

Moving the OH group to the meta (or ortho) position keeps the formula $C_8H_9NO_2$ but changes the bonding order → structural isomer. Removing or adding an OH changes the formula, and turning the drawing upside down does not change the molecule. In the doctor's answer below, the compound on the left is the question compound; the ones on the right are its isomers (OH at meta, and in brackets at ortho). The ≡ sign only separates the question compound from its isomers; it does not mean "identical".



Q12 **A. A and C**

A: $C_4H_{10}O$ and $C_4H_{10}O$, different order → isomers. B: same compound written in reverse. C: both $C_7H_{12}O$, different structures → isomers. D: both 1,2-dichlorobenzene → same compound. (Not solved in the doctor's file. Study guide Q15 repeats this same question.)

Q13 **A. 1-Bromopentane**

The compound is $C_5H_{11}Br$. 1-Bromopentane ($CH_3CH_2CH_2CH_2CH_2Br$) is also $C_5H_{11}Br$. The other choices have fewer carbons. (Not solved in the doctor's file.)

1.2 Stereoisomers & Geometric Isomerism

Q14 **A. Geometric isomerism**

Geometric (cis/trans) isomerism, because rotation around a C=C double bond or within a ring is restricted, so groups can be locked on the same side (cis) or on opposite sides (trans).

Q15 **A. Constitutional isomers are bonded in different orders; stereoisomers are bonded in the same order but oriented differently in space**

Structural isomers have their atoms bonded in different orders, while stereoisomers are bonded in the same order but are oriented in space differently.

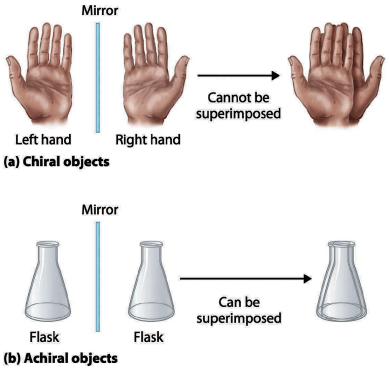
Q16 **D. Constitutional isomers**

Types of isomers: structural (constitutional) isomers, and stereoisomers (geometric isomers, enantiomers, diastereomers, meso compounds). Constitutional isomers have a different bonding order, so they are not stereoisomers.

1.3 Chirality & Chiral Carbon

Q17 **A. A hand**

A chiral object cannot be superposed on its mirror image, e.g. a hand or a foot. A ball, a cup and a conical flask are achiral.

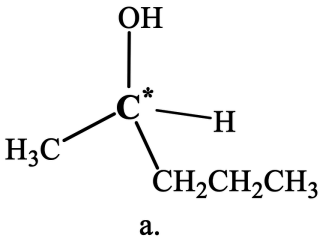


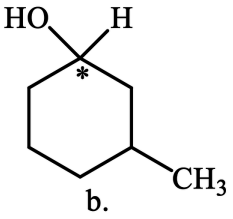
Q18 **True**

An achiral object can be superposed on its mirror image, e.g. a ball, a cup or a conical flask.

Q19 **A. Four different atoms or groups attached**

A chiral carbon (C*) is an sp^3 carbon with four different atoms or groups attached to it. Two examples from the doctor's file are shown below; in each, the carbon marked * carries four different groups.





Q20 **A. Stereocenter**

Also called asymmetric carbon, asymmetric center, stereocenter or stereogenic center.

Q21 **C. c. 2-Butanol**

Only 2-butanol: its central carbon has four different groups (OH, H, CH_3, CH_2CH_3), so it can exist as mirror images.

Q22 **C. c. 2-Butanol**

Only compound c. 2-butanol, because it contains one chiral center, hence it can exist as a pair of mirror images. Using the formula number of stereoisomers = 2^n, where n is the number of chiral carbons: $2^1 = 2$ (one pair of enantiomers).

1.4 Optical Activity & Enantiomers

Q23 **A. To the right (clockwise)**

Dextrorotatory (Latin dexter, right) rotates plane-polarized light to the right (clockwise). Levorotatory rotates it to the left (counterclockwise).

Q24 **A. Polarimeter**

Polarimeter: instrument for measuring the rotation of plane-polarized light by a chiral compound.

Q25 **True**

Optically active indicates that a compound rotates plane-polarized light.

Q26 **A. Interaction with plane-polarized light**

Enantiomers have the same boiling point, molecular weight and molecular formula. They differ only in two properties: chemical, their interactions with other chiral substances; physical, their interaction with plane-polarized light (they rotate it in opposite directions).

Q27 **B. Their interactions with other chiral substances or molecules**

The study guide lists two properties that are different for a pair of enantiomers: i. Chemical property: interactions with other chiral substances or molecules. ii. Physical property: interactions with plane-polarized light.

Q28 **C. They have a plane of symmetry**

Enantiomers do NOT have a plane of symmetry. They are non-superimposable mirror images, rotate light in opposite directions, have almost identical properties and differ with other chiral substances.

Q29 **A. Sour milk**

D(−)-Lactic acid is found in sour milk. The two forms are a pair of enantiomers ($C_3H_6O_3$).

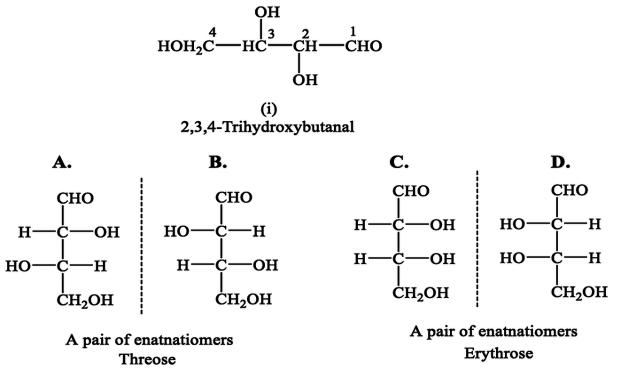
Q30 **C. Enantiomers**

Enantiomers: they have the same bonding order and differ only in the direction in which they rotate plane-polarized light (same angle, opposite directions). (Not solved in the doctor's file.)

1.5 Diastereomers & Meso Compounds

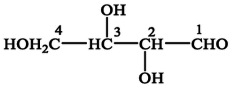
Q31 **C. 4**

It has 2 dissimilar chiral carbons (C-2 and C-3): $2^n = 2^2 = 4$ isomers. The doctor's figure below shows the 4 stereoisomers (A–D) as Fischer projections; the dashed lines are mirrors.

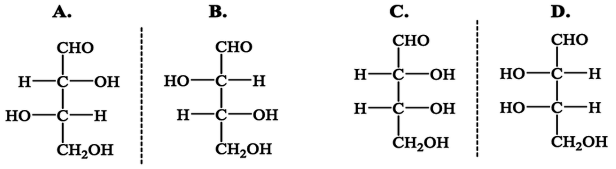


Q32 **B. Diastereomers**

A and B are enantiomers; C and D are enantiomers. A and C, A and D, B and C, B and D are diastereomers (stereoisomers that are not mirror images).



(i)
2,3,4-Trihydroxybutanal

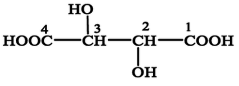


Q33 **A. Enantiomers**

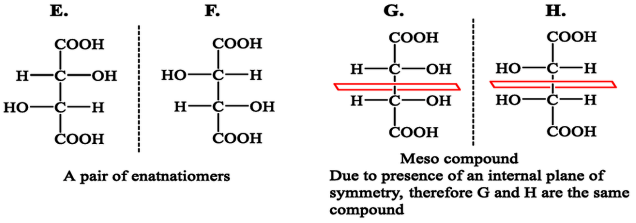
A and B are non-superimposable mirror images, so they are enantiomers (the threose pair). Likewise, C and D are enantiomers (the erythrose pair). All other pairs (A and C, A and D, B and C, B and D) are diastereomers.

Q34 **B. 3**

2 similar chiral carbons → $2^2 = 4$ expected, but G and H are the same meso compound → 2 enantiomers + 1 meso form = 3. In the doctor's figure below, E and F are the pair of enantiomers, and G and H (the red box marks the internal plane of symmetry) are the same meso compound.

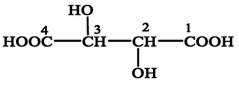


(ii)
2,3-Dihydroxysuccinic acid

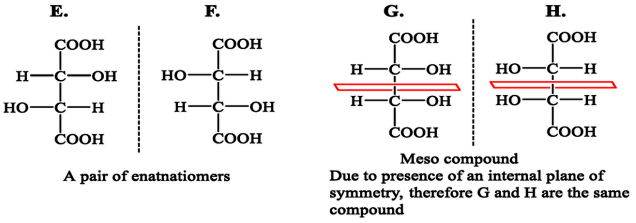


Q35 **C. The same meso compound**

G and H have an internal plane of symmetry, so they are the same compound (meso compound): turning H by 180° in the plane gives G. The red boxes in the doctor's figure mark the plane of symmetry.



(ii)
2,3-Dihydroxysuccinic acid



Q36 **A. Enantiomers**

E and F are a pair of enantiomers: they are non-superimposable mirror images and neither has an internal plane of symmetry.

Q37 **B. Diastereomers**

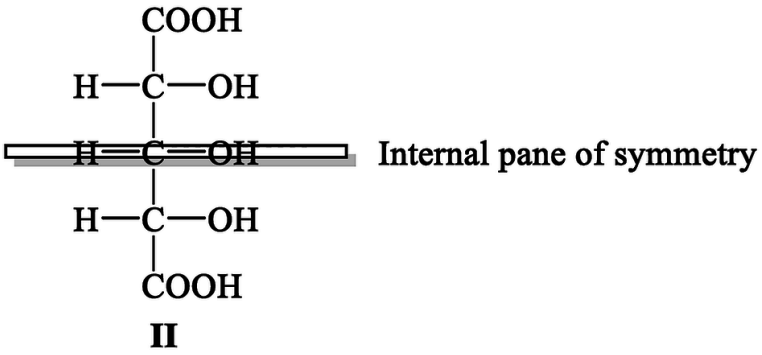
E and the meso form are diastereomers (and so are F and the meso form): they are stereoisomers with more than one chiral carbon that are not mirror images of each other.

Q38 **A. Has an internal plane of symmetry and is optically inactive**

A meso compound requires at least two stereocenters, contains an internal plane of symmetry, and is achiral (does not rotate plane-polarized light).

Q39 **B. II**

II has an internal plane of symmetry: the upper and lower halves are mirror images of each other. In the doctor's figure below, the box through the middle carbon marks this internal plane of symmetry ("pane" is a typo for "plane").



Q40 **B. I and III**

I and III are mirror images → enantiomers. I and II, and II and III, are diastereomers (stereoisomers that are not mirror images).

Q41 **C. I and II, and II and III**

Diastereomers: I and II; II and III – they have more than one chiral carbon and they are not mirror images. (I and III are mirror images, so they are enantiomers, not diastereomers.)

Q42 **A. I**

In I all OH groups are on the same side → internal plane of symmetry between C-3 and C-4 → meso. (Not solved in the doctor's file.)

Q43 **A. Enantiomers**

Every OH is switched between II and III → mirror images → enantiomers. I with II, and I with III, are diastereomers.

Q44 **B. Diastereomers**

I is the meso form (all OH on the same side), while II is chiral. They are stereoisomers with more than one chiral carbon that are not mirror images, so I and II are diastereomers. (The same is true for I and III.) (Not solved in the doctor's file.)

Q45 **C. 8**

3 chiral carbons and no symmetry (COOH at one end, CH_3 at the other): $2^3 = 8$. (Not solved in the doctor's file.)

2 Geometric Isomerism

In this lecture

1. Geometric Isomerism (cis / trans) 10 questions
2. E / Z System & Priority Rules 15 questions
3. Geometric Isomerism in Cyclic Compounds 6 questions
4. Conformations of Open-Chain Compounds 11 questions
5. Conformations of Cyclohexane 8 questions

2.1 Geometric Isomerism (cis / trans)

Q1 Geometric isomerism is a result of:

A. Restricted rotation (rigidity) about a double bond or in a ring

B. The presence of a chiral carbon

C. A different order of bonded atoms

D. Different molecular formulas

Q2 Geometric isomerism occurs in:

A. Alkynes only

B. Alkenes and cyclic compounds

C. Alkanes only

D. Compounds with a chiral carbon only

Q3 Geometric isomers have the same molecular formula and the same order of bonded atoms.

True

False

Q4 A molecule is designated *cis* when the two groups are:

A. On the same carbon atom

B. On opposite sides of the double bond

C. On the same side of the double bond

D. Perpendicular to the double bond

Q5 Which statement about the two isomers of 1,2-dichloroethene is correct?

A. They have different molecular formulas

B. trans-1,2-Dichloroethene boils at 60 °C, higher than the cis isomer

C. Both isomers have the same boiling point

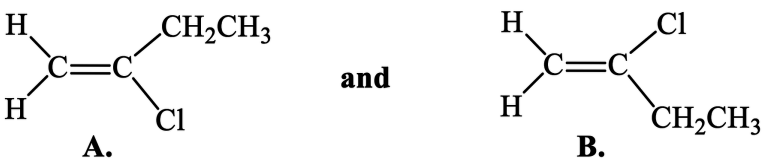
D. cis-1,2-Dichloroethene boils at 60 °C, higher than the trans isomer (48 °C)

Q6 Rotation about a C–C sigma (single) bond is free, but rotation about a C=C double bond is restricted.

True

False

Q7 Molecules A and B are:



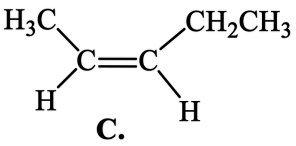
A. The same molecule

B. cis/trans isomers

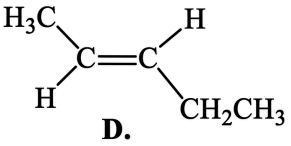
C. Structural isomers

D. Enantiomers

Q8 Molecules C and D are:



C. **and**



D.

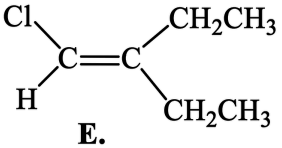
A. Not isomers

B. Geometric isomers: cis-2-pentene and trans-2-pentene

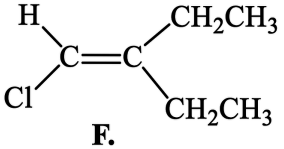
C. The same molecule

D. Structural isomers

Q9 Molecules E and F are:



E. **and**



F.

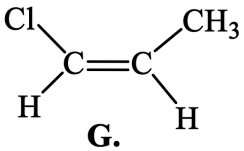
A. cis/trans isomers

B. Enantiomers

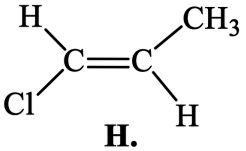
C. Structural isomers

D. The same molecule

Q10 Molecules G and H are:



G. **and**



H.

A. The same molecule

B. Structural isomers

C. cis-1-chloropropene and trans-1-chloropropene

D. trans-1-chloropropene and cis-1-chloropropene

2.2 E / Z System & Priority Rules

Q11 The letter **Z** comes from the German word *zusammen*, which means:

A. Left

B. Across

C. Right

D. Together

Q12 The configuration is **(E)** when the higher-priority atoms or groups are:

A. On opposite sides of the C=C double bond

B. On the same carbon atom

C. Both hydrogen atoms

D. On the same side of the C=C double bond

Q13 By sequence rule 1, which atom has the **highest** priority?

A. F

B. I

C. Br

D. Cl

Q14 By sequence rule 2 (for a group of atoms, look at the atomic number of the atom directly attached to the sp² carbon), which group has the **highest** priority?

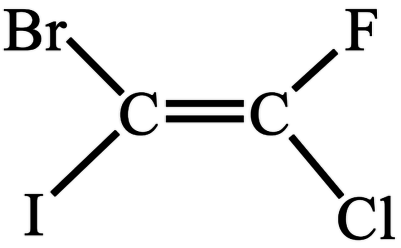
A. –SH

B. –OH

C. $–NH_2$

D. $–CH_3$

Q15 What is the configuration of molecule (ii), 1-bromo-2-chloro-2-fluoro-1-iodoethene?



Molecule (ii)

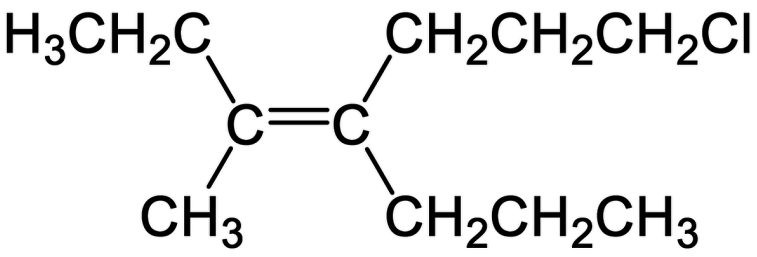
A. (E)

B. It has no E/Z configuration

C. Cannot be decided

D. (Z)

Q16 In molecule (i), which group on the right-hand carbon has the higher priority?



molecule (i)

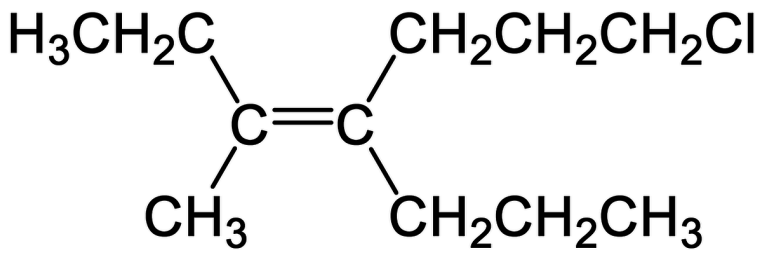
A. It cannot be determined

B. They have equal priority

C. $–CH_2CH_2CH_2Cl$

D. $–CH_2CH_2CH_3$

Q17 What is the configuration of molecule (i), 7-chloro-3-methyl-4-propylhept-3-ene?



molecule (i)

A. cis only, not E/Z

B. It has no E/Z configuration

C. (Z)-7-chloro-3-methyl-4-propylhept-3-ene

D. (E)-7-chloro-3-methyl-4-propylhept-3-ene

Q18 By sequence rule 3 (isotopes), which has the higher priority?

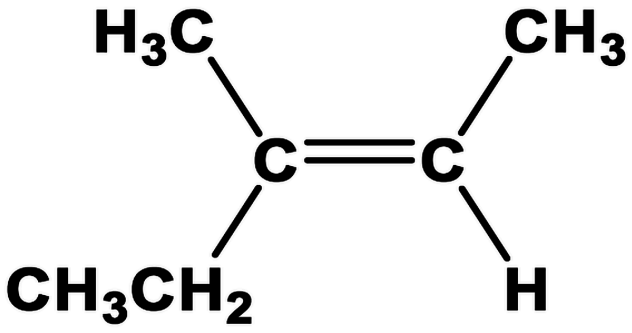
A. Hydrogen (H, ^{1}H)

B. It depends on the position

C. They are equal

D. Deuterium (D, ^{2}H)

Q19 In (E)-3-methyl-2-pentene, why does the ethyl group have a higher priority than the methyl group?



A. Ethyl contains a double bond

B. Its carbon carries (C,H,H) while the methyl carbon carries (H,H,H)

C. Methyl has a higher atomic number

D. Ethyl is attached by an oxygen atom

Q20 By sequence rule 5 (multiple bonds), which group has the **highest** priority?

A. –COOH

B. –CN

C. $–CH_2OH$

D. –CHO

Q21 Which list shows the groups in order of **increasing** priority (lowest to highest)?

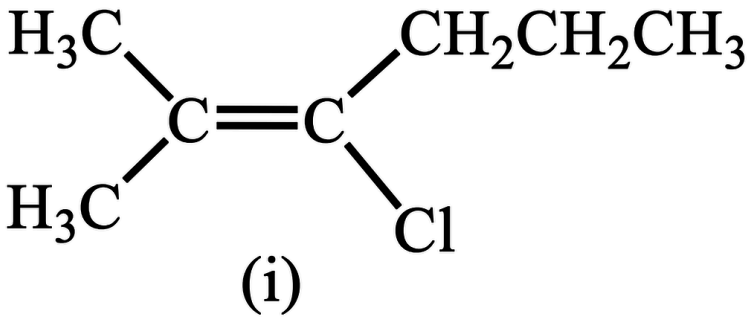
A. $–H < –OH < –CH_2OH < –CO_2H < –CO_2CH_3 < –SCH_3$

B. $–H < –CH_2OH < –CO_2CH_3 < –CO_2H < –OH < –SCH_3$

C. $–H < –CH_2OH < –CO_2H < –CO_2CH_3 < –OH < –SCH_3$

D. $–SCH_3 < –OH < –CO_2CH_3 < –CO_2H < –CH_2OH < –H$

Q22 What is the configuration of compound (i)?



(i)

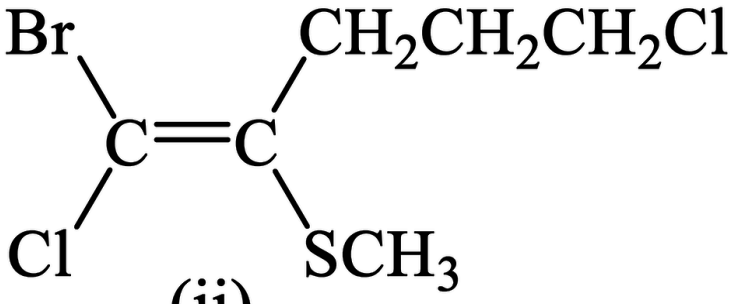
A. (Z)

B. No E/Z configuration

C. cis

D. (E)

Q23 What is the configuration of compound (ii)?



(ii)

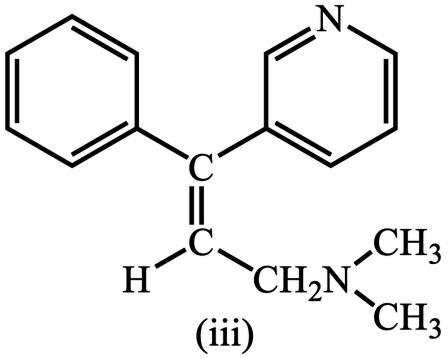
A. cis

B. No E/Z configuration

C. (Z)

D. (E)

Q24 What is the configuration of compound (iii)?



(iii)

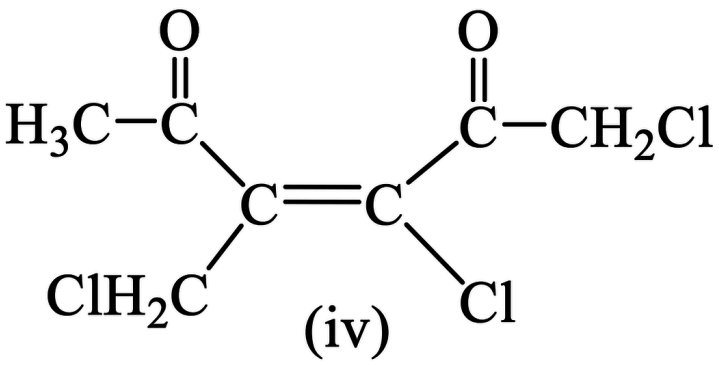
A. (Z)

B. trans

C. (E)

D. No E/Z configuration

Q25 What is the configuration of compound (iv)?



A. No E/Z configuration

B. cis only

C. (E)

D. (Z)

2.3 Geometric Isomerism in Cyclic Compounds

Q26 In a wedge-dash drawing of a ring, a solid wedge bond means the group is:

A. Above the plane of the ring

B. Below the plane of the ring

C. In the plane of the ring

D. Always axial

Q27 A dashed bond represents a group below the plane of the ring.

True

False

Q28 In cis-1,2-dimethylcyclopentane, the two CH_3 groups are:

A. In the plane of the ring

B. On the same carbon

C. On opposite sides of the ring

D. On the same side of the ring (both wedges or both dashes)

Q29 In trans-1,4-dimethylcyclohexane drawn with wedge-dash bonds:

A. Both CH_3 groups are wedges

B. One CH_3 is a wedge and the other is a dash

C. Both CH_3 groups are dashes

D. Both CH_3 groups are on the same carbon

Q30 In cis-1,4-dimethylcyclohexane drawn with wedge-dash bonds:

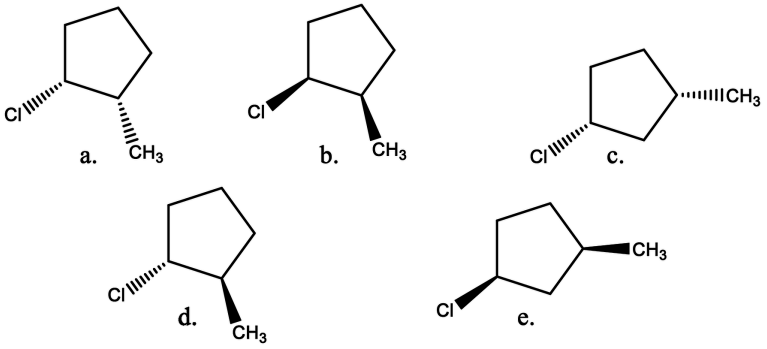
A. One CH_3 is drawn in the plane of the ring and the other above it

B. One CH_3 is a wedge and the other is a dash

C. Both CH_3 groups are on the same side of the ring (both wedges, or both dashes)

D. Both CH_3 groups are on the same carbon

Q31 The structure of *trans*-1-chloro-2-methylcyclopentane is:



A. a
B. b
C. c
D. d
E. e

2.4 Conformations of Open-Chain Compounds

Q32 Which of the following is **NOT** a projection formula used to show conformations?

A. Lewis electron-dot formula
B. Wedge-dash formula
C. Newman projection
D. Sawhorse projection

Q33 Which projection looks straight down the C–C bond, with the front carbon as a point and the back carbon as a circle?

A. Fischer projection
B. Ball-and-stick formula
C. Newman projection
D. Sawhorse projection

Q34 Which of the projection / dimensional formulas in the doctor's answer draws the molecule with crossing horizontal and vertical lines, the carbon atoms being at the crossing points?

A. Fischer projection formula
B. Wedge-dash formula
C. Sawhorse projection formula
D. Ball-and-stick formula

Q35 Which conformer of butane is the most stable?

A. Conformer with eclipsed methyl groups
B. Anti (staggered) conformer
C. Gauche conformer
D. Eclipsed conformer (CH_3 eclipsing H)

Q36 Which conformer of butane is the **least** stable?

A. Eclipsed (CH_3 eclipsing H)
B. Gauche
C. Anti (staggered)
D. Eclipsed methyls (CH_3 eclipsing CH_3)

Q37 What is the order of stability of the butane conformers?

A. Eclipsed methyls > eclipsed > gauche > anti

B. Anti > eclipsed > gauche > eclipsed methyls

C. Anti > gauche > eclipsed > eclipsed methyls

D. Gauche > anti > eclipsed > eclipsed methyls

Q38 In the gauche conformation of butane, the two CH_3 groups are:

A. 60° apart (staggered)

B. 0° apart (eclipsed)

C. 180° apart

D. 120° apart (eclipsed)

Q39 What is the major difference between configuration and conformation?

A. Conformation is fixed in space; configuration changes by rotation

B. Configuration depends only on the molecular formula

C. Configuration is fixed in space; conformation changes by rotation about single bonds

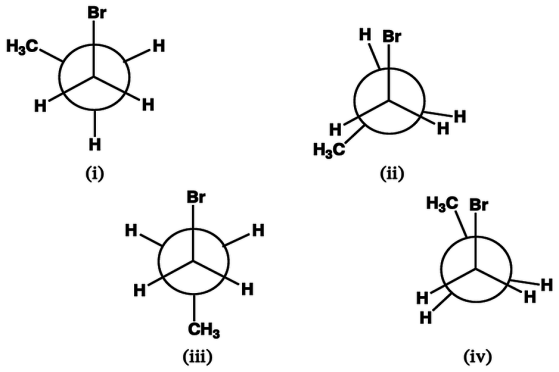
D. They are the same thing

Q40 Conformers can be interconverted by rotation about single bonds without breaking any bonds.

True

False

Q41 Arrange the following Newman projections in order of **decreasing** energy (highest to lowest):



A. (i) > (ii) > (iii) > (iv)

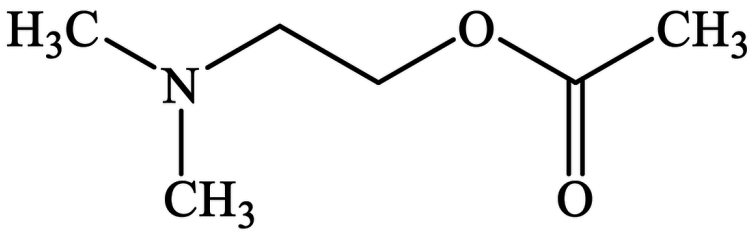
B. (iii) > (i) > (ii) > (iv)

C. (ii) > (iii) > (iv) > (i)

D. (iii) > (iv) > (ii) > (i)

E. (iv) > (ii) > (i) > (iii)

Q42 Draw all the possible conformers of the compound shown below (looking along the central CH_2–CH_2 bond). Which conformer is the **most stable**?



A. All conformers have the same energy

B. The anti (staggered) conformer, with $–N(CH_3)_2$ and $–OC(O)CH_3$ 180° apart

C. The eclipsed conformer, with $–N(CH_3)_2$ and $–OC(O)CH_3$ eclipsing each other

D. The gauche (staggered) conformer, with $–N(CH_3)_2$ and $–OC(O)CH_3$ 60° apart

2.5 Conformations of Cyclohexane

Q43 Which is the most stable conformation of cyclohexane?

A. Planar hexagon

B. Chair

C. Boat

D. Fully eclipsed

Q44 The boat conformation of cyclohexane is more stable than the chair conformation.

True

False

Q45 In the chair conformation, axial bonds are:

A. Always pointing down

B. Always pointing up

C. Parallel to the ring axis, pointing straight up or straight down

D. Roughly in the plane of the ring, pointing outward

Q46 In chair cyclohexane, each carbon carries:

A. Only one hydrogen

B. Two equatorial hydrogens

C. Two axial hydrogens

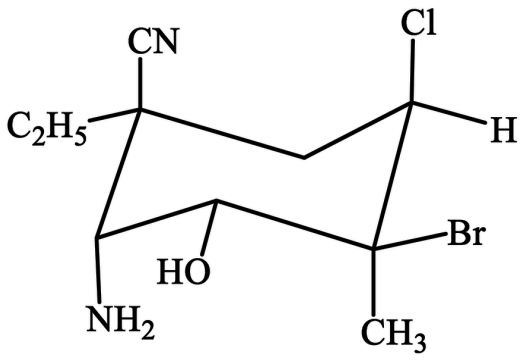
D. One axial and one equatorial hydrogen

Q47 Axial positions alternate up and down around the cyclohexane chair.

True

False

Q48 Which two atoms or groups are in the axial position and above the plane of the molecule?



A. Cl and CH_3

B. CN and NH_2

C. Br and Cl

D. NH_2 and OH

E. CN and Cl

Q49 In the chair conformations of trans-1,2-dimethylcyclohexane, the two CH_3 groups are:

A. Always both equatorial

B. Always one axial and one equatorial

C. Both axial (a,a) in one conformer and both equatorial (e,e) in the other

D. Always both axial

Q50 Which chair conformer of trans-1,2-dimethylcyclohexane is more stable?

A. The boat conformer

B. e,e (both CH_3 equatorial)

C. Both are equally stable

D. a,a (both CH_3 axial)

Answers · Lecture 2

2.1 Geometric Isomerism (cis / trans)

Q1 **A. Restricted rotation (rigidity) about a double bond or in a ring**

Geometric isomerism is a result of rigidity in molecules: there is no free rotation about the C=C double bond or about the bonds of a ring.

Q2 **B. Alkenes and cyclic compounds**

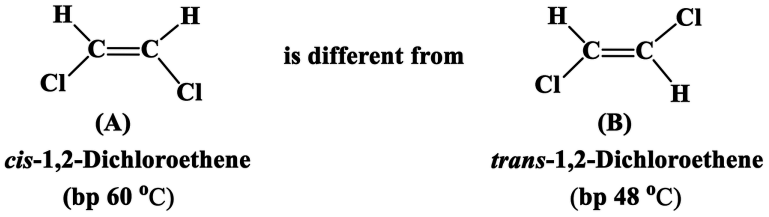
It occurs in two classes of compounds: alkenes (C=C) and cyclic compounds.

Q3 **True**

Geometric isomers have the same molecular formula, the same molecular weight and the same order of bonding; they differ only in the orientation of the groups in space (different configurations).

Q4 **C. On the same side of the double bond**

cis (Latin, “on this side”): the two groups are on the same side of the double bond. *trans* (Latin, “across”): on opposite sides.



(A)
***cis*-1,2-Dichloroethene**
(bp 60 °C)

(B)
***trans*-1,2-Dichloroethene**
(bp 48 °C)

Q5 **D. cis-1,2-Dichloroethene boils at 60 °C, higher than the trans isomer (48 °C)**

cis-1,2-Dichloroethene: bp 60 °C. trans-1,2-Dichloroethene: bp 48 °C.

(A)
***cis*-1,2-Dichloroethene**
(bp 60 °C)

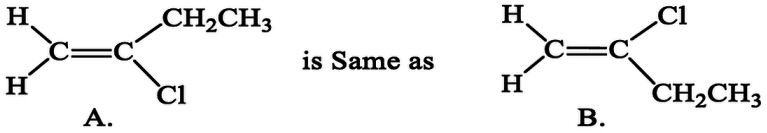
(B)
***trans*-1,2-Dichloroethene**
(bp 48 °C)

Q6 **True**

A sigma bond allows free rotation. Rotating about a double bond would require breaking the pi bond (the side-by-side p orbitals), so rotation is restricted.

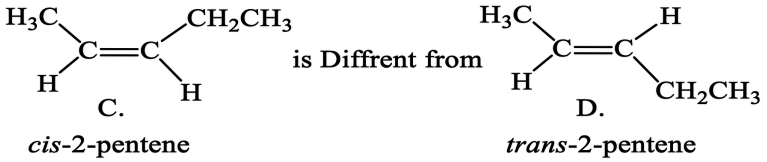
Q7 **A. The same molecule**

A is the same as B.



Q8 **B. Geometric isomers: cis-2-pentene and trans-2-pentene**

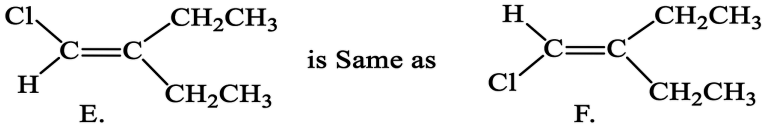
C is different from D: C is cis-2-pentene and D is trans-2-pentene.



cis-2-pentene *trans*-2-pentene

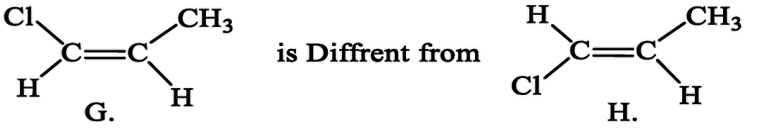
Q9 **D. The same molecule**

E is the same as F.



Q10 **C. cis-1-chloropropene and trans-1-chloropropene**

G is different from H: G is cis-1-chloropropene and H is trans-1-chloropropene.



***cis*-1-chloropropene** ***trans*-1-chloropropene**

2.2 E / Z System & Priority Rules

Q11 **D. Together**

(Z) is from the German *zusammen* (“together”); (E) is from *entgegen* (“across”).

Q12 **A. On opposite sides of the C=C double bond**

The configuration is (Z) if the higher-priority atoms or groups are on the same side of the C=C, and (E) if they are on opposite sides.

Q13 **B. I**

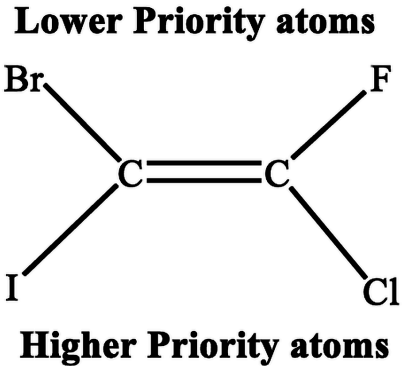
For atoms, the sequence order is by atomic number: I (53) > Br (35) > Cl (17) > F (9).

Q14 **A. –SH**

Rule 2 ranks groups by the atomic number of the atom directly attached: –H (1) < $–CH_3$ (6) < $–NH_2$ (7) < –OH (8) < –SH (16) < –Cl (17) < –Br (35) < –I (53). S (16) is higher than O (8), N (7) and C (6), so –SH has the highest priority.

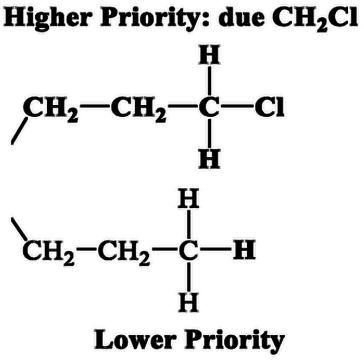
Q15 **D. (Z)**

On the left carbon I > Br; on the right carbon Cl > F. I and Cl are on the same side of the C=C, so the compound is (Z)-1-bromo-2-chloro-2-fluoro-1-iodoethene.



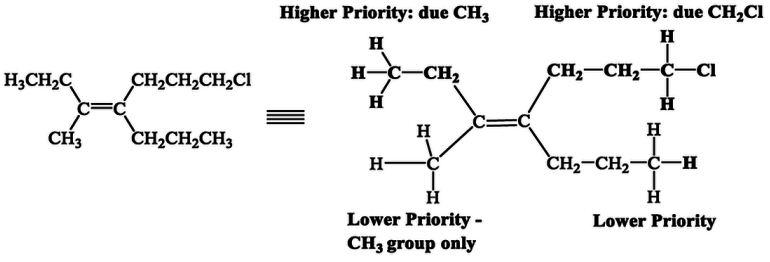
Q16 **C. $–CH_2CH_2CH_2Cl$**

Both groups are attached through a CH_2 carbon (C,H,H), and the next carbon is also CH_2 (C,H,H), so there is still a tie. At the third carbon, $–CH_2Cl$ has (Cl,H,H) while $–CH_3$ has (H,H,H); Cl beats H, so $–CH_2CH_2CH_2Cl$ has the higher priority (rule 4). The figure shows the two right-hand groups written out atom by atom.



Q17 **C. (Z)-7-chloro-3-methyl-4-propylhept-3-ene**

Left carbon: $–CH_2CH_3$ > $–CH_3$. Right carbon: $–CH_2CH_2CH_2Cl$ > $–CH_2CH_2CH_3$. Both higher-priority groups are on the top side, so the configuration is (Z). In the figure, "≡" means the expanded drawing on the right is the same molecule with each group written out; the labels mark the higher- and lower-priority group on each carbon.

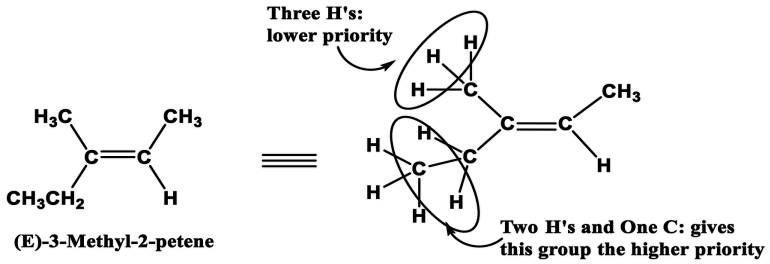


Q18 **D. Deuterium (D, ^2H)**

For isotopes of the same element, the higher mass number has the higher priority: D (^2H) > H (^1H). (The doctor's file says "atomic number", but isotopes have the same atomic number; it is the mass number that differs.)

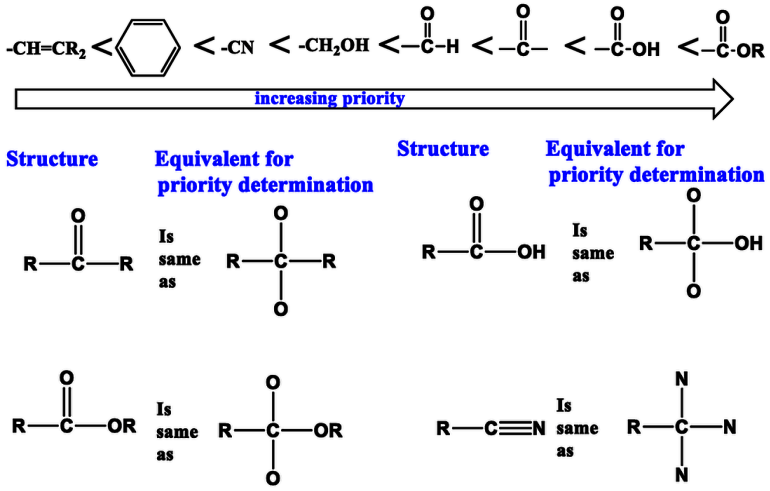
Q19 **B. Its carbon carries (C,H,H) while the methyl carbon carries (H,H,H)**

Both groups are attached through carbon. At the next atoms, ethyl has two H's and one C, while methyl has three H's, so ethyl has the higher priority. In the figure, "≡" means the right-hand drawing is the same molecule with the ethyl and methyl groups written out; the circled groups are the two being compared.



Q20 **A. –COOH**

Atoms attached by double or triple bonds are given single-bond equivalents. Increasing priority: $–CH=CR_2$ < phenyl < –CN < $–CH_2OH$ < –CHO < –COR < –COOH < –COOR.



Q21 **C. –H < $–CH_2OH$ < $–CO_2H$ < $–CO_2CH_3$ < –OH < $–SCH_3$**

Attached atoms: H (1) < C (6) < O (8) < S (16). The three carbon groups: $–CH_2OH$ (O,H,H) < $–CO_2H$ and $–CO_2CH_3$ (O,O,O). For the last tie, the –OH oxygen carries H but the $–OCH_3$ oxygen carries C, so $–CO_2CH_3$ > $–CO_2H$. (Not solved in the doctor's file.)

Q22 **B. No E/Z configuration**

The left carbon carries two identical CH_3 groups, so the compound cannot have E/Z isomers. (Not solved in the doctor's file.)

Q23 **D. (E)**

Left carbon: Br (35) > Cl (17), Br is on top. Right carbon: $–SCH_3$ (S, 16) > $–CH_2CH_2CH_2Cl$ (C, 6), SCH_3 is at the bottom. The higher-priority groups are on opposite sides, so (E). (Not solved in the doctor's file.)

Q24 **A. (Z)**

Bottom carbon: $–CH_2N(CH_3)_2$ > H. Top carbon: phenyl and 3-pyridyl both start with C(C,C,C); one atom further, the pyridyl ring reaches N while phenyl has only C, so pyridyl wins. Pyridyl and $–CH_2N(CH_3)_2$ are on the same side, so (Z). (Not solved in the doctor's file.)

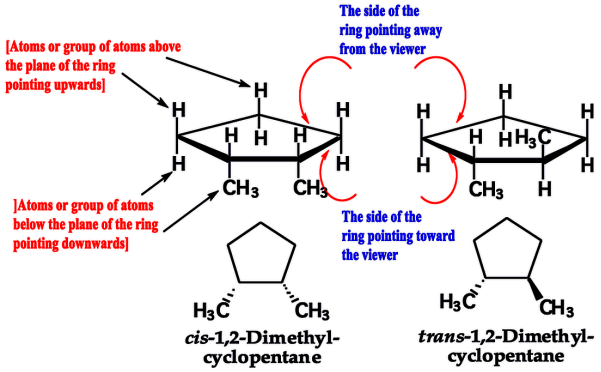
Q25 **D. (Z)**

Left carbon: $–CH_2Cl$ (Cl,H,H) > $–C(=O)CH_3$ (O,O,C), because Cl (17) > O (8) at the first point of difference. Right carbon: –Cl > $–C(=O)CH_2Cl$. $–CH_2Cl$ and –Cl are both at the bottom, so (Z). (Not solved in the doctor's file.)

2.3 Geometric Isomerism in Cyclic Compounds

Q26 **A. Above the plane of the ring**

A solid line (wedge) represents groups or atoms above the plane of the ring; a dashed line represents groups below the plane.

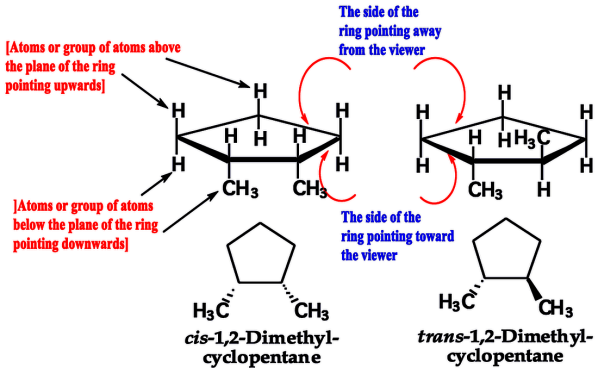


Q27 **True**

Dashed line: below the plane of the ring. Solid line: above the plane of the ring.

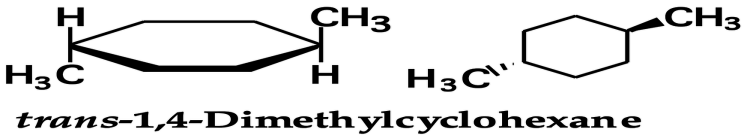
Q28 **D. On the same side of the ring (both wedges or both dashes)**

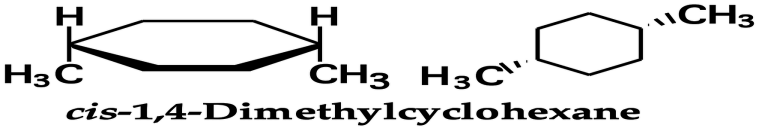
cis-1,2-Dimethylcyclopentane: both CH_3 groups on the same side of the ring. trans: one above and one below.



Q29 **B. One CH_3 is a wedge and the other is a dash**

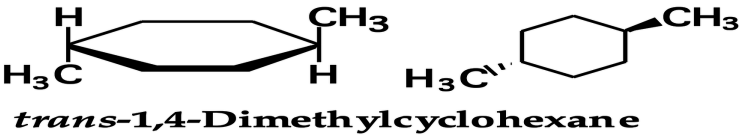
trans-1,4-Dimethylcyclohexane has one CH_3 above and one below the ring; cis-1,4-dimethylcyclohexane has both on the same side.



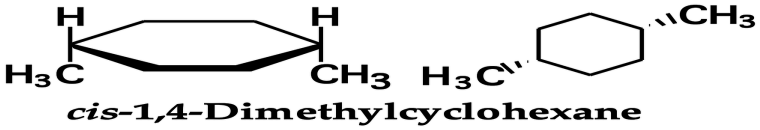


Q30 **C. Both CH_3 groups are on the same side of the ring (both wedges, or both dashes)**

cis-1,4-Dimethylcyclohexane has both CH_3 groups on the same side of the ring (both above or both below the plane); trans-1,4-dimethylcyclohexane has one CH_3 above and one below. The doctor's answer shows both drawings.



trans-1,4-Dimethylcyclohexane



cis-1,4-Dimethylcyclohexane

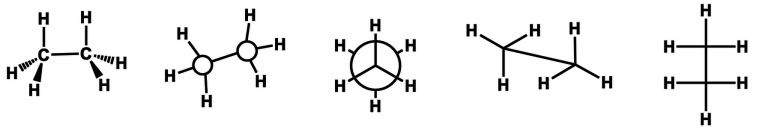
Q31 **D. d**

In structure d, Cl (dash, below) and CH_3 (wedge, above) are on adjacent carbons and on opposite sides of the ring. Structures a and b are cis (both dashes or both wedges); structures c and e are 1,3-compounds, not 1,2. This matches the doctor's answer (d).

2.4 Conformations of Open-Chain Compounds

Q32 **A. Lewis electron-dot formula**

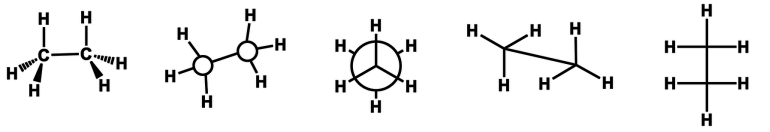
Projection / dimensional formulas: (i) wedge-dash, (ii) ball-and-stick, (iii) Newman projection, (iv) sawhorse projection, (v) Fischer projection.



(i) Wedge-Dash formula | (ii) Ball-and-stick formula | (iii) Newman projection formula | (iv) Sawhorse projection formula | (v) Fischer projection formula

Q33 **C. Newman projection**

This is the Newman projection (iii in the figure).



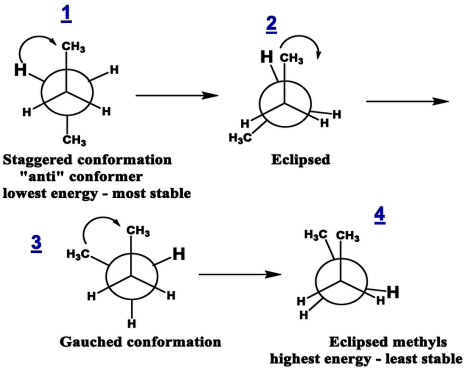
(i) Wedge-Dash formula | (ii) Ball-and-stick formula | (iii) Newman projection formula | (iv) Sawhorse projection formula | (v) Fischer projection formula

Q34 **A. Fischer projection formula**

The doctor lists five types of projection / dimensional formulas for ethane: (i) wedge-dash formula, (ii) ball-and-stick formula, (iii) Newman projection formula, (iv) sawhorse projection formula and (v) Fischer projection formula. The Fischer projection draws each carbon as the crossing point of a horizontal and a vertical line.

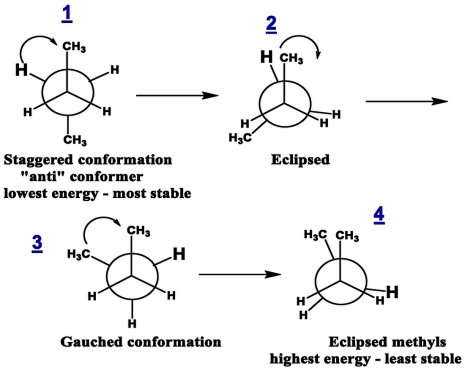
Q35 **B. Anti (staggered) conformer**

In the figure, conformer 1, the staggered “anti” conformer, has the lowest energy and is the most stable.



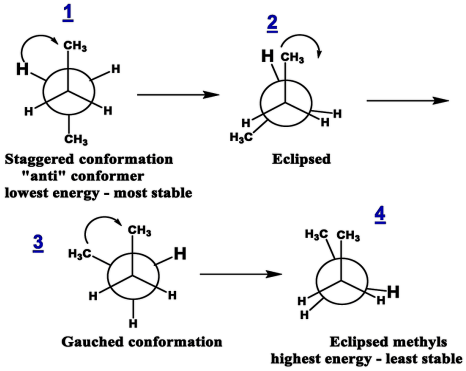
Q36 **D. Eclipsed methyls (CH_3 eclipsing CH_3)**

In the figure, conformer 4, with eclipsed methyl groups, has the highest energy and is the least stable.



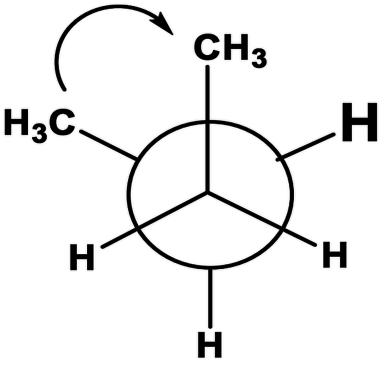
Q37 **C. Anti > gauche > eclipsed > eclipsed methyls**

The numbers 1–4 are the butane conformers in the figure (each curved arrow is a 60° rotation to the next one). Order of energy: 4 > 2 > 3 > 1. Order of stability: 1 > 3 > 2 > 4 (anti > gauche > eclipsed > eclipsed methyls).



Q38 **A. 60° apart (staggered)**

Gauche: staggered conformation with the CH_3 groups 60° apart (shown in the figure). Anti: 180° apart.



Q39 **C. Configuration is fixed in space; conformation changes by rotation about single bonds**

The major difference is in the arrangement of groups around the bonds in space: in configuration the arrangement is fixed, but in conformation it assumes different arrangements in space.

Q40 **True**

Conformations are different arrangements produced by rotation about single (sigma) bonds.

Q41 **E. (iv) > (ii) > (i) > (iii)**

(iv) Br eclipsing CH_3 (highest) > (ii) eclipsed, Br and CH_3 each eclipsing H > (i) gauche, Br and CH_3 60° apart > (iii) anti, Br and CH_3 180° apart (lowest).

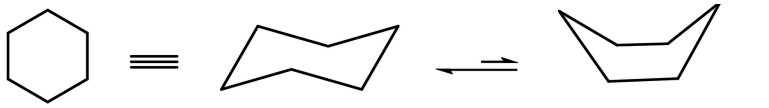
Q42 **B. The anti (staggered) conformer, with $–N(CH_3)_2$ and $–OC(O)CH_3$ 180° apart**

As for butane: the conformers are anti (staggered, the two large groups 180° apart), gauche (staggered, 60° apart) and eclipsed. The anti conformer is the most stable because the two large groups ($–N(CH_3)_2$ and $–OC(O)CH_3$) are as far apart as possible; the eclipsed conformer with the two large groups eclipsing each other is the least stable (highest energy). This answer is based on keeping the two large groups as far apart as possible, as for butane. (Not solved in the doctor's file.)

2.5 Conformations of Cyclohexane

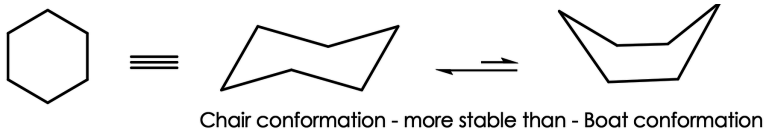
Q43 **B. Chair**

The chair conformation is more stable than the boat conformation. In the figure, "≡" means the flat hexagon and the chair drawing are the same molecule, and the double arrow shows the chair and boat interconverting.



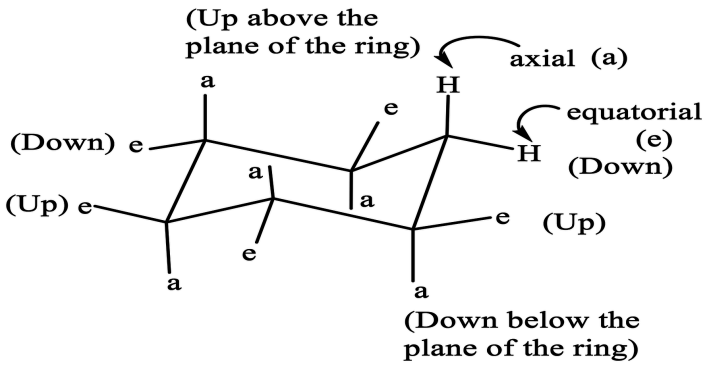
Q44 **False**

The chair conformation is more stable than the boat conformation. In the figure, "≡" means the flat hexagon and the chair drawing are the same molecule, and the double arrow shows the chair and boat interconverting.



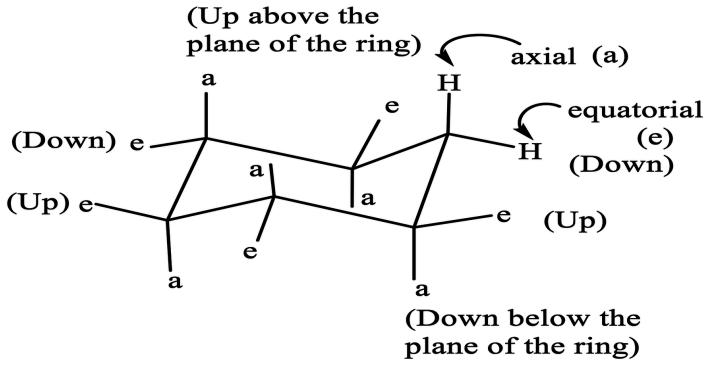
Q45 **C. Parallel to the ring axis, pointing straight up or straight down**

Axial (a) bonds point straight up or down; equatorial (e) bonds point outward around the “equator” of the ring.



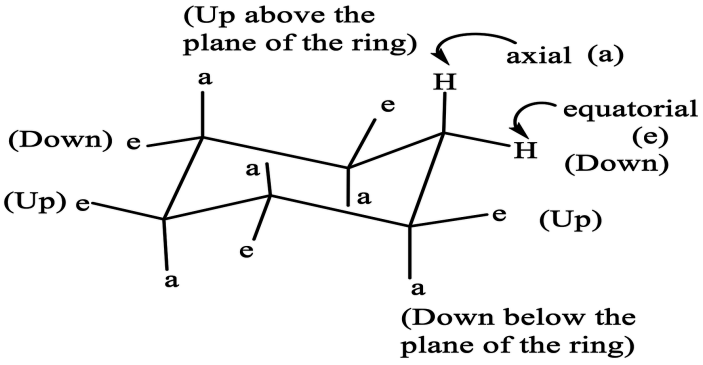
Q46 **D. One axial and one equatorial hydrogen**

Each carbon has one axial (a) and one equatorial (e) H.



Q47 **True**

The axial H atoms alternate: up above the plane, then down below the plane, on neighbouring carbons.

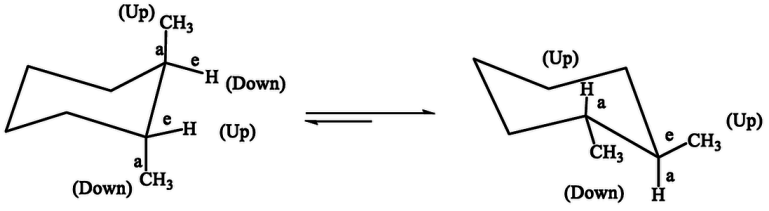


Q48 **E. CN and Cl**

CN and Cl are drawn with vertical bonds pointing up (axial, up). NH_2 and CH_3 are axial but point down; C_2H_5, OH, Br and H are equatorial. (Not solved in the doctor's file.)

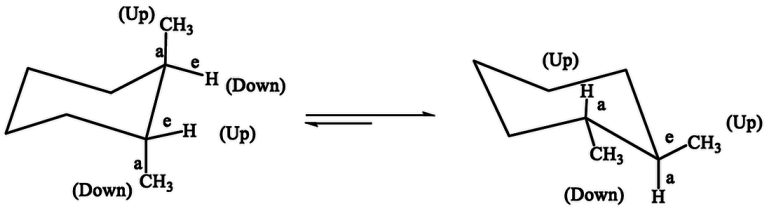
Q49 **C. Both axial (a,a) in one conformer and both equatorial (e,e) in the other**

trans-1,2-Dimethylcyclohexane interconverts by ring flip between the a,a and the e,e conformers.



Q50 **B. e,e (both CH_3 equatorial)**

The e,e conformer is more stable, because axial CH_3 groups are crowded by the axial H atoms on the same side of the ring (1,3-diaxial interactions). The equilibrium arrows in the doctor's figure favour the e,e conformer.



3 Assignment of Configuration

In this lecture

1. Absolute Configuration & Priority Rules 9 questions
2. Drawing Enantiomers: Wedge-Dash & Fischer 9 questions
3. Assigning (R) and (S) 10 questions
4. Stereoisomers, Meso Forms & Relationships 18 questions
5. Racemic Mixtures & Resolution 15 questions
6. Key Terms & Classification of Isomers 14 questions

3.1 Absolute Configuration & Priority Rules

Q1 What does “absolute configuration” around a chiral carbon mean?

A. The order of arrangement in space of the four groups around the chiral carbon

B. The number of chiral carbons in the molecule

C. The direction in which the compound rotates plane-polarized light

D. The molecular formula of the compound

Q2 What is the **first** step in assigning (R) or (S) to a chiral carbon?

A. Measure the optical rotation

B. Draw a curved arrow from group 1 to group 3

C. Determine the molecular formula

D. Identify the chiral carbon and rank its four groups by the Cahn-Ingold-Prelog sequence rules

Q3 After ranking the four groups by the Cahn-Ingold-Prelog rules, what is the next step in assigning (R) or (S)?

A. Count the number of possible stereoisomers using 2^n

B. Project the molecule so that the group with the highest priority is in the rear

C. Project the molecule so that the group with the lowest priority is in the rear

D. Measure the optical rotation of the compound with a polarimeter

Q4 With the lowest-priority group in the rear, how is the curved arrow drawn?

A. From the group of lowest priority to the highest priority

B. From the group of highest priority to the next highest priority, then to the next (1 → 2 → 3)

C. From the lowest-priority group to the chiral carbon

D. From priority 3 → 2 → 1

Q5 With the lowest-priority group pointing away from the viewer, the arrow 1 → 2 → 3 goes **clockwise**. The configuration is:

A. (R)

B. (S)

Q6 By rule 1, which order of priority is correct?

A. I > Br > Cl > S > F > O > N > C > H

B. H > C > N > O > F > S > Cl > Br > I

C. F > Cl > Br > I > O > N > C > H

D. I > Br > Cl > F > S > O > N > C > H

Q7 By rule 2, which group has the **highest** priority?

A. $–CH_2NH_2$

B. $–CH_2OCH_3$

C. $–CH_2SCH_2CH_3$

D. $–CH_2SH$

Q8 By rule 3, how are the atoms of a double or triple bond treated?

A. They are counted only once

B. As duplicated (double bond) or triplicated (triple bond) atoms

C. They always get the lowest priority

D. They are ignored

Q9 The priority order of the groups on the chiral carbon of 1-bromo-1-chloroethane is:

A. Cl > Br > CH_3 > H

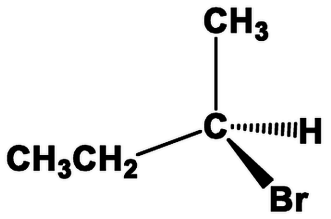
B. CH_3 > Br > Cl > H

C. Br > CH_3 > Cl > H

D. Br > Cl > CH_3 > H

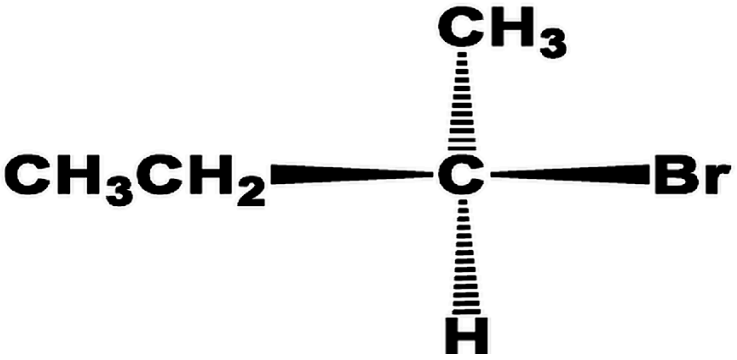
3.2 Drawing Enantiomers: Wedge-Dash & Fischer

Q10 Which structure is the mirror image of 2-bromobutane (a)?

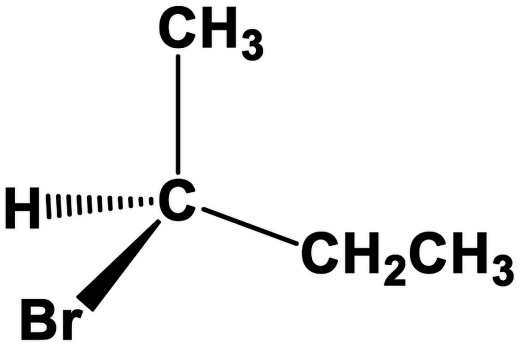


(a)
Tetrahedral structure
of 2-Bromobutane

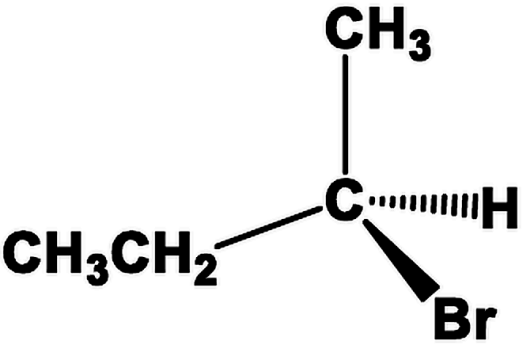
A.



B.

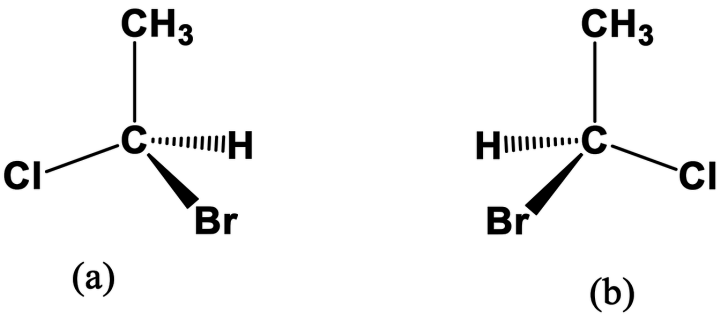


C.



D. None of these

Q11 Structures (a) and (b) of 1-bromo-1-chloroethane shown below are:



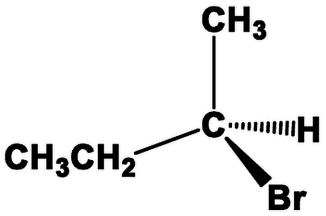
A. The same molecule

B. Structural isomers

C. Diastereomers

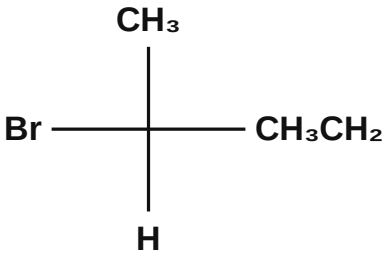
D. Enantiomers

Q12 Which Fischer projection represents 2-bromobutane (a)?

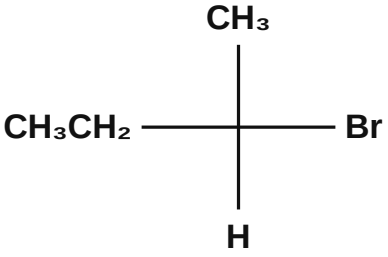


(a)
Tetrahedral structure
of 2-Bromobutane

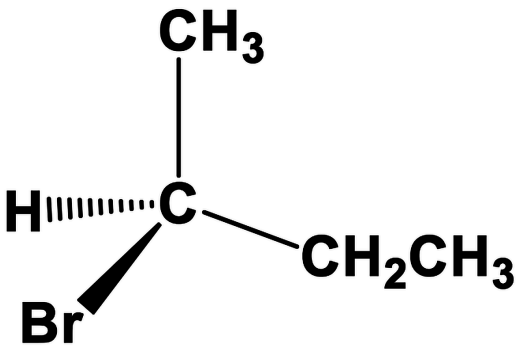
A.



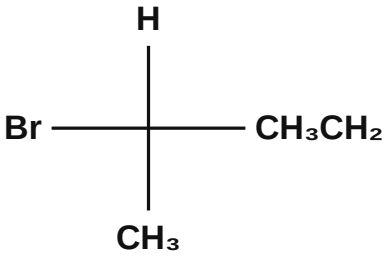
B.



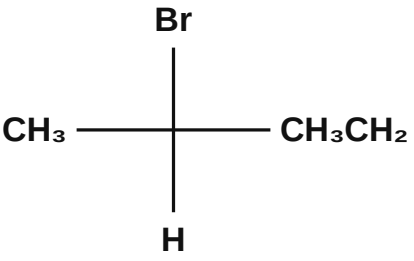
Q13 Which Fischer projection represents 2-bromobutane (b), the mirror image of (a), shown below?



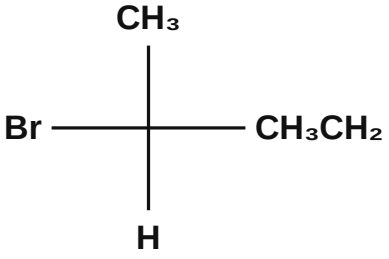
A.



B.



C.



Q14 In a Fischer projection, the vertical bonds point _____ the viewer and the horizontal bonds point _____ the viewer.

A. away from … towards

B. towards … away from

C. away from … away from

D. towards … towards

Q15 A Fischer projection of a chiral carbon has the lowest-priority group on a horizontal bond. To assign (R)/(S) properly, the doctor's method says:

A. Swap it with a vertical group and leave the other two groups unchanged

B. Rotate the whole drawing by 90° so that the group becomes vertical

C. Swap it with a vertical group, then swap the other two groups to keep the same configuration

D. Ignore its position and read the arrow 1 → 2 → 3 directly as drawn

Q16 Interchanging only **one pair** of groups around a chiral carbon gives:

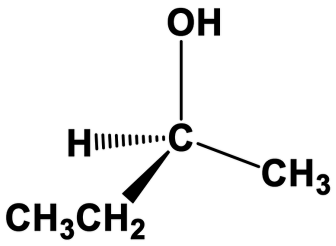
A. A meso compound

B. The enantiomer (mirror image)

C. The same molecule

D. A structural isomer

Q17 What is the configuration of the 2-butanol shown?

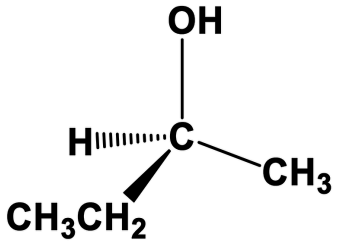


Wedge-Dash (tetrahedral structure) of Butan-2-ol

A. (S)

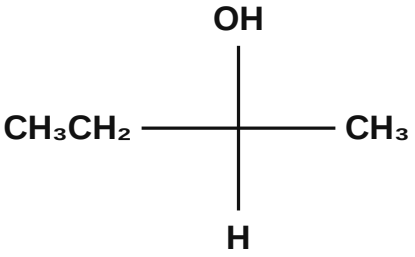
B. (R)

Q18 Which Fischer projection matches the 2-butanol shown?

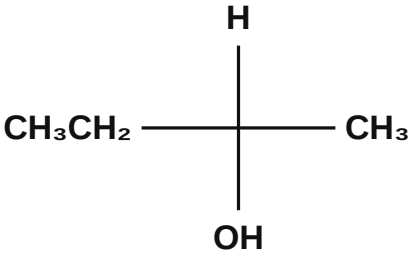


Wedge-Dash (tetrahedral structure) of Butan-2-ol

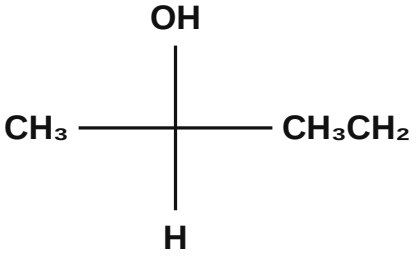
A.



B.

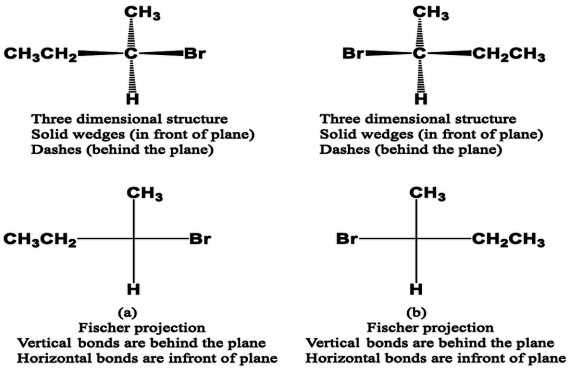


C.



3.3 Assigning (R) and (S)

Q19 What are the configurations of 2-bromobutane (a) and (b)?



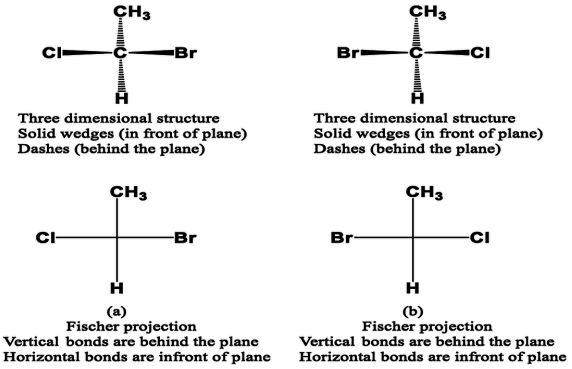
A. (a) is (S) and (b) is (R)

B. (a) is (R) and (b) is (S)

C. Both are (R)

D. Both are (S)

Q20 What are the configurations of 1-bromo-1-chloroethane (a) and (b)?



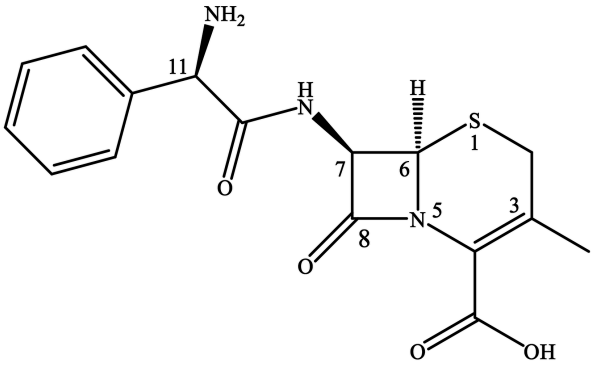
A. Both are (R)

B. Both are (S)

C. (a) is (R) and (b) is (S)

D. (a) is (S) and (b) is (R)

Q21 Assign the configuration at C-6, C-7 and C-11 of the compound shown:



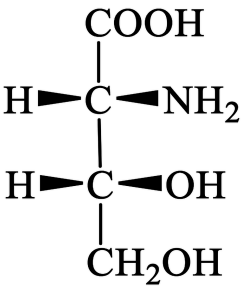
A. (6R, 7R, 11R)

B. (6R, 7S, 11S)

C. (6S, 7R, 11S)

D. (6S, 7S, 11R)

Q22 What is the configuration of molecule (I)?



(I)

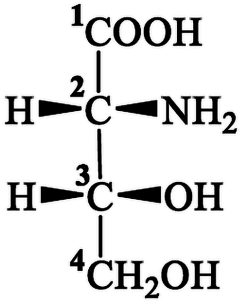
A. (2S, 3R)

B. (2S, 3S)

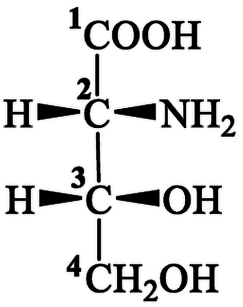
C. (2R, 3R)

D. (2R, 3S)

Q23 Assign the configurations of I and II and comment on their relationship:



(I)



(II)

A. (2R, 3S), I and II are diastereomers

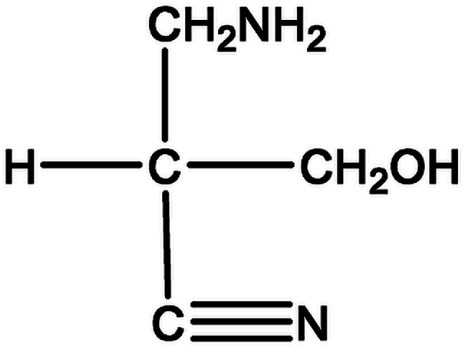
B. (2S, 3S), I and II are meso forms

C. (2S, 3S), I and II are the same molecule

D. (2R, 3S), I and II are enantiomers

E. (2R, 3S), I and II are the same molecule

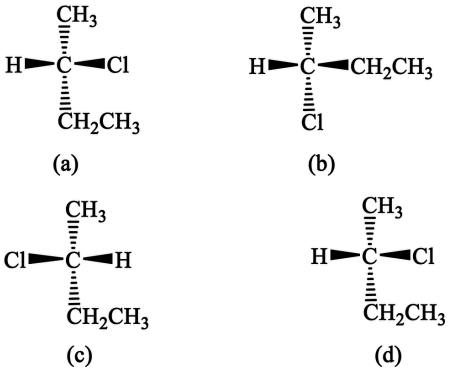
Q24 The absolute configuration of the molecule shown is:



A. (R)

B. (S)

Q25 Which two compounds are the **same** compound?



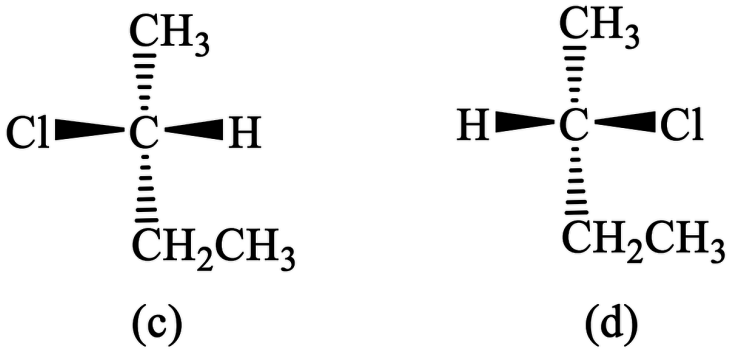
A. (a) and (d)

B. (c) and (d)

C. (a) and (b)

D. (a) and (c)

Q26 Compounds (c) and (d) shown below are:



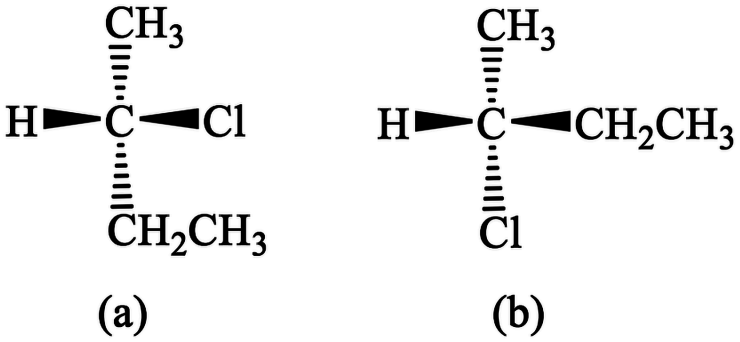
A. Diastereomers

B. The same compound

C. Structural isomers

D. Enantiomers

Q27 Compounds (a) and (b) shown below are:



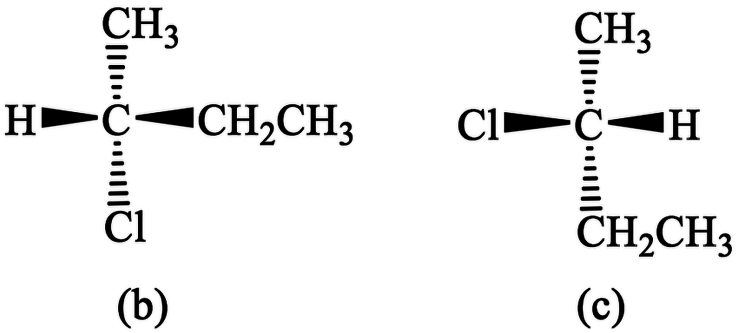
A. Diastereomers

B. Enantiomers

C. The same compound

D. Structural isomers

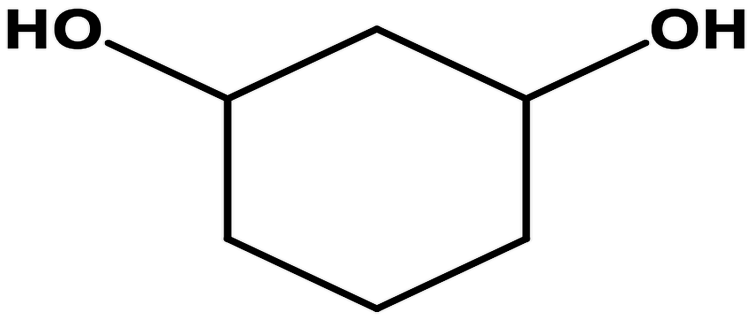
Q28 Compounds (b) and (c) shown below are:



A. Enantiomers
B. Structural isomers
C. The same compound
D. Diastereomers

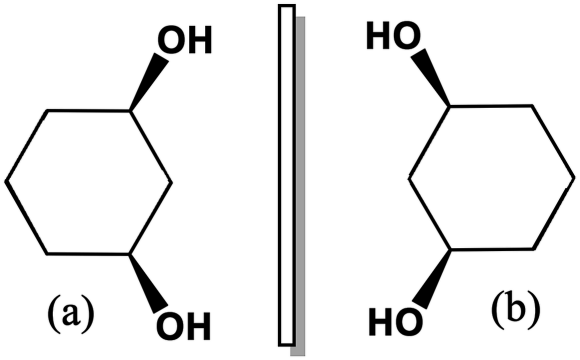
3.4 Stereoisomers, Meso Forms & Relationships

Q29 How many **different** stereoisomers exist for cyclohexane-1,3-diol?



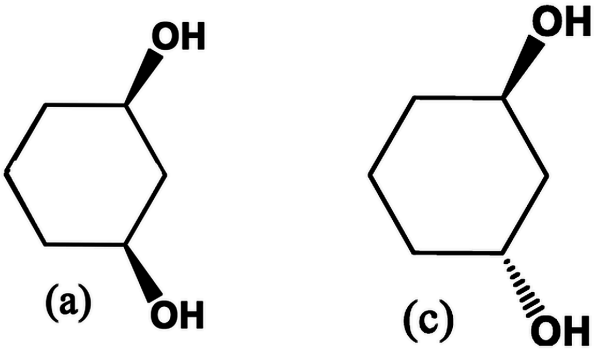
A. 3
B. 8
C. 4
D. 2

Q30 cis-Cyclohexane-1,3-diol, shown below as (a) and its mirror image (b), is:



A. Optically active
B. A pair of enantiomers
C. A structural isomer of the trans form
D. A meso compound

Q31 In cyclohexane-1,3-diol, the trans isomer (c) and the meso form are:



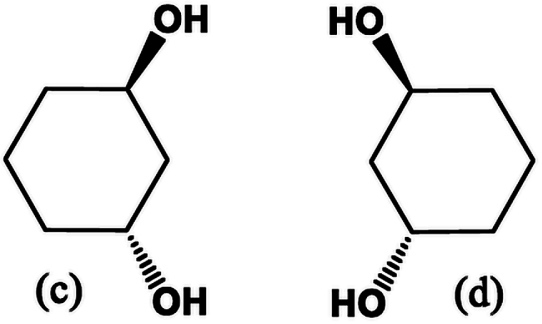
A. Enantiomers

B. Diastereomers

C. Structural isomers

D. The same compound

Q32 In cyclohexane-1,3-diol, the trans isomers (c) and (d) shown below are:



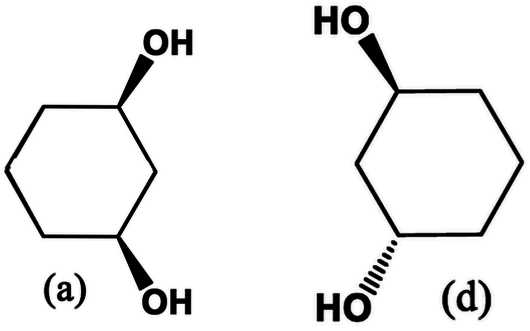
A. The same meso compound

B. Structural isomers

C. Enantiomers

D. Diastereomers

Q33 In cyclohexane-1,3-diol, the meso form (a) and the trans isomer (d) shown below are:



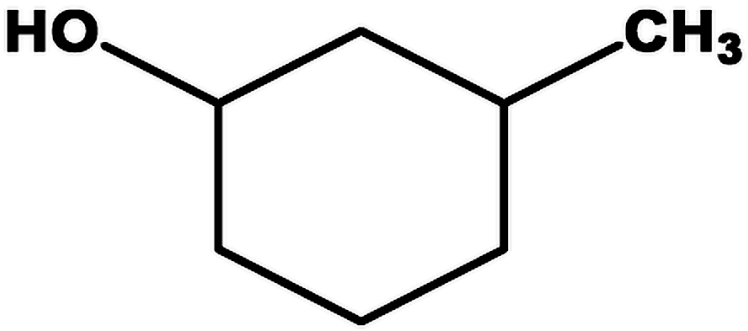
A. Diastereomers

B. The same compound

C. Enantiomers

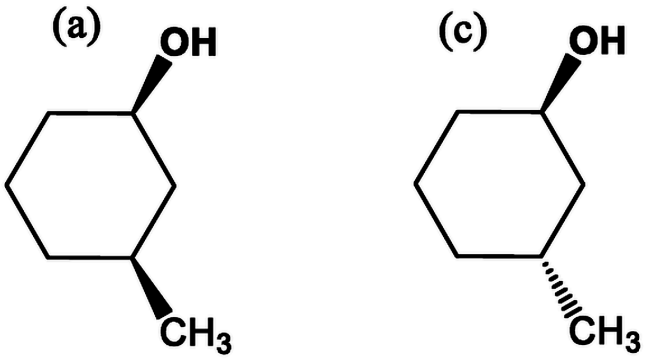
D. Structural isomers

Q34 How many stereoisomers exist for 3-methylcyclohexan-1-ol?



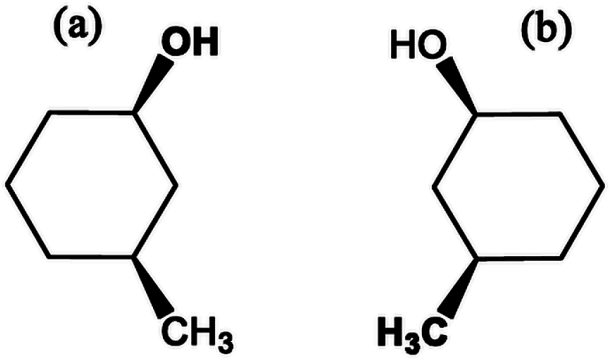
A. 8
B. 3
C. 2
D. 4

Q35 In 3-methylcyclohexan-1-ol, isomers (a) and (c) are:



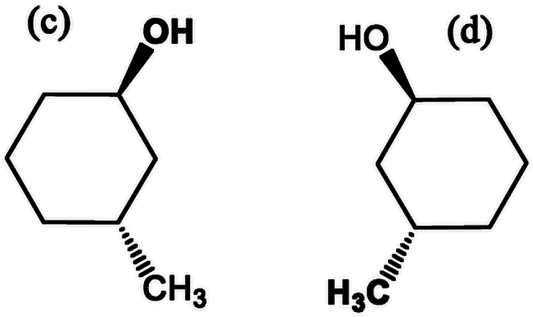
A. The same compound
B. Diastereomers
C. Meso forms
D. Enantiomers

Q36 In 3-methylcyclohexan-1-ol, isomers (a) and (b) are:



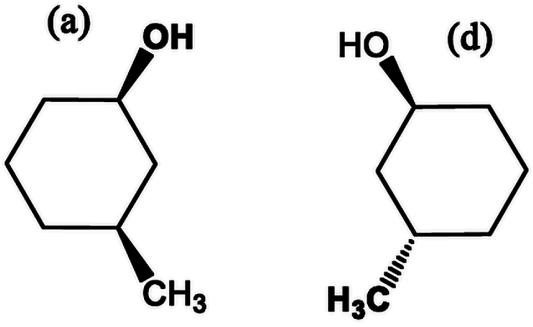
A. Meso forms
B. Enantiomers
C. Diastereomers
D. The same compound

Q37 In 3-methylcyclohexan-1-ol, isomers (c) and (d) are:



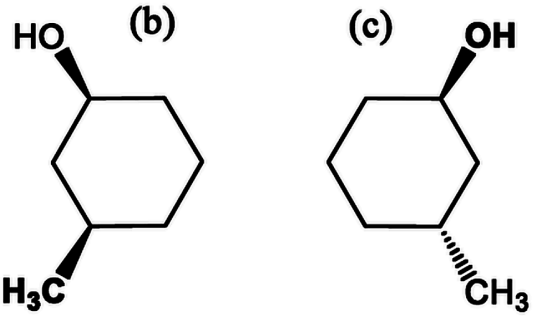
A. The same compound
B. Diastereomers
C. Enantiomers
D. Meso forms

Q38 In 3-methylcyclohexan-1-ol, isomers (a) and (d) are:



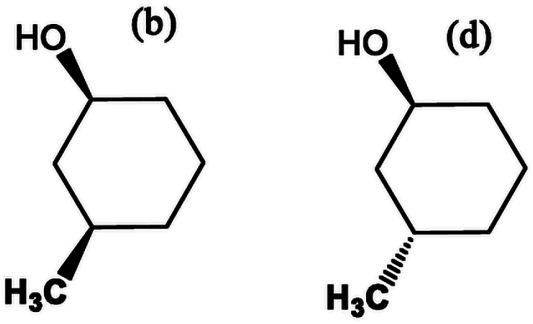
A. Meso forms
B. Enantiomers
C. The same compound
D. Diastereomers

Q39 In 3-methylcyclohexan-1-ol, isomers (b) and (c) are:



A. Diastereomers
B. Meso forms
C. Enantiomers
D. The same compound

Q40 In 3-methylcyclohexan-1-ol, isomers (b) and (d) are:



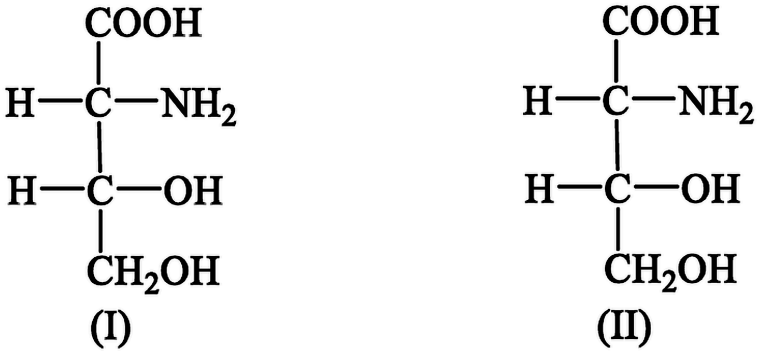
A. Diastereomers

B. The same compound

C. Meso forms

D. Enantiomers

Q41 Molecules (I) and (II) are:



A. The same

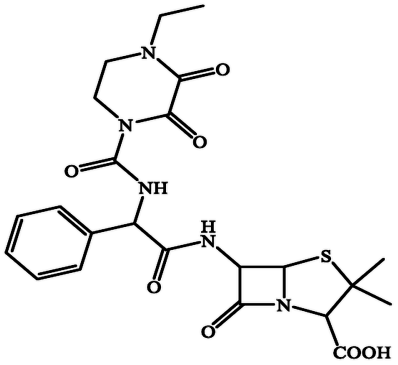
B. Diastereomers

C. Enantiomers

D. Constitutional isomers

E. Optically inactive isomers

Q42 How many chiral carbons are present in the compound shown?



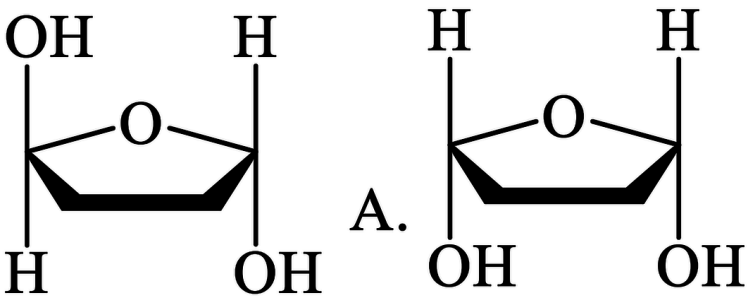
A. 3

B. 4

C. 5

D. 2

Q43 Pair A is best classified as:



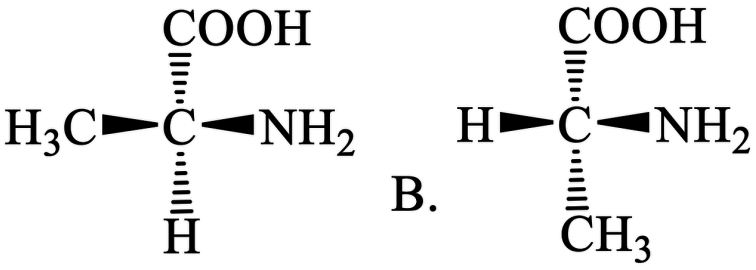
A. Enantiomers

B. Structural isomers

C. Geometric isomers (diastereomers)

D. The same molecule

Q44 Pair B is best classified as:



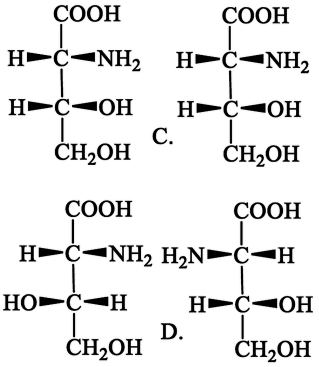
A. The same molecule

B. Diastereomers

C. Geometric isomers

D. Enantiomers

Q45 Pairs C and D are, respectively:



A. Enantiomers; the same molecule

B. Diastereomers; enantiomers

C. The same molecule; enantiomers

D. The same molecule; diastereomers

Q46 In the classification of isomers, geometric isomers are a type of:

A. Structural isomers

B. Conformers

C. Enantiomers

D. Diastereomers

3.5 Racemic Mixtures & Resolution

Q47 A racemic mixture (racemate) is:

A. A mixture of two diastereomers

B. A pure single enantiomer

C. A meso compound

D. A mixture of equal parts of a pair of enantiomers

Q48 Why is a racemic mixture optically inactive?

A. Each of its molecules has an internal plane of symmetry, like a meso form

B. Its molecules contain no chiral carbon, so light passes through unchanged

C. The two enantiomers rotate light equally in opposite directions, so the rotations cancel

D. It absorbs the plane-polarized light instead of rotating it

Q49 Resolution of a racemic mixture means:

A. The physical separation of a racemic mixture into its pure enantiomers

B. Converting one enantiomer into the other

C. Mixing two enantiomers in equal amounts

D. Measuring the specific rotation

Q50 Which of the following is **NOT** one of the three main methods of resolving a racemate?

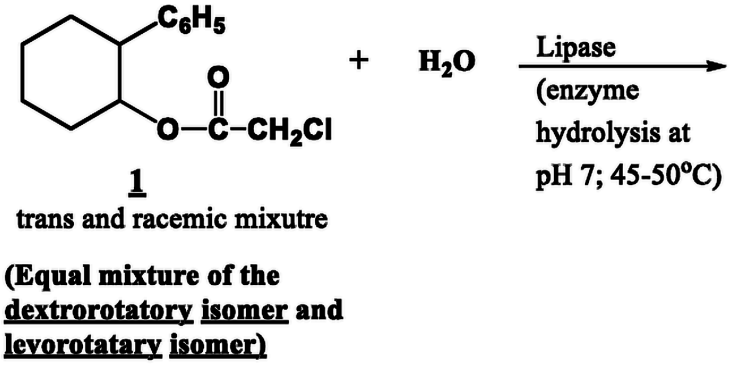
A. Formation of diastereomeric products

B. Simple distillation

C. Chromatography (HPLC) with a chiral stationary phase

D. Enzymatic (biological) resolution

Q51 In the resolution of racemic trans-2-phenylcyclohexyl chloroacetate, the enzyme lipase:



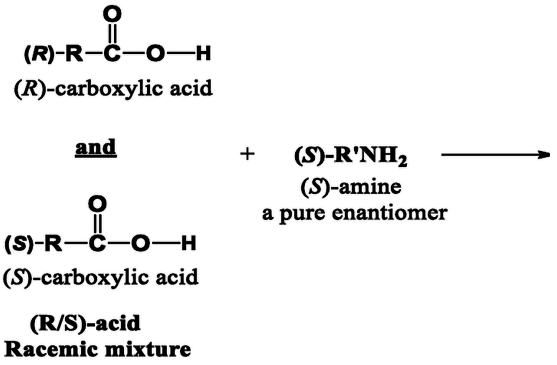
A. Hydrolyzes both enantiomers equally

B. Hydrolyzes the ester of only one enantiomer

C. Does not react with either ester

D. Converts the trans ester into the cis ester

Q52 A racemic (R/S) carboxylic acid is resolved by reacting it with a pure (S)-amine. The products are:



A. A pair of enantiomeric salts that cannot be separated

B. A single meso salt

C. A racemic amine

D. A pair of diastereomeric salts that can be separated by recrystallization

Q53 Which term best describes an enantiomer?

A. Optically active

B. Diastereomer

C. Structural isomer

D. Optically inactive

Q54 A prochiral molecule is:

A. A molecule with chiral carbons and an internal plane of symmetry

B. A molecule that has two or more chiral carbons in its structure

C. A molecule that can undergo a chemical reaction to give a chiral compound

D. A mixture of equal amounts of a pair of enantiomers

Q55 A levorotatory enantiomer rotates plane-polarized light:

A. In both directions

B. It does not rotate light

C. To the right (clockwise), shown as (+)

D. To the left (counterclockwise), shown as (−)

Q56 A dextrorotatory enantiomer rotates plane-polarized light:

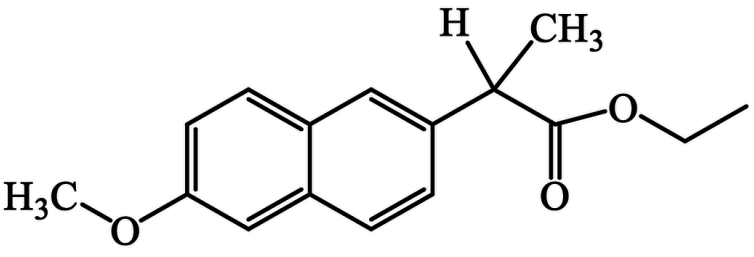
A. To the right (clockwise), shown as (+)

B. It does not rotate light

C. In both directions

D. To the left (counterclockwise), shown as (−)

Q57 Synthesis gave racemic (R/S)-naproxen ethyl ester. Which method gives the active (S)-naproxen?



(R/S)-Naproxen ethyl ester

A. Recrystallization, since the (R)- and (S)-esters have different solubilities

B. Enzymatic hydrolysis with an esterase, which hydrolyzes only the (S)-ester

C. Heating the racemic ester, which converts the (R)-ester into the (S)-ester

D. Simple distillation, since the (R)- and (S)-esters boil at different temperatures

Q58 In the naproxen resolution, how is pure (S)-naproxen (the acid) obtained from its sodium salt?

A. By heating with NaOH

B. By adding more esterase

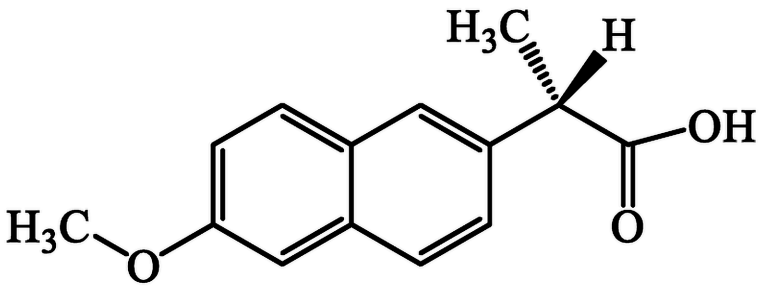
C. By treating the sodium salt with HCl

D. By distillation

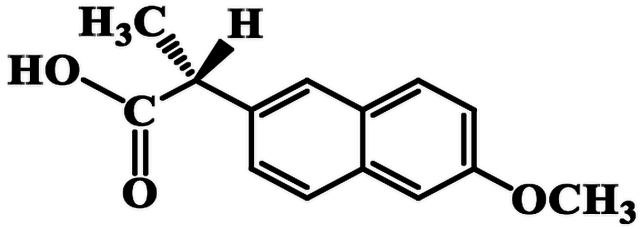
Q59 Which wedge-dash structure is (S)-naproxen?

A. Neither; naproxen has no chiral carbon

B.



C.



Q60 A 50:50 mixture of (+)-glyceraldehyde ([α] = +8.7°) and (−)-glyceraldehyde ([α] = −8.7°) has an observed rotation of:

A. +17.4°

B. −8.7°

C. 0°

D. +8.7°

Q61 The prefix (±) in a name indicates:

A. A racemic mixture

B. A meso compound

C. A structural isomer

D. A pure dextrorotatory enantiomer

3.6 Key Terms & Classification of Isomers

Q62 Stereoisomers are isomers with:

A. Different molecular formulas but the same orientation in space

B. The same connectivity of their atoms but a different orientation in space

C. A different connectivity of their atoms but the same molecular formula

D. The same connectivity and the same orientation of atoms in space

Q63 Structural (constitutional) isomers are isomers with:

A. The same connectivity but a different orientation in space

B. Non-superposable mirror-image molecules

C. Rigidity in their structure

D. A different order of attachment (connectivity) of their atoms

Q64 According to the flow chart, stereoisomers are divided into:

A. Enantiomers and diastereomers

B. Enantiomers and structural isomers

C. Structural and constitutional isomers

D. Conformers and constitutional isomers

Q65 Diastereomers are:

A. Isomers with a different order of attachment of their atoms

B. Stereoisomers whose molecules are not mirror images

C. Stereoisomers whose molecules are non-superposable mirror images

D. Two drawings of the same molecule

Q66 Enantiomers are:

A. Isomers with different molecular formulas

B. Isomers with a different order of attachment of their atoms

C. Stereoisomers whose molecules are non-superposable mirror images

D. Stereoisomers whose molecules are not mirror images

Q67 Geometric isomers are:

A. Stereoisomers whose molecules are non-superposable mirror images

B. Conformations formed by free rotation about single bonds

C. Structural isomers that contain different functional groups

D. Stereoisomers whose molecules possess rigidity in their structure

Q68 A stereogenic center (stereocenter) is:

A. Any carbon atom that is part of a ring

B. A carbon with four different atoms or groups attached (a chiral carbon)

C. A carbon with two identical atoms or groups attached

D. An sp^2 carbon that is part of a C=C double bond

Q69 A chiral molecule is a molecule that:

A. Has an internal plane of symmetry

B. Does not rotate plane-polarized light

C. Can be superposed on its mirror image

D. Cannot be superposed on its mirror image

Q70 A plane of symmetry is:

A. An imaginary plane dividing a molecule into two mirror-image halves

B. The single plane in which plane-polarized light vibrates

C. A plane passing through the four different groups of a chiral carbon

D. The plane of the paper on which a Fischer projection is drawn

Q71 A meso form (meso compound):

A. Has no plane of symmetry, so it rotates plane-polarized light

B. Has only one chiral carbon, so it is optically active

C. Has chiral carbons and an internal plane of symmetry, so it is optically inactive

D. Is a 50:50 mixture of a pair of enantiomers, so it is optically inactive

Q72 Plane-polarized light is:

A. Light of many different colors

B. Light whose waves vibrate in only one plane

C. Light that has passed through a racemic mixture

D. Light whose waves vibrate in all planes

Q73 The specific rotation [α] of an optically active compound is:

A. Its characteristic rotation of plane-polarized light under standard conditions

B. The temperature at which it starts to rotate light

C. The number of chiral carbons present in its molecule

D. The rotation of its racemic mixture, which is always 0°

Q74 Compounds with the same order of bonded atoms that differ only in the direction in which they rotate plane-polarized light are called:

A. Constitutional isomers

B. Conformers

C. Optical isomers (enantiomers)

D. Geometric isomers

Q75 What is the major difference between configuration and conformation?

A. Both are fixed and can only change by breaking bonds

B. Both change freely by rotation about single bonds

C. Conformation is fixed in space; configuration changes by rotation about single bonds

D. Configuration is fixed in space; conformation changes by rotation about single bonds

Answers · Lecture 3

3.1 Absolute Configuration & Priority Rules

Q1 **A. The order of arrangement in space of the four groups around the chiral carbon**

Absolute configuration is the order of arrangement in space of the four groups around a chiral carbon atom.

Q2 **D. Identify the chiral carbon and rank its four groups by the Cahn-Ingold-Prelog sequence rules**

Step (a) of the doctor's four steps: identify the chiral carbon, then rank the four groups (or atoms) attached to it in order of priority by the Cahn-Ingold-Prelog sequence rules.

Q3 **C. Project the molecule so that the group with the lowest priority is in the rear**

Doctor's four steps: (a) identify the chiral carbon and rank the four groups by the Cahn-Ingold-Prelog rules; (b) project the molecule with the lowest-priority group in the rear; (c) draw a curved arrow from the highest-priority group to the next highest, then to the next; (d) clockwise = (R), counterclockwise = (S).

Q4 **B. From the group of highest priority to the next highest priority, then to the next (1 → 2 → 3)**

Step (c): select the group of highest priority and draw a curved arrow to the group of next highest priority, then to the next one. Step (d): clockwise = (R), counterclockwise = (S).

Q5 **A. (R)**

Clockwise = (R) (Latin *rectus*, right). Counterclockwise = (S) (Latin *sinister*, left).

Q6 **A. I > Br > Cl > S > F > O > N > C > H**

Priority is by atomic number: I (53) > Br (35) > Cl (17) > S (16) > F (9) > O (8) > N (7) > C (6) > H (1).

Q7 **C. $–CH_2SCH_2CH_3$**

When the first atoms tie, compare the next atoms and work outward: $–CH_2SCH_2CH_3$ > $–CH_2SCH_3$ > $–CH_2SH$ > $–CH_2OCH_3$ > $–CH_2OH$ > $–CH_2NH_2$.

Q8 **B. As duplicated (double bond) or triplicated (triple bond) atoms**

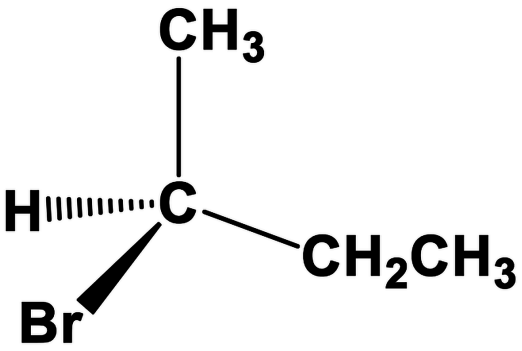
If a double or triple bond is present, the atoms involved are treated as being duplicated or triplicated, respectively (as in geometric isomerism).

Q9 **D. Br > Cl > CH_3 > H**

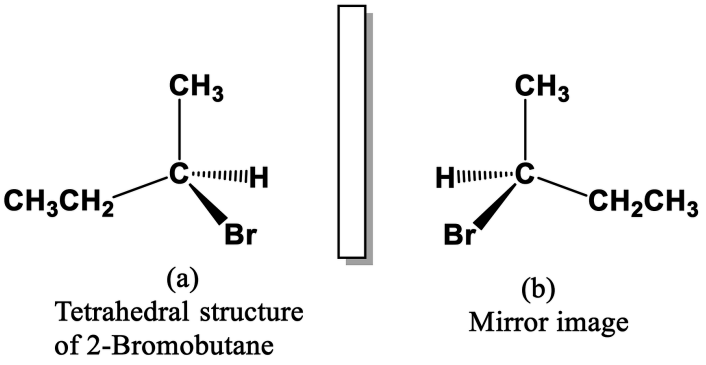
By atomic number of the attached atoms: Br (35) > Cl (17) > C (6) > H (1).

3.2 Drawing Enantiomers: Wedge-Dash & Fischer

Q10 **B.**

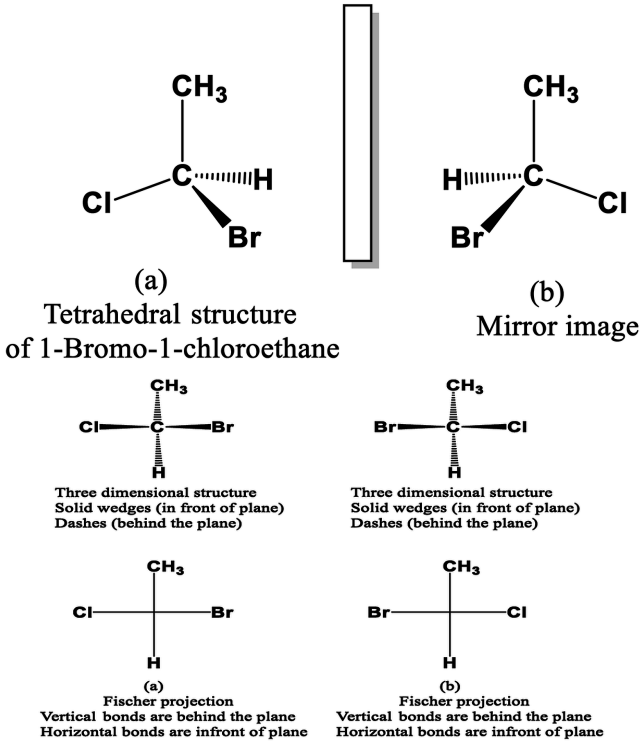


The mirror image (b) has every group reflected: CH_3 stays on top, while H, Br and CH_3CH_2 swap sides. In the figure, the tall rectangle between (a) and (b) is the mirror.

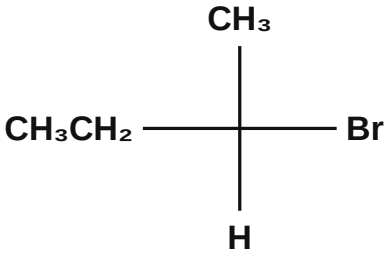


Q11 **D. Enantiomers**

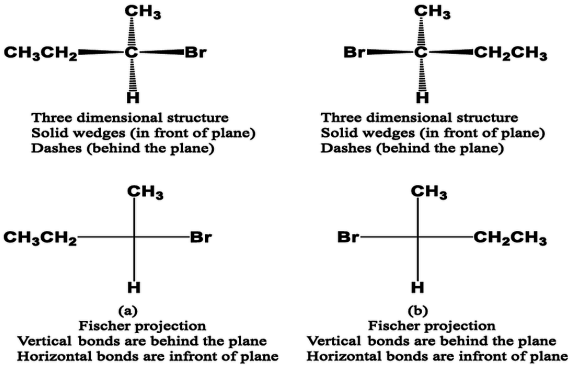
(b) is the mirror image of (a). Since the carbon is chiral (Br, Cl, CH_3, H), they are a pair of enantiomers. The figure shows the same pair drawn as 3D (wedge-dash) structures and as Fischer projections.



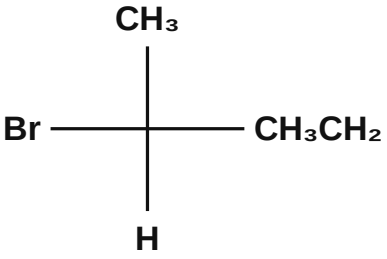
Q12 **B.**



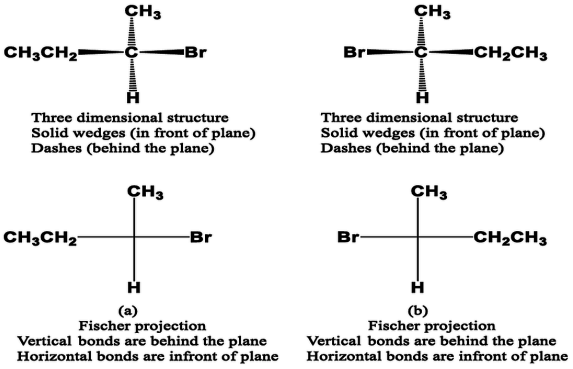
Turn (a) so that CH_3 and H point back (vertical) and CH_3CH_2 and Br point toward you (horizontal), then flatten it.



Q13 **C.**



Turn (b) so that CH_3 and H point back (vertical) and Br and CH_3CH_2 point toward you (horizontal), then flatten it: CH_3 on top, Br on the left, CH_3CH_2 on the right, H at the bottom.



Q14 **A. away from … towards**

The vertical bonds are pointing away from the viewer, while the horizontal bonds are pointing towards the viewer.

Q15 **C. Swap it with a vertical group, then swap the other two groups to keep the same configuration**

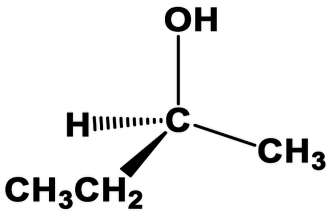
The lowest-priority group is placed on a vertical bond. If it is not, interchange it with a group that is, then interchange the other two groups to restore the original configuration.

Q16 **B. The enantiomer (mirror image)**

When only one pair of groups is interchanged, the result is the enantiomer. When both pairs (all groups) or three groups are interchanged, the result is identical to the original.

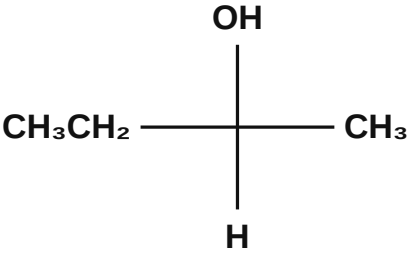
Q17 **A. (S)**

Priority: OH > CH_2CH_3 > CH_3 > H. H is on the dash (away), and OH → CH_2CH_3 → CH_3 runs counterclockwise, so (S)-butan-2-ol.

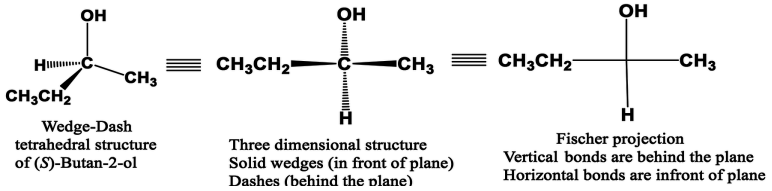


Wedge-Dash
tetrahedral structure
of (*S*)-Butan-2-ol

Q18 **A.**



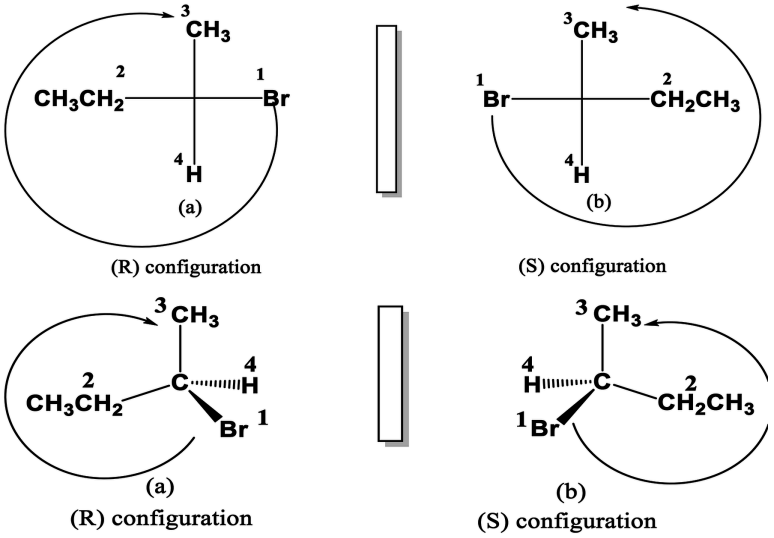
Rotate the molecule so OH and H are vertical (away) and CH_3CH_2 and CH_3 are horizontal (toward you). In the figure, "≡" means the three drawings are the same molecule: wedge-dash, 3D cross and Fischer projection.



3.3 Assigning (R) and (S)

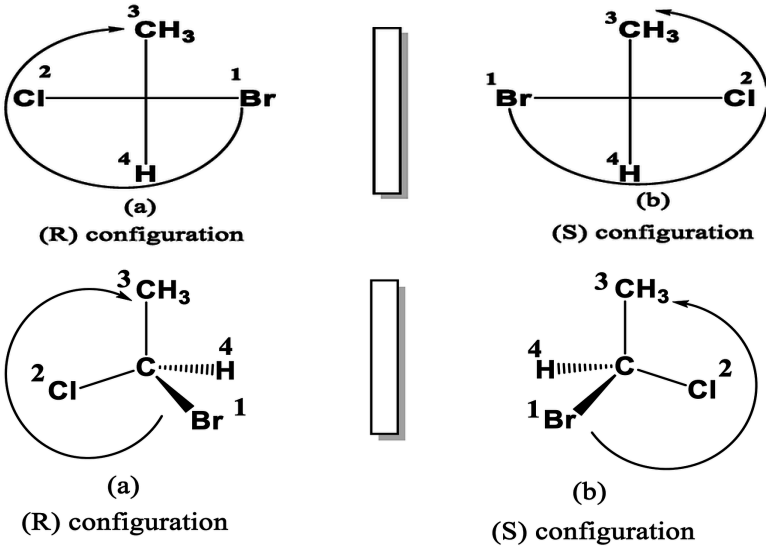
Q19 **B. (a) is (R) and (b) is (S)**

Priority: Br > CH_2CH_3 > CH_3 > H. With H at the bottom (vertical, away), Br → CH_2CH_3 → CH_3 is clockwise in (a) → (R)-2-bromobutane, and counterclockwise in (b) → (S)-2-bromobutane.



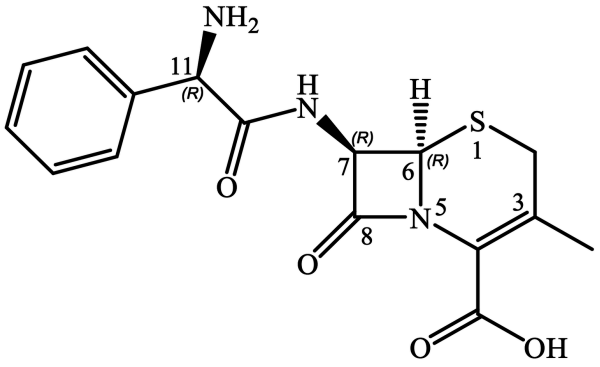
Q20 **C. (a) is (R) and (b) is (S)**

Priority: Br > Cl > CH_3 > H. H is at the bottom; the arrow Br → Cl → CH_3 is clockwise in (a) (R) and counterclockwise in (b) (S).



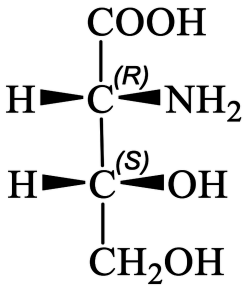
Q21 **A. (6R, 7R, 11R)**

C-6: S > N > C-7 > H; H is on the dash, and the arrow is clockwise → (6R). C-7: N (of NH) > C-6 (S,N,H) > C-8 (O,O,N) > H; H points back, arrow clockwise → (7R). C-11: NH_2 > C=O (O,O,N) > phenyl (C,C,C) > H; H points back, arrow clockwise → (11R). Answer: (6R, 7R, 11R).



Q22 **D. (2R, 3S)**

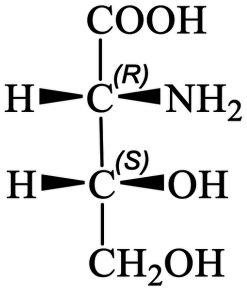
C-2: NH_2 > COOH > C-3 > H → (R). C-3: OH > CH_2OH > C-2 > H → (S). So (I) is (2R, 3S).



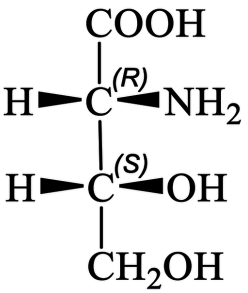
(I)

Q23 **E. (2R, 3S), I and II are the same molecule**

C-2: NH_2 > COOH > C-3 > H → (R). C-3: OH > CH_2OH > C-2 > H → (S). Both I and II are (2R, 3S), so they are the same molecule (not enantiomers or diastereomers).



(I)



(II)

Q24 **B. (S)**

Rank the groups: CH_2OH > CN > CH_2NH_2 > H (CH_2OH (O,H,H) beats CN (N,N,N) because O beats N; CN (N,N,N) beats CH_2NH_2 (N,H,H)). CH_2OH → CN → CH_2NH_2 appears clockwise, but H is on a horizontal bond (toward the viewer), so reverse it: (S).

Q25 **A. (a) and (d)**

Enantiomers: (a) and (b); (c) and (d). Same compound: (a) and (d); (b) and (c).

Q26 **D. Enantiomers**

(c) and (d) are enantiomers: (c) is (R) and (d) is (S).

Q27 **B. Enantiomers**

Doctor's answer: (a) and (b) are enantiomers. (a) is (S) and (b) is (R).

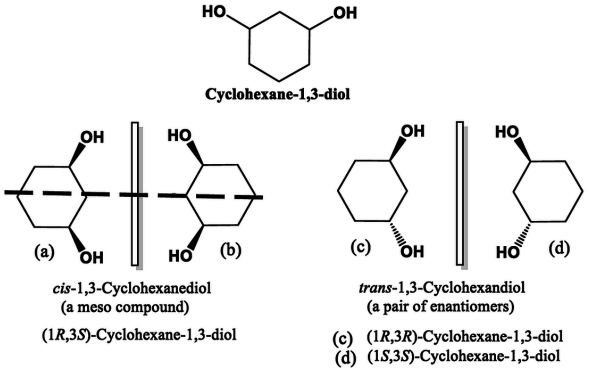
Q28 **C. The same compound**

Doctor's answer: (b) and (c) are the same compound. Both are (R).

3.4 Stereoisomers, Meso Forms & Relationships

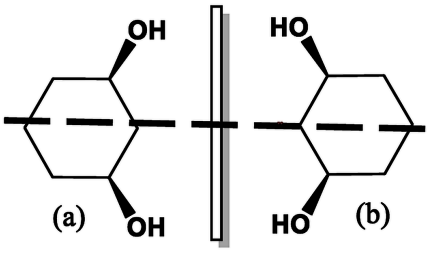
Q29 **A. 3**

2 similar chiral carbons → $2^2 = 4$ drawn, but (a) and (b) are the same meso compound → 1 meso form + 1 pair of enantiomers = 3.



Q30 **D. A meso compound**

(a) and (b) are the same compound, the meso form, because it has an internal plane of symmetry (the dashed line in the figure), so it is optically inactive.

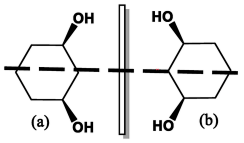


cis-1,3-Cyclohexanediol
(a meso compound)

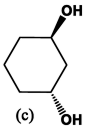
(1*R*,3*S*)-Cyclohexane-1,3-diol

Q31 **B. Diastereomers**

The meso form is (1R,3S) and (c) is (1R,3R): they are stereoisomers but not mirror images, so they are diastereomers.



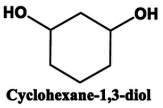
cis-1,3-Cyclohexanediol
(a meso compound)
(1*R*,3*S*)-Cyclohexane-1,3-diol



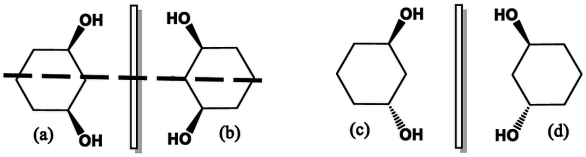
(c) (1*R*,3*R*)-Cyclohexane-1,3-diol

Q32 **C. Enantiomers**

Doctor's answer: (c) and (d) are enantiomers (mirror images). (c) is (1R,3R) and (d) is (1S,3S); neither has an internal plane of symmetry.



Cyclohexane-1,3-diol



cis-1,3-Cyclohexanediol
(a meso compound)
(1*R*,3*S*)-Cyclohexane-1,3-diol

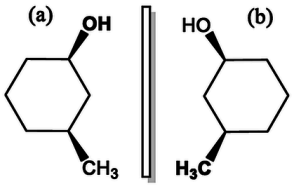
trans-1,3-Cyclohexandiol
(a pair of enantiomers)
(c) (1*R*,3*R*)-Cyclohexane-1,3-diol
(d) (1*S*,3*S*)-Cyclohexane-1,3-diol

Q33 **A. Diastereomers**

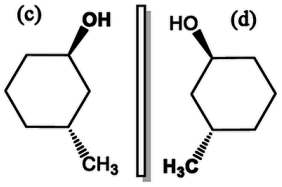
Doctor's answer: (d) and the meso form are diastereomers (as are (c) and the meso form). The meso form is (1R,3S) and (d) is (1S,3S): stereoisomers that are not mirror images.

Q34 **D. 4**

The two chiral carbons (C-1 and C-3) are dissimilar, so there is no meso form: $2^2 = 4$ (two pairs of enantiomers).



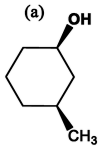
cis-3-Methylcyclohexanol
(a pair of enantiomers)
(a): (1*R*,3*S*)-3-Methylcyclohexan-1-ol
(b): (1*S*,3*R*)-3-Methylcyclohexan-1-ol



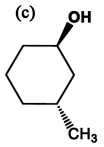
trans-3-Methylcyclohexanol
(a pair of enantiomers)
(c): (1*R*,3*R*)-3-Methylcyclohexan-1-ol
(d): (1*S*,3*S*)-3-Methylcyclohexan-1-ol

Q35 **B. Diastereomers**

(a) is (1R,3S) and (c) is (1R,3R). Only C-3 differs, so they are not mirror images: they are diastereomers.



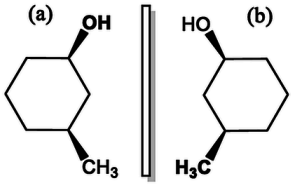
(a): (1*R*,3*S*)-3-Methylcyclohexan-1-ol



(c): (1*R*,3*R*)-3-Methylcyclohexan-1-ol

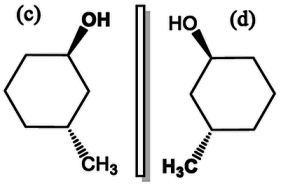
Q36 **B. Enantiomers**

The doctor's relationships: (a) and (b) are enantiomers, (c) and (d) are enantiomers; (a) and (c), (a) and (d), (b) and (c), (b) and (d) are diastereomers. (a) is (1R,3S) and (b) is (1S,3R): both chiral carbons are inverted, so (a) and (b) are non-superimposable mirror images.



cis-3-Methylcyclohexanol
(a pair of enantiomers)

(a): (1*R*,3*S*)-3-Methylcyclohexan-1-ol
(b): (1*S*,3*R*)-3-Methylcyclohexan-1-ol

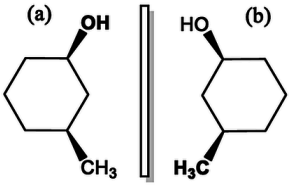


trans-3-Methylcyclohexanol
(a pair of enantiomers)

(c): (1*R*,3*R*)-3-Methylcyclohexan-1-ol
(d): (1*S*,3*S*)-3-Methylcyclohexan-1-ol

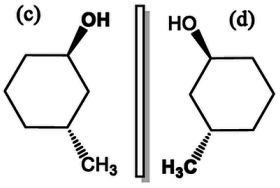
Q37 **C. Enantiomers**

Doctor's answer: (c) and (d) are enantiomers. (c) is (1R,3R) and (d) is (1S,3S): both chiral carbons are inverted.



cis-3-Methylcyclohexanol
(a pair of enantiomers)

(a): (1*R*,3*S*)-3-Methylcyclohexan-1-ol
(b): (1*S*,3*R*)-3-Methylcyclohexan-1-ol



trans-3-Methylcyclohexanol
(a pair of enantiomers)

(c): (1*R*,3*R*)-3-Methylcyclohexan-1-ol
(d): (1*S*,3*S*)-3-Methylcyclohexan-1-ol

Q38 **D. Diastereomers**

Doctor's answer: (a) and (d) are diastereomers. (a) is (1R,3S) and (d) is (1S,3S): only C-1 differs.

Q39 **A. Diastereomers**

Doctor's answer: (b) and (c) are diastereomers. (b) is (1S,3R) and (c) is (1R,3R): only C-1 differs.

Q40 **A. Diastereomers**

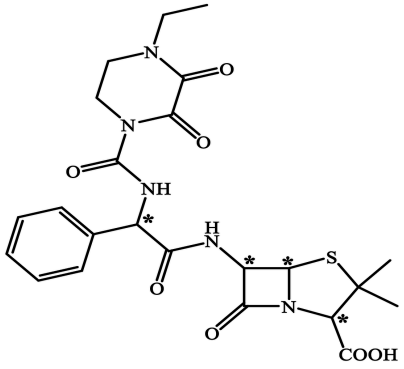
Doctor's answer: (b) and (d) are diastereomers. (b) is (1S,3R) and (d) is (1S,3S): only C-3 differs.

Q41 **A. The same**

The two Fischer projections are identical: every group is in the same position, so they are the same molecule (2R, 3S). (Not solved in the doctor's file.)

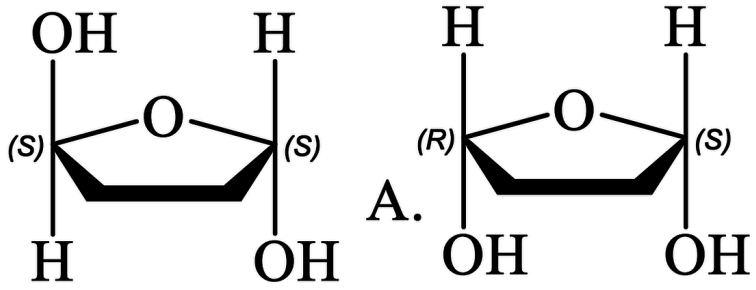
Q42 **B. 4**

4 chiral carbons.



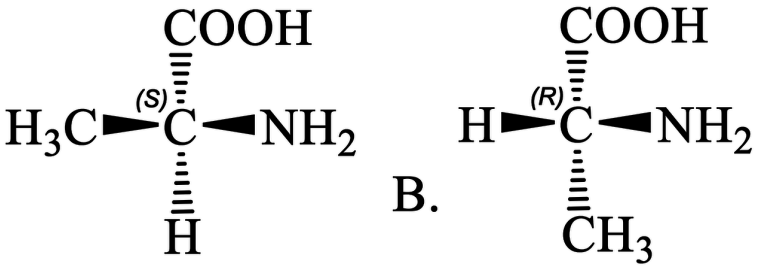
Q43 **C. Geometric isomers (diastereomers)**

A: geometric isomers (cis/trans), which are diastereomers.



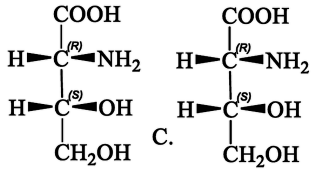
Q44 **D. Enantiomers**

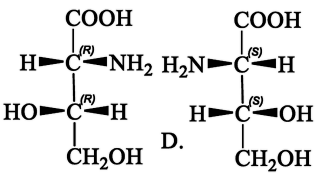
B: (S) and (R) → enantiomers.



Q45 **C. The same molecule; enantiomers**

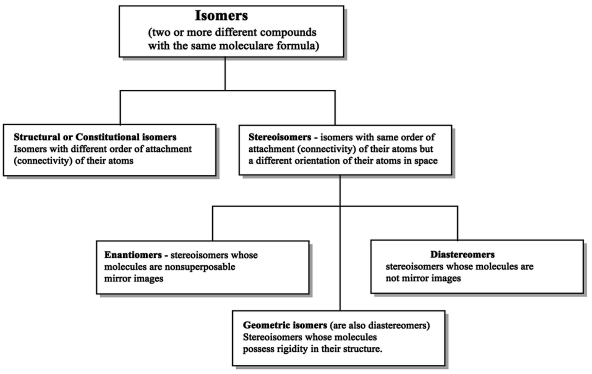
C: both (2R, 3S) → same molecule. D: (2R, 3R) and (2S, 3S) → enantiomers.





Q46 **D. Diastereomers**

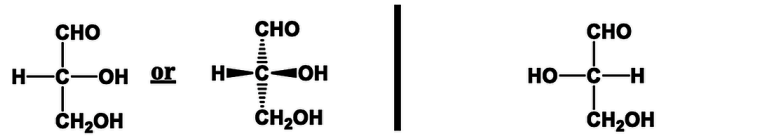
Isomers → structural (constitutional) isomers and stereoisomers. Stereoisomers → enantiomers and diastereomers; geometric isomers are also diastereomers.



3.5 Racemic Mixtures & Resolution

Q47 **D. A mixture of equal parts of a pair of enantiomers**

A mixture of equal parts of any pair of enantiomers is called a racemate, racemic mixture or racemic modification. It may be indicated by the prefix (±).



(+)-Glyceraldehyde
Specific optical rotation: $[\alpha]^{D}_{20} = +8.7^{o}$

(-)-Glyceraldehyde
Specific optical rotation: $[\alpha]^{D}_{20} = -8.7^{o}$

Q48 **C. The two enantiomers rotate light equally in opposite directions, so the rotations cancel**

A racemate does not rotate plane-polarized light because the rotation by each enantiomer is cancelled by the equal and opposite rotation of the other.

Q49 **A. The physical separation of a racemic mixture into its pure enantiomers**

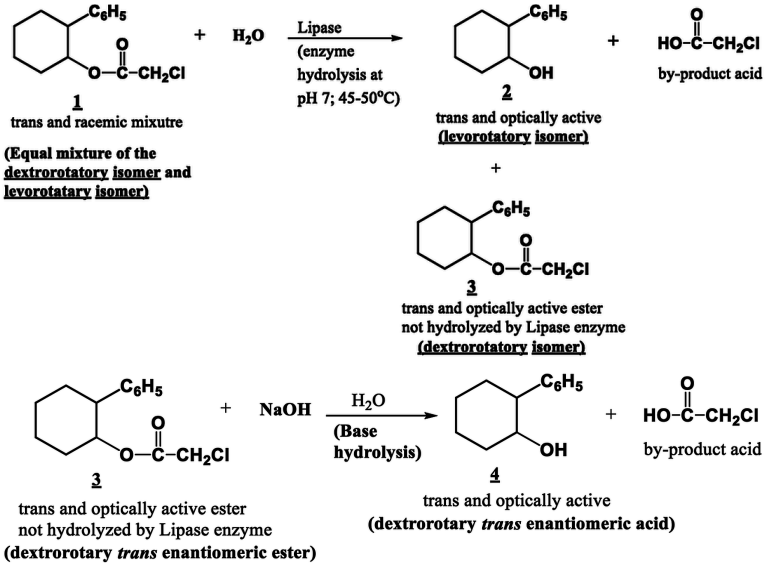
Resolution (resolving): the physical separation of a racemic mixture into its pure enantiomers.

Q50 **B. Simple distillation**

The three methods: (i) formation of diastereomeric products; (ii) enzymatic or biological means; (iii) chromatographic separation on HPLC with chiral stationary phases (expensive and tedious).

Q51 **B. Hydrolyzes the ester of only one enantiomer**

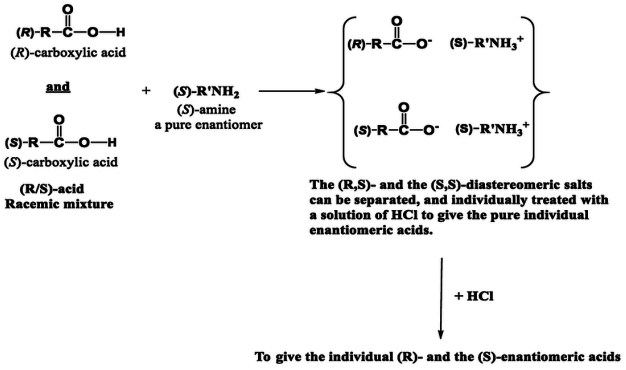
Lipase can only hydrolyze the ester from one enantiomer, giving the levorotatory trans alcohol (2). The dextrorotatory trans ester (3) remains and is later hydrolyzed with NaOH to give the second optically active alcohol (4).



Note: 4 is an alcohol (–OH); the word “acid” in the doctor's label under 4 should read “alcohol”.

Q52 **D. A pair of diastereomeric salts that can be separated by recrystallization**

The (R,S)- and (S,S)-salts are diastereomers with different physical properties, so they can be separated by ordinary physical means such as recrystallization; HCl then gives back the pure (R)- and (S)-acids. In the figure, the curly brackets group the two salts formed together.



Q53 **A. Optically active**

A single enantiomer rotates plane-polarized light, so it is optically active. (Not solved in the doctor's file.)

Q54 **C. A molecule that can undergo a chemical reaction to give a chiral compound**

Prochirality: a molecule can undergo a chemical reaction to give a chiral compound. A prochiral molecule is such a molecule.

Q55 **D. To the left (counterclockwise), shown as (−)**

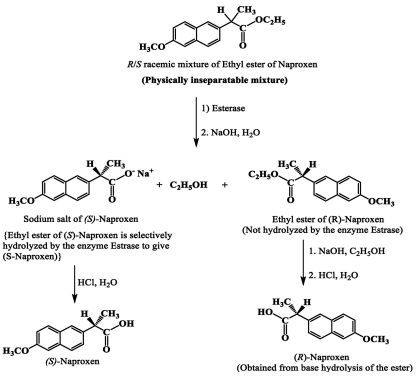
Dextrorotatory (Latin *dexter*, right): rotates to the right (clockwise), (+). Levorotatory (Latin *levo*, left): rotates to the left (counterclockwise), (−).

Q56 **A. To the right (clockwise), shown as (+)**

Dextrorotatory (Latin: dexter, right) refers to an enantiomer that rotates plane-polarized light to the right (clockwise rotation). Levorotatory (Latin: levo, left) refers to an enantiomer that rotates plane-polarized light to the left (counterclockwise rotation).

Q57 **B. Enzymatic hydrolysis with an esterase, which hydrolyzes only the (S)-ester**

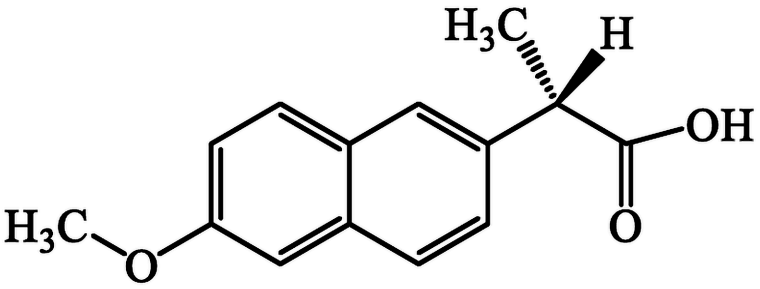
The esterase selectively hydrolyzes the ethyl ester of (S)-naproxen to the (S)-naproxen sodium salt (with NaOH), while the (R)-ester is not hydrolyzed and is separated. Treating the salt with HCl gives pure (S)-naproxen. The (R)-ester can be hydrolyzed with NaOH in ethanol, then HCl, to give (R)-naproxen.



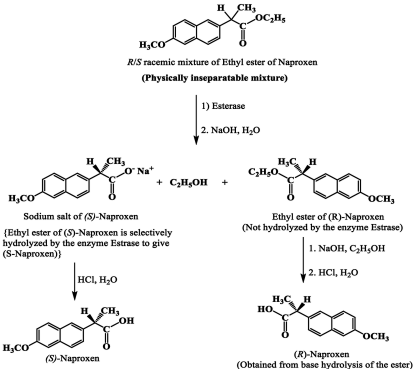
Q58 **C. By treating the sodium salt with HCl**

The pure (S)-naproxen (acid) is obtained by treating the sodium salt with an inorganic acid, HCl.

Q59 **B.**



Priority: COOH > naphthyl (aryl) > CH_3 > H. In both drawings H is on a wedge (toward the viewer). In the structure with COOH on the right of the chiral carbon (aryl on the left), COOH → aryl → CH_3 appears clockwise; reversing gives (S). In the structure with HO–C=O on the left (aryl on the right), the arrow appears counterclockwise; reversing gives (R).



Q60 **C. 0°**

Equal and opposite rotations cancel.

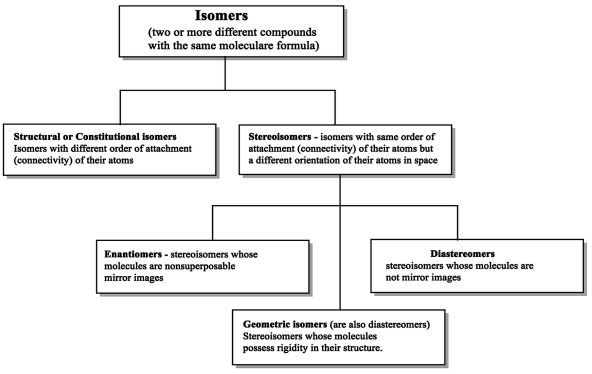
Q61 **A. A racemic mixture**

A racemic mixture may be indicated by the prefix (±).

3.6 Key Terms & Classification of Isomers

Q62 **B. The same connectivity of their atoms but a different orientation in space**

Doctor's flow chart: isomers (same molecular formula) → structural (constitutional) isomers and stereoisomers. Stereoisomers have the same connectivity but a different orientation in space.

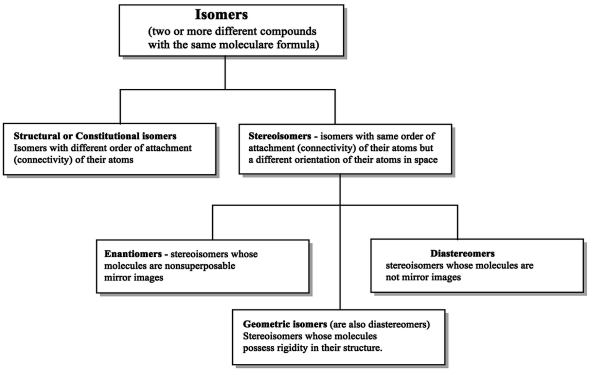


Q63 **D. A different order of attachment (connectivity) of their atoms**

Doctor's flow chart: structural or constitutional isomers are isomers with a different order of attachment (connectivity) of their atoms.

Q64 **A. Enantiomers and diastereomers**

Stereoisomers → enantiomers (non-superposable mirror images) and diastereomers (not mirror images); geometric isomers are also diastereomers.



Q65 **B. Stereoisomers whose molecules are not mirror images**

Doctor's flow chart: diastereomers are stereoisomers whose molecules are not mirror images.

Q66 **C. Stereoisomers whose molecules are non-superposable mirror images**

Doctor's flow chart: enantiomers are stereoisomers whose molecules are non-superposable mirror images.

Q67 **D. Stereoisomers whose molecules possess rigidity in their structure**

Doctor's flow chart: geometric isomers (are also diastereomers) are stereoisomers whose molecules possess rigidity in their structure. (The rigid part is a C=C double bond or a ring.)

Q68 **B. A carbon with four different atoms or groups attached (a chiral carbon)**

Chiral carbon = asymmetric carbon = stereocenter = stereogenic center: an sp^3-hybridized carbon with four different atoms or groups attached.

Q69 **D. Cannot be superposed on its mirror image**

A chiral molecule (like a hand) cannot be superposed on its mirror image; an achiral one can. A chiral molecule is optically active.

Q70 **A. An imaginary plane dividing a molecule into two mirror-image halves**

An internal plane of symmetry cuts the molecule into two mirror-image halves. A molecule with one is achiral, e.g. a meso form.

Q71 **C. Has chiral carbons and an internal plane of symmetry, so it is optically inactive**

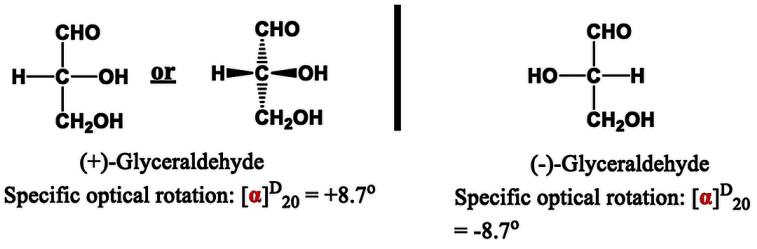
A meso compound requires at least two stereocenters, contains an internal plane of symmetry, and is achiral (does not rotate plane-polarized light), e.g. cis-cyclohexane-1,3-diol.

Q72 **B. Light whose waves vibrate in only one plane**

Ordinary light vibrates in all planes; after passing through a polarizer it vibrates in only one plane. Chiral compounds rotate this plane (measured with a polarimeter).

Q73 **A. Its characteristic rotation of plane-polarized light under standard conditions**

Example: $[\alpha]_D^{20} = +8.7°$ for (+)-glyceraldehyde. Specific rotation is a physical constant of an optically active compound: the rotation measured under standard conditions (wavelength, temperature, concentration and tube length). Enantiomers have equal and opposite values: (+)-glyceraldehyde +8.7°, (−)-glyceraldehyde −8.7°.



Q74 **C. Optical isomers (enantiomers)**

Optical isomers (enantiomers) have identical physical properties except the direction in which they rotate plane-polarized light.

Q75 **D. Configuration is fixed in space; conformation changes by rotation about single bonds**

In configuration the arrangement of groups around the bonds in space is fixed, but in conformation it assumes different arrangements in space by rotation about single bonds.

4 Alkanes

In this lecture

1. Hydrocarbons & Classes of Alkanes 5 questions
2. Types of Carbon Atoms 4 questions
3. Physical Properties of Alkanes 2 questions
4. Nomenclature (IUPAC) of Alkanes 4 questions
5. Reactions of Alkanes 1 questions

4.1 Hydrocarbons & Classes of Alkanes

Q1 Which of the following is **NOT** a source of alkanes?

A. Seawater

B. Petroleum

C. Natural gas

D. Coal

Q2 Alkanes can be classified into 3 classes, namely:

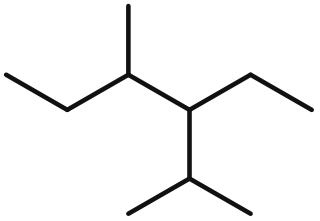
A. Alkenes, alkynes and aromatics

B. Straight-chain, branched-chain and cyclic alkanes

C. Primary, secondary and tertiary alkanes

D. Gases, liquids and solids

Q3 The following compound can be classified as a:



A. Cycloalkane

B. Straight-chain alkane

C. Substituted cycloalkane

D. Branched alkane

Q4 Alkanes are hydrocarbons that have …….:

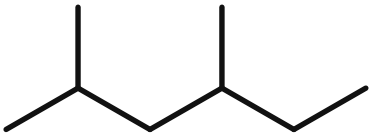
A. Single ω bonds

B. Single α bonds

C. Single σ (sigma) bonds

D. Single π (pi) bonds

Q5 Give the molecular formula for the following compound:



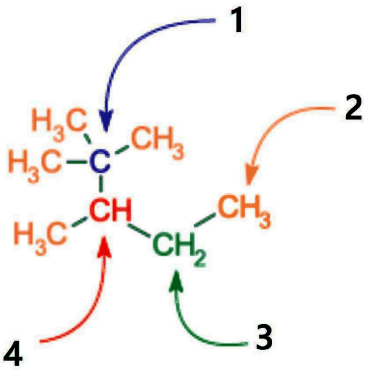
A. C_6H_{14}
B. C_7H_{16}
C. C_8H_{16}
D. C_8H_{18}

4.2 Types of Carbon Atoms

Q6 Carbon atoms present in an alkane can be classified into 4 types of carbons, namely:

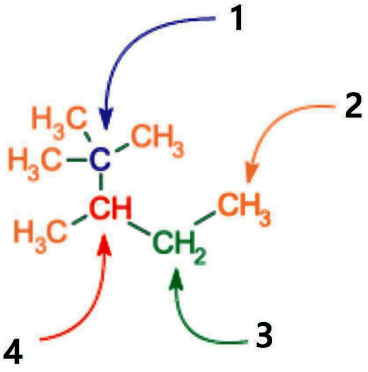
A. sp, sp^2, sp^3 and sp^3d
B. Alpha, beta, gamma and delta
C. Axial, equatorial, cis and trans
D. Primary, secondary, tertiary and quaternary

Q7 In the compound shown, carbon **1** is:



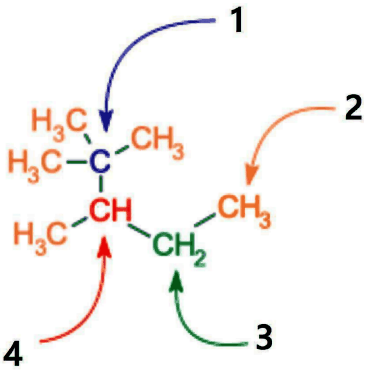
A. Primary (1°)
B. Secondary (2°)
C. Tertiary (3°)
D. Quaternary (4°)

Q8 In the same compound, carbon **4** (the CH) is:



A. Secondary (2°)
B. Primary (1°)
C. Tertiary (3°)
D. Quaternary (4°)

Q9 In the same compound, carbons **2** and **3** are, respectively:



A. Primary (1°) and secondary (2°)
B. Primary (1°) and tertiary (3°)
C. Tertiary (3°) and secondary (2°)
D. Secondary (2°) and primary (1°)

4.3 Physical Properties of Alkanes

Q10 Alkanes are insoluble in water because:

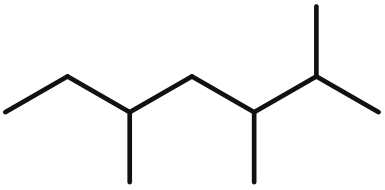
A. They are non-polar and cannot form hydrogen bonds with water
B. They are too heavy
C. They react with water
D. They are ionic compounds

Q11 Which of the following is **NOT** one of the physical properties of alkanes?

A. Lower boiling point than other organic compounds
B. Boiling point increases with the molecular weight of the compound
C. Boiling point increases with branching
D. Water-insoluble (insoluble in water)

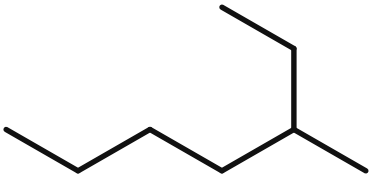
4.4 Nomenclature (IUPAC) of Alkanes

Q12 What is the name of this alkane (find the longest chain)?



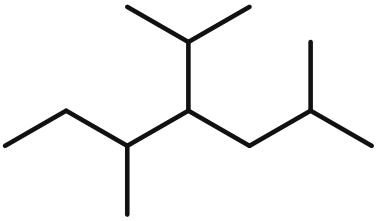
A. 2,3,5-Trimethylheptane
B. 2,3,5-Trimethylhexane
C. 3,5,6-Trimethylheptane
D. Trimethyloctane

Q13 What is the name of this alkane (find the longest chain)?



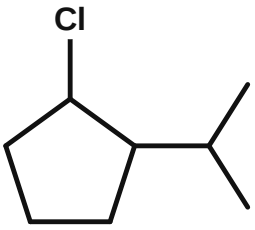
A. 5-Methylheptane
B. 3-Methylheptane
C. 2-Ethylhexane
D. 3-Methylhexane

Q14 Give the correct IUPAC name for this compound:



A. 2,5-Dimethyl-4-isopropylheptane
B. 4-Isopropyl-3,6-dimethylheptane
C. 4-Isopropyl-2,5-dimethylheptane
D. 3-Ethyl-4-isopropyl-2-methylpentane

Q15 Give the correct IUPAC name for this compound:



A. 1-Chloro-2-isopropylcyclopentane
B. 1-Isopropyl-2-chlorocyclopentane
C. 2-Chloro-1-isopropylcyclopentane
D. Chloroisopropylpentane

4.5 Reactions of Alkanes

Q16 The 2 chemical reactions of alkanes are:

A. Hydration and hydrolysis
B. Esterification and saponification
C. Acid–base neutralization and reduction
D. Combustion and free-radical halogenation

Answers · Lecture 4

4.1 Hydrocarbons & Classes of Alkanes

Q1 **A. Seawater**

Three sources of alkanes: natural gas, petroleum and coal.

Q2 **B. Straight-chain, branched-chain and cyclic alkanes**

i. Straight-chain alkanes. ii. Branched-chain alkanes. iii. Cyclic alkanes (cycloalkanes).

Q3 **D. Branched alkane**

It is a branched alkane: side groups (a methyl and an isopropyl group) branch off the carbon chain, and there is no ring.

Q4 **C. Single σ (sigma) bonds**

All C–C and C–H bonds in alkanes are single bonds, and every single bond is a sigma (σ) bond.

Q5 **D. C_8H_{18}**

8 carbons (a 6-carbon chain + 2 methyl groups). Using C_nH_{2n+2}: C_8H_{18}.

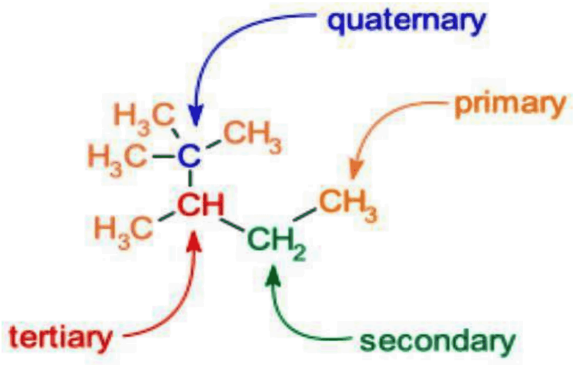
4.2 Types of Carbon Atoms

Q6 **D. Primary, secondary, tertiary and quaternary**

Primary (1°), secondary (2°), tertiary (3°) and quaternary (4°) carbons.

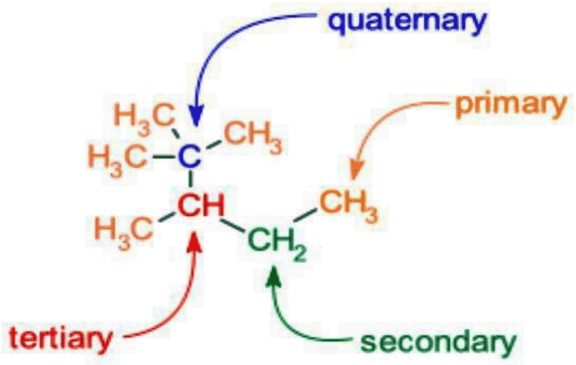
Q7 **D. Quaternary (4°)**

1 = 4°, 2 = 1°, 3 = 2°, 4 = 3°.



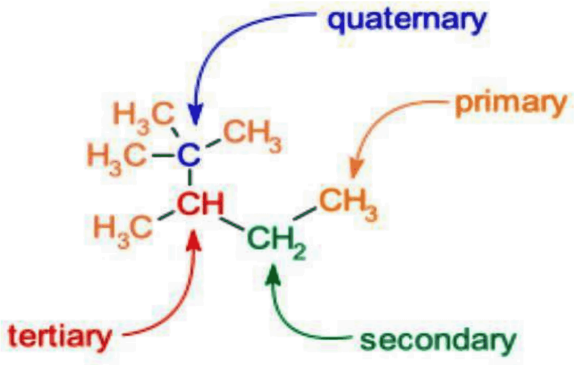
Q8 **C. Tertiary (3°)**

4 = 3°: it is bonded to three carbons (the quaternary C, a CH_3 and the CH_2).



Q9 **A. Primary (1°) and secondary (2°)**

2 = CH_3 (one carbon neighbour) = 1°. 3 = CH_2 (two carbon neighbours) = 2°.



4.3 Physical Properties of Alkanes

Q10 **A. They are non-polar and cannot form hydrogen bonds with water**

Alkanes are water-insoluble because they are non-polar compounds: all C–C and C–H bonds are non-polar covalent bonds, so they do not form hydrogen bonds with water molecules.

Q11 **C. Boiling point increases with branching**

The study guide lists four physical properties of alkanes: i. Water-insoluble, because they are non-polar (all C–C and C–H bonds are covalent), so they do not form hydrogen bonds with water. ii. Lower boiling point than other organic compounds. iii. Boiling point increases with molecular weight. iv. Boiling point **decreases** with branching. So "boiling point increases with branching" is wrong.

4.4 Nomenclature (IUPAC) of Alkanes

Q12 **A. 2,3,5-Trimethylheptane**

Longest chain: 7 carbons (heptane). Numbering from the end nearest a branch gives methyls at C-2, C-3 and C-5: 2,3,5-trimethylheptane.

Q13 **B. 3-Methylheptane**

The longest chain turns up into the ethyl branch: 7 carbons, with a methyl at C-3 → 3-methylheptane.

Q14 **C. 4-Isopropyl-2,5-dimethylheptane**

Parent: heptane. Numbering from the right gives {2,4,5} instead of {3,4,6}. Groups in alphabetical order (isopropyl before methyl; di- is ignored): 4-isopropyl-2,5-dimethylheptane.

Q15 **A. 1-Chloro-2-isopropylcyclopentane**

With more than one substituent on a ring, the one first in alphabetical order (chloro) gets C-1; isopropyl is then at C-2: 1-chloro-2-isopropylcyclopentane.

4.5 Reactions of Alkanes

Q16 **D. Combustion and free-radical halogenation**

i. Combustion reactions. ii. Halogenation reactions via a free-radical reaction.

5 Alkenes

In this lecture

1. Structure & Classification of Alkenes 8 questions
2. Nomenclature of Alkenes 6 questions
3. Physical Properties & C–C vs C=C 2 questions
4. Addition Reactions & Markovnikov's Rule 3 questions
5. Oxidation of Alkenes 3 questions
6. Diels–Alder Reaction 2 questions

5.1 Structure & Classification of Alkenes

Q1 What is the general formula for an alkene?

A. C_nH_{2n}

B. C_nH_{2n-2}

C. C_nH_{2n+2}

D. C_nH_n

Q2 Which of the following molecular formulas represents an alkene?

A. C_5H_{12}

B. C_6H_6

C. C_5H_{10}

D. C_6H_8

Q3 In alkenes, the σ bond of the C=C is formed between two _____ hybridized carbons.

A. $sp^2 – sp^3$

B. $sp^3 – sp$

C. $sp – sp^2$

D. $sp^2 – sp^2$

Q4 Which statement about the C=C double bond is correct?

A. It rotates freely like a C–C single bond

B. It is planar with 120° angles, and rotation about it is restricted

C. The π bond is less reactive than the σ bond

D. It is longer than a C–C single bond

Q5 Why is rotation about the C=C double bond restricted?

A. Hydrogen bonding holds the carbons in place

B. The carbons are sp^3 hybridized

C. The σ bond is too strong to rotate

D. The π bond would have to break

Q6 Does isomerism occur in alkenes?

A. Yes — cis/trans (E/Z) geometric isomerism

B. No, never

C. Only in alkenes with triple bonds

D. Only optical isomerism

Q7 Alkenes with more than one double bond are classified into three types:

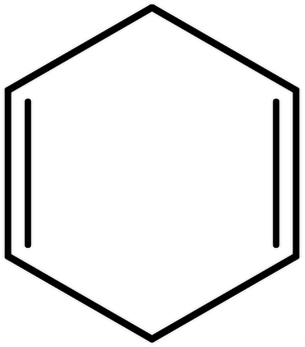
A. Cis, trans and meso

B. Terminal, internal and cyclic

C. Cumulated, conjugated and non-conjugated (isolated)

D. Primary, secondary and tertiary

Q8 The following compound can be classified as:



A. Non-conjugated

B. Cumulated

C. Conjugated

D. Aromatic

5.2 Nomenclature of Alkenes

Q9 Give the IUPAC name of the following compound: $CH_3–CH_2=CH_2–CH_3$

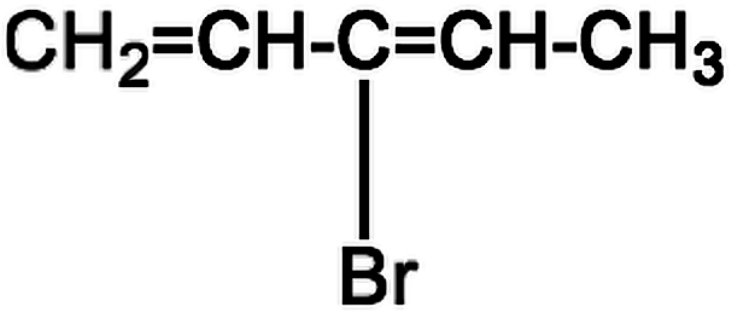
A. 2-Butyne

B. 1-Butene

C. Butane

D. 2-Butene (but-2-ene)

Q10 Give the IUPAC name of the following compound:



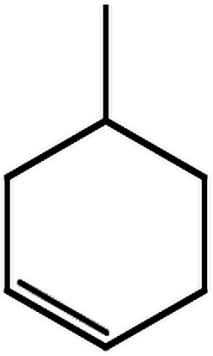
A. 3-Bromo-1-pentene

B. 3-Bromo-2,4-pentadiene

C. 3-Bromo-1,3-pentadiene (3-bromopenta-1,3-diene)

D. 2-Bromo-1,3-pentadiene

Q11 Give the IUPAC name of the following compound:



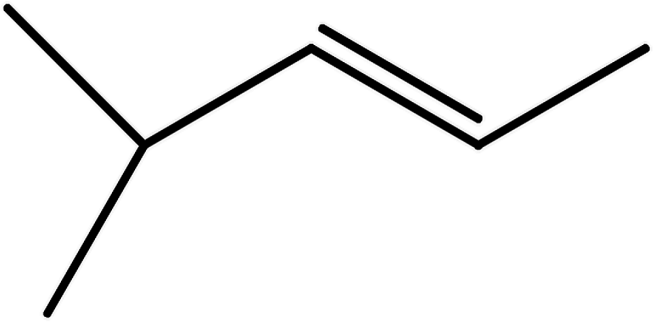
A. 1-Methylcyclohexene

B. 4-Methylcyclohexene (4-methyl-1-cyclohexene)

C. 3-Methylcyclohexene

D. 4-Methylcyclohexane

Q12 Give the IUPAC name of the following compound:



A. 2-Methyl-2-pentene

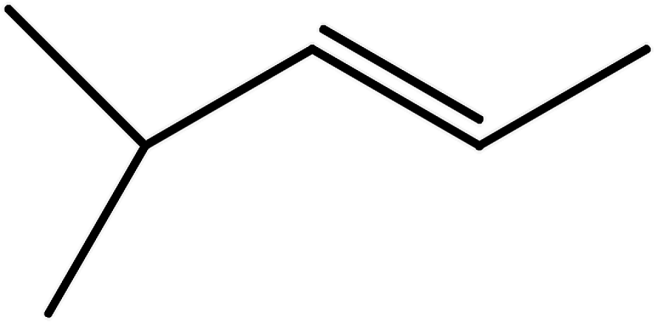
B. 4-Methyl-3-pentene

C. 2-Methyl-3-pentene

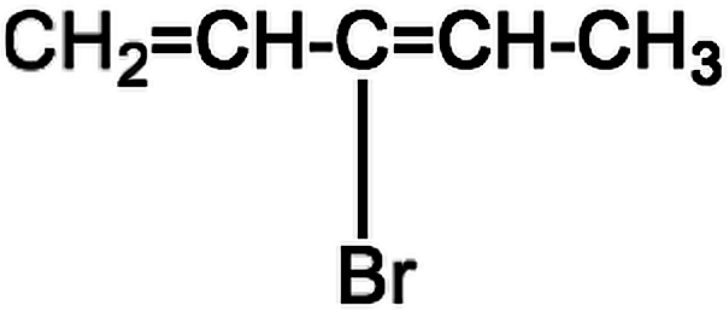
D. 4-Methyl-2-pentene (4-methylpent-2-ene)

Q13 Which structure is 1-bromo-3-heptene?

A.



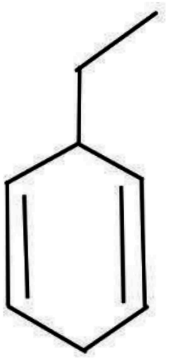
B.



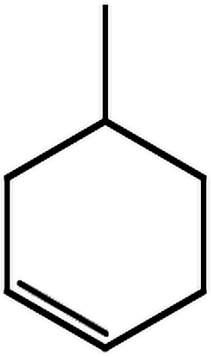
C.

Q14 Which structure is 3-ethyl-1,4-cyclohexadiene?

A.



B.



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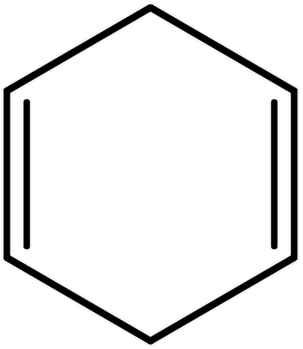


5.3 Physical Properties & C-C vs C=C

Q15 Compare C–C and C=C: which pair is correct?

A. C–C: bond angle 109.5°, 1.54 Å; C=C: bond angle 120°, 1.34 Å

B. Both 109.5°

C. C=C is longer than C–C

D. C–C: 120°, 1.34 Å; C=C: 109.5°, 1.54 Å

Q16 How many atoms are attached to each carbon of a C=C?

A. 2

B. 4

C. 1

D. 3

5.4 Addition Reactions & Markovnikov's Rule

Q17 Which is **NOT** one of the four types of addition reactions of alkenes listed?

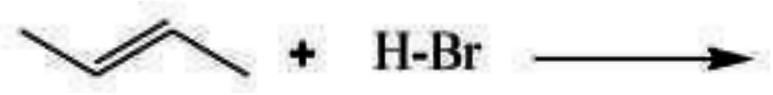
A. Addition of halogens

B. Addition of $NaNH_2$

C. Addition of borane (BH_3)

D. Addition of water (H_2O)

Q18 Complete: 2-butene + HBr →



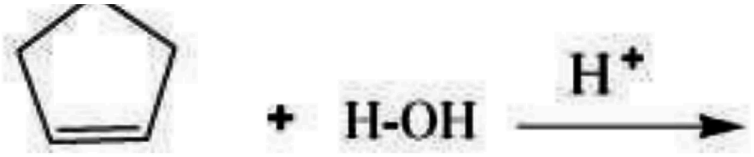
A. Butane

B. 1-Bromobutane

C. 2-Bromobutane

D. 2,3-Dibromobutane

Q19 Complete: cyclopentene + H_2O (H^+) →



A. Cyclopentane-1,2-diol

B. Cyclopentanone

C. Cyclopentene oxide

D. Cyclopentanol

5.5 Oxidation of Alkenes

Q20 Which reagent oxidizes an alkene to an **epoxide**?

A. A peroxycarboxylic acid (3-chloroperbenzoic acid)

B. OsO_4

C. Cold alkaline $KMnO_4$

D. Ozone then Zn/H^+

Q21 Which two reagents oxidize an alkene to a cis-1,2-diol?

A. BH_3 and H_2O_2

B. Cold alkaline $KMnO_4$ and OsO_4

C. Hot $KMnO_4$ and HCl

D. O_3 and H_2O_2

Q22 Ozone (O_3) reacts with alkenes to give which type of products?

A. Carbonyl compounds

B. Alkanes

C. Epoxides

D. Diols

5.6 Diels–Alder Reaction

Q23 The Diels–Alder reaction is:

A. A cycloaddition between a diene and a dienophile giving a cyclic product

B. An elimination reaction

C. An oxidation with $KMnO_4$

D. A free-radical substitution

Q24 The simplest Diels–Alder reaction, 1,3-butadiene + ethylene, gives:

A. Hexane

B. Cyclohexene

C. Cyclobutane

D. Benzene

Answers · Lecture 5

5.1 Structure & Classification of Alkenes

Q1 **A. C_nH_{2n}**

Alkenes (one C=C): C_nH_{2n}.

Q2 **C. C_5H_{10}**

C_5H_{10} fits C_nH_{2n} (n = 5).

Q3 **D. $sp^2 – sp^2$**

Both carbons of a C=C are sp^2 hybridized (trigonal planar, 120°).

Q4 **B. It is planar with 120° angles, and rotation about it is restricted**

Alkenes: planar, trigonal, bond angle 120°; rotation around the double bond is restricted; C=C is shorter than C–C; π bonds are more reactive than σ bonds.

Q5 **D. The π bond would have to break**

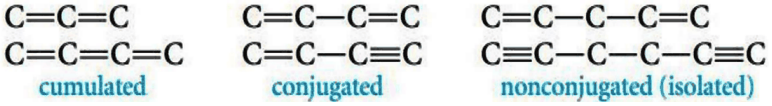
The π bond prevents rotation because of the planar character of the C=C double bond.

Q6 **A. Yes — cis/trans (E/Z) geometric isomerism**

Yes (cis & trans, E & Z). No rotation around the double bond (without breaking the π bond) results in geometric isomers, which cannot be interconverted.

Q7 **C. Cumulated, conjugated and non-conjugated (isolated)**

i. Cumulated (C=C=C). ii. Conjugated (C=C–C=C, alternating). iii. Non-conjugated / isolated (double bonds separated by two or more single bonds).



Q8 **A. Non-conjugated**

Non-conjugated: in cyclohexa-1,4-diene the two C=C are separated on each side by a CH_2 group (two single bonds), so they are isolated.

5.2 Nomenclature of Alkenes

Q9 **D. 2-Butene (but-2-ene)**

2-Butene. Note: the formula as printed has a typo — a carbon in a double bond cannot carry CH_2 and still have 4 bonds; the intended structure is CH_3–CH=CH–CH_3.

Q10 **C. 3-Bromo-1,3-pentadiene (3-bromopenta-1,3-diene)**

3-Bromo-1,3-pentadiene: number from the CH_2= end to give the double bonds 1 and 3; Br on C-3.

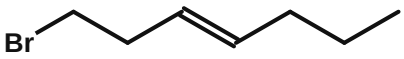
Q11 **B. 4-Methylcyclohexene (4-methyl-1-cyclohexene)**

Ring carbons of the C=C are C-1 and C-2; continuing round gives the CH_3 at C-4: 4-methyl-1-cyclohexene.

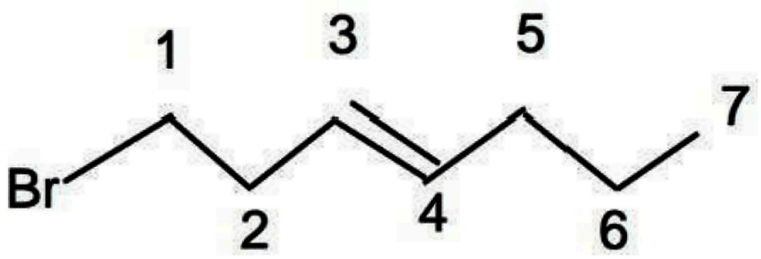
Q12 **D. 4-Methyl-2-pentene (4-methylpent-2-ene)**

4-Methyl-2-pentene: number from the end nearest the double bond (C-2), methyl at C-4.

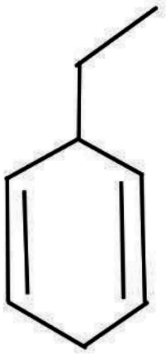
Q13 **C.**



Br on C-1 and C=C between C-3 and C-4 of a 7-carbon chain.



Q14 **A.**



A six-membered ring with double bonds at C1=C2 and C4=C5, and an ethyl group on C-3 (between them).

5.3 Physical Properties & C-C vs C=C

Q15 **A. C–C: bond angle 109.5°, 1.54 Å; C=C: bond angle 120°, 1.34 Å**

C–C: 4 atoms attached to carbon, free rotation, many conformations, 109.5°, 1.54 Å. C=C: 3 atoms attached, restricted rotation, planar, 120°, 1.34 Å.

	Property	C-C	C=C
1	Number of atoms attached to carbon	4	3
2	Rotation	free	restricted
3	Geometry	Many conformations	Planar
4	Bond angle	109.5	120
5	Bond length	1.54 Angstrom	1.34 Angstrom

Q16 **D. 3**

sp^2 carbons have 3 attached atoms (trigonal planar); sp^3 carbons of C–C have 4.

	Property	C-C	C=C
1	Number of atoms attached to carbon	4	3

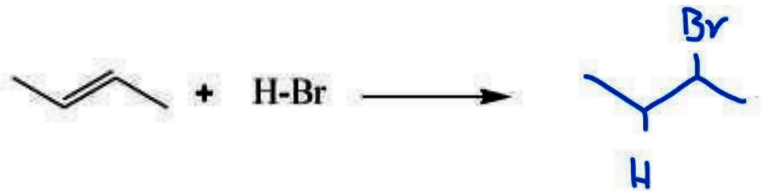
5.4 Addition Reactions & Markovnikov's Rule

Q17 **B. Addition of $NaNH_2$**

i. Addition of halogens. ii. Addition of water. iii. Addition of acids. iv. Addition of borane (BH_3).

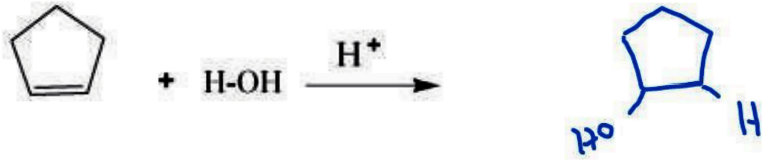
Q18 **C. 2-Bromobutane**

H adds to one C=C carbon and Br to the other → 2-bromobutane (symmetric alkene, one product). The figure shows the doctor's handwritten answer: H on one carbon, Br on the next.



Q19 **D. Cyclopentanol**

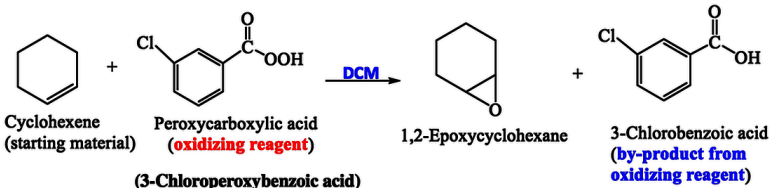
H and OH add across the C=C → cyclopentanol. In the doctor's handwritten answer, "HO" is the OH group.



5.5 Oxidation of Alkenes

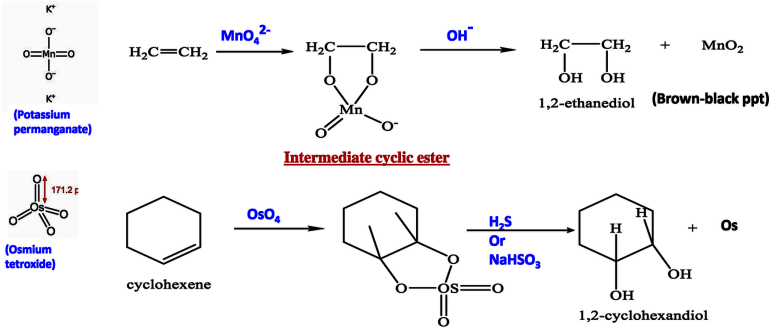
Q20 **A. A peroxycarboxylic acid (3-chloroperbenzoic acid)**

Oxidation reagents: $KMnO_4$ – diols and/or carbonyl compounds; OsO_4 – diols; O_3 – carbonyl compounds; peroxycarboxylic acid (3-chloroperbenzoic acid) – epoxides.



Q21 **B. Cold alkaline $KMnO_4$ and OsO_4**

Both MnO_4^- and OsO_4 add syn (cis) to the π bond to form an intermediate cyclic ester, finally giving a cis-diol (e.g. ethene → 1,2-ethanediol; cyclohexene → cis-1,2-cyclohexanediol).



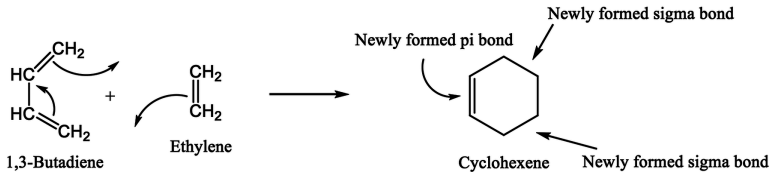
Q22 **A. Carbonyl compounds**

The study guide lists four oxidation reagents and their products: i. Potassium permanganate ($KMnO_4$) – diols and/or carbonyl compounds. ii. Osmium tetroxide (OsO_4) – diols. iii. Ozone (O_3) – carbonyl compounds. iv. Peroxycarboxylic acid (3-chloroperbenzoic acid) – epoxides.

5.6 Diels–Alder Reaction

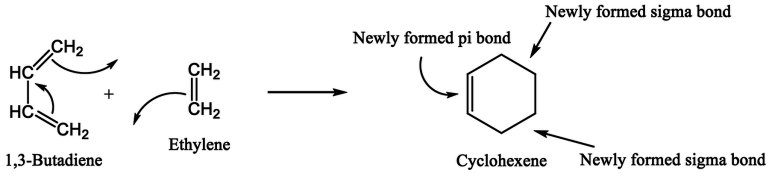
Q23 **A. A cycloaddition between a diene and a dienophile giving a cyclic product**

It is an example of a cycloaddition reaction of dienes, which results in a cyclic product; it involves a diene and a dienophile (“diene lover”), which is an alkene or alkyne.



Q24 **B. Cyclohexene**

1,3-Butadiene + ethylene → cyclohexene, with two newly formed σ bonds and one newly formed π bond.



6 Alkynes

In this lecture

1. Structure & Classification of Alkynes 3 questions
2. Nomenclature of Alkynes 5 questions
3. Physical Properties & Acidity 2 questions
4. Addition of HX and X_2 3 questions
5. Hydration & Hydroboration 4 questions
6. Acetylide Anions: Alkylation 1 questions
7. Review Questions: Alkenes & Alkynes 10 questions

6.1 Structure & Classification of Alkynes

Q1 What is the general formula for alkynes?

A. C_nH_{2n}

B. C_nH_n

C. C_nH_{2n+2}

D. C_nH_{2n-2}

Q2 Which of the following molecular formulas represents an alkyne?

A. C_6H_6

B. C_5H_8

C. C_5H_{12}

D. C_5H_{10}

Q3 Alkynes can be classified generally into 2 types:

A. Cyclic and aromatic alkynes

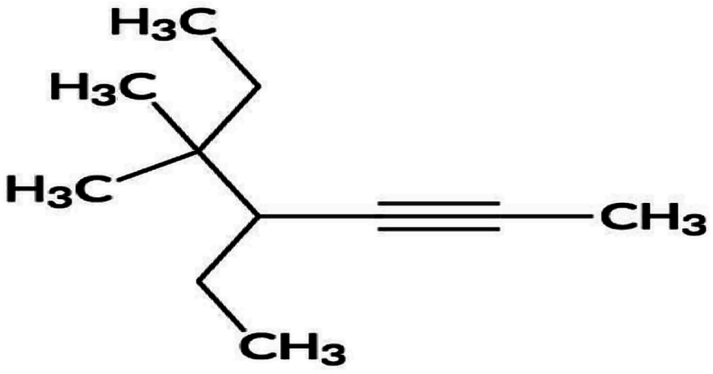
B. Cis and trans alkynes

C. Terminal and internal alkynes

D. Primary and secondary alkynes

6.2 Nomenclature of Alkynes

Q4 Give the IUPAC name of the following compound:



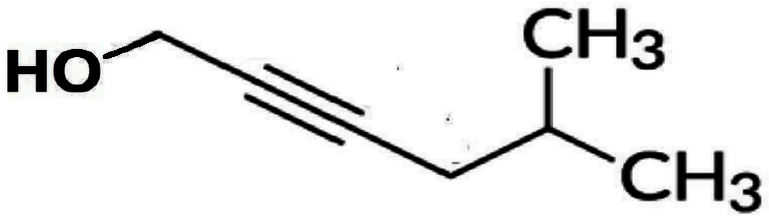
A. 4-Ethyl-3,3-dimethylhept-5-yne

B. 3,3-Dimethyl-4-ethylhept-5-yne

C. 4-Ethyl-5,5-dimethyl-2-hexyne

D. 4-Ethyl-5,5-dimethylhept-2-yne

Q5 Give the IUPAC name of the following compound:

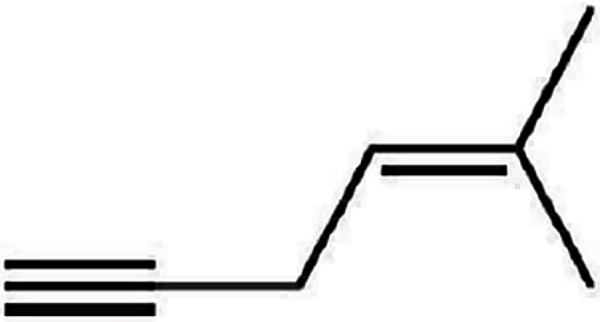


A. 4-Methylhex-2-yn-1-ol
B. 2-Methylhex-4-yn-6-ol
C. 5-Methylhex-2-yn-1-ol
D. 5-Methylhex-3-yn-1-ol

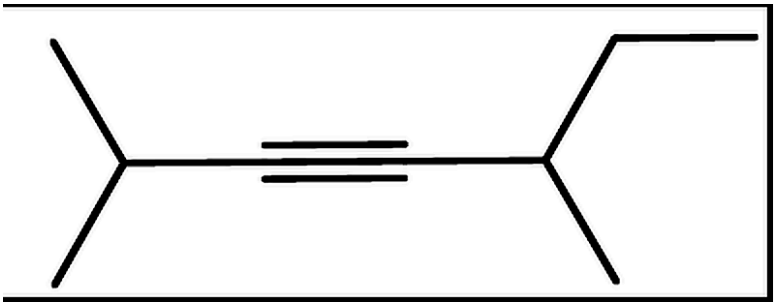
Q6 Which structure is 2,5-dimethylhept-3-yne?

A.

B.

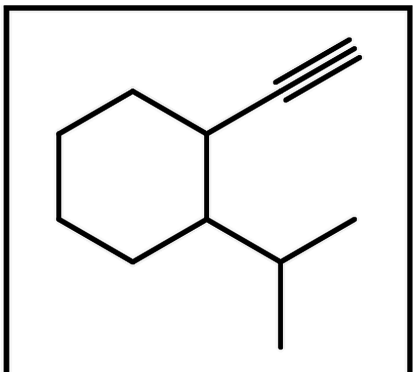


C.

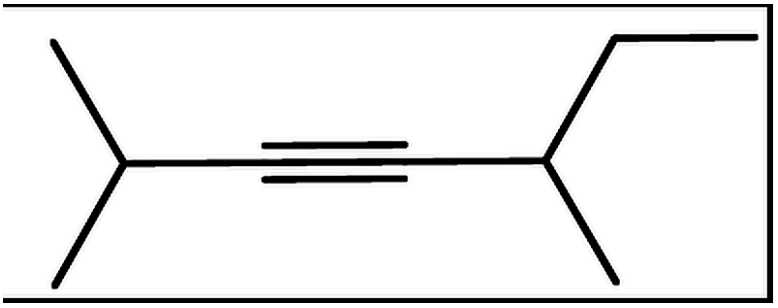


Q7 Which structure is 1-ethynyl-2-isopropylcyclohexane?

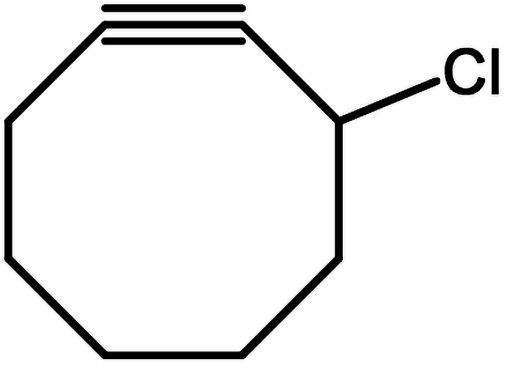
A.



B.

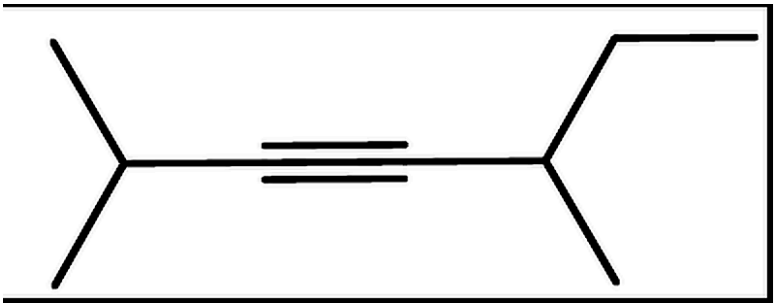


C.

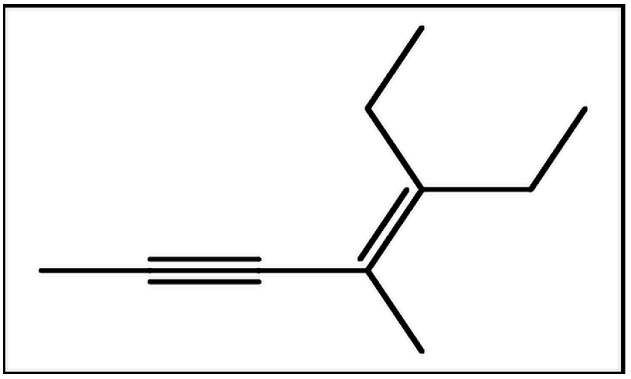


Q8 Which structure is 3-ethyl-4-methylhept-3-en-5-yne?

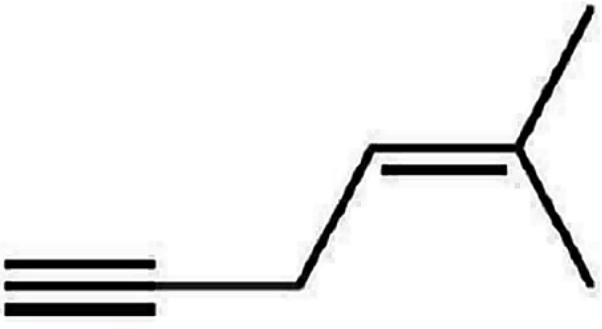
A.



B.



C.



6.3 Physical Properties & Acidity

Q9 Which of the following is **NOT** one of the physical properties of alkynes?

A. Soluble in organic solvents

B. Highly soluble in water

C. Low melting and boiling points

D. Melting and boiling points increase with the number of carbons

Q10 Complete the equation: R–C≡C–H + $NaNH_2$ (liquid NH_3) →

A. A sodium acetylide (R–C≡C:$^-$ Na^+) + NH_3

B. No reaction

C. An alkene + NaH

D. An aldehyde

6.4 Addition of HX and X_2

Q11 Complete the equation: R–C≡C–H + Br_2 (1 equiv.) →

A. A ketone

B. A tetrabromide

C. A geminal dibromide

D. A 1,2-dibromoalkene, R–CBr=CHBr (dihalide)

Q12 Complete the equation: R–C≡C–H + Br_2 (4 equiv.) →

A. An alkyl bromide

B. A 1,2-dibromoalkene

C. No reaction

D. A 1,1,2,2-tetrabromoalkane

Q13 Complete the equation: R–C≡C–H + HCl (1 equiv.) →

A. A geminal dichloride

B. A vinyl chloride with Cl on the more substituted carbon (R–CCl=CH_2)

C. An alkane

D. R–CH=CHCl (Cl on the terminal carbon)

6.5 Hydration & Hydroboration

Q14 Complete the equation: H–C≡C–CH_3 + H_2O (H_2SO_4, $HgSO_4$) →

A. An alkane

B. A methyl ketone, acetone (via an enol)

C. A primary alcohol

D. An aldehyde

Q15 Complete the equation: R–C≡C–CH_3 + H_2O (H_2SO_4) →

A. A carboxylic acid

B. A ketone

C. An ether

D. An aldehyde

Q16 Complete the equation: Ph–C≡C–H + 1) THF, BH_3 2) H_2O_2, OH^- →

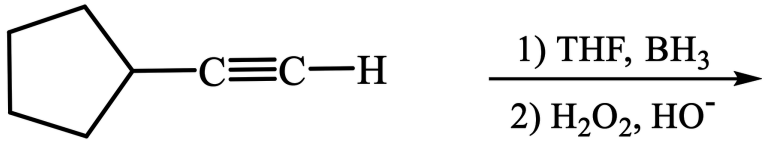
A. Acetophenone (Ph–CO–CH_3)

B. Benzoic acid

C. Styrene

D. Phenylacetaldehyde (Ph–CH_2–CHO)

Q17 The reaction of ethynylcyclopentane with BH_3 in THF, followed by basic H_2O_2/OH^-, gives:



Ethynylcyclopentane

A. Cyclopentanecarboxylic acid

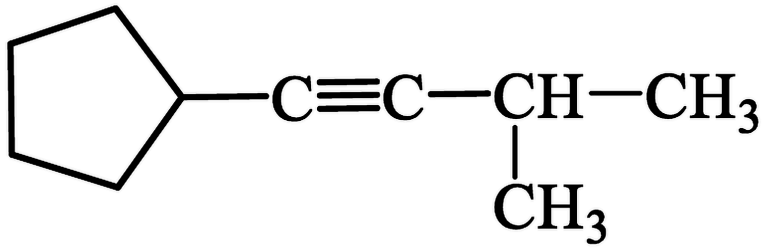
B. 2-Cyclopentylethanol

C. 1-Cyclopentylethanone (cyclopentyl–CO–CH_3)

D. Cyclopentylacetaldehyde (cyclopentyl–CH_2–CHO)

6.6 Acetylide Anions: Alkylation

Q18 Suggest starting materials and reagents for the alkyne shown:



A. Ethynylcyclopentane + $NaNH_2$, then 2-bromopropane (acetylide alkylation)

B. Cyclopentene + Br_2

C. Ethynylcyclopentane + $H_2O/HgSO_4$

D. Cyclopentanol + H_2SO_4

6.7 Review Questions: Alkenes & Alkynes

Q19 What is the major organic product of the acid-catalyzed hydration of 1-methylcyclohexene?

A. 1-Methylcyclohexene oxide

B. Cyclohexylmethanol

C. 2-Methylcyclohexanol

D. 1-Methylcyclohexanol

Q20 The addition of water to an alkene in the presence of a strong acid catalyst is called:

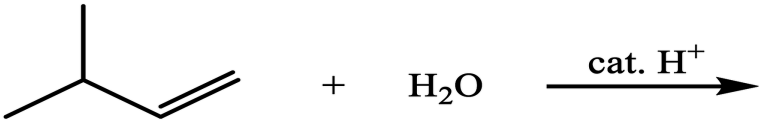
A. Acid-catalyzed hydration

B. Hydrogenation

C. Hydroboration

D. Dehydration

Q21 Predict the major product of this reaction:



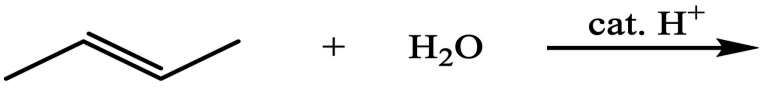
A. 3-Methylbutan-2-ol

B. 2-Methylbutan-1-ol

C. 3-Methylbutan-1-ol

D. 2-Methylbutan-2-ol

Q22 Predict the major product of this reaction:



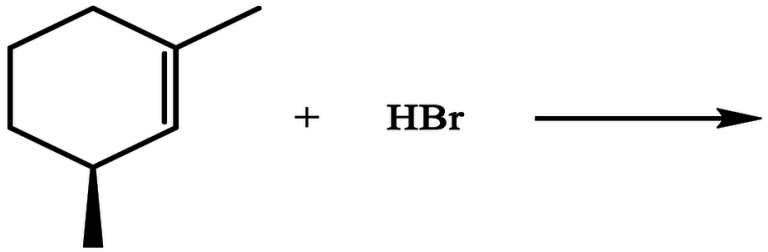
A. Butan-2-ol

B. Butan-1-ol

C. Butane

D. 2-Methylpropan-2-ol

Q23 Predict the major product of this reaction:



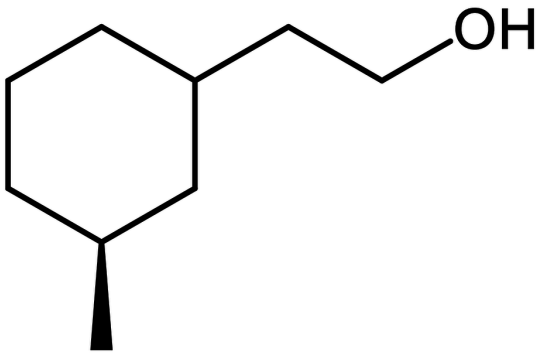
A. 1,3-Dimethylcyclohexane

B. 1-Bromo-1,3-dimethylcyclohexane

C. 2-Bromo-1,3-dimethylcyclohexane

D. 1-Bromo-2,4-dimethylcyclohexane

Q24 Which alkene and reagents are needed to prepare this alcohol?



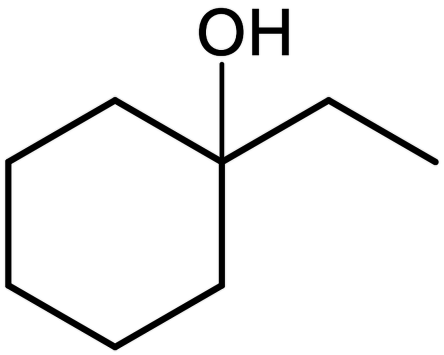
A. Ethylidene-3-methylcyclohexane + HBr

B. 3-Methylcyclohexene + H_2O, H_2SO_4

C. 1-Methyl-3-vinylcyclohexane, then 1) BH_3/THF 2) H_2O_2, OH^-

D. 1-Methyl-3-vinylcyclohexane + H_2O, H_2SO_4

Q25 Which alkene and reagents are needed to prepare 1-ethylcyclohexanol?



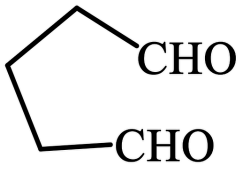
A. Vinylcyclohexane, then 1) BH_3 2) H_2O_2, OH^-

B. 1-Ethylcyclohexene + H_2O, H_2SO_4 (acid-catalyzed hydration)

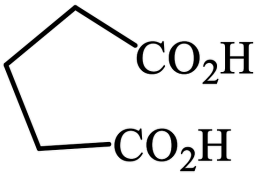
C. Cyclohexene + ethanol

D. 1-Ethylcyclohexene, then 1) BH_3 2) H_2O_2, OH^-

Q26 Starting with cyclopentene, which reagents give pentanedial (i) and pentanedioic acid (ii)?



(i)



(ii)

A. (i) HBr; (ii) Br_2

B. (i) H_2O, H_2SO_4; (ii) BH_3 then H_2O_2

C. (i) H_2, Pd; (ii) $NaNH_2$

D. (i) 1) O_3 2) Zn/H_2O (or $(CH_3)_2S$); (ii) hot $KMnO_4$ (or O_3 then H_2O_2)

Q27 The reaction of 1-methylcyclopentene with BH_3 in THF, followed by basic H_2O_2/OH^-, gives:

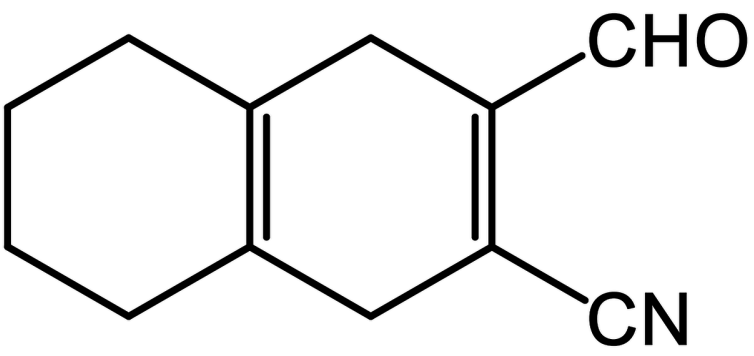
A. 1-Methylcyclopentan-1-ol

B. cis-2-Methylcyclopentan-1-ol

C. trans-2-Methylcyclopentan-1-ol

D. Cyclopentylmethanol

Q28 Which diene and dienophile react (Diels–Alder) to give the bicyclic molecule shown?



A. 1,2-Dimethylenecyclohexane + NC–C≡C–CHO

B. Buta-1,3-diene + cyclohexyne

C. 1,3-Cyclohexadiene + acrylonitrile

D. Cyclohexene + NC–CH=CH–CHO

Answers · Lecture 6

6.1 Structure & Classification of Alkynes

Q1 **D. C_nH_{2n-2}**

Alkynes are hydrocarbons with a triple bond: C_nH_{2n-2}.

Q2 **B. C_5H_8**

C_5H_8 fits C_nH_{2n-2} (n = 5 → 2(5) − 2 = 8).

Q3 **C. Terminal and internal alkynes**

i. Terminal (the triple bond is at the end of the chain, C≡C–H). ii. Internal (the triple bond is inside the chain).

6.2 Nomenclature of Alkynes

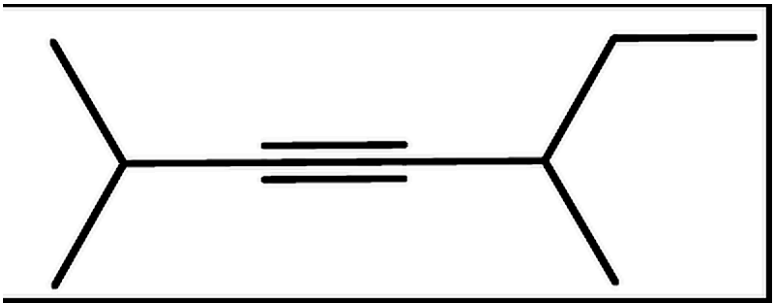
Q4 **D. 4-Ethyl-5,5-dimethylhept-2-yne**

4-Ethyl-5,5-dimethyl-2-heptyne OR 4-ethyl-5,5-dimethylhept-2-yne.

Q5 **C. 5-Methylhex-2-yn-1-ol**

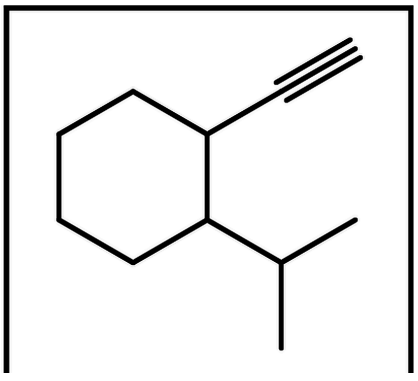
5-Methylhex-2-yn-1-ol OR 5-methyl-2-hexyn-1-ol.

Q6 **C.**



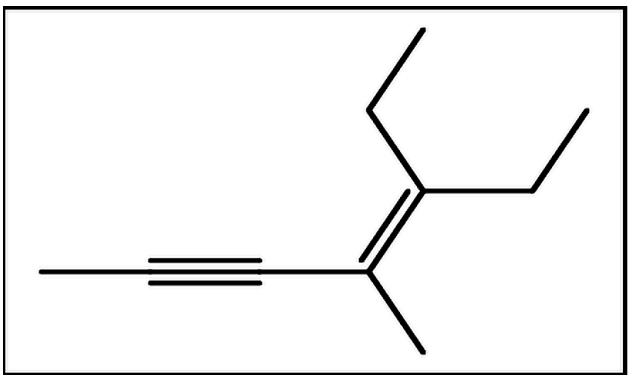
CH_3–$CH(CH_3)$–C≡C–$CH(CH_3)$–CH_2–CH_3: a 7-carbon chain, triple bond at C-3, methyls at C-2 and C-5.

Q7 **A.**



A cyclohexane ring carrying an ethynyl group (–C≡CH) at C-1 and an isopropyl group at C-2.

Q8 **B.**



CH_3–C≡C–C(CH_3)=C(CH_2CH_3)–CH_2CH_3: a 7-carbon chain with a C=C and a C≡C (an enyne), an ethyl group and a methyl group on the double-bond carbons. (Note: this is the name written in the study guide; by current IUPAC rules the multiple bonds as a set get the lowest locants, giving the preferred name 5-ethyl-4-methylhept-4-en-2-yne.)

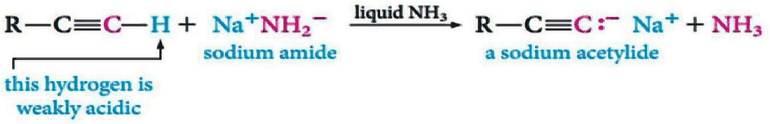
6.3 Physical Properties & Acidity

Q9 **B. Highly soluble in water**

i. Low melting and boiling points. ii. Melting and boiling points increase with increasing number of carbon atoms. iii. Water-insoluble. iv. Soluble in organic solvents.

Q10 **A. A sodium acetylide (R–C≡C:$^-$ Na^+) + NH_3**

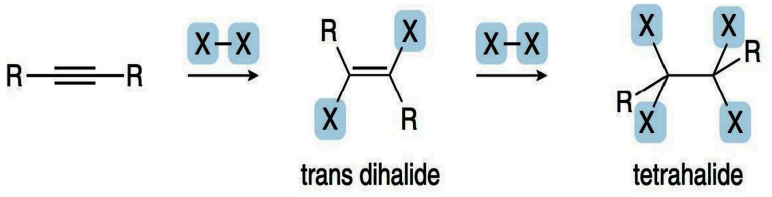
R–C≡C–H + $NaNH_2$ → (liquid NH_3) R–C≡C:$^-$ Na^+ + NH_3. The hydrogen on the triple-bonded carbon is weakly acidic and is removed by a very strong base.



6.4 Addition of HX and X_2

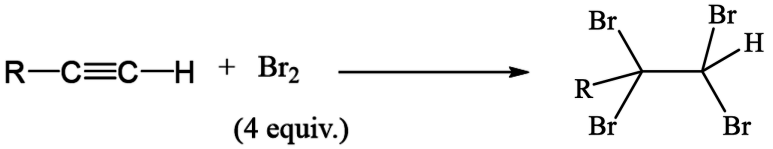
Q11 **D. A 1,2-dibromoalkene, R–CBr=CHBr (dihalide)**

With one equivalent of X_2 the alkyne gives a dihalide (mainly the trans isomer; the figure shows the general scheme for R–C≡C–R). With a second equivalent a tetrahalide forms. (Note: the study guide answer key draws both Br atoms on the same side of the double bond; the lecture shows the product as mainly the trans dihalide.)



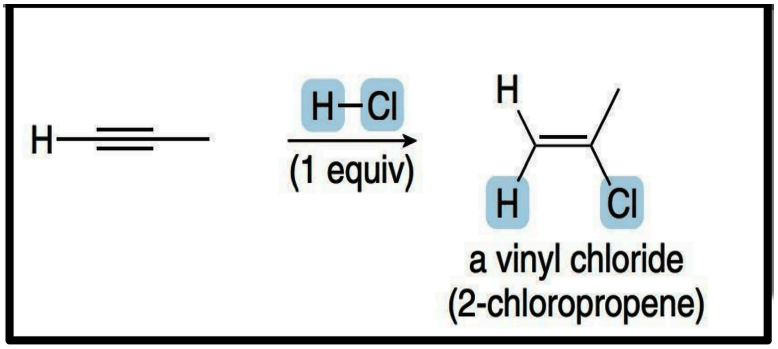
Q12 **D. A 1,1,2,2-tetrabromoalkane**

R–C≡C–H + excess Br_2 → R–CBr_2–$CHBr_2$. Two equivalents of X_2 add, and four σ bonds are formed. (The doctor's sheet shows “4 equiv.”; two moles of Br_2 are consumed per triple bond.)



Q13 **B. A vinyl chloride with Cl on the more substituted carbon (R–CCl=CH_2)**

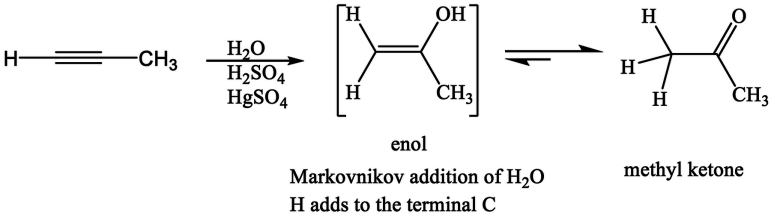
With only 1 equivalent of HX the reaction stops at the vinyl halide. Markovnikov: H adds to the terminal carbon and Cl to the internal carbon, e.g. propyne + HCl → 2-chloropropene. (Note: the study guide answer key writes Br on the product by mistake; with HCl the halogen added is Cl.)



6.5 Hydration & Hydroboration

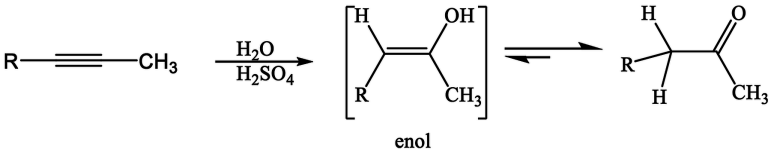
Q14 **B. A methyl ketone, acetone (via an enol)**

H_2O, H_2SO_4 (often catalyzed by $HgSO_4$) gives ketones via Markovnikov addition and enol–keto tautomerism; H adds to the terminal carbon, so a methyl ketone forms. (In the figure, the square brackets mark the enol as a short-lived intermediate.)



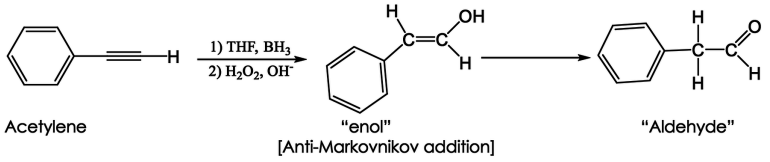
Q15 **B. A ketone**

R–C≡C–CH_3 + H_2O/H_2SO_4 → enol → ketone (R–CH_2–CO–CH_3). (When R is not CH_3, the OH can add to either triple-bond carbon, so a mixture of two ketones usually forms; both are ketones.)



Q16 **D. Phenylacetaldehyde (Ph–CH_2–CHO)**

Anti-Markovnikov addition of H_2O: the enol Ph–CH=CH–OH tautomerizes to the aldehyde Ph–CH_2–CHO. (In the doctor's figure the starting alkyne is labelled “Acetylene”; it is phenylacetylene, Ph–C≡C–H.)



Q17 **D. Cyclopentylacetaldehyde (cyclopentyl–CH_2–CHO)**

Terminal alkyne + hydroboration–oxidation → anti-Markovnikov enol → aldehyde: 2-cyclopentylethanal (cyclopentylacetaldehyde). (Not solved in the doctor's file.)

6.6 Acetylide Anions: Alkylation

Q18 **A. Ethynylcyclopentane + $NaNH_2$, then 2-bromopropane (acetylide alkylation)**

Alkylation of an acetylide: deprotonate ethynylcyclopentane with $NaNH_2$, then react the acetylide with an isopropyl halide (or deprotonate 3-methylbut-1-yne and use a cyclopentyl halide). Note: both possible halides are 2°, so E2 elimination competes and the yield is low. (Not solved in the doctor's file.)

6.7 Review Questions: Alkenes & Alkynes

Q19 **D. 1-Methylcyclohexanol**

Markovnikov addition: H^+ adds to the CH carbon, giving the tertiary carbocation on the carbon bearing CH_3; water attacks there → 1-methylcyclohexanol. (Not solved in the doctor's file.)

Q20 **A. Acid-catalyzed hydration**

Adding H_2O across C=C with H^+ catalysis is (acid-catalyzed) hydration. (Not solved in the doctor's file.)

Q21 **D. 2-Methylbutan-2-ol**

H^+ adds to C-1 (Markovnikov) giving a 2° carbocation at C-2; a 1,2-hydride shift moves the positive charge to the more stable 3° carbon (C-3), and water attacks there → 2-methylbutan-2-ol. (Without the shift, the Markovnikov product would be 3-methylbutan-2-ol.) (Not solved in the doctor's file.)

Q22 **A. Butan-2-ol**

But-2-ene is symmetrical; adding H and OH across C2=C3 gives butan-2-ol (as a racemic mixture). (Not solved in the doctor's file.)

Q23 **B. 1-Bromo-1,3-dimethylcyclohexane**

Markovnikov addition of HBr: H adds to the CH alkene carbon and Br to the alkene carbon bearing CH_3 (tertiary carbocation) → 1-bromo-1,3-dimethylcyclohexane (as a mixture of stereoisomers). (Not solved in the doctor's file.)

Q24 **C. 1-Methyl-3-vinylcyclohexane, then 1) BH_3/THF 2) H_2O_2, OH^-**

The OH is on the terminal (primary) carbon of the side chain, so anti-Markovnikov hydroboration–oxidation of the vinyl group (–CH=CH_2) is needed. (Not solved in the doctor's file.)

Q25 **B. 1-Ethylcyclohexene + H_2O, H_2SO_4 (acid-catalyzed hydration)**

The OH is on the tertiary (more substituted) carbon, so Markovnikov acid-catalyzed hydration of 1-ethylcyclohexene is used. (Not solved in the doctor's file.)

Q26 **D. (i) 1) O_3 2) Zn/H_2O (or $(CH_3)_2S$); (ii) hot $KMnO_4$ (or O_3 then H_2O_2)**

Oxidative cleavage of the C=C opens the ring. Ozonolysis with a reductive work-up stops at the dialdehyde (i); strong oxidation (hot $KMnO_4$, or ozonolysis with oxidative work-up) gives the diacid (ii). (Not solved in the doctor's file.)

Q27 **C. trans-2-Methylcyclopentan-1-ol**

Hydroboration–oxidation is anti-Markovnikov and syn: H and OH add to the same face, with OH on the less substituted carbon. The H on C-1 and the OH on C-2 are cis, so the CH_3 and OH end up trans. (Not solved in the doctor's file.)

Q28 **A. 1,2-Dimethylenecyclohexane + NC–C≡C–CHO**

The new ring contains two C=C bonds (a cyclohexa-1,4-diene), which comes from an alkyne dienophile. The diene is 1,2-dimethylenecyclohexane (its two =CH_2 groups become the ring CH_2 carbons); the dienophile carries the CHO and CN groups: NC–C≡C–CHO. (Not solved in the doctor's file.)

7 Ethers & Epoxides

In this lecture

1. Ethers: Structure & Nomenclature 8 questions
2. Physical Properties of Ethers 3 questions
3. Preparation of Ethers 5 questions
4. Reactions of Ethers with HX 10 questions
5. Epoxides: Structure & Preparation 5 questions
6. Ring-Opening Reactions of Epoxides 9 questions

7.1 Ethers: Structure & Nomenclature

Q1 The general structure of an ether is:

A. R–CO–R′

B. R–COOH

C. R–OH

D. R–O–R′

Q2 In common nomenclature, ethers are named by:

A. Naming the alkyl groups by size, largest first, followed by the word “oxide”

B. Naming the alkyl groups on oxygen in alphabetical order, followed by the word “ether”

C. Naming the longest chain as the parent and adding the suffix -ol to it

D. Naming the longer alkyl group as an alkoxy substituent on the shorter one

Q3 In IUPAC nomenclature of ethers, the –OR group is named as:

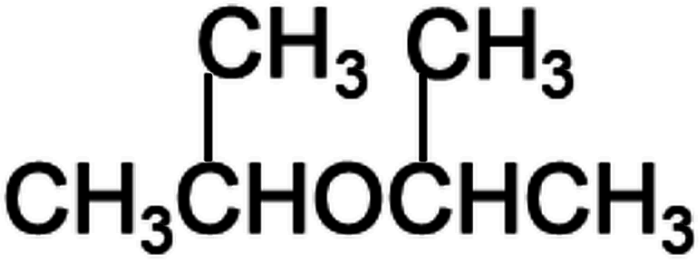
A. An alkoxy substituent on the longest carbon chain

B. An -ol suffix added to the longest chain

C. A suffix “-ether” added to the longest chain

D. The parent chain, with the longer alkyl group as a substituent

Q4 What is the common name of this ether?



A. Isopropyl propyl ether

B. Dimethyl ether

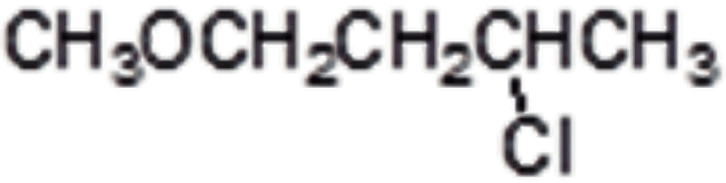
C. Diisopropyl ether

D. Dipropyl ether

Q5 What is the common name of $CH_3CH_2OCH_3$?

A. Methoxyethene
B. Diethyl ether
C. Ethyl propyl ether
D. Ethyl methyl ether

Q6 What is the IUPAC name of this compound?



A. 3-Chloro-1-methoxybutane
B. 2-Chloro-4-methoxybutane
C. 2-Chloro-4-methoxypentane
D. 1-Methoxy-3-chlorobutane

Q7 What is the IUPAC name of $CH_3OCH_2CH_2CH_2OH$?

A. 1-Methoxypropane
B. 1-Methoxy-3-propanol
C. 3-Methoxy-1-propanol
D. 1-Hydroxy-3-methoxypropane

Q8 What is the IUPAC name of $CH_3CH_2CH_2OCH_2(CH_2)_5CH_3$?

A. 1-Heptoxypropane
B. 1-Propoxyheptane
C. 1-Propoxyoctane
D. 2-Propoxyheptane

7.2 Physical Properties of Ethers

Q9 Using the table, why does 1-butanol (118 °C) boil much higher than diethyl ether (35 °C)?

Compound	BP (°C)	MW	Water sol. (g/100 mL, 20 °C)
1-butanol $CH_3CH_2CH_2CH_2CH_2OH$	118	74	7.9
Diethyl ether $CH_3CH_2OCH_2CH_3$	35	74	7.5

A. 1-Butanol is non-polar, so its molecules pack more tightly

B. Ether molecules form stronger hydrogen bonds with each other

C. Diethyl ether has a much higher molecular weight than 1-butanol

D. Alcohol molecules form hydrogen bonds with each other; ether molecules cannot

Q10 Why does diethyl ether have a boiling point similar to pentane?

Compound	BP (°C)	MW	Water sol. (g/100 mL, 20 °C)
Diethyl ether $CH_3CH_2OCH_2CH_3$	35	74	7.5
pentane $CH_3CH_2CH_2CH_2CH_3$	36	72	0.03

A. There is no hydrogen bonding between ether molecules

B. Ether molecules form strong hydrogen bonds with each other

C. Both compounds are ionic and have similar charges

D. Both compounds contain an OH group in their molecules

Q11 Why is diethyl ether much more soluble in water (7.5 g/100 mL) than pentane (0.03 g/100 mL)?

Compound	BP (°C)	MW	Water sol. (g/100 mL, 20 °C)
Diethyl ether $CH_3CH_2OCH_2CH_3$	35	74	7.5
pentane $CH_3CH_2CH_2CH_2CH_3$	36	72	0.03

A. Ethers can donate H-bonds to water through an O–H group

B. Ethers react with water to form water-soluble alcohols

C. Pentane is more polar than ether, so it repels water

D. Ethers can accept H-bonds from water through the oxygen lone pairs

7.3 Preparation of Ethers

Q12 The two methods used to prepare ethers are:

A. Ozonolysis of alkenes and hydroboration of alkynes

B. Air oxidation of alkenes and peroxyacid oxidation

C. Williamson synthesis and acid-catalyzed dehydration of alcohols

D. Combustion of alkanes and free-radical halogenation

Q13 In the Williamson synthesis, the alkyl halide should be:

A. Methyl, primary or benzylic

B. An aryl halide

C. Tertiary only

D. Secondary or tertiary

Q14 In the Williamson synthesis (R–X + NaOR′), the R′ group of the alkoxide can be:

A. 2° or 3° only

B. CH_3, 1°, 2°, 3° or aryl

C. CH_3, 1° or benzylic only

D. Aryl only

Q15 In the doctor's Q7 table, the dehydration-of-alcohols row gives the reagent H_2SO_4 and the product $CH_3CH_2OCH_2CH_3$ (a symmetrical ether). The starting material is:

A. Methanol (CH_3OH)

B. Bromoethane (CH_3CH_2Br)

C. Ethanol (CH_3CH_2OH)

D. Ethene ($CH_2=CH_2$)

Q16 To make an unsymmetrical ether from $CH_3CH_2CH_2Br$ by the Williamson synthesis, a suitable reagent and product are:

A. $H_2SO_4 \rightarrow CH_3CH_2CH_2OCH_2CH_2CH_3$

B. $NaOH \rightarrow CH_3CH_2CH_2OH$

C. $HBr \rightarrow CH_3CH_2CH_2OCH_3$

D. $NaOCH_3 \rightarrow CH_3CH_2CH_2OCH_3$

7.4 Reactions of Ethers with HX

Q17 The reaction sites in an ether R–O–R are:

A. A polar C=O double bond and the O–H bond

B. One polar O–H bond and one C–O bond

C. Only the C–H bonds of the two alkyl groups

D. Two polar C–O bonds and two non-bonding electron pairs on O

Q18 An ether reacts with **one** mole of HX (X = Cl, Br, I) to give:

A. R–X + R–OH

B. No reaction

C. R–OH + R–OH

D. 2 R–X + H_2O

Q19 An ether reacts with **two** moles of HX (X = Cl, Br, I) to give:

A. 2 R–OH

B. R–X + R–OH

C. 2 R–X + H_2O

D. R–O–X + R–H

Q20 When an ether is cleaved by HX, the first step is:

A. Loss of a carbocation from the ether

B. Formation of an alkyl radical

C. Attack of X^- on the ether oxygen

D. Protonation of the ether oxygen by H^+

Q21 Ether cleavage by HX follows SN1 when the carbon attached to oxygen is:

A. Primary or methyl

B. Secondary or tertiary

C. Aromatic (aryl)

D. Any carbon, whatever its type

Q22 Ether cleavage by HX through the SN2 mechanism occurs at a:

A. Tertiary (3°) carbon, in a 2-step process

B. Secondary (2°) carbon, in a 2-step process

C. Primary (1°) carbon, in a 1-step process

D. Tertiary (3°) carbon, in a 1-step process

Q23 Complete: $CH_3OC(CH_3)_3$ + HBr →

A. CH_3OH + $(CH_3)_3CBr$, by SN1

B. CH_3Br + $(CH_3)_3COH$, by SN2

C. CH_3Br + $(CH_3)_3CBr$ + H_2O

D. No reaction

Q24 Complete: $CH_3OC_2H_5$ + 2 HI →

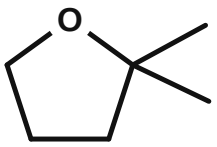
A. C_2H_5I + CH_3OH, by SN2

B. CH_3I + C_2H_5OH, by SN1

C. CH_3I + C_2H_5I + H_2O, by SN2

D. CH_3I + C_2H_5I + H_2O, by SN1

Q25 Complete the reaction of this cyclic ether with HCl:



A. HO–$CH_2CH_2CH_2$–$C(CH_3)_2Cl$, by SN1

B. HO–$CH_2CH_2CH_2$–$C(CH_3)_2Cl$, by SN2

C. Cl–$CH_2CH_2CH_2$–$C(CH_3)_2OH$, by SN2

D. Cl–$CH_2CH_2CH_2$–$C(CH_3)_2OH$, by SN1

Q26 Hint from the doctor: the halide attacks the R group that gives the more stable carbocation. Which order of carbocation stability is correct?

A. 1° > 2° > 3° > CH_3

B. 3° > 2° > 1° > CH_3

C. 2° > 3° > 1° > CH_3

D. CH_3 > 1° > 2° > 3°

7.5 Epoxides: Structure & Preparation

Q27 Epoxides are:

A. Six-membered cyclic ethers containing one oxygen atom

B. Five-membered cyclic ethers containing two oxygen atoms

C. Three-membered cyclic ethers containing one oxygen atom

D. Three-membered cyclic alcohols containing one OH group

Q28 The simplest epoxide is:

A. Tetrahydrofuran

B. Ethylene oxide

C. Diethyl ether

D. Dioxane

Q29 How do epoxides differ from other cyclic ethers (such as those shown below) in structure?

A. Other cyclic ethers have rings larger than the three-membered epoxide ring

B. Epoxides contain two oxygen atoms in the ring

C. Epoxides have a larger ring than other cyclic ethers

D. Other cyclic ethers have no oxygen in the ring

Q30 Ethylene oxide is made industrially from ethene by:

A. Addition of HBr followed by heating

B. Hydroboration followed by oxidation

C. Williamson synthesis with sodium ethoxide

D. Air oxidation with O_2 over a silver catalyst

Q31 Cyclohexene reacts with a peroxycarboxylic acid (RCO_3H) to give:

A. Cyclohexane-1,2-diol only

B. Cyclohexanol + RCOOH

C. Cyclohexane + RCO_3H

D. Cyclohexene oxide + RCOOH

7.6 Ring-Opening Reactions of Epoxides

Q32 Why are epoxides more reactive than ordinary ethers?

A. Their bond angles are widened from ~109° to ~120°

B. Their bond angles are distorted from ~109° to ~60°, causing ring strain

C. Their oxygen has no lone pairs, so it is very electron-poor

D. They contain a C=C double bond next to the oxygen

Q33 In acid-catalyzed ring opening of ethylene oxide with water, what happens in the **last** step of the mechanism?

A. Nucleophilic attack of H_2O on a ring carbon

B. Loss of a water molecule, giving an alkene

C. Loss of a hydrogen ion, giving the 1,2-diol (ethylene glycol)

D. Protonation of the epoxide oxygen

Q34 In acid-catalyzed ring opening of an epoxide with water, what is the correct order of the mechanism steps?

A. Protonation of the epoxide O → nucleophilic attack of H_2O → loss of a hydrogen ion

B. Protonation of the epoxide O → loss of a hydrogen ion → nucleophilic attack of H_2O

C. Nucleophilic attack of H_2O → protonation of the epoxide O → loss of a hydrogen ion

D. Loss of a hydrogen ion → protonation of the epoxide O → nucleophilic attack of H_2O

Q35 Acid-catalyzed hydrolysis of ethylene oxide gives:

A. Diethyl ether ($CH_3CH_2OCH_2CH_3$)

B. Ethylene glycol ($HOCH_2CH_2OH$)

C. Ethanol (CH_3CH_2OH)

D. Acetaldehyde (CH_3CHO)

Q36 Complete the reaction:

A. $HOCH_2CH_2OH$

B. $CH_3OCH_2CH_2OCH_3$

C. $CH_3OCH_2CH_3$

D. $HOCH_2CH_2OCH_3$

Q37 Complete the reaction:

A. $H_2NCH_2CH_2OH$

B. $HOCH_2CH_2OH$

C. $H_2NCH_2CH_2NH_2$

D. $CH_3CH_2NH_2$

Q38 A Grignard reagent (R–MgX) reacts with ethylene oxide, followed by water, to give:

A. $R–O–CH_2CH_3$

B. RCH_2CH_2OR

C. RCH_2CH_2OH

D. RCH_2OH

Q39 An organolithium reagent (R–Li) reacts with ethylene oxide. Before water is added, the intermediate formed is:

A. $R–O–CH_2CH_2Li$ (a lithium alkyl)

B. RCH_2CH_2OLi (a lithium alkoxide)

C. RCH_2CH_2Li (a lithium alkyl)

D. RCH_2OLi (a lithium alkoxide)

Q40 Epoxides undergo ring-opening reactions with which nucleophiles (HA)?

A. O_2, O_3, N_2, CO_2

B. Alkenes, alkynes, H_2, CH_4

C. H_2O, ROH, R_2NH, HX

D. H_2, CH_4, C_2H_6, N_2

Answers · Lecture 7

7.1 Ethers: Structure & Nomenclature

Q1 **D. R–O–R′**

Ethers: R–O–R, R–O–R′, Ar–O–R, Ar–O–Ar (an oxygen atom bonded to two carbon groups, alkyl or aryl).

Q2 **B. Naming the alkyl groups on oxygen in alphabetical order, followed by the word “ether”**

Common nomenclature: the alkyl groups connected to oxygen are named first in alphabetical order, followed by the word “ether”.

Q3 **A. An alkoxy substituent on the longest carbon chain**

1. Find the longest continuous carbon chain. 2. Identify the substituents (alkyl groups containing oxygen = alkoxy, e.g. methoxy). 3. Number the chain and list substituents alphabetically. Note: alcohol (OH) gets priority over ether in naming and numbering.

Q4 **C. Diisopropyl ether**

Both groups on oxygen are isopropyl, $(CH_3)_2CH–$: di-isopropyl ether.

Q5 **D. Ethyl methyl ether**

Ethyl and methyl, in alphabetical order, then “ether”: ethyl methyl ether.

Q6 **A. 3-Chloro-1-methoxybutane**

3-Chloro-1-methoxybutane: butane chain, methoxy at C-1, Cl at C-3; substituents listed alphabetically (chloro before methoxy).

Q7 **C. 3-Methoxy-1-propanol**

3-Methoxy-1-propanol (3-methoxypropan-1-ol): the OH has priority over the ether, so it gets C-1 and the -ol suffix.

Q8 **B. 1-Propoxyheptane**

1-Propoxyheptane: the longest chain (7 C, heptane) is the parent; the shorter group with oxygen is the propoxy substituent.

7.2 Physical Properties of Ethers

Q9 **D. Alcohol molecules form hydrogen bonds with each other; ether molecules cannot**

1-Butanol and diethyl ether have the same molecular weight (74), so mass is not the reason. Alcohols have higher boiling points than ethers because of H-bonding: alcohol molecules H-bond to each other (like water), while ether molecules have no O–H and cannot. (Note: the table writes the 1-butanol formula with one CH_2 too many; 1-butanol is $CH_3CH_2CH_2CH_2OH$, $C_4H_{10}O$.)

Q10 **A. There is no hydrogen bonding between ether molecules**

Ethers have no O–H, so there is no H-bonding between ether molecules; their boiling points are similar to alkanes of similar weight (35 °C vs 36 °C).

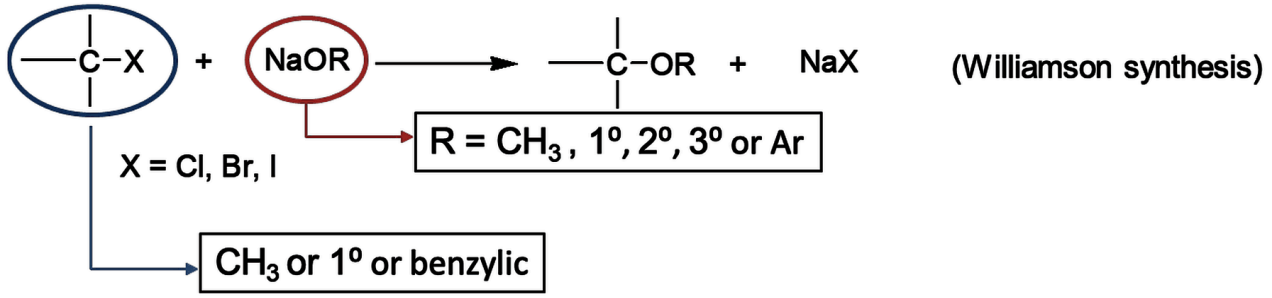
Q11 **D. Ethers can accept H-bonds from water through the oxygen lone pairs**

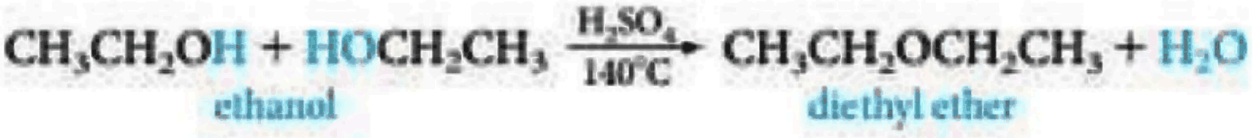
Ethers can accept H-bonds from water (through the O lone pairs) but cannot donate them. This makes them soluble in water. Pentane has no oxygen, so it cannot H-bond with water at all.

7.3 Preparation of Ethers

Q12 **C. Williamson synthesis and acid-catalyzed dehydration of alcohols**

Nucleophilic substitution (Williamson synthesis, SN2) and acid-catalyzed dehydration of alcohols.

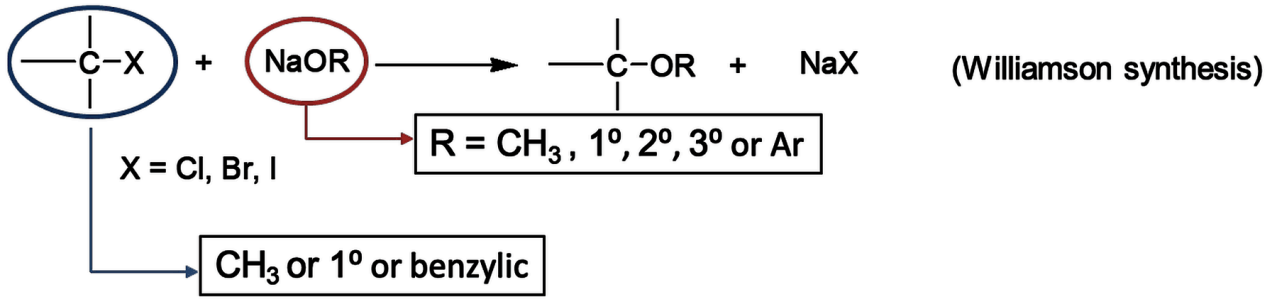




(Top: Williamson synthesis; bottom: dehydration of ethanol.)

Q13 **A. Methyl, primary or benzylic**

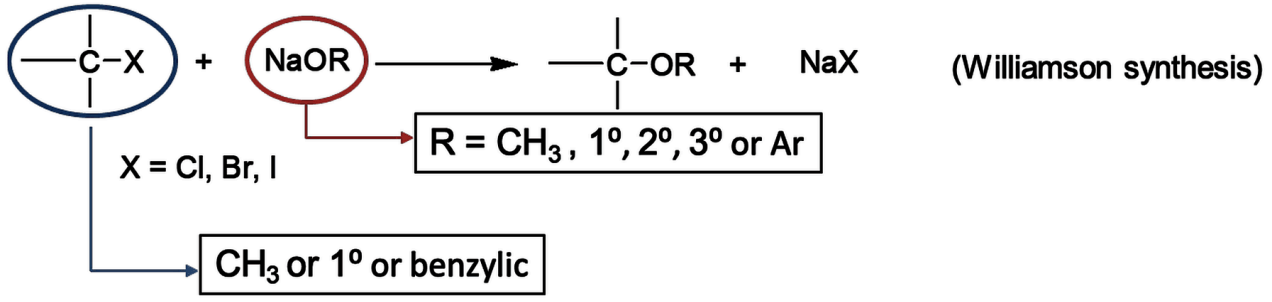
R–X + NaOR′ → R–OR′ + NaX (SN2). The alkyl halide must be CH_3, 1° or benzylic; the alkoxide R′ can be CH_3, 1°, 2°, 3° or aryl.



(In the figure, each box shows which groups are allowed on that reactant.)

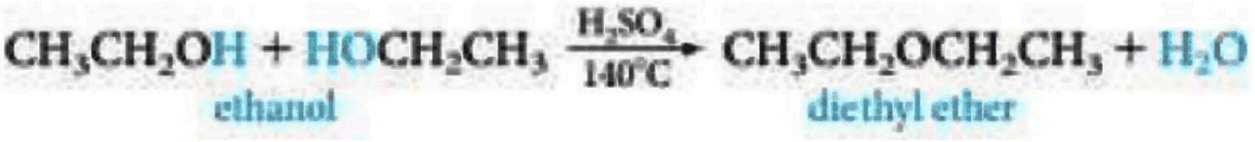
Q14 **B. CH_3, 1°, 2°, 3° or aryl**

From the doctor's figure: the alkoxide group (called R in the figure, R′ here) can be CH_3, 1°, 2°, 3° or Ar, while the alkyl halide must be CH_3, 1° or benzylic.



Q15 **C. Ethanol (CH_3CH_2OH)**

$CH_3CH_2OH + HOCH_2CH_3 \rightarrow (H_2SO_4, 140\ °C)\ CH_3CH_2OCH_2CH_3 + H_2O$. Starting material: ethanol; product: a symmetrical ether (diethyl ether).



(The doctor's model answer leaves this table empty; the answer comes from the dehydration equation in the doctor's Q6, preparation methods.)

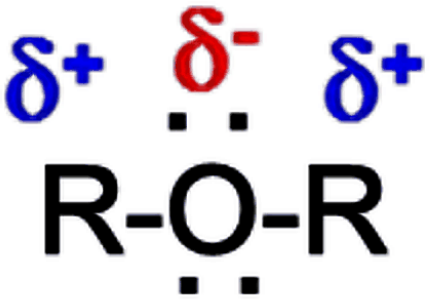
Q16 **D. $NaOCH_3 \rightarrow CH_3CH_2CH_2OCH_3$**

A sodium alkoxide (e.g. $NaOCH_3$) attacks the primary bromide by SN2: $CH_3CH_2CH_2Br + NaOCH_3 \rightarrow CH_3CH_2CH_2OCH_3 + NaBr$ (1-methoxypropane, an asymmetrical ether). (Not solved in the doctor's file.)

7.4 Reactions of Ethers with HX

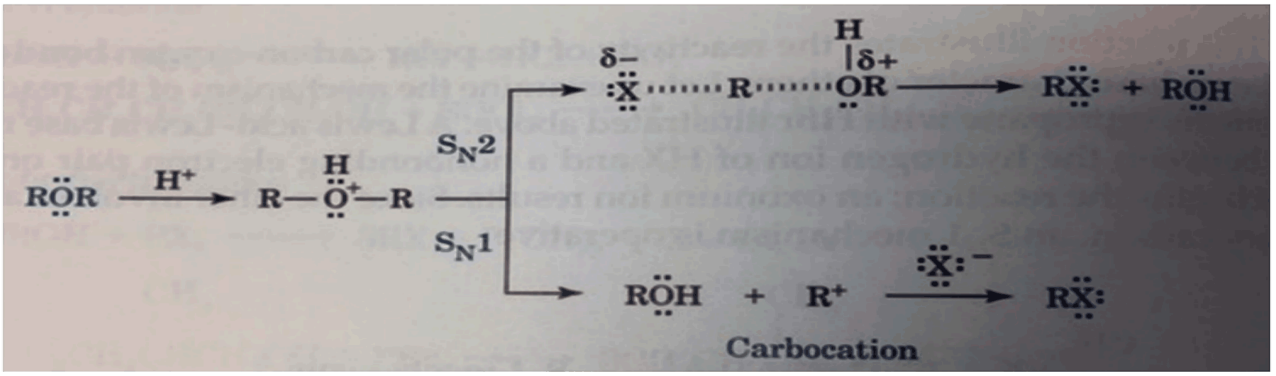
Q17 **D. Two polar C–O bonds and two non-bonding electron pairs on O**

1) Two polar bonds (C–O). 2) Two non-bonding electron pairs on the oxygen.



Q18 **A. R–X + R–OH**

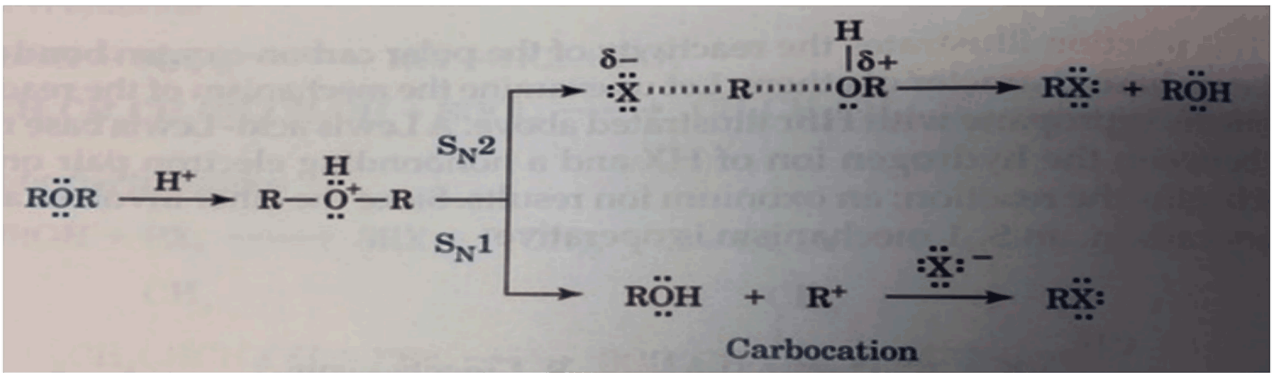
R–O–R + HX → R–X + R–OH. These are nucleophilic substitution reactions (SN1 and SN2).



In the figure, both paths end with R–X (RX) + R–OH (ROH).

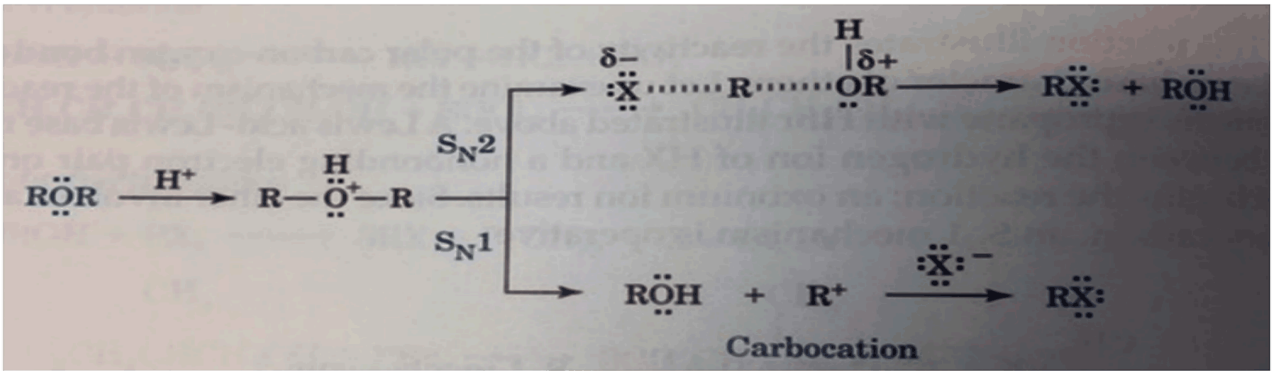
Q19 **C. 2 R–X + H_2O**

R–O–R + 2 HX → 2 R–X + H_2O (nucleophilic substitution, SN1 and SN2). The second HX converts the alcohol R–OH formed by the first HX into another R–X.



Q20 **D. Protonation of the ether oxygen by H^+**

H^+ protonates the oxygen (R–O^+H–R), creating a good leaving group; then X^- attacks by SN2, or the C–O bond breaks first to form a carbocation (SN1).



In the figure, the dotted lines in the SN2 branch are bonds being formed/broken at the same time.

Q21 **B. Secondary or tertiary**

SN1: cleavage occurs at the 2° or 3° carbon of the ether (2-step process). SN2: cleavage occurs at the 1° carbon (1-step process).

Q22 **C. Primary (1°) carbon, in a 1-step process**

SN2: cleavage of the ether occurs at the 1° carbon; 1-step process. SN1: cleavage occurs at the 2° or 3° carbon; 2-step process (via a carbocation).

Q23 **A. CH_3OH + $(CH_3)_3CBr$, by SN1**

Cover the O and compare: the tert-butyl group gives the more stable (3°) carbocation, so the halide attacks the tert-butyl carbon: CH_3OH + $BrC(CH_3)_3$. Doctor's answer: CH_3OH + $BrC(CH_3)_3$, SN1 (cleavage at the 3° carbon).

Q24 **C. CH_3I + C_2H_5I + H_2O, by SN2**

With two moles of HI both C–O bonds are cleaved: CH_3I + C_2H_5I (+ H_2O). Both carbons are methyl/1°, so the mechanism is SN2 / SN2.

Q25 **A. HO–$CH_2CH_2CH_2$–$C(CH_3)_2Cl$, by SN1**

The ring opens at the C–O bond to the tertiary carbon (more stable carbocation), so Cl ends up on the tertiary carbon and the oxygen stays as an OH on the CH_2 end: HO–$CH_2CH_2CH_2$–$C(CH_3)_2Cl$, 4-chloro-4-methylpentan-1-ol (SN1).

Q26 **B. 3° > 2° > 1° > CH_3**

More alkyl groups stabilize the positive charge.

7.5 Epoxides: Structure & Preparation

Q27 **C. Three-membered cyclic ethers containing one oxygen atom**

Epoxides are three-membered cyclic ethers (also known as oxiranes) containing one oxygen atom. The simplest is ethylene oxide. Other cyclic ethers (e.g. THF, 5-membered; dioxane, 6-membered) have larger rings and are not epoxides.

Q28 **B. Ethylene oxide**

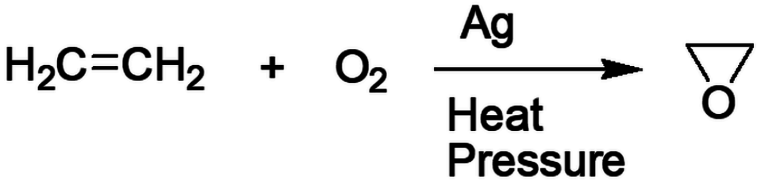
The simplest form is ethylene oxide (C_2H_4O).

Q29 **A. Other cyclic ethers have rings larger than the three-membered epoxide ring**

Epoxides are three-membered cyclic ethers. Cyclic ethers have rings that are larger than three-membered epoxides (e.g. the six- and five-membered rings shown).

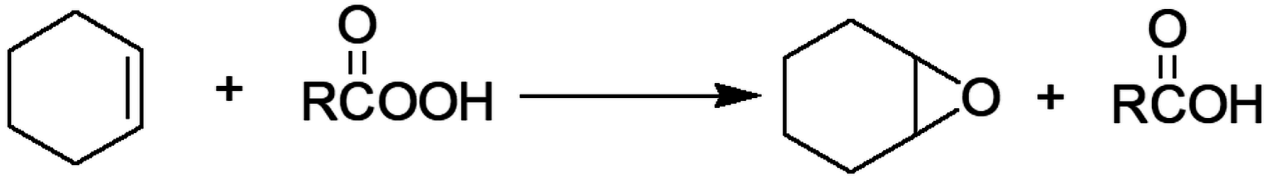
Q30 **D. Air oxidation with O_2 over a silver catalyst**

From alkenes: air oxidation of ethene, $H_2C{=}CH_2 + O_2$ (Ag catalyst, heat, pressure) → ethylene oxide.



Q31 **D. Cyclohexene oxide + RCOOH**

From alkenes using peroxycarboxylic acids: the peracid transfers one oxygen to the C=C, forming the epoxide and a carboxylic acid.



(RC(=O)OOH is the peroxy acid; RC(=O)OH is the acid left over.)

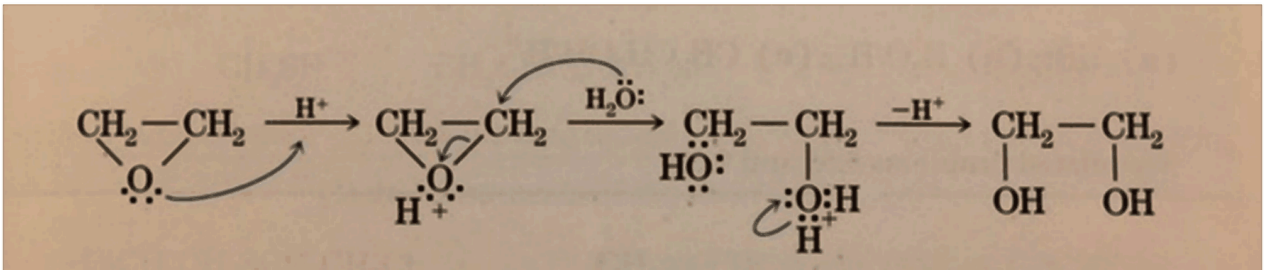
7.6 Ring-Opening Reactions of Epoxides

Q32 **B. Their bond angles are distorted from ~109° to ~60°, causing ring strain**

Epoxides suffer from acute angle strain because of distortion of normal bond angles from 109° to ~60°. This strain is relieved by cleavage of a polar C–O bond under acidic or basic conditions.

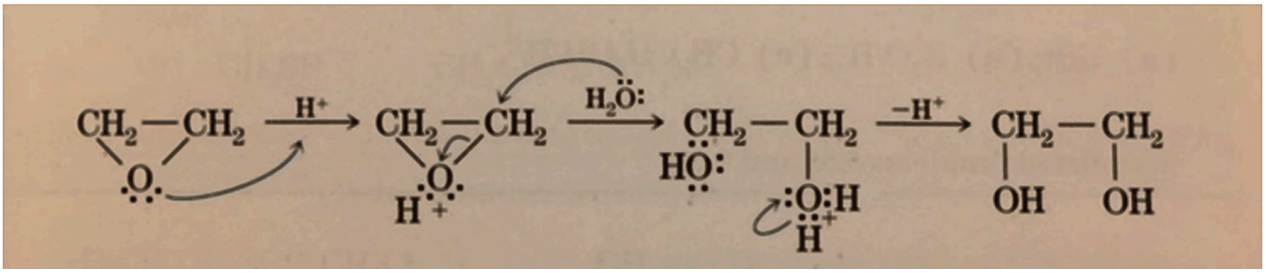
Q33 **C. Loss of a hydrogen ion, giving the 1,2-diol (ethylene glycol)**

Mechanism: 1. The epoxide oxygen is protonated. 2. Nucleophilic attack of an H_2O molecule. 3. Loss of a hydrogen ion → 1,2-diol (ethylene glycol from ethylene oxide).



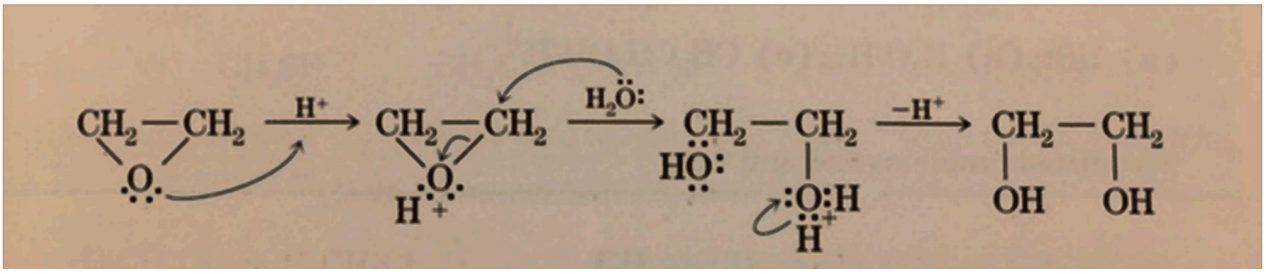
Q34 **A. Protonation of the epoxide O → nucleophilic attack of H_2O → loss of a hydrogen ion**

Doctor's mechanism: 1. The epoxide oxygen is protonated. 2. Nucleophilic attack of an H_2O molecule. 3. Loss of a hydrogen ion.



Q35 **B. Ethylene glycol ($HOCH_2CH_2OH$)**

Water opens the ring: HO–CH_2–CH_2–OH (1,2-ethanediol).



Q36 **D. $HOCH_2CH_2OCH_3$**

CH_3OH + ethylene oxide (H^+) → HO–CH_2–CH_2–OCH_3, 2-methoxyethanol: methanol is the nucleophile that opens the ring.

Q37 **A. $H_2NCH_2CH_2OH$**

NH_3 opens the ring: H_2N–CH_2–CH_2–OH, ethanolamine (2-aminoethanol).

Q38 **C. RCH_2CH_2OH**

R–MgX + ethylene oxide → RCH_2CH_2OMgX (a magnesium alkoxide); then H–OH → RCH_2CH_2OH + Mg(OH)X.

Q39 **B. RCH_2CH_2OLi (a lithium alkoxide)**

R–Li + ethylene oxide → RCH_2CH_2OLi (a lithium alkoxide); then H–OH → RCH_2CH_2OH + LiOH. It reacts the same way as a Grignard reagent.

Q40 **C. H_2O, ROH, R_2NH, HX**

General reaction: epoxide + H–A → A–C–C–OH, where HA = H_2O, ROH, R_2NH, HX (X = Cl, Br, I). NH_3 reacts the same way (ethanolamine).

8 Alcohols

In this lecture

1. Structure & Classification of Alcohols 11 questions
2. Nomenclature of Alcohols 13 questions
3. Physical Properties of Alcohols 5 questions
4. Preparation of Alcohols 14 questions
5. Reactions of Alcohols 26 questions

8.1 Structure & Classification of Alcohols

Q1 An alcohol (ROH) is:

A. An alkane in which one CH_2 is replaced by an O atom

B. An alkane in which one hydrogen is replaced by an OH group

C. An alkane in which one CH_2 is replaced by a C=O group

D. An alkane in which one CH_3 is replaced by a COOH group

Q2 Alcohols are classified as 1°, 2° or 3° according to:

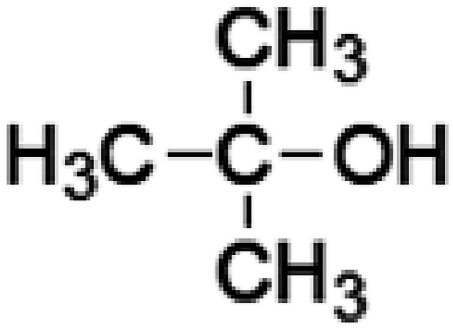
A. The number of OH groups attached to the carbon chain

B. The number of carbon atoms in the longest chain

C. The number of alkyl (R) groups bonded directly to the carbon carrying the OH

D. The number of hydrogen atoms bonded to the carbon carrying the OH

Q3 Classify this alcohol:



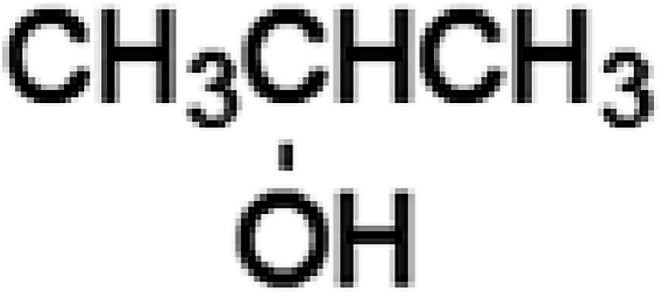
A. Tertiary (3°)

B. Primary (1°)

C. Secondary (2°)

D. Quaternary

Q4 Classify this alcohol:

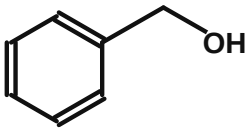


A. Quaternary
B. Tertiary (3°)
C. Primary (1°)
D. Secondary (2°)

Q5 Classify this alcohol:

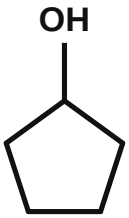
A. Secondary (2°)
B. Phenol
C. Tertiary (3°)
D. Primary (1°)

Q6 Classify this alcohol:



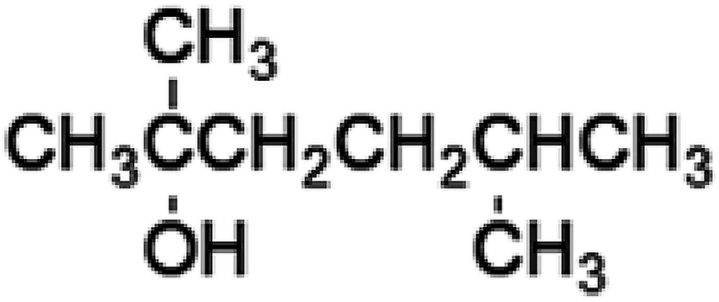
A. Primary (1°)
B. A phenol
C. Tertiary (3°)
D. Secondary (2°)

Q7 Classify this alcohol:



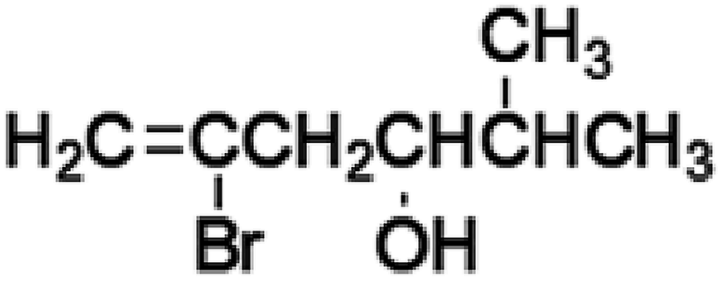
A. Primary (1°)
B. Quaternary
C. Secondary (2°)
D. Tertiary (3°)

Q8 Classify this alcohol:



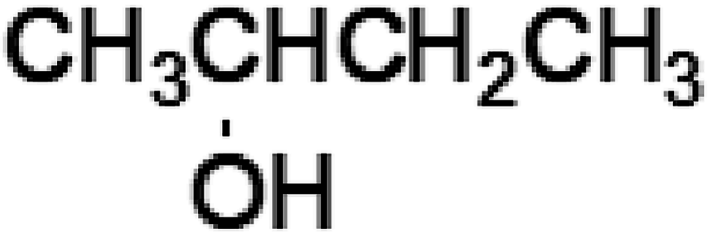
A. Primary (1°)
B. Tertiary (3°)
C. Quaternary
D. Secondary (2°)

Q9 Classify this alcohol:



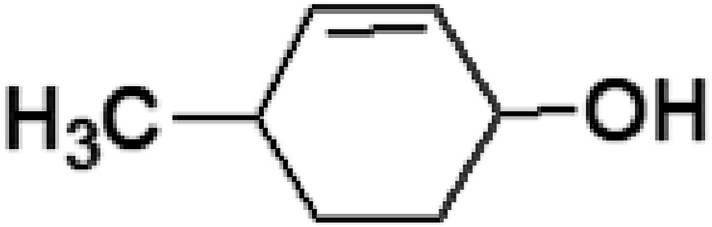
A. Secondary (2°)
B. Quaternary
C. Primary (1°)
D. Tertiary (3°)

Q10 Classify this alcohol:



A. Primary (1°)
B. Secondary (2°)
C. Quaternary
D. Tertiary (3°)

Q11 Classify this alcohol:



A. Tertiary (3°)
B. Quaternary
C. Secondary (2°)
D. Primary (1°)

8.2 Nomenclature of Alcohols

Q12 In IUPAC naming of alcohols, the parent chain is:

A. The chain with the most branches attached to it
B. The longest chain, even if it does not contain the OH
C. The shortest chain that contains the OH group
D. The longest continuous carbon chain containing the OH group

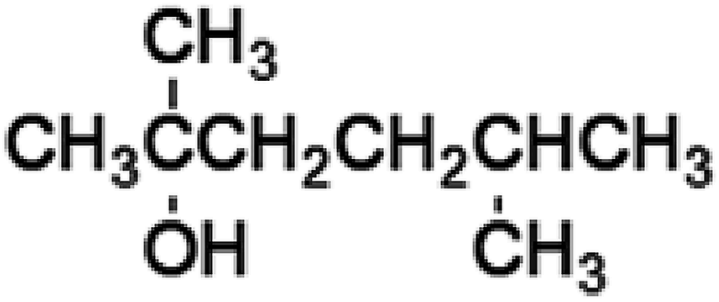
Q13 When numbering an alcohol chain, the lowest number goes first to:

A. The triple bond
B. The halogen
C. The carbon carrying the OH group
D. The double bond

Q14 What is the IUPAC name of this compound?

A. 2-Bromo-3-hexyn-6-ol
B. 5-Bromo-3-hexyn-1-ol
C. 5-Bromo-3-hexen-1-ol
D. 2-Bromo-3-hexen-6-ol

Q15 What is the IUPAC name of this compound?



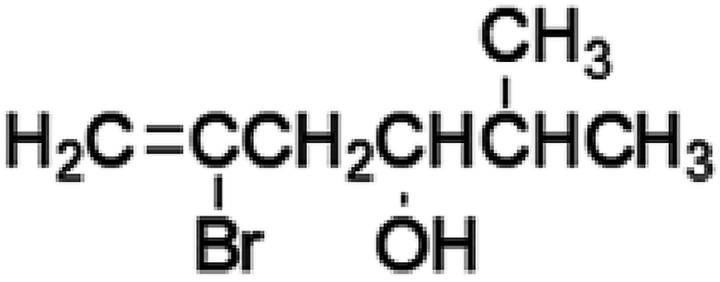
A. 2,5-Dimethylhexan-2-ol

B. 2-Methyl-5-methylhexan-2-ol

C. Octan-2-ol

D. 2,5-Dimethylhexan-5-ol

Q16 What is the IUPAC name of this compound?



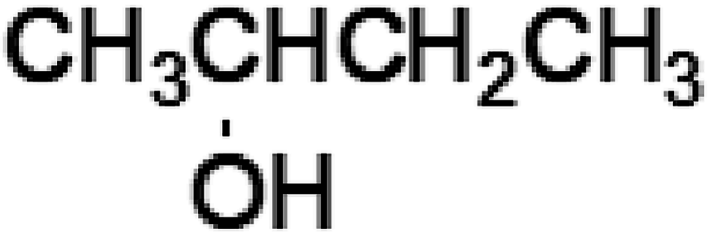
A. 2-Bromo-5-methylhex-1-ene

B. 2-Bromo-5-methyl-1-hexen-4-ol

C. 5-Bromo-2-methylhexan-3-ol

D. 5-Bromo-2-methyl-5-hexen-3-ol

Q17 What is the IUPAC name of $CH_3CH(OH)CH_2CH_3$?



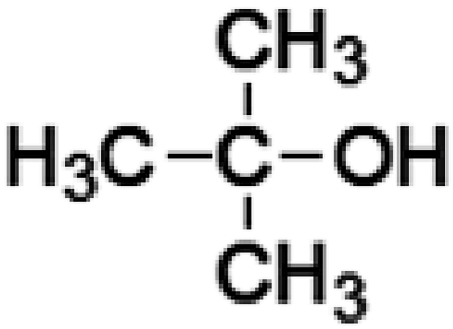
A. 2-Butanone

B. 2-Butanol

C. 1-Butanol

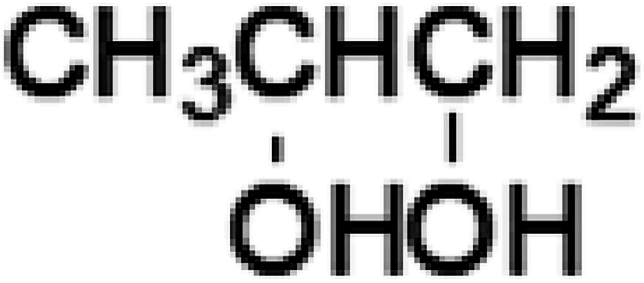
D. 1-Butanone

Q18 What is the IUPAC name of this compound?



A. 2-Methyl-1-propanol
B. 2,2-Dimethylethanol
C. 2-Methyl-2-propanol
D. 2-Butanol

Q19 What is the IUPAC name of this compound?

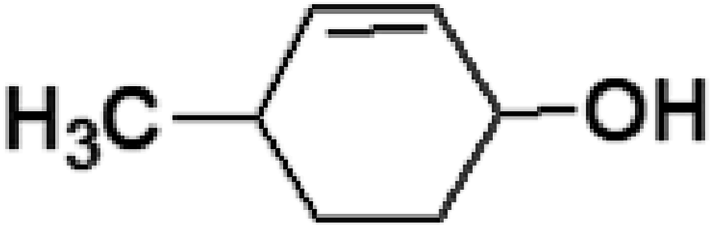


A. 1,2-Propanediol
B. 2,3-Propanediol
C. 1,2-Propanetriol
D. 1,2-Propanol

Q20 What is the IUPAC name of this compound?

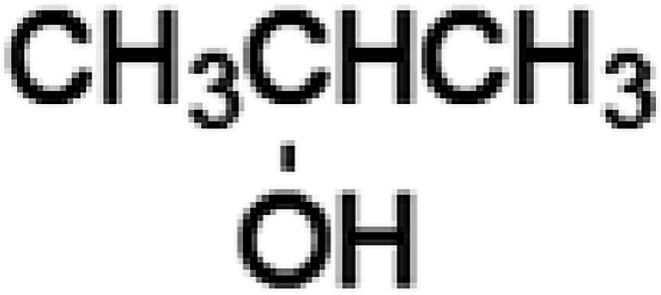
A. 1,2,3-Propanol
B. 1,2,3-Propanetriol
C. 2-Propanol
D. 1,3-Propanediol

Q21 What is the IUPAC name of this compound?



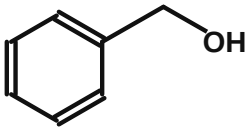
A. 4-Methyl-2-cyclohexen-1-ol
B. 4-Methylcyclohexanol
C. 4-Methyl-3-cyclohexen-1-ol
D. 1-Methyl-3-cyclohexen-4-ol

Q22 What is the IUPAC name of this compound?



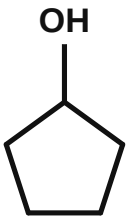
A. 2-Propanone
B. Propan-1,2-ol
C. 1-Propanol
D. 2-Propanol

Q23 What is the IUPAC name of this compound?



A. Methylphenol
B. Phenylethanol
C. Phenylmethanol
D. Phenol

Q24 What is the IUPAC name of this compound?



A. Cyclopentanone
B. Cyclohexanone
C. Cyclopentanol
D. Cyclohexanol

8.3 Physical Properties of Alcohols

Q25 Which molecule has the highest boiling point?

A. $CH_3(CH_2)_5CH_2OH$

B. $CH_3OCH_2CH_3$

C. $CH_3CH_2CH_3$

D. $CH_3CH_2CH_2OH$

Q26 What can be concluded from the table?

	mwt	bp °C
CH_3OH	32	65
CH_3CH_2OH	46	78.5
$CH_3CH_2CH_2OH$	60	97
$CH_3(CH_2)_3CH_2OH$	80*	137
$CH_3(CH_2)_5CH_2OH$	116	176

*As written in the doctor's table.

A. Boiling point stays the same, whatever the molecular weight

B. Boiling point decreases with increasing molecular weight within a homologous series

C. Boiling point increases with increasing molecular weight within a homologous series

D. Boiling point depends only on the position of the OH group

Q27 Why do alcohols boil higher than alkanes and ethers of similar molecular weight?

	mwt	bp °C
CH_3OH	32	65
CH_3CH_3	30	−89
$CH_3CH_2CH_2OH$	60	97
$CH_3OCH_2CH_3$	60	11

A. Alcohols form hydrogen bonds between their molecules

B. Ethers and alkanes form stronger hydrogen bonds

C. Alcohols are ionic compounds with strong charges

D. Alcohols have a much higher molecular weight

Q28 Which statement about the water solubility of alcohols is correct?

A. Low-Mwt alcohols are insoluble; solubility rises as the hydrocarbon part grows

B. Low-Mwt alcohols are soluble; solubility falls as the hydrocarbon part grows

C. Solubility in water rises as the hydrocarbon part grows

D. All alcohols, whatever their size, mix fully with water

Q29 Hydrogen bonding occurs when hydrogen is bonded to:

A. A metal atom (Na, K)

B. A halogen only (Cl, Br, I)

C. A carbon atom (C)

D. A strongly electronegative atom (N, O or F)

8.4 Preparation of Alcohols

Q30 Which is **NOT** a method of preparing alcohols?

A. Hydration of alkenes

B. Reduction of aldehydes and ketones

C. Nucleophilic substitution of alkyl halides

D. Oxidation of alcohols with PCC

Q31 Ethene + H_2O / H^+ gives:

A. CH_3CH_2OH

B. $HOCH_2CH_2OH$

C. CH_3OH

D. CH_3CHO

Q32 $CH_3CH_2CH_2Br$ + NaOH gives:

A. $CH_3CH_2CH_2OH$, by electrophilic addition

B. $CH_3CH_2CH_2OH$, by nucleophilic substitution

C. $CH_3CH_2CH_2ONa$, by nucleophilic substitution

D. $CH_3CH_2CH_2ONa$, by electrophilic addition

Q33 Catalytic hydrogenation (H_2, Ni, pressure) of a ketone gives:

A. A tertiary alcohol

B. A secondary alcohol

C. A primary alcohol

D. A carboxylic acid

Q34 In the doctor's table, acetone (CH_3COCH_3) is reduced with $NaBH_4$ or $LiAlH_4$. The product is:

A. $(CH_3)_3COH$ (a 3° alcohol)

B. CH_3COOH (a carboxylic acid)

C. $CH_3CH(OH)CH_3$ (a 2° alcohol)

D. $CH_3CH_2CH_2OH$ (a 1° alcohol)

Q35 In the doctor's table, the aldehyde RCH_2CHO is reduced with $NaBH_4$ or $LiAlH_4$. The product is:

A. RCH_2CH_2OH (a 1° alcohol)

B. $RCH_2CH(OH)CH_3$ (a 2° alcohol)

C. RCH_2COOH (a carboxylic acid)

D. $RCH{=}CH_2$ (an alkene)

Q36 $NaBH_4$ and $LiAlH_4$ reduce aldehydes and ketones because they act as:

A. Free-radical sources

B. Oxidizing agents

C. Proton (H^+) donors

D. Hydride ($H:^-$) donors

Q37 A Grignard reagent (RMgX) reacts with formaldehyde, then H_2O/H^+, to give:

A. A secondary alcohol

B. A primary alcohol

C. A tertiary alcohol

D. A ketone

Q38 A Grignard reagent reacts with an aldehyde other than formaldehyde (HCOR′), then H_2O/H^+, to give:

A. A primary alcohol

B. A tertiary alcohol

C. A carboxylic acid

D. A secondary alcohol

Q39 A Grignard reagent reacts with a ketone (R″COR′), then H_2O/H^+, to give:

A. A tertiary alcohol

B. A primary alcohol

C. A secondary alcohol

D. An aldehyde

Q40 Grignard reagents are prepared from:

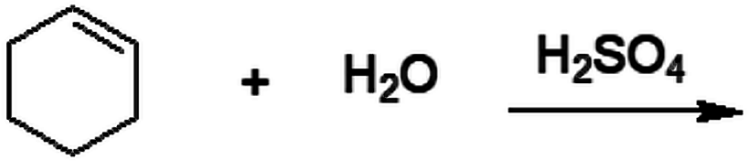
A. An alkene + H_2O with an acid catalyst

B. A ketone + $NaBH_4$ in water

C. An alkyl or aryl halide + Mg in diethyl ether

D. An alcohol + Na metal in dry diethyl ether

Q41 Complete: cyclohexene + H_2O / H_2SO_4 →



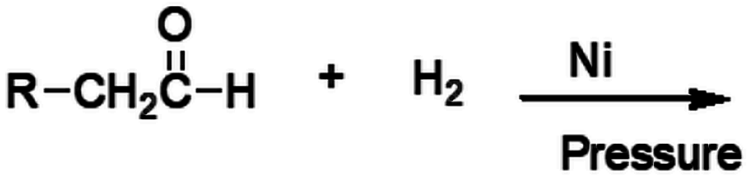
A. Cyclohexane-1,2-diol

B. Cyclohexanone

C. Cyclohexanol

D. Cyclohexane

Q42 Complete: $RCH_2CHO + H_2$ (Ni, pressure) →



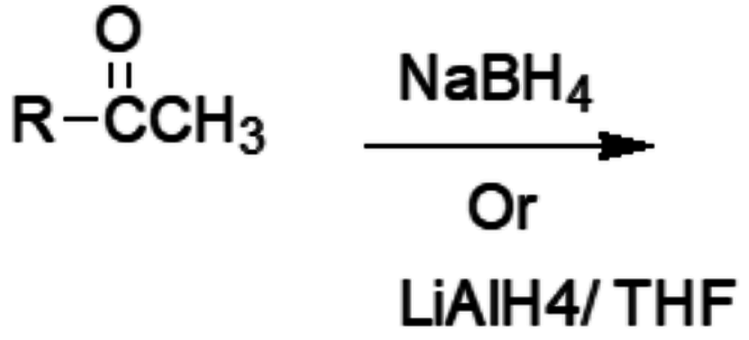
A. $RCH_2CH(OH)CH_3$

B. $RCH{=}CH_2$

C. RCH_2COOH

D. RCH_2CH_2OH

Q43 Complete: $RCOCH_3 + NaBH_4$ (or $LiAlH_4$/THF) →



A. RCH_2CH_3

B. $RCH(OH)CH_3$

C. RCOOH

D. RCH_2CH_2OH

8.5 Reactions of Alcohols

Q44 The reaction sites in an alcohol are:

A. One polar C–O bond and two non-bonding electron pairs on O only

B. Two polar bonds (C–O and O–H) and two non-bonding electron pairs on O

C. Only the C–H bonds of the alkyl group attached to O

D. A polar C=O bond, the O–H bond and one lone pair on O

Q45 Alcohols react with sodium metal to give:

A. An alkene (C=C) + NaOH

B. Sodium hydroxide (NaOH) + RH

C. Sodium alkoxide (RONa) + H_2

D. Alkylsodium (RNa) + H_2O

Q46 By which mechanism do alcohols react with hydrogen halides to produce alkyl halides?

A. Nucleophilic substitution

B. Elimination

C. Electrophilic substitution

D. Addition

Q47 Which statement about SN1 vs SN2 for alcohols + HX is correct?

A. SN1: 1 step, 2° and 3° alcohols; SN2: 2 steps, methyl and 1° alcohols, carbocation

B. SN1: 1 step, methyl and 1° alcohols; SN2: 2 steps, 2° and 3° alcohols, carbocation

C. SN1: 2 steps, methyl and 1° alcohols, carbocation; SN2: 1 step, 2° and 3° alcohols

D. SN1: 2 steps, 2° and 3° alcohols, carbocation; SN2: 1 step, methyl and 1° alcohols

Q48 The Lucas test (HCl in $ZnCl_2$) distinguishes 1°, 2° and 3° alcohols because:

A. Tertiary alcohols do not react with the reagent

B. They give different colours with $ZnCl_2$

C. They react at different rates, depending on carbocation stability

D. Only primary alcohols react, as they form the most stable carbocation

Q49 Besides HX, which reagents convert alcohols into alkyl halides?

A. $SOCl_2$ and PX_3

B. H_2SO_4 and H_3PO_4

C. $NaBH_4$ and $LiAlH_4$

D. PCC and CrO_3

Q50 By which mechanism do alcohols react with H_2SO_4 to produce alkenes (dehydration)?

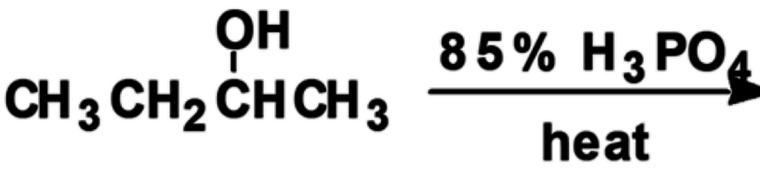
A. Addition reaction

B. Nucleophilic substitution

C. Electrophilic substitution

D. Elimination reaction

Q51 Dehydration of the alcohol shown (85% H_3PO_4, heat). What is the major product?



A. Butane

B. 2-Butene

C. 1-Butene

D. 2-Butanone

Q52 Oxidation of alcohols with Jones' reagent (CrO_3 in H_2SO_4) distinguishes them because:

A. 1° give aldehydes, 2° do not react, 3° give ketones

B. 1° do not react, 2° give ketones, 3° give carboxylic acids

C. 1° give carboxylic acids, 2° give ketones, 3° do not react

D. 1° give ketones, 2° give carboxylic acids, 3° do not react

Q53 Oxidizing a primary alcohol with PCC gives:

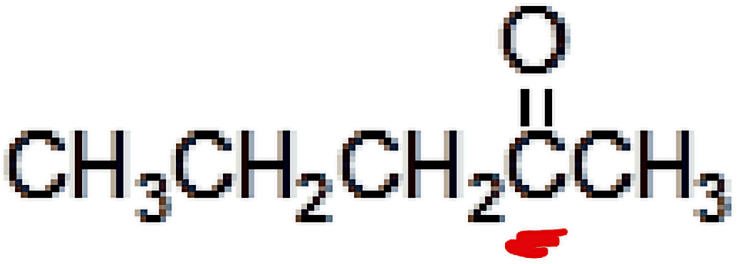
A. An aldehyde

B. No reaction

C. A ketone

D. A carboxylic acid

Q54 Which alcohol and oxidizing agent give $CH_3CH_2CH_2COCH_3$ (2-pentanone)?



A. 1-Pentanol + PCC

B. 1-Pentanol + CrO_3

C. 2-Methyl-2-butanol + CrO_3

D. 2-Pentanol + CrO_3

Q55 Which alcohol and reagent give $CH_3CH_2CH_2CH_2CHO$ (pentanal)?

A. Pentanoic acid + PCC

B. 1-Pentanol + PCC

C. 2-Pentanol + PCC

D. 1-Pentanol + CrO_3/H^+

Q56 Which alcohol and oxidizing agent give $CH_3CH_2CH_2CH_2COOH$ (pentanoic acid)?

A. 2-Pentanol + PCC

B. 1-Pentanol + CrO_3

C. 2-Pentanol + CrO_3

D. 1-Pentanol + PCC

Q57 A secondary alcohol (R_2CHOH) is oxidized with PCC or CrO_3/H^+. The product is:

A. A carboxylic acid

B. An aldehyde

C. A ketone

D. No reaction

Q58 Alcohols react with carboxylic acids (H_2SO_4 catalyst) to form:

A. Alkenes + water

B. Ketones + water

C. Ethers + water

D. Esters + water

Q59 Complete: 2-propanol + HBr → (mechanism?)

A. 2-Bromopropane + H_2O, SN1

B. 1-Bromopropane + H_2O, SN1

C. 2-Bromopropane + H_2O, SN2

D. 1-Bromopropane + H_2O, SN2

Q60 Complete: $CH_3CH_2CH_2OH + SOCl_2 \rightarrow$

A. $CH_3CH{=}CH_2 + SO_2 + HCl$

B. $CH_3CH_2CH_2SO_2Cl + HCl$

C. $CH_3CH_2CHO + SO_2 + HCl$

D. $CH_3CH_2CH_2Cl + SO_2 + HCl$

Q61 Complete: $3\ CH_3CH(OH)CH_2CH_3 + PBr_3 \rightarrow$

A. $3\ CH_3CH{=}CHCH_3 + H_3PO_3$

B. $3\ CH_3CHBrCH_2CH_3 + H_3PO_3$

C. $3\ CH_3CH_2CH_2CH_2Br + H_3PO_3$

D. $3\ CH_3COCH_2CH_3 + H_3PO_3$

Q62 Complete: $CH_3CH(OH)CH(CH_3)CH_3 + CrO_3/H^+ \rightarrow$

A. $CH_3COCH(CH_3)_2$

B. $(CH_3)_2C{=}CHCH_3$

C. $(CH_3)_2CHCH_2COOH$

D. No reaction

Q63 Complete: $CH_3(CH_2)_{10}CH_2OH + PCC \rightarrow$

A. $CH_3(CH_2)_9CH{=}CH_2$

B. No reaction

C. $CH_3(CH_2)_{10}CHO$

D. $CH_3(CH_2)_{10}COOH$

Q64 Complete: $CH_3CH(OH)CH_2CH_3 + Na \rightarrow$

A. $CH_3CH{=}CHCH_3$

B. $CH_3CH(Na)CH_2CH_3$

C. $CH_3COCH_2CH_3$

D. $CH_3CH(ONa)CH_2CH_3$

Q65 Complete: $CH_3CH(OH)CH_2CH_3 + H_2SO_4 \rightarrow$. Which alkene is the **minor** product?

A. $CH_3CH{=}CHCH_3$

B. $CH_2{=}CHCH_2CH_3$

C. $CH_2{=}C(CH_3)_2$

D. $CH_3CH_2CH_2CH_3$

Q66 Complete: $CH_3CH(OH)CH_2CH_3 + HCl,\ ZnCl_2 \rightarrow$

A. $CH_3CH(Cl)CH_2CH_3$

B. $CH_3COCH_2CH_3$

C. $CH_3CH{=}CHCH_3$

D. $CH_3CH_2CH_2CH_2Cl$

Q67 Complete: $CH_3CH(OH)CH_2CH_3 + CrO_3/H^+ \rightarrow$

A. $CH_3CH_2CH_2COOH$

B. $CH_3CH_2CH_2CHO$

C. $CH_3COCH_2CH_3$

D. No reaction

Q68 Dehydration of 3-methyl-2-butanol with H_2SO_4 gives mainly:

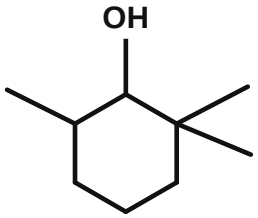
A. $(CH_3)_2CHCH{=}CH_2$

B. $(CH_3)_2CHCH_2CH_3$

C. $CH_3COCH(CH_3)_2$

D. $(CH_3)_2C{=}CHCH_3$

Q69 What is the major product of dehydration of this alcohol with H_2SO_4?



A. 1,1,2-Trimethylcyclohexane + H_2O

B. 1,3,3-Trimethylcyclohexene + H_2O

C. 2,2,6-Trimethylcyclohexanone + H_2O

D. No reaction; the alcohol is recovered

Answers · Lecture 8

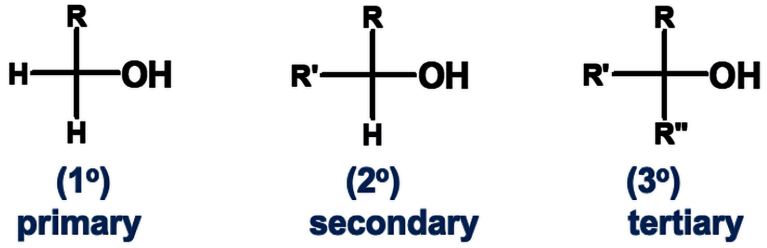
8.1 Structure & Classification of Alcohols

Q1 **B. An alkane in which one hydrogen is replaced by an OH group**

ROH: an alkane in which a hydrogen is replaced with OH; structurally similar to water with one H replaced by an alkyl (R) group.

Q2 **C. The number of alkyl (R) groups bonded directly to the carbon carrying the OH**

1°, 2°, 3° according to the number of alkyl (R) groups directly bonded to the alcohol carbon.



Q3 **A. Tertiary (3°)**

The OH carbon is bonded to three CH_3 groups → 3°.

Q4 **D. Secondary (2°)**

The OH carbon carries two CH_3 groups → 2°.

Q5 **D. Primary (1°)**

The OH is on a CH_2 at the end of the chain (one carbon neighbour) → 1°.

Q6 **A. Primary (1°)**

The OH carbon (CH_2) is bonded to one carbon (the ring) → 1°.

Q7 **C. Secondary (2°)**

The OH ring carbon is bonded to two ring carbons → 2°.

Q8 **B. Tertiary (3°)**

The OH carbon (C-2) is bonded to three carbons (two CH_3 and the chain) → 3°.

Q9 **A. Secondary (2°)**

The OH carbon (C-3) is bonded to two carbons (C-2 and C-4) → 2°.

Q10 **B. Secondary (2°)**

The OH carbon is bonded to CH_3 and CH_2CH_3 → 2°.

Q11 **C. Secondary (2°)**

The OH ring carbon is bonded to two ring carbons → 2°.

8.2 Nomenclature of Alcohols

Q12 **D. The longest continuous carbon chain containing the OH group**

1. Longest chain containing the OH. 2. Suffix -an / -en / -yn for the chain. 3. Add -ol. 4. Number giving the lowest number to the OH, then double, then triple bonds, then prefix groups. 5. Name the other groups with numbers.

Q13 **C. The carbon carrying the OH group**

Number the C chain giving the lowest number to the OH group, followed by double then triple bonds, and finally groups named by prefixes.

Q14 **B. 5-Bromo-3-hexyn-1-ol**

Six carbons with a triple bond (-yn); OH on C-1, triple bond C-3, Br on C-5: 5-bromo-3-hexyn-1-ol (5-bromohex-3-yn-1-ol). It is a 1° alcohol.

Q15 **A. 2,5-Dimethylhexan-2-ol**

Six-carbon chain; OH on C-2 (lowest number); methyl groups on C-2 and C-5.

Q16 **D. 5-Bromo-2-methyl-5-hexen-3-ol**

OH gets the lowest number (C-3) when numbering from the right; double bond at C-5, Br at C-5, CH_3 at C-2.

Q17 **B. 2-Butanol**

Four carbons, OH on C-2 → 2-butanol.

Q18 **C. 2-Methyl-2-propanol**

Three-carbon chain with OH and CH_3 on C-2 → 2-methyl-2-propanol (tert-butyl alcohol).

Q19 **A. 1,2-Propanediol**

Two OH groups → “-diol”; on C-1 and C-2.

Q20 **B. 1,2,3-Propanetriol**

Three OH groups on three carbons → 1,2,3-propanetriol (glycerol).

Q21 **A. 4-Methyl-2-cyclohexen-1-ol**

The OH carbon is C-1; number toward the double bond so it gets C-2/C-3; the CH_3 is on C-4. It is a 2° alcohol.

Q22 **D. 2-Propanol**

Three carbons with the OH on C-2 → 2-propanol (propan-2-ol). It is a 2° alcohol.

Q23 **C. Phenylmethanol**

A CH_2OH (methanol) carrying a phenyl group → phenylmethanol (benzyl alcohol). It is a 1° alcohol.

Q24 **C. Cyclopentanol**

A five-membered ring with an OH → cyclopentanol. It is a 2° alcohol.

8.3 Physical Properties of Alcohols

Q25 **A. $CH_3(CH_2)_5CH_2OH$**

$CH_3(CH_2)_5CH_2OH$: alcohols boil higher than the alkane ($CH_3CH_2CH_3$) and the ether ($CH_3OCH_2CH_3$), and it is heavier than $CH_3CH_2CH_2OH$.

Q26 **C. Boiling point increases with increasing molecular weight within a homologous series**

Melting point and boiling point increase with increasing molecular weight within a homologous series: 32 → 46 → 60 → 88 → 116 g/mol gives 65 → 78.5 → 97 → 137 → 176 °C. (The doctor's table lists 80 for $CH_3(CH_2)_3CH_2OH$; the correct molecular weight of 1-pentanol, $C_5H_{12}O$, is 88.)

Q27 **A. Alcohols form hydrogen bonds between their molecules**

Alcohols have higher bp compared to alkanes and ethers with the same Mwt because of H-bonding between alcohol molecules.

Q28 **B. Low-Mwt alcohols are soluble; solubility falls as the hydrocarbon part grows**

Alcohols hydrogen-bond with water, so low-Mwt alcohols are water soluble. As Mwt increases, the hydrocarbon portion grows and the alcohol becomes more like an alkane → less soluble.

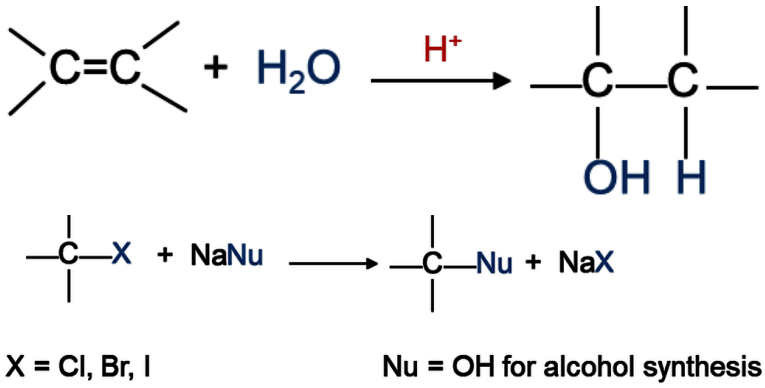
Q29 **D. A strongly electronegative atom (N, O or F)**

H-bonding occurs in molecules where H is bonded to an electronegative atom (N, O, F). (The doctor's file writes H_2; it means a hydrogen atom, H.)

8.4 Preparation of Alcohols

Q30 **D. Oxidation of alcohols with PCC**

Oxidation of alcohols with PCC is a reaction OF alcohols (it turns them into aldehydes or ketones); it does not make alcohols. The three methods are: 1) hydration of alkenes; 2) nucleophilic substitution (R–X + OH^-); 3) reduction of aldehydes and ketones (catalytic hydrogenation, $NaBH_4$ or $LiAlH_4$, Grignard addition). The figures show hydration and substitution (where Nu = OH gives an alcohol).



Q31 **A. CH_3CH_2OH**

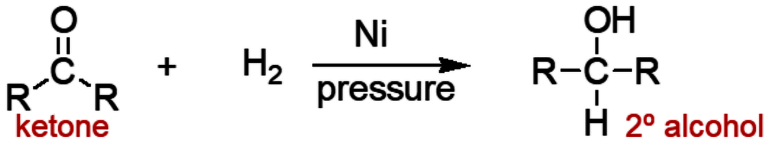
Hydration of alkenes: $H_2C{=}CH_2 + H_2O/H^+ \rightarrow CH_3CH_2OH$ (1°): H adds to one carbon and OH to the other.

Q32 **B. $CH_3CH_2CH_2OH$, by nucleophilic substitution**

Reaction of an alkyl halide with hydroxide ion is a nucleophilic substitution: $CH_3CH_2CH_2Br + NaOH \rightarrow CH_3CH_2CH_2OH$ (1°) + NaBr.

Q33 **B. A secondary alcohol**

H_2 adds to aldehydes → 1° alcohols, and to ketones → 2° alcohols. The figure shows the ketone case.

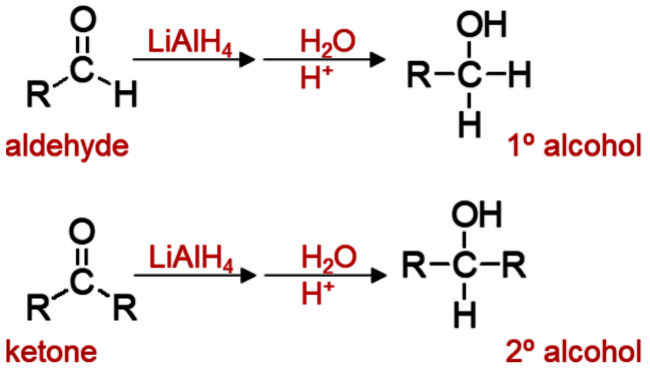


Q34 **C. $CH_3CH(OH)CH_3$ (a 2° alcohol)**

Reduction of a ketone gives a secondary alcohol: $CH_3COCH_3 \rightarrow CH_3CH(OH)CH_3$, 2-propanol. (The doctor's table writes the product as R–$CH(OH)CH_3$; for acetone, R = CH_3.)

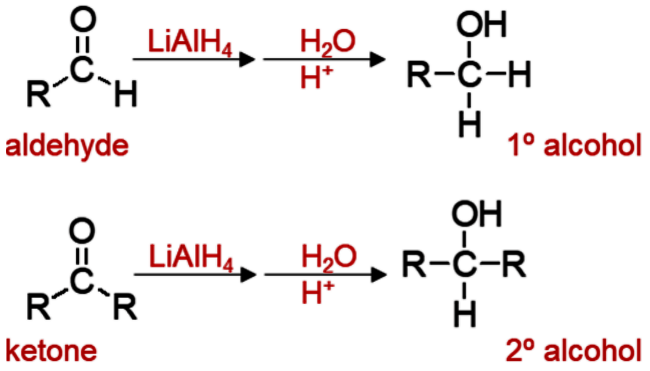
Q35 **A. RCH_2CH_2OH (a 1° alcohol)**

Doctor's table: RCH_2CHO (aldehyde) + $NaBH_4$ or $LiAlH_4$ → $R–CH_2CH_2OH$ (1°). Hydride adds to the carbonyl carbon and water/acid protonates the oxygen.



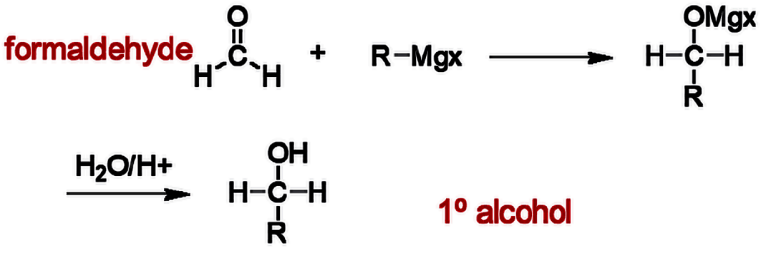
Q36 **D. Hydride ($H:^-$) donors**

Hydrogen atoms are added to the carbonyl group by $NaBH_4$ or $LiAlH_4$ followed by H_2O or dilute acid; both reagents act as hydride ($H:^-$) donors; the hydride ion is a nucleophile and a very strong base.



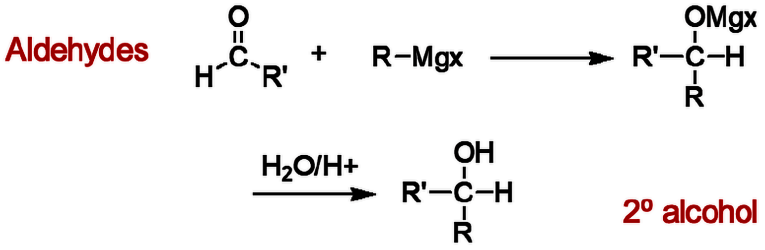
Q37 **B. A primary alcohol**

Formaldehyde → 1° alcohol. The R of the Grignard acts as a nucleophile. (In the doctor's figure, Mgx means MgX, the magnesium halide part.)



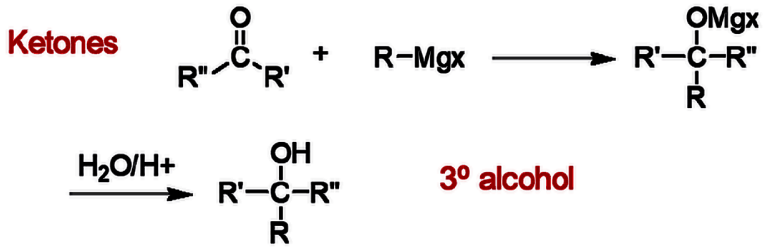
Q38 **D. A secondary alcohol**

Aldehyde (HCOR′) + RMgX → R′CH(OH)R, a 2° alcohol. (In the doctor's figure, Mgx means MgX.)



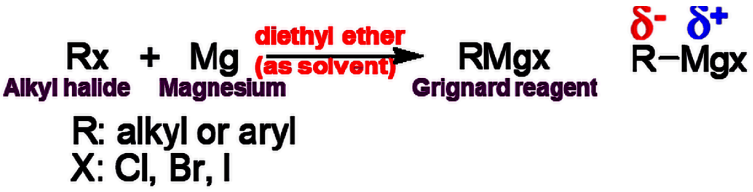
Q39 **A. A tertiary alcohol**

Ketone + RMgX → 3° alcohol. (In the doctor's figure, Mgx means MgX, the magnesium halide part.)



Q40 **C. An alkyl or aryl halide + Mg in diethyl ether**

R–X + Mg (diethyl ether as solvent) → RMgX (R = alkyl or aryl; X = Cl, Br, I). (In the doctor's figure, Rx means RX and Mgx means MgX, the magnesium halide part.)



Q41 **C. Cyclohexanol**

Acid-catalysed hydration adds H and OH → cyclohexanol.

Q42 **D. RCH_2CH_2OH**

Hydrogenation of an aldehyde gives a primary alcohol, RCH_2CH_2OH.

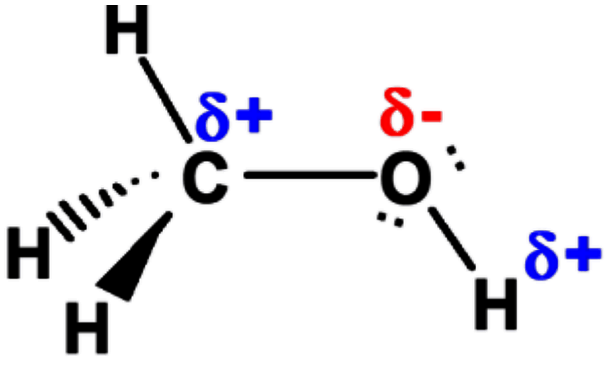
Q43 **B. $RCH(OH)CH_3$**

Reduction of a ketone gives a secondary alcohol, R–CH(OH)–CH_3.

8.5 Reactions of Alcohols

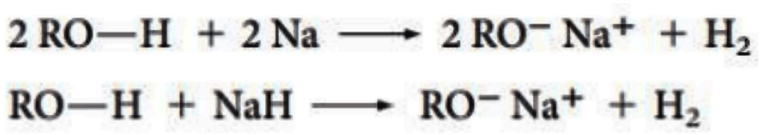
Q44 **B. Two polar bonds (C–O and O–H) and two non-bonding electron pairs on O**

Two polar bonds and two non-bonding electron pairs on oxygen; reactions of alcohols involve the C–O and O–H bonds. In the figure, δ+ and δ− show the partial charges and the dots on O are its lone pairs.



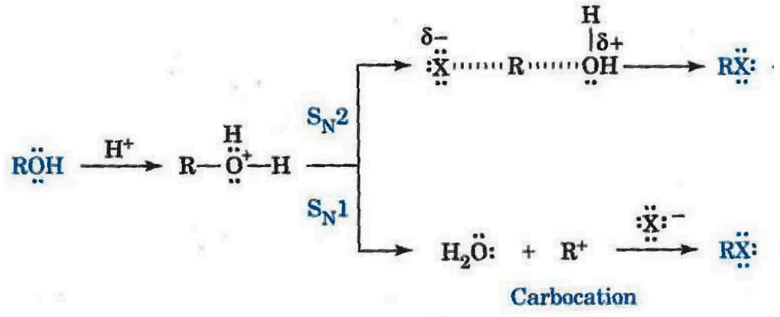
Q45 **C. Sodium alkoxide (RONa) + H_2**

2 ROH + 2 Na → 2 RO^-Na^+ + H_2 (also ROH + NaH → RO^-Na^+ + H_2).



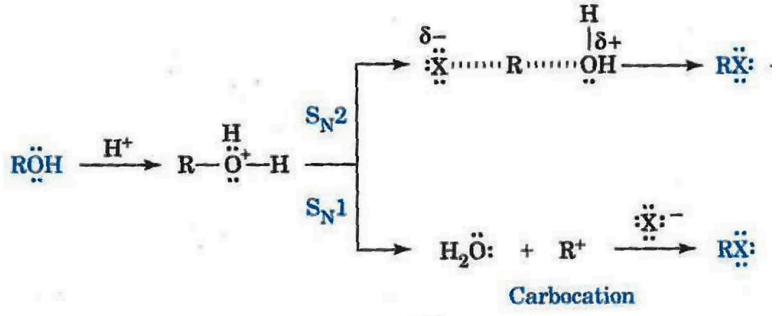
Q46 **A. Nucleophilic substitution**

Nucleophilic substitution reaction: the OH is protonated, then X^- replaces it.



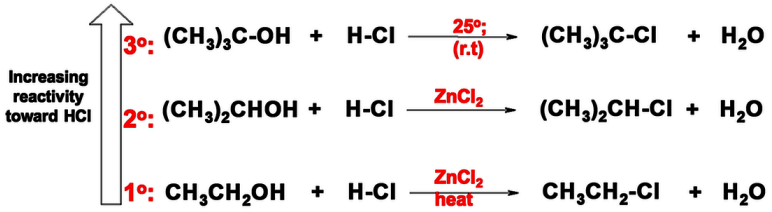
Q47 **D. SN1: 2 steps, 2° and 3° alcohols, carbocation; SN2: 1 step, methyl and 1° alcohols**

SN1: 2 steps, secondary & tertiary alcohols, stable 2°/3° carbocation. SN2: 1 step, methyl & primary alcohols.



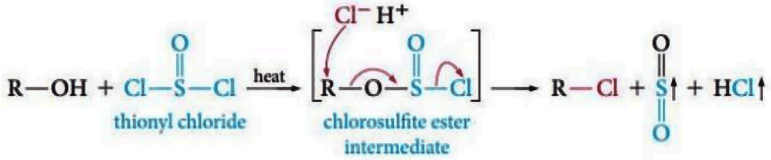
Q48 **C. They react at different rates, depending on carbocation stability**

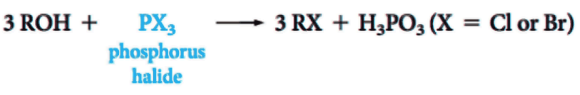
3°, 2°, 1° alcohols react with HCl/$ZnCl_2$ at different rates; the rate depends on the stability of the carbocation intermediate in SN1. 3° reacts fastest (room temperature), 2° needs $ZnCl_2$, 1° needs $ZnCl_2$ + heat.



Q49 **A. $SOCl_2$ and PX_3**

$SOCl_2$ (thionyl chloride) and PX_3 (phosphorus trihalide): R–OH + $SOCl_2$ → R–Cl + SO_2 + HCl; 3 R–OH + PX_3 → 3 R–X + H_3PO_3 (X = Cl, Br).



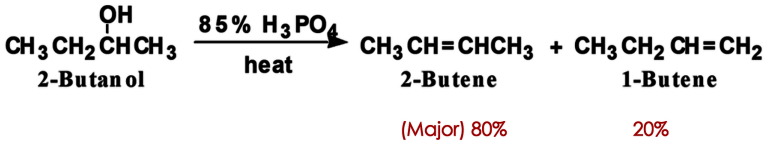


Q50 **D. Elimination reaction**

Elimination: H and OH are removed from neighbouring carbons as water.

Q51 **B. 2-Butene**

When isomeric alkenes are obtained, the more stable alkene (more substituents on the double bond) predominates: 2-butene 80%, 1-butene 20%.

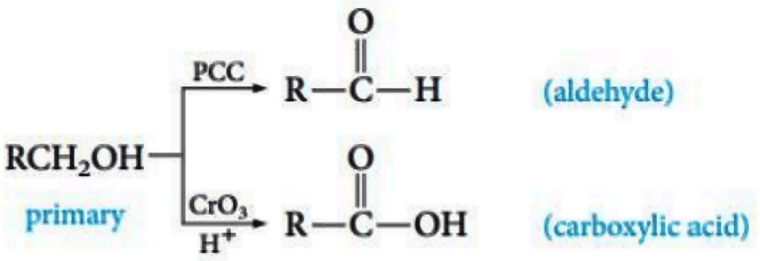


Q52 **C. 1° give carboxylic acids, 2° give ketones, 3° do not react**

Jones' reagent: 1° → carboxylic acid; 2° → ketone; 3° → no reaction.

Q53 **A. An aldehyde**

RCH_2OH → (PCC) RCHO (aldehyde); → (CrO_3, H^+) RCOOH (carboxylic acid).

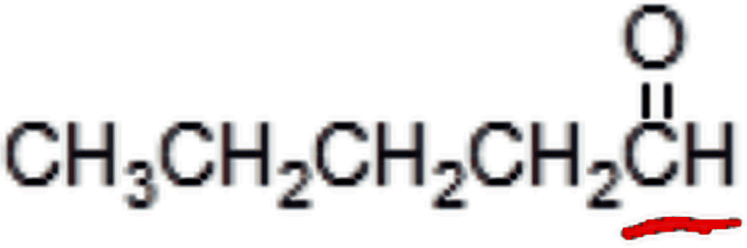


Q54 **D. 2-Pentanol + CrO_3**

A ketone comes from a 2° alcohol: $CH_3CH_2CH_2CH(OH)CH_3$ + CrO_3 → 2-pentanone. (The red mark in the figure only highlights the C=O carbon.)

Q55 **B. 1-Pentanol + PCC**

A 1° alcohol with mild PCC stops at the aldehyde. The figure shows pentanal; the red mark highlights the CHO (aldehyde) group.

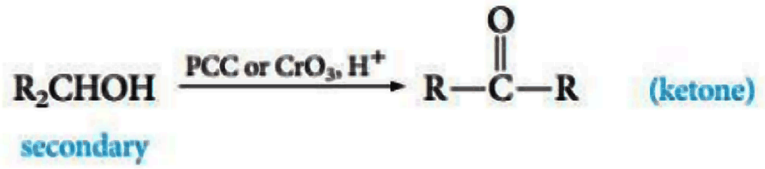


Q56 **B. 1-Pentanol + CrO_3**

Doctor's answer: $CH_3CH_2CH_2CH_2CH_2OH$ + CrO_3 → $CH_3CH_2CH_2CH_2COOH$. A 1° alcohol with CrO_3 is oxidized all the way to the carboxylic acid; PCC would stop at the aldehyde.

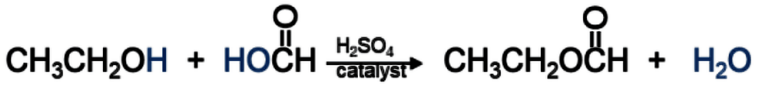
Q57 **C. A ketone**

R_2CHOH → (PCC or CrO_3, H^+) R–CO–R, a ketone. A 1° alcohol gives an aldehyde (PCC) or a carboxylic acid (CrO_3).



Q58 **D. Esters + water**

$CH_3CH_2OH + HCOOH$ → (H_2SO_4) $HCOOCH_2CH_3 + H_2O$ (ester formation).



Q59 **A. 2-Bromopropane + H_2O, SN1**

$CH_3CH(OH)CH_3$ is a 2° alcohol → 2° carbocation → SN1: $CH_3CHBrCH_3 + H_2O$.

Q60 **D. $CH_3CH_2CH_2Cl + SO_2 + HCl$**

Thionyl chloride converts the 1° alcohol to 1-chloropropane, releasing SO_2 and HCl gases.

Q61 **B. 3 $CH_3CHBrCH_2CH_3 + H_3PO_3$**

3 ROH + PBr_3 → 3 RBr + H_3PO_3: 2-bromobutane.

Q62 **A. $CH_3COCH(CH_3)_2$**

3-Methyl-2-butanol is a 2° alcohol → ketone: $CH_3COCH(CH_3)_2$, 3-methyl-2-butanone.

Q63 **C. $CH_3(CH_2)_{10}CHO$**

A 1° alcohol with PCC gives the aldehyde, $CH_3(CH_2)_{10}CHO$ (dodecanal).

Q64 **D. $CH_3CH(ONa)CH_2CH_3$**

Doctor's answer: $CH_3CH(ONa)CH_2CH_3$ (sodium 2-butoxide); H_2 gas is also released.

Q65 **B. $CH_2=CHCH_2CH_3$**

Doctor's answer: $CH_2=CHCH_2CH_3 + CH_3CH=CHCH_3$ (major). The less substituted alkene, 1-butene, is the minor product.

Q66 **A. $CH_3CH(Cl)CH_2CH_3$**

Doctor's answer: $CH_3CH(Cl)CH_2CH_3$ (2-chlorobutane). The Cl replaces the OH on the same carbon (Lucas reagent).

Q67 **C. $CH_3COCH_2CH_3$**

Doctor's answer: $CH_3COCH_2CH_3$ (2-butanone). A 2° alcohol is oxidized to a ketone.

Q68 **D. $(CH_3)_2C=CHCH_3$**

Doctor's answer: $CH_3CH(OH)CH(CH_3)_2$ → (H_2SO_4) $(CH_3)_2C=CHCH_3$ (major, 2-methyl-2-butene, more substituted) + $CH_2=CHCH(CH_3)_2$.

Q69 **B. 1,3,3-Trimethylcyclohexene + H_2O**

The OH carbon has one neighbour with H (the CH carrying CH_3); the other neighbour ($C(CH_3)_2$) has no H. Elimination toward the CH–CH_3 gives the trisubstituted ring alkene, 1,3,3-trimethylcyclohexene (the doctor's answer, with no rearrangement). In practice, a methyl shift from the neighbouring $C(CH_3)_2$ carbon can also give rearranged alkenes, but none of those is among the options.

9 Amines

In this lecture

1. Structure & Classification of Amines 7 questions
2. Nomenclature of Amines 8 questions
3. Physical Properties of Amines 4 questions
4. Basicity of Amines 8 questions
5. Preparation of Amines 5 questions
6. Reactions of Amines 7 questions
7. Aromatic Diazonium Compounds 16 questions

9.1 Structure & Classification of Amines

Q1 Amines are:

A. Oxygen-containing organic acids derived from water

B. Nitrogen-containing organic bases derived from ammonia

C. Nitrogen-containing organic acids derived from nitric acid

D. Carbon–nitrogen compounds that contain a C=O group

Q2 $(CH_3)_4N^+ Cl^-$ (R_4N^+) is a:

A. 4° alkylammonium salt

B. 4° amine

C. 1° alkylammonium salt

D. 3° alkylammonium salt

Q3 Amines are classified as 1°, 2° or 3° according to:

A. The number of hydrogen atoms on the carbon next to N

B. The type of carbon (1°, 2°, 3°) that carries the N

C. The number of nitrogen atoms in the molecule

D. The number of carbon groups bonded to the nitrogen

Q4 Classify $(CH_3)_3CCH_2NH_2$:

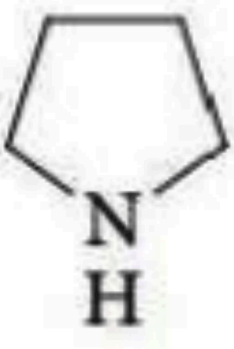
A. Quaternary

B. Secondary (2°)

C. Primary (1°)

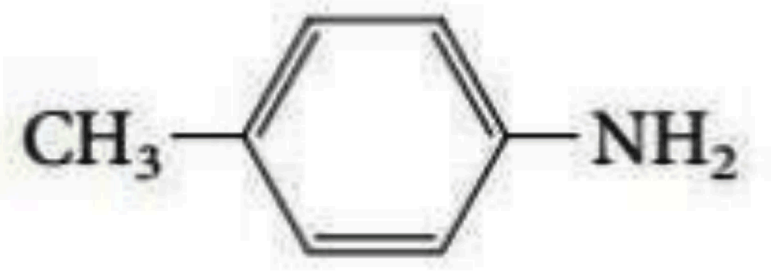
D. Tertiary (3°)

Q5 Classify this amine (pyrrolidine):



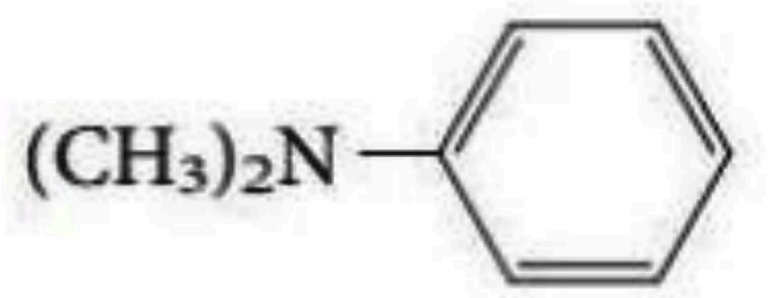
A. Primary (1°)
B. Secondary (2°)
C. Tertiary (3°)
D. Quaternary

Q6 Classify this amine:



A. Tertiary (3°)
B. Secondary (2°)
C. Quaternary
D. Primary (1°)

Q7 Classify this amine:



A. Quaternary
B. Primary (1°)
C. Tertiary (3°)
D. Secondary (2°)

9.2 Nomenclature of Amines

Q8 In common nomenclature, simple amines are named as:

A. Alkylamines
B. Alkanols
C. Alkanamides
D. Alkylammonium salts

Q9 In systematic (IUPAC) naming of amines, how is the chain numbered?

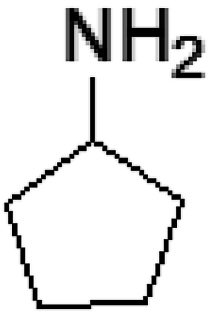
A. So that the alkyl substituents get the lowest numbers

B. From whichever end is nearer to the longest branch

C. So that the double bond always gets number 1

D. So that the carbon containing the amine group gets the lowest number

Q10 What is the name of this compound?



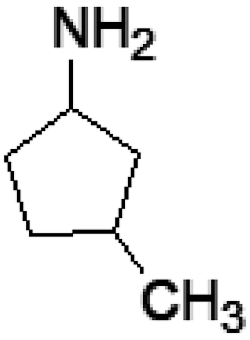
A. Aminocyclohexane

B. Cyclopentanamine

C. Cyclopentylamide

D. Pentanamine

Q11 What is the name of this compound?



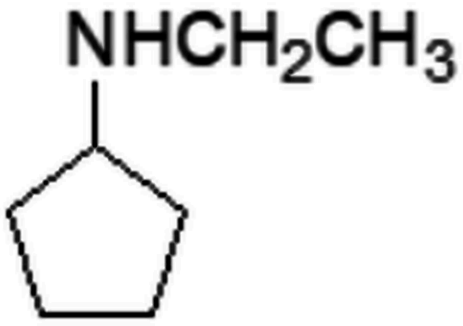
A. 4-Methylcyclopentanamine

B. 1-Methylcyclopentanamine

C. 3-Methylcyclopentanamine

D. 2-Methylcyclopentanamine

Q12 What is the name of this compound?



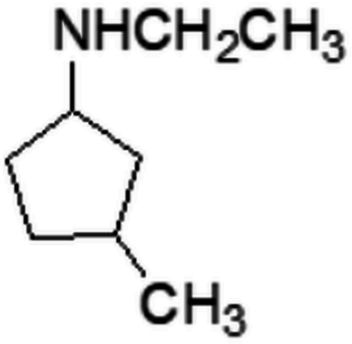
A. N-Ethylcyclopentanamine

B. Ethylcyclopentylamide

C. 2-Ethylcyclopentanamine

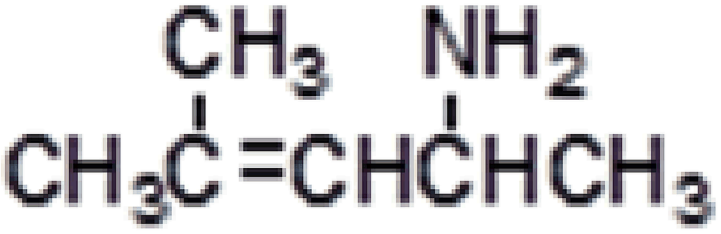
D. N-Methylcyclopentanamine

Q13 What is the name of this compound?



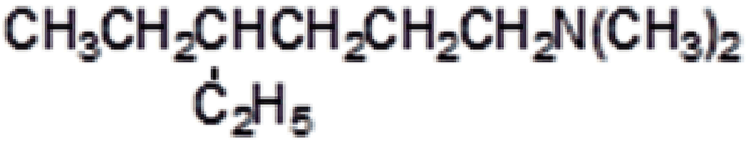
A. 1-Ethyl-3-methylcyclopentane
B. N-Ethyl-3-methylcyclopentanamine
C. 3-Ethyl-N-methylcyclopentanamine
D. N-Ethyl-2-methylcyclopentanamine

Q14 What is the IUPAC name of this compound?



A. 4-Methylpent-2-en-3-amine
B. 2-Methylpent-2-en-4-amine
C. 4-Methylpent-3-en-2-amine
D. 2-Aminopent-3-ene

Q15 What is the IUPAC name of this compound?



A. 4-Ethyl-N,N-dimethylhexan-1-amine
B. 3-Ethyl-N,N-dimethylhexan-6-amine
C. 4-Propyl-N,N-dimethylbutanamine
D. N,N-Dimethyl-4-ethylheptan-1-amine

9.3 Physical Properties of Amines

Q16 What can be concluded from the table?

Name	BP (°C)	Water sol. (mg/L)
Propane $CH_3CH_2CH_3$	−42	40
Ethanamine $CH_3CH_2NH_2$	17	Miscible
Ethanol CH_3CH_2OH	78	Miscible

A. Alcohol < amine < alkane in bp; small amines are insoluble

B. Alkane < amine < alcohol in bp; small amines are water soluble

C. Amine < alkane < alcohol in bp; small amines are insoluble

D. Alkane < alcohol < amine in bp; small amines are water soluble

Q17 Why do alcohols boil higher than amines of similar molecular weight?

A. Alcohols always have much higher molecular weights

B. Amines cannot form any hydrogen bonds with each other

C. O is more electronegative than N, so O–H hydrogen bonds are stronger

D. N is more electronegative than O, so N–H hydrogen bonds are weaker

Q18 Rank in order of **increasing** boiling point: $H_2NCH_2CH_2NH_2$, $(CH_3)_3N$, $CH_3NHCH_2CH_3$, $CH_3CH_2CH_2NH_2$

A. $H_2NCH_2CH_2NH_2 < CH_3CH_2CH_2NH_2 < CH_3NHCH_2CH_3 < (CH_3)_3N$

B. $CH_3NHCH_2CH_3 < (CH_3)_3N < H_2NCH_2CH_2NH_2 < CH_3CH_2CH_2NH_2$

C. All have the same boiling point

D. $(CH_3)_3N < CH_3NHCH_2CH_3 < CH_3CH_2CH_2NH_2 < H_2NCH_2CH_2NH_2$

Q19 Tertiary amines cannot form hydrogen bonds with each other.

True

False

9.4 Basicity of Amines

Q20 Amines are Lewis bases because:

A. The nitrogen has a non-bonding electron pair that it can share with H^+

B. The nitrogen has a positive charge that attracts electron pairs

C. The N–H bond breaks easily, so amines release H^+

D. The nitrogen has an empty orbital that accepts an electron pair

Q21 Which order of basicity is correct?

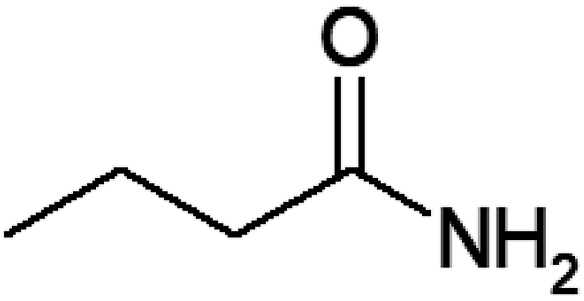
A. $NH_3 > CH_3NH_2 > (CH_3)_2NH$

B. $(CH_3)_2NH > CH_3NH_2 > NH_3$

C. $CH_3NH_2 > NH_3 > (CH_3)_2NH$

D. All are equally basic

Q22 Why is an amide (e.g. butanamide) much less basic than ammonia?



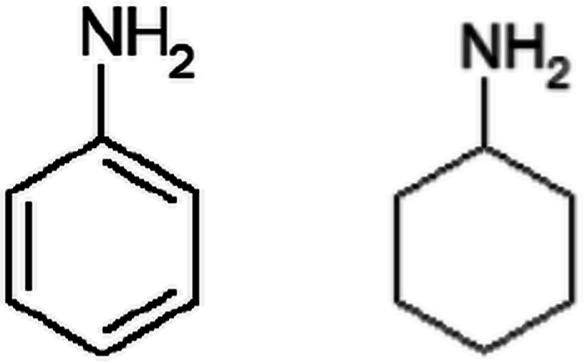
A. The C=O makes the amide ionic, so it cannot accept H^+

B. The N of an amide has no non-bonding electron pair at all

C. The alkyl chain (electron-releasing) pushes electrons away from N

D. The C=O (electron-withdrawing) delocalizes the N lone pair by resonance

Q23 Which is more basic, aniline (left) or cyclohexylamine (right)?



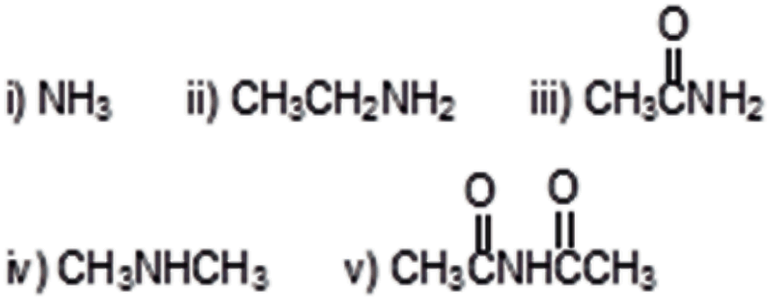
A. Neither is basic

B. Equal basicity

C. Cyclohexylamine

D. Aniline

Q24 Rank in order of **increasing** basicity:



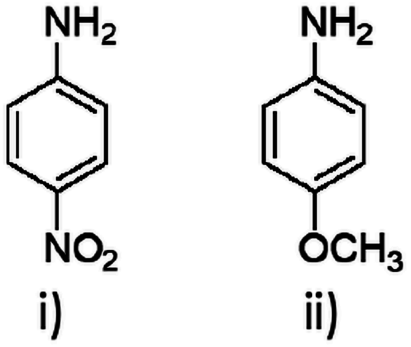
A. iii < v < i < iv < ii

B. v < iii < i < ii < iv

C. iv < ii < i < iii < v

D. i < ii < iii < iv < v

Q25 Rank in order of **increasing** basicity:



A. i = ii

B. ii < i

C. Neither is basic

D. i < ii

Q26 Why do substituents at the ortho and para positions of aniline have a larger effect on its basicity?

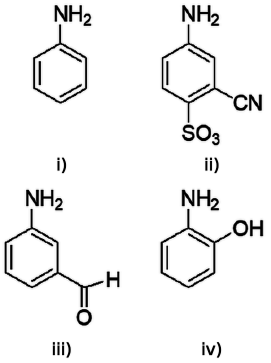
A. Ortho and para groups are always electron-releasing

B. Groups at ortho and para positions remove the lone pair from N completely

C. Resonance has special significance at the ortho and para positions

D. Inductive effects act only at the ortho and para positions

Q27 Rank in order of **increasing** basicity:



A. ii < iii < i < iv

B. iv < i < iii < ii

C. i < ii < iii < iv

D. iii < ii < iv < i

9.5 Preparation of Amines

Q28 Nitrobenzene is converted to aniline by:

A. Reduction with 3 H_2/Ni

B. Reaction with $NaNO_2$/HCl

C. Hydration with H_2O/H^+

D. Oxidation with $KMnO_4$

Q29 Butanamide ($CH_3CH_2CH_2CONH_2$) is reduced with $LiAlH_4$, then H_3O^+. What happens to the C=O group?

A. It is lost with one carbon, giving $CH_3CH_2CH_2NH_2$

B. It becomes CH=N, giving $CH_3CH_2CH_2CH{=}NH$

C. It becomes CH_2, giving $CH_3CH_2CH_2CH_2NH_2$

D. It becomes CH–OH, giving $CH_3CH_2CH_2CH(OH)NH_2$

Q30 Reduction of a nitrile (R–C≡N) with 2 H_2/Ni gives:

A. A secondary amine, $(RCH_2)_2NH$

B. A primary amine, $R–CH_2–NH_2$

C. A tertiary amine, $(RCH_2)_3N$

D. An amide, $R–CO–NH_2$

Q31 Why does alkylation of ammonia with alkyl halides give a mixture of products?

A. The alkyl halide is eliminated to an alkene before NaOH is added

B. The NaOH reacts with R–X to form alcohols instead of amines

C. Ammonia does not react with alkyl halides, so it stays unchanged

D. Each new amine is also a nucleophile, so it reacts again with the alkyl halide

Q32 Reductive amination of an aldehyde/ketone with a primary amine ($NaBH_3CN$) gives:

A. A primary amine

B. A secondary amine

C. An amide

D. A nitro compound

9.6 Reactions of Amines

Q33 Most reactions of amines are due to the non-bonding electron pair on N, which makes amines:

A. Acidic and nucleophilic

B. Acidic and electrophilic

C. Basic and electrophilic

D. Basic and nucleophilic

Q34 CH_3NH_2 + HCl gives:

A. $CH_3NHCl + H_2$

B. $CH_3Cl + NH_3$

C. $CH_3NH_3^+Cl^-$

D. No reaction

Q35 A useful application of forming amine salts with strong acids is:

A. Separating amines from non-basic organic compounds

B. Making amines insoluble in water so they precipitate

C. Converting amines into alkyl halides

D. Oxidizing amines to nitro compounds

Q36 Amines react with acid chlorides (RCOCl). Which statement is correct?

A. 1° amines give 3° amides, 2° give 2° amides, 3° do not react

B. 1° amines give 2° amides, 2° give 3° amides, 3° do not react

C. 1° amines do not react, 2° give 2° amides, 3° give 3° amides

D. 1° amines give 1° amides, 2° give 2° amides, 3° give 3° amides

Q37 Alkylation of amines with alkyl halides occurs by:

A. Free-radical substitution

B. Elimination (E2)

C. SN2 nucleophilic substitution

D. SN1 nucleophilic substitution (via a carbocation)

Q38 Complete: $(CH_3)_3NH^+Cl^- + CH_3NH_2 \rightarrow$

A. No reaction

B. $(CH_3)_3N + CH_3NH_3^+Cl^-$

C. $(CH_3)_3N + CH_3Cl + NH_3$

D. $(CH_3)_4N^+Cl^- + NH_3$

Q39 Complete: CH_3CONH_2 + $LiAlH_4$, then H_2O →

A. $CH_3CH_2NH_2$

B. CH_3COOH

C. CH_3CH_2OH

D. CH_3CN

9.7 Aromatic Diazonium Compounds

Q40 Aromatic diazonium salts are prepared from:

A. 3° aromatic amines + nitrous acid (HNO_2)

B. Aryl halides + sodium nitrite ($NaNO_2$)

C. Phenols + ammonia (NH_3)

D. 1° aromatic amines + nitrous acid (HNO_2)

Q41 The process that forms an aryl diazonium salt is called:

A. Reductive amination

B. Azo coupling

C. Diazotization

D. Acylation

Q42 How many important resonance forms does the aryl diazonium ion ($Ar–N_2^+$) have?

A. One

B. Four

C. Three

D. Two

Q43 Which resonance form of the aryl diazonium ion is important in the **substitution** (replacement) reactions?

A. $Ar–N=N^+$

B. $Ar–N^+\equiv N$

C. Ar–N=N–Ar′

D. $Ar–NH_2$

Q44 Which resonance form and mechanism are involved in the **coupling** reaction of aryl diazonium salts?

A. $Ar–N=N^+$ form; electrophilic aromatic substitution

B. $Ar–N=N^+$ form; nucleophilic substitution (SN2)

C. $Ar–N^+\equiv N$ form; electrophilic aromatic substitution

D. $Ar–N^+\equiv N$ form; free-radical substitution

Q45 In substitution (replacement) reactions of aryl diazonium salts:

A. N_2 is evolved and the ring is reduced

B. N_2 is retained and the ring is reduced

C. N_2 is retained and a nucleophile adds to the ring

D. N_2 is evolved and replaced by a nucleophile

Q46 Which reagent replaces the diazonium group by H?

A. HBF_4, heat

B. H_3O^+

C. H_3PO_2

D. KI

Q47 Benzenediazonium chloride + KI gives:

A. Iodobenzene + N_2

B. Aniline + I_2

C. Phenol + N_2

D. Benzene + N_2

Q48 Benzenediazonium chloride + H_3O^+ gives:

A. Chlorobenzene

B. Phenol

C. Aniline

D. Benzene

Q49 Which reagent converts benzenediazonium chloride into benzonitrile (C_6H_5CN)?

A. HCl, Cu_2Cl_2

B. KCN, $Cu_2(CN)_2$

C. HBF_4, heat

D. H_3PO_2

Q50 Benzenediazonium chloride + HCl, Cu_2Cl_2 gives:

A. Benzonitrile + N_2

B. Benzene + N_2

C. Chlorobenzene + N_2

D. Phenol + N_2

Q51 Benzenediazonium chloride + HBr, Cu_2Br_2 gives:

A. Chlorobenzene + N_2

B. Benzene + N_2

C. Phenol + N_2

D. Bromobenzene + N_2

Q52 Which reagent converts benzenediazonium chloride into fluorobenzene?

A. HBF_4, heat

B. HCl, Cu_2Cl_2

C. KI

D. H_3O^+

Q53 In coupling reactions, diazonium ions react with which kind of aromatic ring?

A. Deactivated rings carrying an EWG (NO_2, CN, CHO)

B. Any benzene ring, whatever its substituents

C. Highly activated rings carrying an ERG (OH, NH_2, NHR, NR_2, OR)

D. Only aryl halides (Ar–Cl, Ar–Br, Ar–I)

Q54 Nitrobenzene → 1) H_2 → 2) $NaNO_2$, HCl, 0 °C → 3) H_3PO_2 gives:

A. Chlorobenzene

B. Aniline

C. Phenol

D. Benzene

Q55 p-Nitrobenzenediazonium chloride + phenol gives:

A. HO–C_6H_4–N=N–C_6H_4–NO_2

B. O_2N–C_6H_4–O–C_6H_5 + N_2

C. O_2N–C_6H_4–OH + N_2

D. HO–C_6H_4–NH–NH–C_6H_4–NO_2

Answers · Lecture 9

9.1 Structure & Classification of Amines

Q1 **B. Nitrogen-containing organic bases derived from ammonia**

Nitrogen-containing compounds / organic bases derived from ammonia. They can be aliphatic, aromatic (arylamines) or heterocyclic.

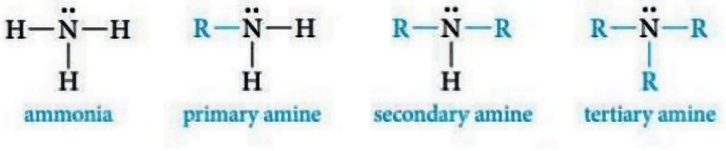
Q2 **A. 4° alkylammonium salt**

Four carbon groups on a positive nitrogen = 4° ammonium salt.

Q3 **D. The number of carbon groups bonded to the nitrogen**

1°, 2°, 3° refer to the degree of substitution on nitrogen, not on the carbon: RNH_2 (1°), R_2NH (2°), R_3N (3°); R_4N^+ is a 4° ammonium salt.

NH_3	ammonia
$R\text{-}NH_2$	1° amine
$R_2\text{-}NH$	2° amine
$R_3\text{-}N$	3° amine
NH_4^+	ammonium
R_4N^+	4° alkylammonium (salt)



Q4 **C. Primary (1°)**

The N carries one carbon group (and two H) → 1°. The bulky $(CH_3)_3C–$ group is on the neighbouring carbon, not on N, so it does not change the class. (Not solved in the doctor's file.)

Q5 **B. Secondary (2°)**

The ring N is bonded to two carbons and one H → 2°. (Not solved in the doctor's file.)

Q6 **D. Primary (1°)**

p-Toluidine: $–NH_2$ on the ring → one carbon on N → 1° (an arylamine). (Not solved in the doctor's file.)

Q7 **C. Tertiary (3°)**

N,N-Dimethylaniline: N bonded to two CH_3 and the ring → 3°. (Not solved in the doctor's file.)

9.2 Nomenclature of Amines

Q8 **A. Alkylamines**

Simple amines are named as “alkylamines” (the alkyl group name + “amine”): “amine” is added to the name of the organic group (e.g. methylamine).

Q9 **D. So that the carbon containing the amine group gets the lowest number**

Doctor's rules: identify the longest continuous chain containing the amine group; use -an/-en/-yn for the chain; name the amine group with the suffix -amine; number the chain giving priority to the amine group (the C containing the amine group gets the lowest number); then name and locate all other substituents.

Q10 **B. Cyclopentanamine**

Cyclopentanamine (cyclopentylamine).

Q11 **C. 3-Methylcyclopentanamine**

The ring carbon carrying NH_2 is C-1; the methyl is at C-3: 3-methylcyclopentanamine.

Q12 **A. N-Ethylcyclopentanamine**

The ethyl group is on the nitrogen, so it gets the locant “N”: N-ethylcyclopentanamine.

Q13 **B. N-Ethyl-3-methylcyclopentanamine**

Ethyl on N, methyl on ring C-3; alphabetical order: ethyl before methyl.

Q14 **C. 4-Methylpent-3-en-2-amine**

Five-carbon chain containing NH_2 and C=C; the amine gets the lowest number (C-2), C=C at C-3, CH_3 at C-4. (Not solved in the doctor's file.)

Q15 **A. 4-Ethyl-N,N-dimethylhexan-1-amine**

Number the longest chain from the CH_2 bonded to N: the three CH_2 groups are C-1 to C-3, the CH is C-4, and the CH_2CH_3 on the left end is C-5 and C-6 → a 6-carbon chain (hexan-1-amine). The C_2H_5 hanging from C-4 is a 4-ethyl branch, and the two CH_3 groups on N give “N,N-dimethyl” → 4-ethyl-N,N-dimethylhexan-1-amine. (Not solved in the doctor's file.)

9.3 Physical Properties of Amines

Q16 **B. Alkane < amine < alcohol in bp; small amines are water soluble**

Doctor's answer: higher Mwt, higher boiling point; amines can form H-bonds; alcohols have higher boiling points than amines. Similar to alcohols, H-bonding makes the smaller amines soluble in water. Conclusion: amines boil higher than alkanes of similar size.

Q17 **C. O is more electronegative than N, so O–H hydrogen bonds are stronger**

Doctor's answer: OH is more polar than NH (O more electronegative than N), so the H-bonds are stronger.

Q18 **D. $(CH_3)_3N < CH_3NHCH_2CH_3 < CH_3CH_2CH_2NH_2 < H_2NCH_2CH_2NH_2$**

All four have almost the same molecular weight (59–60), so hydrogen bonding decides: $(CH_3)_3N$ (no N–H) < $CH_3NHCH_2CH_3$ (one N–H) < $CH_3CH_2CH_2NH_2$ (two N–H) < $H_2NCH_2CH_2NH_2$ (two NH_2 groups = four N–H, most H-bonding).

Q19 **True**

Amines (except 3°) can form H-bonds; 3° amines have no H on N.

9.4 Basicity of Amines

Q20 **A. The nitrogen has a non-bonding electron pair that it can share with H^+**

A Lewis base is an electron-pair donor. Amines have a non-bonding electron pair on nitrogen; they share it with H^+ from strong acids to form ammonium salts. The more available the electron pair, the stronger the base.

Q21 **B. $(CH_3)_2NH > CH_3NH_2 > NH_3$**

Electron-releasing groups (alkyl) increase basicity: alkyl amines are more basic than ammonia, $(CH_3)_2NH$ > CH_3NH_2 > NH_3.

Q22 **D. The C=O (electron-withdrawing) delocalizes the N lone pair by resonance**

Electron-withdrawing groups decrease basicity: in amides the N lone pair is delocalized onto the C=O oxygen, so amides are less basic than ammonia.

Q23 **C. Cyclohexylamine**

Aromatic amines are less basic than alkyl amines: in aniline the N lone pair is delocalized into the ring (resonance); in cyclohexylamine it is localized on N.

Q24 **B. v < iii < i < ii < iv**

Doctor's answer: v < iii < i < ii < iv. Reason (alkyl groups raise and C=O groups lower basicity): v has two C=O, iii has one C=O (amide), i is NH_3, ii has one alkyl group, iv has two alkyl groups.

Q25 **D. i < ii**

p-Nitroaniline (i, EWG NO_2) is less basic than p-methoxyaniline (ii, ERG OCH_3). ERG on aromatic amines increases basicity; EWG reduces it, especially at ortho/para positions.

Q26 **C. Resonance has special significance at the ortho and para positions**

Doctor's answer: the presence of an ERG on aromatic amines increases the basicity, while an EWG reduces it. Resonance has special significance on the ortho (o) and para (p) positions, so placing EWG/ERG groups there has a larger effect on basicity.

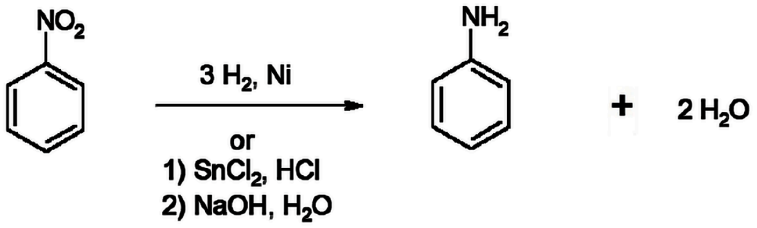
Q27 **A. ii < iii < i < iv**

ii (two EWGs: CN and SO_3) < iii (one EWG: CHO) < i (aniline) < iv (ERG: OH).

9.5 Preparation of Amines

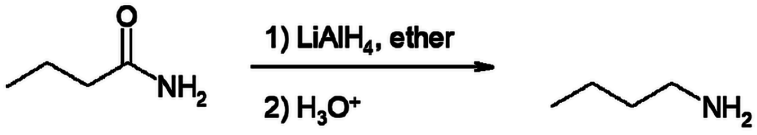
Q28 **A. Reduction with 3 H_2/Ni**

Reduction of nitro aromatic compounds gives primary amines: nitrobenzene → aniline (3 H_2/Ni, or 1) $SnCl_2$, HCl 2) NaOH, H_2O).



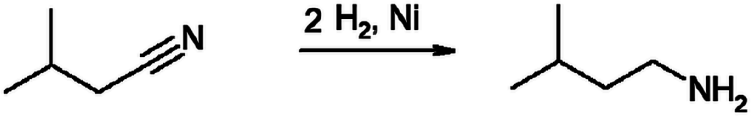
Q29 **C. It becomes CH_2, giving $CH_3CH_2CH_2CH_2NH_2$**

Reduction of amides gives primary, secondary or tertiary amines: C=O becomes CH_2. Example: butanamide → butan-1-amine.



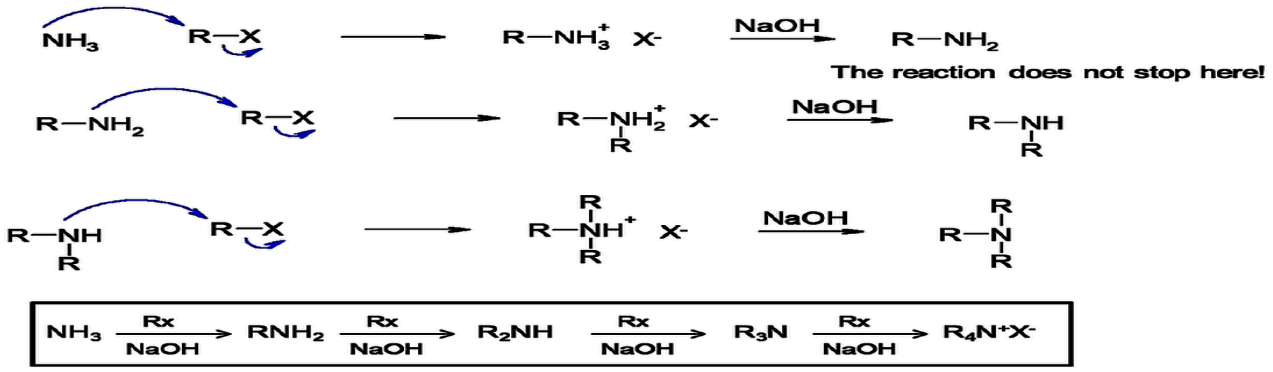
Q30 **B. A primary amine, R–CH_2–NH_2**

Reduction of nitriles gives primary amines: the C≡N carbon becomes CH_2–NH_2.



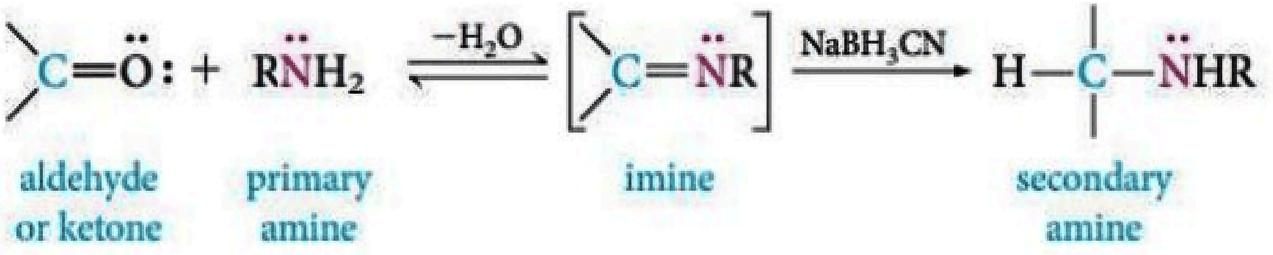
Q31 **D. Each new amine is also a nucleophile, so it reacts again with the alkyl halide**

Alkylation of ammonia: $NH_3 \rightarrow RNH_2 \rightarrow R_2NH \rightarrow R_3N \rightarrow R_4N^+X^-$. The reaction does not stop at RNH_2; repeated alkylation gives 2°, 3° amines and the 4° ammonium salt. (Blue curved arrows = the N lone pair attacking the carbon and X leaving.)



Q32 **B. A secondary amine**

Reductive amination: C=O + RNH_2 (1° amine) → (−H_2O) imine C=NR → ($NaBH_3CN$) secondary amine. The square brackets around the imine mean it is an intermediate that is reduced directly, not isolated.



9.6 Reactions of Amines

Q33 **D. Basic and nucleophilic**

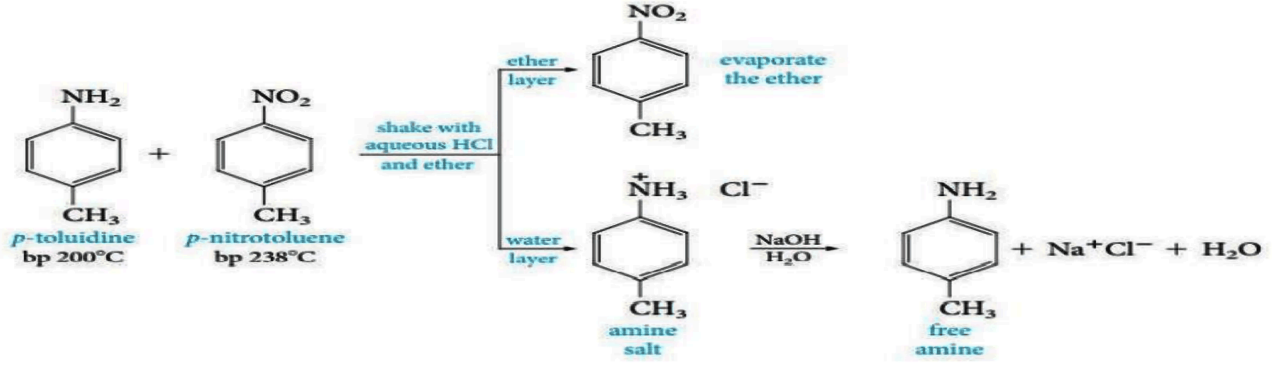
Basic (Lewis base, react with acids) and nucleophilic (react with alkyl halides).

Q34 **C. $CH_3NH_3^+Cl^-$**

Salt formation: the amine (a base) takes H^+ from the acid, forming an ammonium salt.

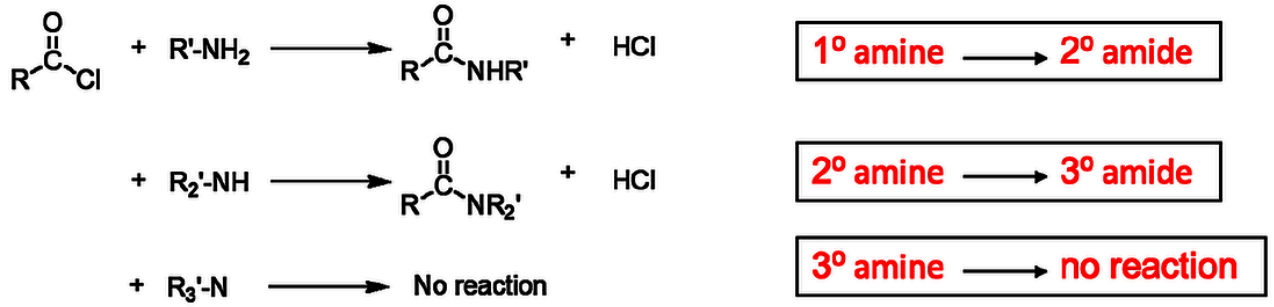
Q35 **A. Separating amines from non-basic organic compounds**

Shaking a mixture (e.g. p-toluidine + p-nitrotoluene) with aqueous HCl and ether: the amine salt goes into the water layer, the non-basic compound stays in ether; NaOH then frees the amine.



Q36 **B. 1° amines give 2° amides, 2° give 3° amides, 3° do not react**

Amide formation (acylation): 1° amine → 2° amide; 2° amine → 3° amide; 3° amine → no reaction (no H on N). In the figure the same acid chloride R–COCl (top left) reacts with each amine; R′ is the amine's carbon group.



Q37 **C. SN2 nucleophilic substitution**

The amine (a nucleophile) attacks 1° or 2° alkyl halides (RX) by SN2, giving higher amines and finally a 4° ammonium salt.

Q38 **B. $(CH_3)_3N + CH_3NH_3^+Cl^-$**

Doctor's answer: $(CH_3)_3N + CH_3NH_3^+Cl^-$. This is an acid–base (proton transfer) reaction: the stronger base, methylamine, takes the H^+ from the trimethylammonium salt, giving free trimethylamine and methylammonium chloride. In water, CH_3NH_2 is a stronger base than $(CH_3)_3N$ because the crowded $(CH_3)_3NH^+$ ion is less stabilized by water (fewer N–H bonds to H-bond with water).

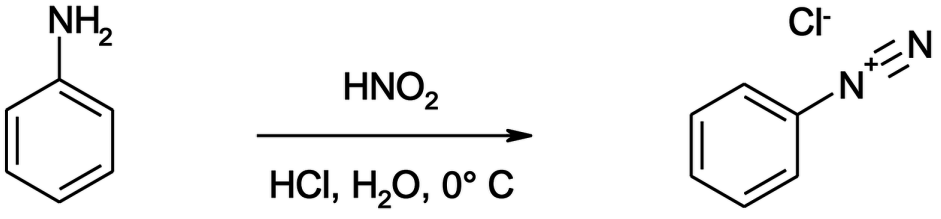
Q39 **A. $CH_3CH_2NH_2$**

Reduction of the amide gives ethylamine.

9.7 Aromatic Diazonium Compounds

Q40 **D. 1° aromatic amines + nitrous acid (HNO_2)**

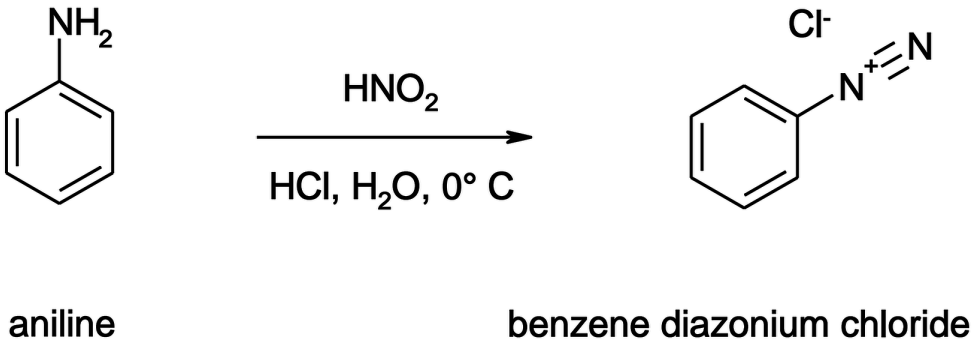
Diazonium salts are compounds in which an N_2 group is attached to an aromatic ring. 1° aromatic amine + nitrous acid ($NaNO_2$/HCl, H_2O, 0 °C) → diazonium salt.



aniline benzene diazonium chloride

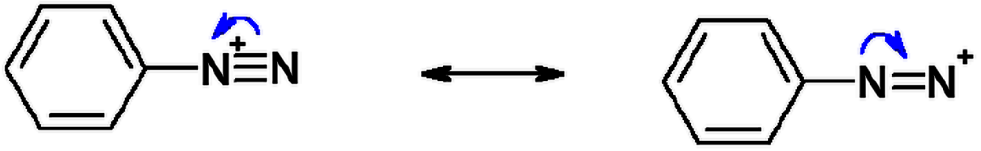
Q41 **C. Diazotization**

Doctor's answer: 1° aromatic amine + nitrous acid = diazonium salt; the process is called diazotization (e.g. aniline + HNO_2, HCl, H_2O, 0 °C → benzenediazonium chloride).



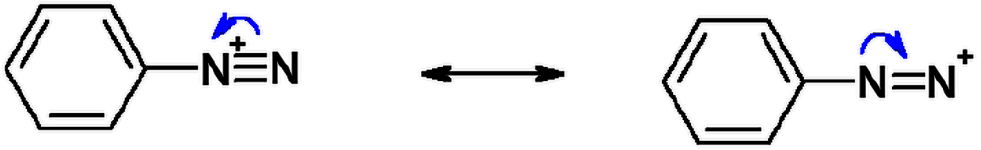
Q42 **D. Two**

Two resonance forms: Ar–N^+≡N and Ar–N=N^+.



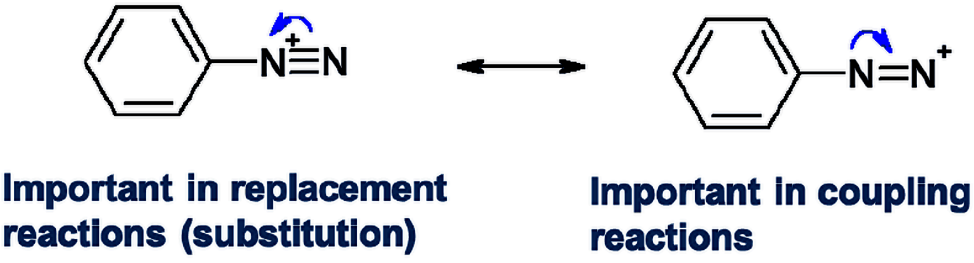
Q43 **B. Ar–N^+≡N**

Doctor's answer: Ar–N^+≡N is important in replacement (substitution) reactions.



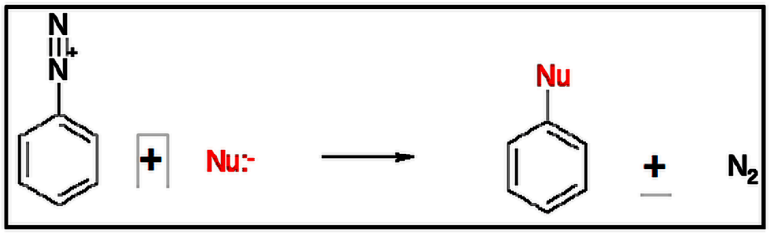
Q44 **A. Ar–N=N^+ form; electrophilic aromatic substitution**

Doctor's answer: the Ar–N=N^+ resonance form is important in coupling reactions (N_2 is retained); the mechanism is electrophilic aromatic substitution. The Ar–N^+≡N form is important in replacement (substitution) reactions, where N_2 is evolved.



Q45 **D. N_2 is evolved and replaced by a nucleophile**

General reaction: ArN_2^+ + $Nu:^-$ → Ar–Nu + N_2 (the N_2 is evolved).

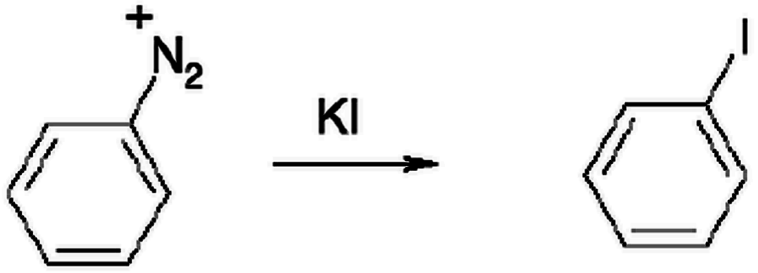


Q46 **C. H_3PO_2**

Doctor's figure: $ArN_2^+ + H_3PO_2 \rightarrow Ar–H$ (benzene) + N_2.

Q47 **A. Iodobenzene + N_2**

Doctor's figure: $ArN_2^+ + KI \rightarrow ArI + N_2$ (substitution with loss of N_2).



Q48 **B. Phenol**

Doctor's figure: $ArN_2^+ + H_3O^+ \rightarrow ArOH + N_2$.

Q49 **B. KCN, $Cu_2(CN)_2$**

Doctor's figure: $ArN_2^+ + KCN/Cu_2(CN)_2 \rightarrow ArCN + N_2$.

Q50 **C. Chlorobenzene + N_2**

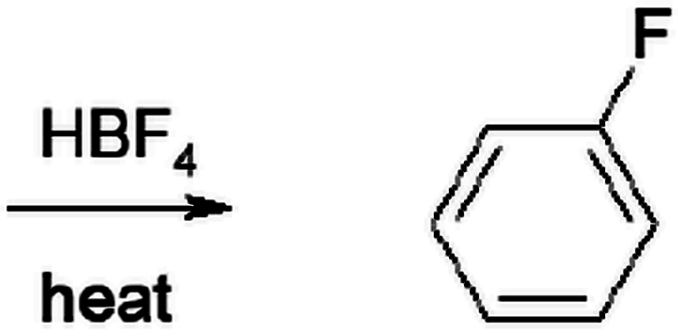
Doctor's figure: $ArN_2^+ + HCl/Cu_2Cl_2 \rightarrow ArCl + N_2$.

Q51 **D. Bromobenzene + N_2**

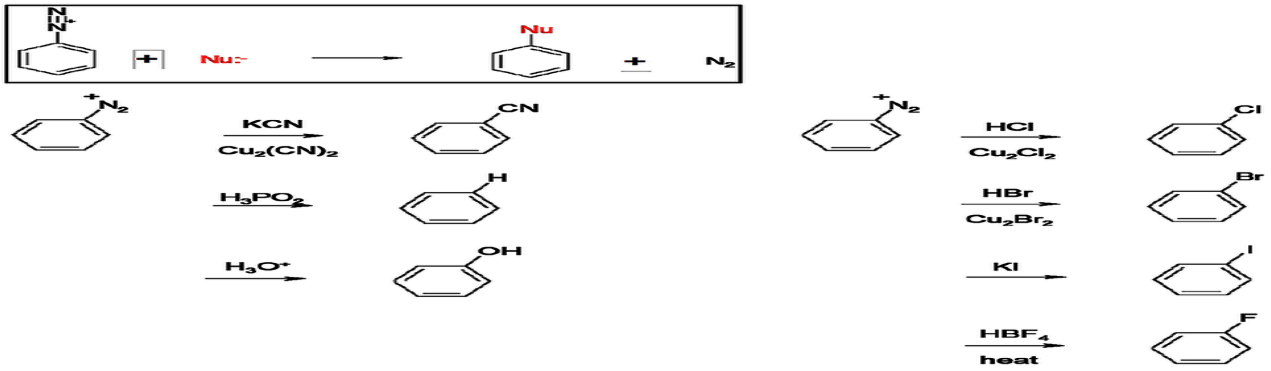
Doctor's figure: $ArN_2^+ + HBr/Cu_2Br_2 \rightarrow ArBr + N_2$.

Q52 **A. HBF_4, heat**

Doctor's figure: $ArN_2^+ + HBF_4$ (heat) → $ArF + N_2$.

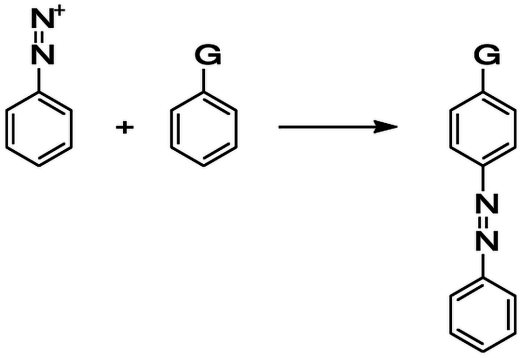


The full set of substitution reactions:



Q53 **C. Highly activated rings carrying an ERG (OH, NH_2, NHR, NR_2, OR)**

Azo compounds: Ar–N=N–Ar′. Diazonium ions are electrophiles; they react with highly activated aromatic rings (G = OH, NH_2, NHR, NR_2, OR) by electrophilic aromatic substitution to form azo compounds. In the figure G is the activating group, and the new –N=N– bond forms para to G.



Q54 **D. Benzene**

H_2 reduces NO_2 to aniline; diazotization gives benzenediazonium chloride; H_3PO_2 replaces N_2^+ with H → benzene.

Q55 **A. $HO–C_6H_4–N=N–C_6H_4–NO_2$**

Doctor's answer: an azo compound, $HO–C_6H_4–N=N–C_6H_4–NO_2$ (coupling at the para position of the activated phenol ring); N_2 is retained.

10 Aldehydes & Ketones

In this lecture

1. Structure of Aldehydes & Ketones 4 questions
2. Nomenclature of Aldehydes & Ketones 15 questions
3. Physical Properties 4 questions
4. Preparation of Aldehydes & Ketones 8 questions
5. Reaction Sites & Oxidation 5 questions
6. Reduction & Nucleophilic Addition 11 questions
7. Nitrogen Derivatives 4 questions
8. Ozonolysis Practice 1 questions
9. Tautomerism & α-Hydrogens 5 questions
10. Aldol Condensation 4 questions

10.1 Structure of Aldehydes & Ketones

Q1 What is the difference between aldehydes and ketones?

A. Aldehydes have at least one H on the carbonyl carbon; ketones have two carbon groups on the carbonyl carbon

B. Aldehydes have a C=O in the middle of the chain; ketones have it at the end

C. Aldehydes contain an O–H group; ketones contain a C=O group

D. Aldehydes have two carbon groups on the carbonyl carbon; ketones have at least one H

Q2 Both aldehydes and ketones contain which functional group?

A. The carboxyl group (–COOH)

B. The ether group (C–O–C)

C. The carbonyl group (C=O)

D. The hydroxyl group (–OH)

Q3 CH_3CH_2CHO is:

A. An aldehyde

B. A carboxylic acid

C. An alcohol

D. A ketone

Q4 CH_3COCH_3 is:

A. A ketone

B. An ester

C. An aldehyde

D. An ether

10.2 Nomenclature of Aldehydes & Ketones

Q5 In IUPAC names, the suffix used for an aldehyde is:

A. -ol

B. -one

C. -al

D. -oxo

Q6 Which order shows the priority for choosing the suffix (highest first)?

A. Amine > alcohol > ketone > aldehyde

B. Alcohol > aldehyde > ketone > amine

C. Aldehyde > ketone > alcohol > amine

D. Ketone > aldehyde > alcohol > amine

Q7 When a ketone is NOT the highest-priority group, it is named with the prefix:

A. oxo

B. hydroxy

C. keto-al

D. carbonyl

Q8 When numbering the chain of an aldehyde or ketone, the lowest number goes to:

A. The substituents in alphabetical order

B. The C=C double bond

C. The functional group named by the suffix (C=O)

D. The C≡C triple bond

Q9 The root name of the chain is followed by -an, -en or -yn depending on:

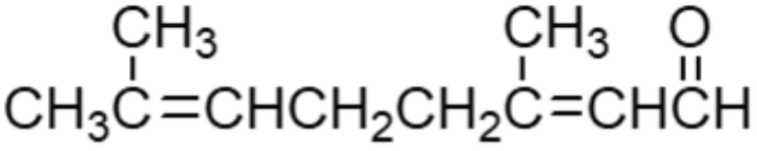
A. Whether the chain has only single bonds, a double bond, or a triple bond

B. Whether the compound is an aldehyde or a ketone

C. The number of substituents

D. The number of oxygen atoms

Q10 Name this compound:



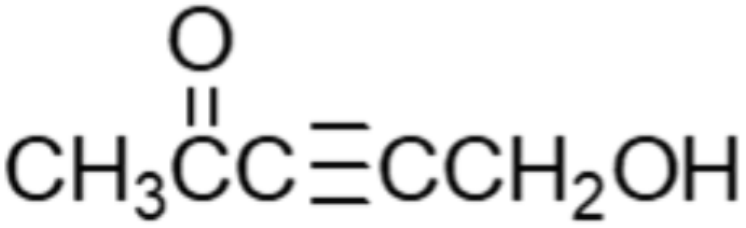
A. 2,6-Dimethyl-3,7-octadienal

B. 2,6-Dimethyl-2,6-octadienal

C. 3,7-Dimethyl-2,6-octadienal

D. 3,7-Dimethyl-2,6-octadienone

Q11 Name this compound:



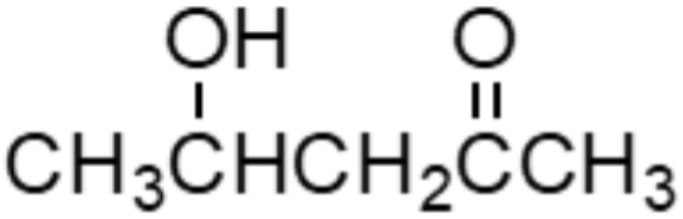
A. 4-Oxo-2-pentyn-1-ol

B. 5-Hydroxy-3-pentyn-2-one

C. 5-Hydroxy-2-pentyn-4-one

D. 1-Hydroxy-2-pentyn-4-one

Q12 Name this compound:



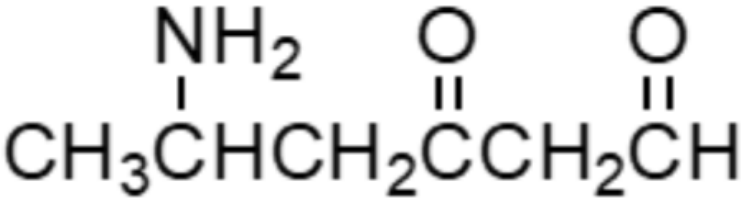
A. 2-Hydroxy-4-pentanone

B. 4-Hydroxy-2-pentanone

C. 2-Hydroxy-4-pentanal

D. 4-Oxo-2-pentanol

Q13 Name this compound:



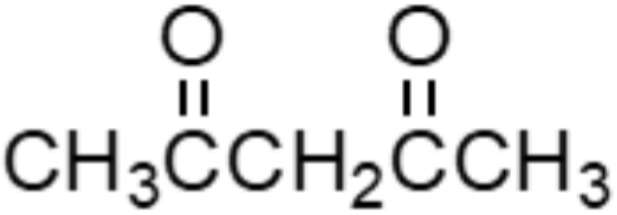
A. 1-Amino-3-oxo-5-hexanal

B. 2-Amino-4-oxohexanal

C. 5-Amino-3-oxohexanal

D. 5-Amino-3-hexanone-1-al

Q14 Name this compound:



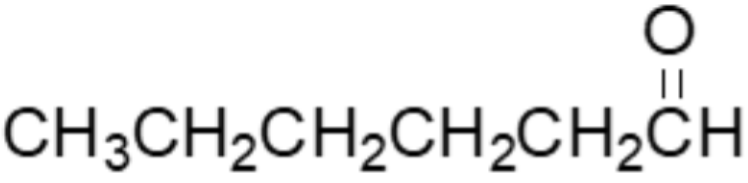
A. 2,4-Pentanedial

B. 1,3-Pentanedione

C. 2,4-Pentanedione

D. 2-Oxo-4-pentanone

Q15 Name this compound:



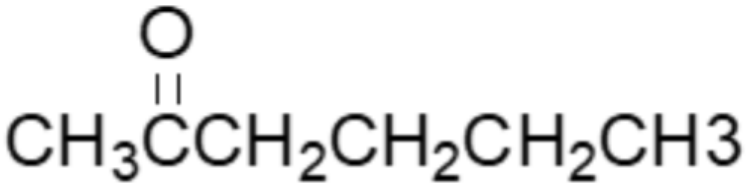
A. 1-Hexanol

B. Pentanal

C. Hexanone

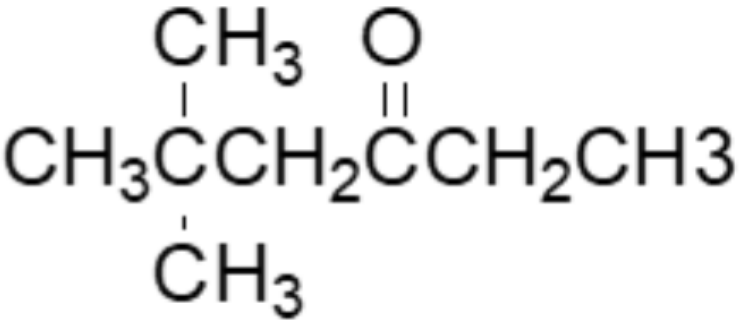
D. Hexanal

Q16 Name this compound:



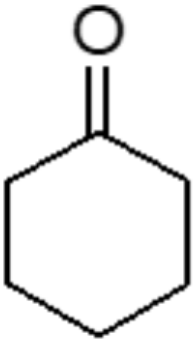
A. 2-Pentanone
B. Hexanal
C. 5-Hexanone
D. 2-Hexanone

Q17 Name this compound:



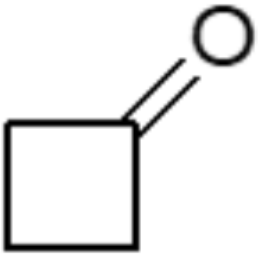
A. 2,2-Dimethyl-4-hexanone
B. 5,5-Dimethyl-3-hexanone
C. 2,2-Dimethyl-3-hexanone
D. 5,5-Dimethyl-4-hexanone

Q18 Name this compound:



A. Cyclohexanol
B. Cyclopentanone
C. Cyclohexanal
D. Cyclohexanone

Q19 Name this compound:



A. Cyclobutanol

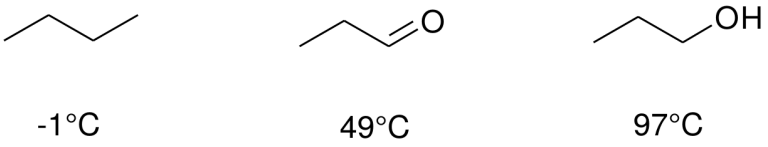
B. Cyclobutanal

C. Cyclobutanone

D. Cyclopropanone

10.3 Physical Properties

Q20 Butane (−1 °C), propanal (49 °C), 1-propanol (97 °C). The boiling points of aldehydes & ketones are:



A. Higher than alcohols

B. Between alkanes and alcohols of similar molecular weight

C. The same as alkanes

D. Lower than alkanes

Q21 Why do aldehydes & ketones boil higher than alkanes of similar size?

A. The C=O group is polar, so there are dipole–dipole attractions between molecules

B. They are ionic compounds

C. They form hydrogen bonds with each other

D. They have a higher molecular weight

Q22 Why do aldehydes & ketones boil lower than alcohols?

A. They have more carbon atoms

B. They form stronger dipole–dipole forces

C. They are more polar than alcohols

D. They have no O–H, so they cannot hydrogen bond with each other

Q23 Why are small aldehydes & ketones soluble in water?

A. They are H-bond donors

B. The carbonyl O is an H-bond acceptor, so it forms H-bonds with water

C. They are non-polar

D. They react with water to form salts

10.4 Preparation of Aldehydes & Ketones

Q24 Which methods are used to prepare aldehydes & ketones?

A. Hydrogenation of alkynes, dehydration of alcohols, halogenation of alkanes, hydrolysis of ethers

B. Reduction of carboxylic acids only

C. Hydration of alkenes, reduction of alcohols, nitration of benzene, ozonolysis of alkanes

D. Hydration of alkynes, ozonolysis of alkenes, oxidation of alcohols, Friedel–Crafts acylation of benzene

Q25 Hydration of an alkyne needs which reagents?

A. H_2O, H_2SO_4 and $HgSO_4$

B. O_3 then Zn/H_2O

C. $NaBH_4$ then H_3O^+

D. RCOCl and $AlCl_3$

Q26 Hydration of phenylacetylene ($C_6H_5C{\equiv}CH$) with H_2O, H_2SO_4, $HgSO_4$ gives:

A. Styrene

B. Phenylacetaldehyde ($C_6H_5CH_2CHO$)

C. Acetophenone ($C_6H_5COCH_3$), through an enol

D. Benzoic acid

Q27 Ozonolysis of an alkene (O_3, then Zn/H_2O) gives:

A. An alcohol

B. An epoxide

C. An alkyne

D. Two carbonyl compounds (aldehydes and/or ketones) — the C=C is cut

Q28 Oxidation of a 2° alcohol gives:

A. A carboxylic acid

B. A ketone

C. An aldehyde

D. No reaction

Q29 Oxidation of a 3° alcohol gives:

A. A ketone

B. No reaction

C. An aldehyde

D. A carboxylic acid

Q30 A 1° alcohol can be oxidized to:

A. A ketone only

B. An aldehyde or a carboxylic acid, depending on the reagent

C. No reaction

D. An ether

Q31 Friedel–Crafts acylation (benzene + RCOCl, $AlCl_3$) is used to prepare:

A. Aliphatic aldehydes

B. Aromatic ketones

C. Phenols

D. Benzoic acid

10.5 Reaction Sites & Oxidation

Q32 In the carbonyl group, nucleophiles attack:

A. The α-carbon only

B. The C–H bond of the aldehyde

C. The carbonyl oxygen (δ−)

D. The carbonyl carbon (δ+)

Q33 Electrophiles and acids (H^+) attack the carbonyl group at:

A. The carbon

B. The oxygen, which has two non-bonding electron pairs

C. The R group

D. The α-hydrogen

Q34 How can aldehydes and ketones be distinguished in the lab?

A. Aldehydes are easily oxidized under mild conditions, ketones are not (e.g. Tollens' test)

B. Only aldehydes are soluble in water

C. Ketones are easily oxidized, aldehydes are not

D. Only ketones react with $NaBH_4$

Q35 Tollens' reagent is:

A. Copper sulfate in NaOH

B. $KMnO_4$ in water

C. Silver nitrate in ammonium hydroxide, $Ag(NH_3)_2OH$

D. CrO_3 in H_2SO_4

Q36 A positive Tollens' test (silver mirror) means:

A. The ketone is reduced to an alcohol

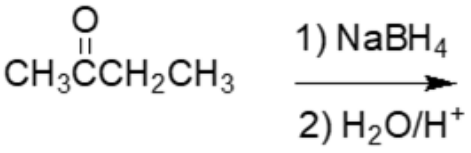
B. Ag is oxidized to Ag^+

C. The aldehyde is reduced to an alcohol

D. The aldehyde is oxidized to a carboxylate and Ag^+ is reduced to metallic silver

10.6 Reduction & Nucleophilic Addition

Q37 Complete the reaction:



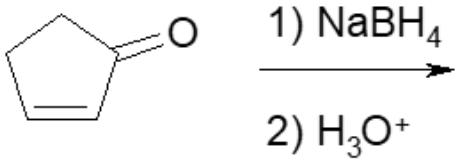
A. 2-Butanol, $CH_3CH(OH)CH_2CH_3$

B. Butane

C. Butanal

D. Butanoic acid

Q38 Complete the reaction:



A. Cyclopentanol — both C=O and C=C are reduced

B. Cyclopentanone — only the C=C is reduced

C. A cyclopentenol — only the C=O is reduced, the C=C stays

D. No reaction

Q39 Base-initiated nucleophilic addition — which step happens first?

A. The nucleophile (Nu^-) attacks the carbonyl carbon

B. The α-H is removed

C. Water leaves

D. H^+ bonds to the carbonyl oxygen

Q40 In the base-initiated mechanism, the intermediate formed after the Nu^- attack is:

A. A trigonal planar enol

B. A free radical

C. A tetrahedral alkoxide ion

D. A carbocation

Q41 Acid-initiated nucleophilic addition — which step happens first?

A. The C=C forms

B. Nu^- attacks the carbonyl carbon

C. The alkoxide takes a proton

D. H^+ bonds to the partially negative carbonyl oxygen

Q42 Addition of an alcohol (R'OH) to an aldehyde first gives a ____, and with a second R'OH and H^+ gives a ____.

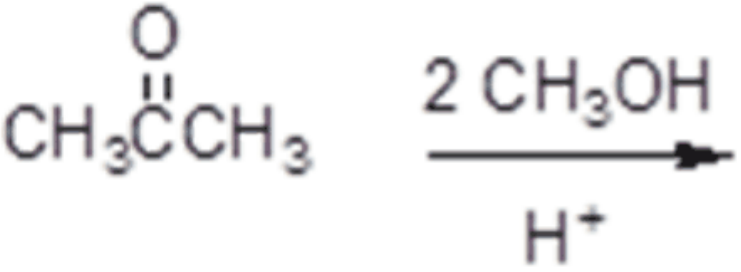
A. Imine; oxime

B. Hydrate; cyanohydrin

C. Acetal; hemiacetal

D. Hemiacetal; acetal

Q43 Complete the reaction:



A. Methyl acetate

B. Isopropanol

C. 2-Methoxy-2-propanol only

D. 2,2-Dimethoxypropane, $(CH_3)_2C(OCH_3)_2$ — an acetal

Q44 Addition of water to an aldehyde or ketone gives a:

A. Hydrate (gem-diol), $R_2C(OH)_2$

B. Enol

C. Carboxylic acid

D. Hemiacetal

Q45 Addition of a Grignard reagent (R'MgX) to a ketone, then H_3O^+, gives:

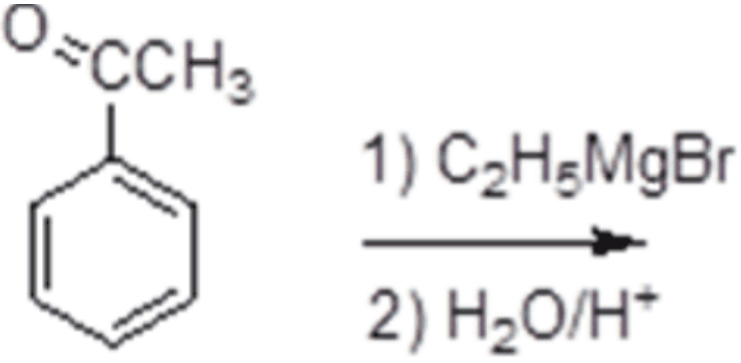
A. A 3° alcohol

B. An aldehyde

C. A 2° alcohol

D. A 1° alcohol

Q46 Complete the reaction:



A. 1-Phenyl-1-propanol

B. 2-Phenylbutane

C. Propiophenone

D. 2-Phenyl-2-butanol, $C_6H_5C(OH)(CH_3)CH_2CH_3$

Q47 Addition of HCN to an aldehyde or ketone gives a:

A. Nitrile, RCN

B. Cyanohydrin, $R_2C(OH)CN$

C. Imine

D. Amide

10.7 Nitrogen Derivatives

Q48 An aldehyde or ketone + a 1° amine (RNH_2) gives:

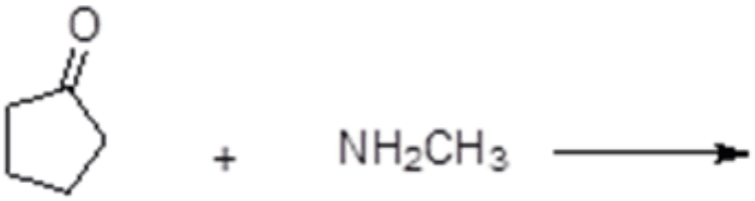
A. An amide

B. A cyanohydrin

C. A 2° amine

D. An imine (Schiff base), $R_2C{=}NR' + H_2O$

Q49 Complete the reaction:



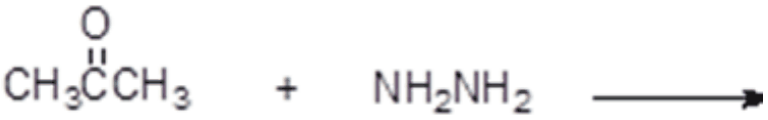
A. Cyclopentanol

B. N-Methylcyclopentylamine

C. N-Methylcyclopentanimine ($C{=}N{-}CH_3$) + H_2O

D. Cyclopentanone oxime

Q50 Complete the reaction:



A. Acetone oxime, $(CH_3)_2C{=}NOH$

B. Acetone hydrazone, $(CH_3)_2C{=}N{-}NH_2 + H_2O$

C. Isopropanol

D. 2-Propanamine

Q51 Which pair is correct?

A. Phenylhydrazine → imine

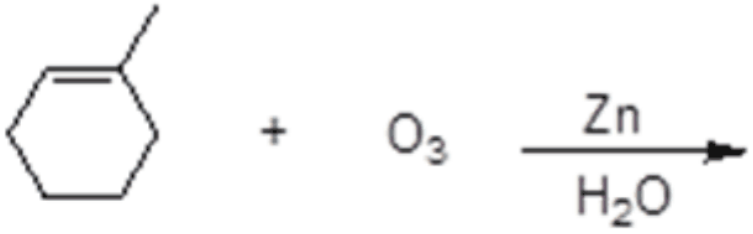
B. Hydroxylamine (NH_2OH) → oxime ($C{=}N{-}OH$)

C. Hydrazine (NH_2NH_2) → oxime

D. 1° amine → hydrazone

10.8 Ozonolysis Practice

Q52 Complete the reaction:



A. 1-Methylcyclohexanol

B. Two molecules: cyclohexanone + formaldehyde

C. Heptanoic acid

D. 6-Oxoheptanal, $CH_3CO(CH_2)_4CHO$ — one molecule

10.9 Tautomerism & α-Hydrogens

Q53 Aldehydes & ketones can exist as a mixture of two forms called:

A. Keto and enol forms

B. Chair and boat forms

C. Cis and trans forms

D. R and S forms

Q54 What is the difference between the keto and enol forms?

A. The molecular formula

B. The position of a hydrogen and of the double bond

C. The number of carbon atoms

D. The arrangement in space only (stereoisomers)

Q55 Tautomers are:

A. Resonance structures of the same molecule

B. Structural isomers that differ in the position of a hydrogen (and a double bond) and rapidly interconvert

C. Stereoisomers that are mirror images

D. Molecules with different molecular formulas

Q56 An α-hydrogen is:

A. A hydrogen on the carbon next to the carbonyl group

B. The hydrogen of an O–H group

C. Any hydrogen on a benzene ring

D. The hydrogen on the carbonyl carbon of an aldehyde

Q57 Why are α-hydrogens acidic?

A. The anion formed (enolate) is stabilized by resonance — the negative charge is shared with the oxygen

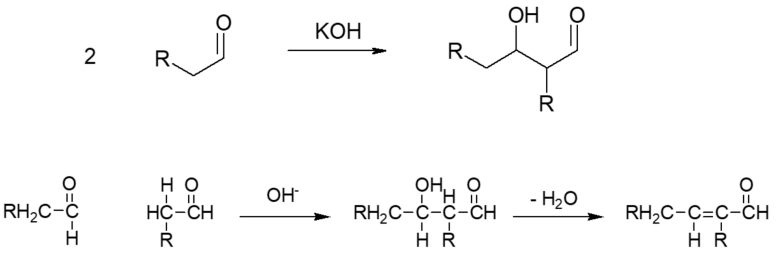
B. Because they are bonded to oxygen

C. Because they are part of an O–H group

D. Because the α-carbon is sp hybridized

10.10 Aldol Condensation

Q58 What are the three steps of the aldol reaction (base, KOH)?



A. 1) Oxidation · 2) reduction · 3) hydration

B. 1) Nu^- attacks · 2) H_2O is added · 3) the C=C is cut

C. 1) OH^- removes an α-H → enolate · 2) enolate attacks the C=O of a second molecule → alkoxide · 3) alkoxide takes H^+ from water → β-hydroxy aldehyde

D. 1) H^+ adds to O · 2) water leaves · 3) carbocation forms

Q59 The product of the aldol reaction is a:

A. Diol

B. Ester

C. α-Hydroxy acid

D. β-Hydroxy aldehyde (or ketone)

Q60 If the aldol product loses water ($-H_2O$), the product is:

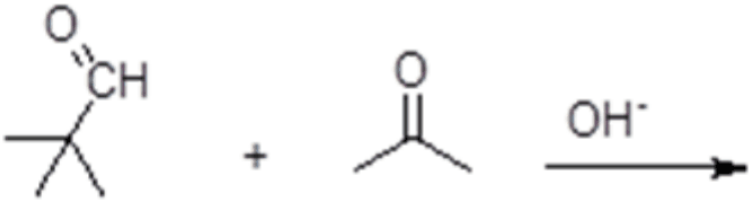
A. A carboxylic acid

B. A saturated alcohol

C. An α,β-unsaturated aldehyde (C=C next to the C=O)

D. An alkyne

Q61 Complete the reaction (crossed aldol):



A. $(CH_3)_3C–CH(OH)–CH_2–CO–CH_3$ (4-hydroxy-5,5-dimethyl-2-hexanone)

B. No reaction — neither compound has α-H

C. $(CH_3)_3C–CO–CH_3$

D. $(CH_3)_3C–COOH$

Answers · Lecture 10

10.1 Structure of Aldehydes & Ketones

Q1 **A. Aldehydes have at least one H on the carbonyl carbon; ketones have two carbon groups on the carbonyl carbon**

General structure R1–CO–R2: aldehyde → R1 or R2 = H; ketone → R1 and R2 are both carbon groups.

Q2 **C. The carbonyl group (C=O)**

Aldehydes and ketones are both carbonyl compounds.

Q3 **A. An aldehyde**

–CHO = H on the carbonyl carbon → aldehyde (propanal).

Q4 **A. A ketone**

The carbonyl carbon has two CH_3 groups → ketone (propanone, acetone).

10.2 Nomenclature of Aldehydes & Ketones

Q5 **C. -al**

Aldehyde → suffix **-al** (prefix oxo). Ketone → **-one** (prefix oxo). Alcohol → -ol (prefix hydroxy). Amine → -amine (prefix amino).

Q6 **C. Aldehyde > ketone > alcohol > amine**

Aldehyde > ketone > alcohol > amine. The highest one gets the suffix; the others are named as prefixes.

Q7 **A. oxo**

Both aldehyde and ketone use the prefix **oxo** when they are not the main group.

Q8 **C. The functional group named by the suffix (C=O)**

Number the chain to give the lowest number to the suffix group. Next come C–C multiple bonds (C=C before C≡C when there is a tie), then the prefixes (alphabetical order).

Q9 **A. Whether the chain has only single bonds, a double bond, or a triple bond**

-an = all C–C single bonds, -en = double bond, -yn = triple bond.

Q10 **C. 3,7-Dimethyl-2,6-octadienal**

Longest chain = 8 C (**oct**) · two C=C (**octadien**) · aldehyde (**octadienal**) · number from the CHO: C=C at 2 and 6 · methyls at 3 and 7 → **3,7-dimethyl-2,6-octadienal**.

Q11 **B. 5-Hydroxy-3-pentyn-2-one**

Chain = 5 C (**pent**) · triple bond (**pentyn**) · ketone is the main group (**pentynone**) · number from the ketone end: C=O at 2, C≡C at 3 · OH at 5 → **5-hydroxy-3-pentyn-2-one**.

Q12 **B. 4-Hydroxy-2-pentanone**

Ketone has priority over alcohol → suffix -one, OH → hydroxy. Number from the end nearest the C=O: C=O at 2, OH at 4 → **4-hydroxy-2-pentanone**.

Q13 **C. 5-Amino-3-oxohexanal**

Aldehyde is the highest priority → suffix -al, CHO = C1. Ketone at C3 → **oxo**, NH_2 at C5 → **amino**. Prefixes in alphabetical order → **5-amino-3-oxohexanal**.

Q14 **C. 2,4-Pentanedione**

5-carbon chain with two ketone groups at C2 and C4 → **2,4-pentanedione**.

Q15 **D. Hexanal**

6 carbons including the CHO carbon → **hexanal**.

Q16 **D. 2-Hexanone**

6-carbon chain, C=O at C2 (lowest number) → **2-hexanone**.

Q17 **B. 5,5-Dimethyl-3-hexanone**

Longest chain = 6 C. Number from the right so the C=O gets the lowest number (3, not 4) → two methyls on C5 → **5,5-dimethyl-3-hexanone**.

Q18 **D. Cyclohexanone**

6-membered ring with a ketone → **cyclohexanone**.

Q19 **C. Cyclobutanone**

4-membered ring with a ketone → **cyclobutanone**.

10.3 Physical Properties

Q20 **B. Between alkanes and alcohols of similar molecular weight**

Intermediate between alkanes and alcohols.

Q21 **A. The C=O group is polar, so there are dipole–dipole attractions between molecules**

The polar C=O gives dipole–dipole attractions; alkanes are non-polar.

Q22 **D. They have no O–H, so they cannot hydrogen bond with each other**

Alcohols have O–H and form hydrogen bonds between molecules; aldehydes & ketones cannot (no H on O).

Q23 **B. The carbonyl O is an H-bond acceptor, so it forms H-bonds with water**

Smaller compounds are soluble in water because the C=O oxygen accepts H-bonds from water.

10.4 Preparation of Aldehydes & Ketones

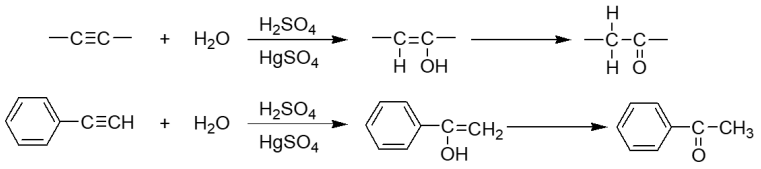
Q24 **D. Hydration of alkynes, ozonolysis of alkenes, oxidation of alcohols, Friedel–Crafts acylation of benzene**

The 4 methods: (A) hydration of alkynes, (B) ozonolysis of alkenes, (C) oxidation of alcohols, (D) acylation of benzene (Friedel–Crafts).

Q25 **A. H_2O, H_2SO_4 and $HgSO_4$**

Alkyne + H_2O with H_2SO_4/$HgSO_4$ → enol → ketone (aldehyde only from ethyne).

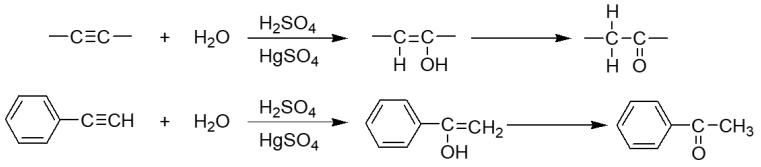
A) Hydration of alkynes (—C≡C—):



Q26 **C. Acetophenone ($C_6H_5COCH_3$), through an enol**

Markovnikov addition: OH goes on the carbon attached to the ring → enol → tautomerizes to the ketone $C_6H_5COCH_3$.

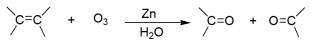
A) Hydration of alkynes (—C≡C—):



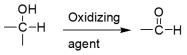
Q27 **D. Two carbonyl compounds (aldehydes and/or ketones) — the C=C is cut**

The C=C breaks and each carbon becomes C=O.

B) Ozonolysis of alkenes ($H_2C=CH_2$):

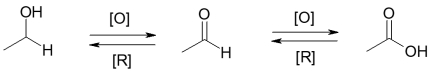


C) Oxidation of alcohols (OH):



The nature of the alcohol and oxidizing agent determine the product.

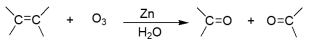
1º Alcohol	Aldehyde or carboxylic acid (depends on the reagent)
2º Alcohol	Ketone
3º Alcohol	No reaction



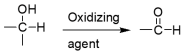
Q28 **B. A ketone**

1° alcohol → aldehyde or carboxylic acid (depends on the reagent) · 2° alcohol → **ketone** · 3° alcohol → no reaction.

B) Ozonolysis of alkenes ($H_2C=CH_2$):

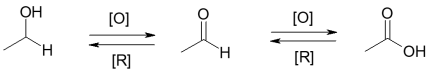


C) Oxidation of alcohols (OH):



The nature of the alcohol and oxidizing agent determine the product.

1º Alcohol	Aldehyde or carboxylic acid (depends on the reagent)
2º Alcohol	Ketone
3º Alcohol	No reaction



Q29 **B. No reaction**

3° alcohols have no H on the C–OH carbon, so they are not oxidized.

Q30 **B. An aldehyde or a carboxylic acid, depending on the reagent**

Mild oxidizing agent → aldehyde; strong oxidizing agent → oxidized further to the carboxylic acid.

Q31 **B. Aromatic ketones**

Benzene + acyl chloride with $AlCl_3$ → aryl ketone (C_6H_5COR) + HCl.

D) Acylation of benzene (Friedel-Crafts Reaction): (prepare aromatic ketones)

10.5 Reaction Sites & Oxidation

Q32 **D. The carbonyl carbon (δ+)**

O is more electronegative → polar C=O. Nucleophiles and bases are attracted to the C (δ+); electrophiles and acids are attracted to the O (δ−), which has two non-bonding electron pairs.

Q33 **B. The oxygen, which has two non-bonding electron pairs**

The O is δ− and has 2 lone pairs, so E^+ and H^+ go to the O.

Q34 **A. Aldehydes are easily oxidized under mild conditions, ketones are not (e.g. Tollens' test)**

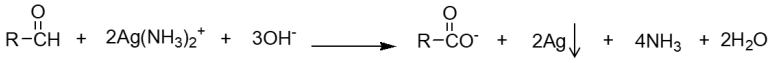
They differ in their susceptibility to oxidation: aldehydes are oxidized by mild oxidizing agents; ketones do not react.

Q35 **C. Silver nitrate in ammonium hydroxide, $Ag(NH_3)_2OH$**

Tollens' reagent = $AgNO_3$ in NH_4OH → $[Ag(NH_3)_2]OH$.

Q36 **D. The aldehyde is oxidized to a carboxylate and Ag^+ is reduced to metallic silver**

$R–CHO + 2Ag(NH_3)_2^+ + 3OH^- \rightarrow R–COO^- + 2Ag\downarrow + 4NH_3 + 2H_2O$. The silver deposits on the clean glass of the test tube.



10.6 Reduction & Nucleophilic Addition

Q37 **A. 2-Butanol, $CH_3CH(OH)CH_2CH_3$**

$NaBH_4$ reduces the ketone C=O to a 2° alcohol → **2-butanol**.

Q38 **C. A cyclopentenol — only the C=O is reduced, the C=C stays**

$NaBH_4$ reduces the C=O to C–OH but does not reduce the C=C.

Q39 **A. The nucleophile (Nu^-) attacks the carbonyl carbon**

1) Nu^- attacks the carbonyl C and gives both electrons for the new Nu–C bond. 2) The π electrons move to the O → alkoxide anion (tetrahedral). 3) The O^- takes H^+ from H–Nu → alcohol product.

Q40 **C. A tetrahedral alkoxide ion**

The carbonyl C goes from trigonal planar to tetrahedral; the O carries the negative charge (alkoxide).

Q41 **D. H^+ bonds to the partially negative carbonyl oxygen**

1) H^+ bonds to the δ− O → the carbon becomes a carbocation (C^+). 2) The C^+ attracts the Nu to the carbon.

Q42 **D. Hemiacetal; acetal**

RCHO + R'OH ⇌ hemiacetal RCH(OH)OR'. Hemiacetal + R'OH (H^+) → acetal $RCH(OR')_2 + H_2O$.

Q43 **D. 2,2-Dimethoxypropane, $(CH_3)_2C(OCH_3)_2$ — an acetal**

Ketone + 2 CH_3OH (H^+) → acetal $(CH_3)_2C(OCH_3)_2 + H_2O$.

Q44 **A. Hydrate (gem-diol), $R_2C(OH)_2$**

$C=O + H_2O \rightleftharpoons R_2C(OH)_2$ — two OH groups on the same carbon.

Q45 **A. A 3° alcohol**

Formaldehyde → 1° alcohol · other aldehydes → 2° alcohol · ketones → 3° alcohol.

Q46 **D. 2-Phenyl-2-butanol, $C_6H_5C(OH)(CH_3)CH_2CH_3$**

Acetophenone (ketone) + C_2H_5MgBr → the ethyl adds to the carbonyl C; H_3O^+ gives the 3° alcohol **2-phenyl-2-butanol**.

Q47 **B. Cyanohydrin, $R_2C(OH)CN$**

CN^- attacks the C, the O takes H^+ → OH and CN on the same carbon (cyanohydrin).

10.7 Nitrogen Derivatives

Q48 **D. An imine (Schiff base), $R_2C{=}NR' + H_2O$**

Addition of RNH_2 then loss of water → C=N (imine).

Q49 **C. N-Methylcyclopentanimine (C=N–CH_3) + H_2O**

Cyclopentanone + CH_3NH_2 → imine: the O is replaced by =N–CH_3, and H_2O is lost.

Q50 **B. Acetone hydrazone, $(CH_3)_2C{=}N{-}NH_2 + H_2O$**

Ketone + hydrazine (NH_2NH_2) → hydrazone (C=N–NH_2) + H_2O.

Q51 **B. Hydroxylamine (NH_2OH) → oxime (C=N–OH)**

1° amine → imine (C=NR) · hydroxylamine → oxime (C=NOH) · hydrazine → hydrazone ($C{=}NNH_2$) · phenylhydrazine → phenylhydrazone ($C{=}NNHC_6H_5$).

10.8 Ozonolysis Practice

Q52 **D. 6-Oxoheptanal, $CH_3CO(CH_2)_4CHO$ — one molecule**

The C=C is inside the ring, so cutting it opens the ring into one chain: the methyl carbon becomes a ketone and the other carbon becomes an aldehyde → **6-oxoheptanal**.

10.9 Tautomerism & α-Hydrogens

Q53 **A. Keto and enol forms**

Keto form: C=O with an H on the α-carbon. Enol form: C=C–OH (ene + ol). The keto form is usually the major one.

Q54 **B. The position of a hydrogen and of the double bond**

The H moves from the α-carbon to the O, and the double bond moves from C=O to C=C.

Q55 **B. Structural isomers that differ in the position of a hydrogen (and a double bond) and rapidly interconvert**

Tautomers are real, different compounds in equilibrium — not resonance forms.

Q56 **A. A hydrogen on the carbon next to the carbonyl group**

The carbon next to C=O is the α-carbon; its H atoms are α-hydrogens.

Q57 **A. The anion formed (enolate) is stabilized by resonance — the negative charge is shared with the oxygen**

Removing an α-H gives an enolate ion; the negative charge is delocalized onto the electronegative O, so α-H are much more acidic than normal C–H.

10.10 Aldol Condensation

Q58 **C. 1) OH^- removes an α-H → enolate · 2) enolate attacks the C=O of a second molecule → alkoxide · 3) alkoxide takes H^+ from water → β-hydroxy aldehyde**

Enolate formation → nucleophilic attack on the second carbonyl → protonation. The product is an aldol (ald + ol): a β-hydroxy aldehyde. With heat it loses H_2O → α,β-unsaturated aldehyde.

Q59 **D. β-Hydroxy aldehyde (or ketone)**

"Aldol" = aldehyde + alcohol in the same molecule; the OH is on the β-carbon.

Q60 **C. An α,β-unsaturated aldehyde (C=C next to the C=O)**

Dehydration forms a C=C between the α and β carbons → condensation product.

Q61 **A. $(CH_3)_3C–CH(OH)–CH_2–CO–CH_3$ (4-hydroxy-5,5-dimethyl-2-hexanone)**

2,2-Dimethylpropanal has **no α-H**, so only acetone forms the enolate. The acetone enolate attacks the aldehyde C=O → β-hydroxy ketone. (With heat it can lose H_2O → $(CH_3)_3C–CH=CH–CO–CH_3$.)

11 Carboxylic Acids & Derivatives

In this lecture

1. Structure & Nomenclature of Carboxylic Acids 10 questions
2. Physical Properties 4 questions
3. Acidity of Carboxylic Acids 9 questions
4. Preparation of Carboxylic Acids 4 questions
5. Carboxylic Acid Derivatives: Structure & Names 5 questions
6. Preparation of Acid Derivatives 4 questions
7. Nucleophilic Acyl Substitution 13 questions
8. Reactions with Grignard Reagents & $LiAlH_4$ 5 questions

11.1 Structure & Nomenclature of Carboxylic Acids

Q1 What is the general structure of a carboxylic acid?

A. R–CHO — a carbonyl with an H

B. R–COOH — a carbonyl (C=O) and a hydroxyl (–OH) on the same carbon

C. R–O–R — oxygen between two carbons

D. R–CO–R — a carbonyl between two carbons

Q2 In IUPAC names of carboxylic acids, the -e of the alkane is replaced by:

A. -oic acid

B. -ol

C. -al

D. -one

Q3 When numbering the chain of a carboxylic acid, the COOH carbon is:

A. Given the highest number

B. Not counted in the chain

C. C2 (the α-carbon)

D. Always C1

Q4 In common names, the carbon next to the COOH is called:

A. γ (gamma)

B. ω (omega)

C. α (alpha)

D. β (beta)

Q5 In a molecule with a COOH and a ketone, the ketone is named with the prefix:

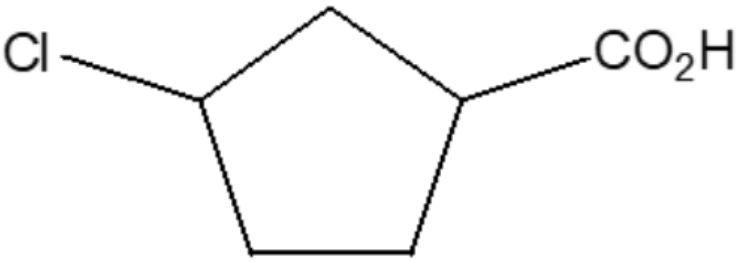
A. hydroxy

B. keto-one

C. carbonyl

D. oxo

Q6 Name this compound:



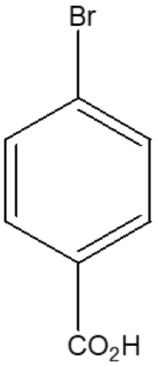
A. 4-Chlorocyclopentanecarboxylic acid

B. 1-Chloro-3-cyclopentanoic acid

C. 3-Chlorocyclopentanecarboxylic acid

D. 3-Chlorocyclopentanoic acid

Q7 Name this compound:



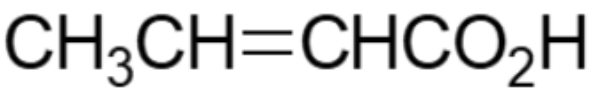
A. 3-Bromobenzoic acid (m-bromobenzoic acid)

B. 4-Bromophenol

C. 4-Bromobenzoic acid (p-bromobenzoic acid)

D. 2-Bromobenzoic acid (o-bromobenzoic acid)

Q8 Name this compound:



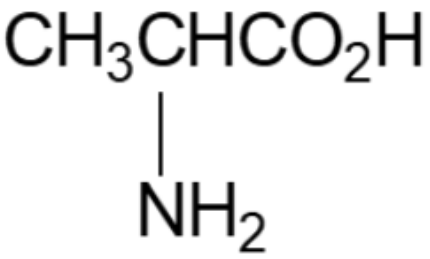
A. 2-Butenal

B. 2-Butenoic acid

C. Butanoic acid

D. 3-Butenoic acid

Q9 Name this compound:



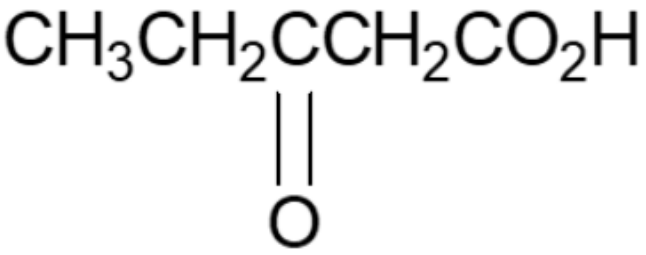
A. 2-Carboxypropanamine

B. 1-Aminopropanoic acid

C. 2-Aminopropanoic acid

D. 3-Aminopropanoic acid

Q10 Name this compound:



A. 3-Oxopentanoic acid

B. 3-Oxopentanal

C. 4-Oxopentanoic acid

D. 3-Pentanone-1-oic acid

11.2 Physical Properties

Q11 How do the boiling points of carboxylic acids compare with alcohols of similar molecular weight?

A. Carboxylic acids do not have boiling points

B. They are the same

C. Carboxylic acids boil higher

D. Carboxylic acids boil lower

Q12 Carboxylic acids have high boiling points mainly because:

A. They form hydrogen-bonded dimers (two H-bonds between two molecules)

B. They are non-polar

C. They cannot form hydrogen bonds

D. They are ionic in the pure liquid

Q13 What about the solubility of carboxylic acids in water?

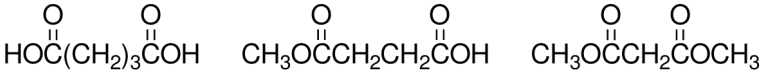
A. They dissolve only because they are ionic

B. All carboxylic acids are insoluble in water

C. Solubility increases as the chain gets longer

D. Small acids (about 1–4 carbons) are water-soluble because of H-bonding; solubility decreases as the carbon chain gets longer

Q14 These three compounds have the same molecular formula ($C_5H_8O_4$). Arrange them in order of increasing boiling point.



A. $CH_3OCOCH_2COOCH_3$ < $CH_3OCOCH_2CH_2COOH$ < $HOOC(CH_2)_3COOH$

B. $CH_3OCOCH_2CH_2COOH$ < $CH_3OCOCH_2COOCH_3$ < $HOOC(CH_2)_3COOH$

C. $HOOC(CH_2)_3COOH$ < $CH_3OCOCH_2CH_2COOH$ < $CH_3OCOCH_2COOCH_3$

D. All three have the same boiling point

11.3 Acidity of Carboxylic Acids

Q15 Order of acidity (most → least):

A. Phenol > carboxylic acid > alcohol

B. Alcohol > phenol > carboxylic acid

C. Carboxylic acid > phenol > alcohol

D. Carboxylic acid > alcohol > phenol

Q16 Why are carboxylic acids acidic?

A. Because the alkyl group donates electrons

B. Because they contain a C=C bond

C. The carboxylate ion ($RCOO^-$) is stabilized by resonance — the negative charge is shared equally by two oxygens

D. Because the O–H bond is very strong

Q17 Which reagent reacts with a carboxylic acid but NOT with a phenol?

A. NaOH

B. HCl

C. $NaHCO_3$ (sodium bicarbonate, weak base) — the acid gives CO_2 bubbles

D. H_2O

Q18 Which statement about NaOH is correct?

A. NaOH (strong base) reacts with both carboxylic acids and phenols to form water-soluble salts

B. NaOH does not react with any of them

C. NaOH reacts with carboxylic acids only

D. NaOH reacts only with alcohols

Q19 What can you conclude from these pKa values?

	HCO_2H	CH_3CO_2H	$O_2N\text{-}CH_2CO_2H$
Pka	3.75	4.76	1.68

	FCH_2CO_2H	$ClCH_2CO_2H$	$BrCH_2CO_2H$	ICH_2CO_2H	CH_3CO_2H
Pka	2.59	2.87	2.90	3.17	4.75

	$CH_3CH_2CHCO_2H$ (Cl)	$CH_3CHCH_2CO_2H$ (Cl)	$CH_2CH_2CH_2CO_2H$ (Cl)	$CH_3CH_2CH_2CO_2H$
Pka	2.86	4.05	4.52	4.82

A. The position of the substituent has no effect

B. Alkyl groups increase acidity

C. Electron-withdrawing groups decrease acidity

D. Electron-withdrawing groups increase acidity; the closer they are to the COOH (and the more there are), the stronger the acid

Q20 Which is the strongest acid?

A. $BrCH_2COOH$

B. FCH_2COOH

C. ICH_2COOH

D. $ClCH_2COOH$

Q21 Why is formic acid (HCOOH, pKa 3.75) stronger than acetic acid (CH_3COOH, pKa 4.76)?

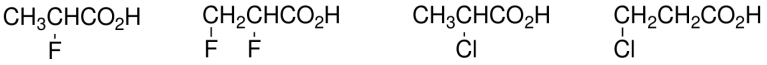
A. Formic acid has more oxygen atoms

B. The CH_3 group donates electrons and destabilizes the carboxylate ion

C. Acetic acid forms more hydrogen bonds

D. The CH_3 group withdraws electrons

Q22 Arrange from least acidic to most acidic:



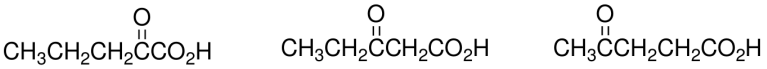
A. $CH_2FCHFCOOH < CH_3CHFCOOH < CH_3CHClCOOH < ClCH_2CH_2COOH$

B. $ClCH_2CH_2COOH < CH_3CHClCOOH < CH_3CHFCOOH < CH_2FCHFCOOH$

C. $CH_3CHFCOOH < CH_3CHClCOOH < ClCH_2CH_2COOH < CH_2FCHFCOOH$

D. $CH_3CHClCOOH < ClCH_2CH_2COOH < CH_3CHFCOOH < CH_2FCHFCOOH$

Q23 Arrange from least acidic to most acidic:



A. 3-Oxopentanoic acid < 2-oxopentanoic acid < 4-oxopentanoic acid

B. 4-Oxopentanoic acid < 3-oxopentanoic acid < 2-oxopentanoic acid

C. They are all equally acidic

D. 2-Oxopentanoic acid < 3-oxopentanoic acid < 4-oxopentanoic acid

11.4 Preparation of Carboxylic Acids

Q24 Which methods are used to prepare carboxylic acids?

A. Oxidation of alkylbenzenes, oxidation of 1° alcohols, hydrolysis of nitriles

B. Dehydration of alcohols only

C. Reduction of esters, hydration of alkenes, ozonolysis of alkynes

D. Oxidation of 3° alcohols, hydrogenation of ketones, nitration of benzene

Q25 Oxidation of toluene ($C_6H_5CH_3$) with $KMnO_4$ (heat) gives:

A. Benzoic acid (C_6H_5COOH)

B. Benzyl alcohol

C. Benzaldehyde

D. Phenol

Q26 Oxidation of a 1° alcohol with a strong oxidizing agent (e.g. CrO_3/H^+ or $KMnO_4$) gives:

A. A ketone

B. An ether

C. No reaction

D. A carboxylic acid

Q27 Hydrolysis of a nitrile (R–C≡N) with H_3O^+ and heat gives:

A. R–COOH + NH_4^+

B. R–OH + HCN

C. R–CHO + NH_3

D. R–CH_2NH_2

11.5 Carboxylic Acid Derivatives: Structure & Names

Q28 Which are the carboxylic acid derivatives?

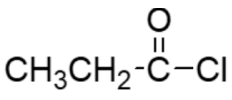
A. Alkenes, alkynes, epoxides, acetals

B. Aldehydes, ketones, alcohols, ethers

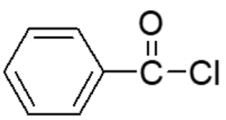
C. Alkyl halides, amines, nitriles, phenols

D. Acid chlorides (RCOCl), acid anhydrides (RCO–O–COR), esters (RCOOR'), amides ($RCONH_2$)

Q29 How do you name an acid chloride?



Propanoyl chloride



Benzoyl chloride

A. Change -oic acid to -amide

B. Change -oic acid to -oate

C. Add the word chloride after the acid name

D. Change -ic acid to -yl chloride (propanoic acid → propanoyl chloride)

Q30 How do you name an acid anhydride (from propanoic or benzoic acid)?

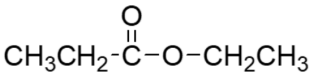
A. Propanoyl anhydride, benzoyl anhydride

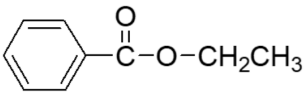
B. Propanoate anhydride

C. Dipropanoic acid

D. Replace the word acid with anhydride: propanoic anhydride, benzoic anhydride

Q31 Name these esters:





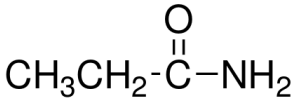
A. Ethyl propanoate and ethyl benzoate

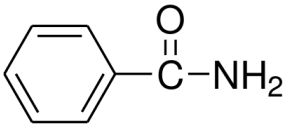
B. Ethyl propanoic acid and ethyl benzoic acid

C. Propanoyl ethane and benzoyl ethane

D. Propyl ethanoate and benzyl ethanoate

Q32 Name these amides:





A. Aminopropanoic acid and aminobenzoic acid

B. Propanoyl amine and benzoyl amine

C. Propanamide and benzamide

D. Propanamine and benzamine

11.6 Preparation of Acid Derivatives

Q33 Acid chlorides are prepared from carboxylic acids with:

A. H_2O

B. $SOCl_2$ (thionyl chloride) or PCl_5

C. $NaBH_4$

D. NaOH

Q34 An acid anhydride can be prepared from:

A. An amide + $LiAlH_4$

B. An alcohol + $KMnO_4$

C. An acid chloride + a carboxylate salt ($RCOCl + RCOO^-Na^+$)

D. An ester + water

Q35 Fischer esterification is:

A. $RCOCl + NH_3 \rightarrow RCONH_2$

B. $RCOOR' + H_2O \rightarrow RCOOH + R'OH$ only in base

C. $RCOOH + R'OH$ with an acid catalyst (H^+) $\rightleftharpoons RCOOR' + H_2O$

D. $RCOOH + NaOH \rightarrow RCOO^-Na^+ + H_2O$

Q36 Amides are best prepared from:

A. An alkene + NH_3

B. An acid chloride + ammonia or an amine

C. An ester + $LiAlH_4$

D. An alcohol + NH_3

11.7 Nucleophilic Acyl Substitution

Q37 The general equation of nucleophilic acyl substitution is:

A. R–CO–L + Nu^- → R–C(OH)(Nu)–L

B. R–CO–L + Nu^- → R–CO–Nu + L^-

C. R–CO–L → R–L + CO

D. R–CO–L + H_2 → R–CH_2–L

Q38 Order of reactivity of the acid derivatives (highest → lowest):

A. Amide > ester > anhydride > acid chloride

B. Acid anhydride > acid chloride > amide > ester

C. Ester > acid chloride > amide > anhydride

D. Acid chloride > acid anhydride > ester > amide

Q39 Acid chloride + H_2O gives:

A. Amide + HCl

B. No reaction

C. Ester + HCl

D. Carboxylic acid + HCl

Q40 Acid chloride + R'OH gives:

A. Amide + HCl

B. Anhydride + HCl

C. Ester + HCl

D. Carboxylic acid + R'Cl

Q41 Acid chloride + R'NH_2 gives:

A. Amine + CO_2

B. Amide (RCONHR') + HCl

C. Ester + HCl

D. Carboxylic acid + R'NH_3^+

Q42 Acid anhydride + H_2O gives:

A. No reaction

B. Amide + acid

C. Ester + acid

D. Two carboxylic acids

Q43 Acid anhydride + R'OH gives:

A. Ester + carboxylic acid

B. Two esters

C. No reaction

D. Ester + HCl

Q44 Acid anhydride + R'NH_2 gives:

A. No reaction

B. Ester + amine

C. Amide + carboxylic acid

D. Two amides

Q45 Ester + H_2O (acid or base) gives:

A. Carboxylic acid (or carboxylate in base) + alcohol

B. Two alcohols

C. Amide + alcohol

D. Ketone + alcohol

Q46 Ester + another alcohol (R''OH) gives:

A. No reaction

B. A carboxylic acid + an ether

C. An amide

D. A new ester + R'OH (transesterification)

Q47 Ester + $R''NH_2$ gives:

A. Amide + alcohol (R'OH)

B. Carboxylic acid + amine

C. Ester + amine

D. No reaction

Q48 Amide + H_2O (acid or base, heat) gives:

A. No reaction under any conditions

B. Carboxylic acid (or its salt) + amine / NH_3

C. Ester + NH_3

D. Acid chloride

Q49 Amide + R'OH or $R'NH_2$ gives:

A. No reaction — amides are the least reactive derivative

B. A new amide

C. An ester

D. A carboxylic acid

11.8 Reactions with Grignard Reagents & $LiAlH_4$

Q50 Ester + 2 R''MgX, then H_3O^+, gives:

A. A 1° alcohol

B. A 3° alcohol with two R'' groups + R'OH

C. A ketone only

D. A carboxylic acid

Q51 Ester + $LiAlH_4$, then H_3O^+, gives:

A. An aldehyde only

B. A 1° alcohol (RCH_2OH) + R'OH

C. An amine

D. A 2° alcohol

Q52 Amide ($RCONR_2$) + $LiAlH_4$, then H_2O/H^+, gives:

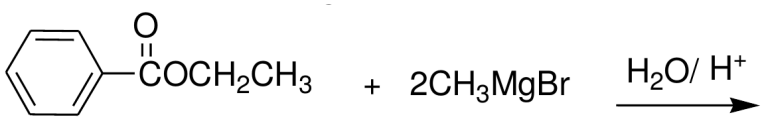
A. A 1° alcohol

B. An amine, RCH_2NR_2

C. A carboxylic acid

D. An aldehyde

Q53 Complete the reaction:



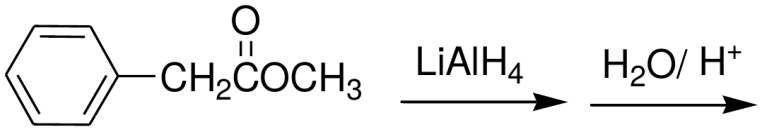
A. Benzoic acid + ethane

B. Acetophenone, $C_6H_5COCH_3$ + CH_3CH_2OH

C. 1-Phenylethanol

D. 2-Phenyl-2-propanol, $C_6H_5C(OH)(CH_3)_2$ + CH_3CH_2OH

Q54 Complete the reaction:



A. 2-Phenylethanol ($C_6H_5CH_2CH_2OH$) + CH_3OH

B. 2-Phenylethylamine

C. Phenylacetic acid + CH_3OH

D. Phenylacetaldehyde + CH_3OH

Answers · Lecture 11

11.1 Structure & Nomenclature of Carboxylic Acids

Q1 **B. R–COOH — a carbonyl (C=O) and a hydroxyl (–OH) on the same carbon**

Carboxylic acids contain the **carboxyl group** –COOH (also written –CO_2H): carb**o**nyl + hydr**oxyl**.

Q2 **A. -oic acid**

Choose the longest chain that contains the COOH, drop the final -e of the alkane name and add **-oic acid** (e.g. propane → propanoic acid).

Q3 **D. Always C1**

The carboxyl carbon is always C1, so no number is needed for the COOH. Other groups get numbers from it.

Q4 **C. α (alpha)**

Common names use Greek letters: the carbon next to COOH = α, then β, γ, δ … (IUPAC C2 = α).

Q5 **D. oxo**

The carboxylic acid has the highest priority, so it gets the suffix (-oic acid); the ketone becomes **oxo**, OH becomes hydroxy, NH_2 becomes amino.

Q6 **C. 3-Chlorocyclopentanecarboxylic acid**

COOH on a ring → **cycloalkanecarboxylic acid**. The ring carbon with COOH = C1, Cl on C3 → **3-chlorocyclopentanecarboxylic acid**.

Q7 **C. 4-Bromobenzoic acid (p-bromobenzoic acid)**

Benzene + COOH = benzoic acid; the COOH carbon is C1, Br is opposite → C4 (para) → **4-bromobenzoic acid**.

Q8 **B. 2-Butenoic acid**

4 carbons with a C=C between C2 and C3 → but**en**oic acid, numbered from the COOH → **2-butenoic acid**.

Q9 **C. 2-Aminopropanoic acid**

3-carbon chain, COOH = C1, NH_2 on C2 → **2-aminopropanoic acid** (the amino acid alanine).

Q10 **A. 3-Oxopentanoic acid**

5 carbons, COOH = C1, ketone C=O on C3 → **3-oxopentanoic acid**.

11.2 Physical Properties

Q11 **C. Carboxylic acids boil higher**

Carboxylic acids boil **higher** than alcohols of similar size because they form strong hydrogen bonds — two molecules join as a **hydrogen-bonded dimer**.

Q12 **A. They form hydrogen-bonded dimers (two H-bonds between two molecules)**

Each COOH has an H-bond donor (O–H) and an acceptor (C=O), so two molecules hold each other with two H-bonds (dimer).

Q13 **D. Small acids (about 1–4 carbons) are water-soluble because of H-bonding; solubility decreases as the carbon chain gets longer**

The COOH group H-bonds with water, so small acids mix with water. A long non-polar R group makes the acid less soluble.

Q14 **A. $CH_3OCOCH_2COOCH_3$ < $CH_3OCOCH_2CH_2COOH$ < $HOOC(CH_2)_3COOH$**

Same molecular weight, so H-bonding decides: the diester has **no O–H** (lowest bp) < the ester-acid has **one COOH** < the diacid has **two COOH** (highest bp).

11.3 Acidity of Carboxylic Acids

Q15 **C. Carboxylic acid > phenol > alcohol**

pKa ≈ 4–5 (carboxylic acid) · ≈ 10 (phenol) · ≈ 16 (alcohol). Lower pKa = stronger acid.

Q16 **C. The carboxylate ion ($RCOO^-$) is stabilized by resonance — the negative charge is shared equally by two oxygens**

After losing H^+, the negative charge of $RCOO^-$ is delocalized over both oxygens (two equivalent resonance structures). A more stable anion = a stronger acid.

Q17 **C. $NaHCO_3$ (sodium bicarbonate, weak base) — the acid gives CO_2 bubbles**

$RCOOH + NaHCO_3 \rightarrow RCOO^-Na^+ + H_2O + CO_2\uparrow$. Phenols are too weak to react with $NaHCO_3$, but they do react with the strong base NaOH.

Q18 **A. NaOH (strong base) reacts with both carboxylic acids and phenols to form water-soluble salts**

$RCOOH + NaOH \rightarrow RCOO^-Na^+ + H_2O$ · $ArOH + NaOH \rightarrow ArO^-Na^+ + H_2O$. Alcohols are too weak to react with aqueous NaOH.

Q19 **D. Electron-withdrawing groups increase acidity; the closer they are to the COOH (and the more there are), the stronger the acid**

Inductive effect: EWG (NO_2, F, Cl, Br, I) pull electrons and stabilize $RCOO^-$ → stronger acid. The more electronegative the group, the stronger the effect (F > Cl > Br > I). The effect gets weaker with distance (2-Cl 2.86 → 3-Cl 4.05 → 4-Cl 4.52). Alkyl groups donate electrons → weaker acid (HCOOH 3.75 vs CH_3COOH 4.76).

Q20 **B. FCH_2COOH**

pKa: F 2.59 < Cl 2.87 < Br 2.90 < I 3.17. F is the most electronegative, so FCH_2COOH is the strongest.

Q21 **B. The CH_3 group donates electrons and destabilizes the carboxylate ion**

Alkyl groups are electron-donating; they increase the negative charge on COO^- → less stable → weaker acid.

Q22 **B. $ClCH_2CH_2COOH$ < $CH_3CHClCOOH$ < $CH_3CHFCOOH$ < $CH_2FCHFCOOH$**

Least: Cl far from COOH (C3) < Cl on C2 < F on C2 (F more electronegative) < two F atoms (most).

Q23 **B. 4-Oxopentanoic acid < 3-oxopentanoic acid < 2-oxopentanoic acid**

The C=O is electron-withdrawing; the closer it is to the COOH, the stronger the acid: C=O on C4 (least) < C3 < C2 (most).

11.4 Preparation of Carboxylic Acids

Q24 **A. Oxidation of alkylbenzenes, oxidation of 1° alcohols, hydrolysis of nitriles**

The 3 methods: (1) oxidation of alkylbenzenes, (2) oxidation of primary alcohols, (3) hydrolysis of nitriles.

Q25 **A. Benzoic acid (C_6H_5COOH)**

Strong oxidizing agents ($KMnO_4$ or $K_2Cr_2O_7/H^+$) oxidize the alkyl side chain on a benzene ring to –COOH. The side chain needs a benzylic H.

Q26 **D. A carboxylic acid**

$R–CH_2OH$ → [R–CHO] → R–COOH. A strong oxidant continues past the aldehyde to the acid.

Q27 **A. R–COOH + NH_4^+**

$R–C{\equiv}N + 2\ H_2O$ (H^+, heat) → R–COOH + NH_4^+. (Base hydrolysis gives $RCOO^-$, then H_3O^+ gives RCOOH.)

11.5 Carboxylic Acid Derivatives: Structure & Names

Q28 **D. Acid chlorides (RCOCl), acid anhydrides (RCO–O–COR), esters (RCOOR'), amides ($RCONH_2$)**

In each derivative the –OH of the acid is replaced: by Cl (acid chloride), by –O–COR (anhydride), by –OR' (ester), by $–NH_2$/–NHR/$–NR_2$ (amide).

Q29 **D. Change -ic acid to -yl chloride (propanoic acid → propanoyl chloride)**

Propanoic acid → **propanoyl chloride**; benzoic acid → **benzoyl chloride**.

Q30 **D. Replace the word acid with anhydride: propanoic anhydride, benzoic anhydride**

Two acid molecules joined with loss of water → name the acid and change **acid** to **anhydride**.

Q31 **A. Ethyl propanoate and ethyl benzoate**

Ester = **alkyl group on O** first (ethyl) + acid name with **-ic acid → -ate** (propanoate, benzoate).

Q32 **C. Propanamide and benzamide**

Amide: change **-oic acid / -ic acid → -amide**: propanoic acid → propanamide, benzoic acid → benzamide.

11.6 Preparation of Acid Derivatives

Q33 **B. $SOCl_2$ (thionyl chloride) or PCl_5**

$RCOOH + SOCl_2$ → $RCOCl + SO_2 + HCl$ (or with PCl_5 / PCl_3).

Q34 **C. An acid chloride + a carboxylate salt ($RCOCl + RCOO^-Na^+$)**

$RCOCl + R'COO^-Na^+$ → $RCO–O–COR' + NaCl$. (Anhydrides can also form by removing water from two COOH groups with heat.)

Q35 **C. RCOOH + R'OH with an acid catalyst (H^+) ⇌ RCOOR' + H_2O**

Esters are made from a carboxylic acid + an alcohol with H^+ (reversible), or from an acid chloride + an alcohol.

Q36 **B. An acid chloride + ammonia or an amine**

$RCOCl + 2\ NH_3$ → $RCONH_2 + NH_4Cl$ (with RNH_2 → $RCONHR$; with R_2NH → $RCONR_2$).

11.7 Nucleophilic Acyl Substitution

Q37 **B. $R–CO–L + Nu^-$ → $R–CO–Nu + L^-$**

The group L attached to the acyl group (RC=O) is replaced by the nucleophile. It goes through a tetrahedral intermediate (addition, then elimination of L).

Q38 **D. Acid chloride > acid anhydride > ester > amide**

The better the leaving group (Cl^- is best, NH_2^- is worst), the more reactive the derivative. A more reactive derivative can be changed into a less reactive one, not the reverse.

Q39 **D. Carboxylic acid + HCl**

$RCOCl + H_2O$ → $RCOOH + HCl$ (hydrolysis).

Q40 **C. Ester + HCl**

RCOCl + R'OH → RCOOR' + HCl (alcoholysis).

Q41 **B. Amide (RCONHR') + HCl**

RCOCl + R'NH_2 → RCONHR' + HCl (aminolysis; a second amine removes the HCl).

Q42 **D. Two carboxylic acids**

$(RCO)_2O + H_2O$ → 2 RCOOH.

Q43 **A. Ester + carboxylic acid**

$(RCO)_2O$ + R'OH → RCOOR' + RCOOH.

Q44 **C. Amide + carboxylic acid**

$(RCO)_2O$ + R'NH_2 → RCONHR' + RCOOH.

Q45 **A. Carboxylic acid (or carboxylate in base) + alcohol**

RCOOR' + H_2O (H^+) ⇌ RCOOH + R'OH. In base (saponification): RCOOR' + NaOH → $RCOO^-Na^+$ + R'OH.

Q46 **D. A new ester + R'OH (transesterification)**

RCOOR' + R''OH (H^+ or base) ⇌ RCOOR'' + R'OH.

Q47 **A. Amide + alcohol (R'OH)**

RCOOR' + R''NH_2 → RCONHR'' + R'OH (ester → less reactive amide).

Q48 **B. Carboxylic acid (or its salt) + amine / NH_3**

Amides are the least reactive; they hydrolyze only with strong acid or base and heat: $RCONH_2 + H_2O$ → RCOOH + NH_4^+ (acid) or $RCOO^-$ + NH_3 (base).

Q49 **A. No reaction — amides are the least reactive derivative**

An amide cannot be converted into a more reactive derivative (ester) and does not normally react with amines.

11.8 Reactions with Grignard Reagents & $LiAlH_4$

Q50 **B. A 3° alcohol with two R'' groups + R'OH**

The first R''MgX substitutes the OR' (gives a ketone), the second adds to the ketone → after H_3O^+: R–C(OH)R''_2 (3° alcohol) + R'OH.

Q51 **B. A 1° alcohol (RCH_2OH) + R'OH**

$LiAlH_4$ (strong reducing agent) reduces the ester to two alcohols: RCH_2OH (from the acyl part) + R'OH.

Q52 **B. An amine, RCH_2NR_2**

$LiAlH_4$ changes the C=O of the amide into CH_2; the nitrogen stays → amine.

Q53 **D. 2-Phenyl-2-propanol, $C_6H_5C(OH)(CH_3)_2$ + CH_3CH_2OH**

Ethyl benzoate + 2 CH_3MgBr → the two CH_3 groups add to the carbonyl C, the OCH_2CH_3 leaves as ethanol → 3° alcohol **$C_6H_5C(OH)(CH_3)_2$**.

Q54 **A. 2-Phenylethanol ($C_6H_5CH_2CH_2OH$) + CH_3OH**

Methyl phenylacetate + $LiAlH_4$ → the acyl part becomes the 1° alcohol **$C_6H_5CH_2CH_2OH$**; the OCH_3 part becomes methanol.

12 Alkyl Halides

In this lecture

1. Types, Uses & Classification 9 questions
2. Physical Properties 3 questions
3. Nomenclature of Alkyl Halides 14 questions
4. Preparation of Alkyl Halides 5 questions
5. Reactions: Substitution & Elimination 8 questions
6. Nucleophilic Substitution Practice 14 questions
7. Safety of Organohalogen Compounds 1 questions

12.1 Types, Uses & Classification

Q1 Which halogens are found in alkyl halides (organohalogen compounds)?

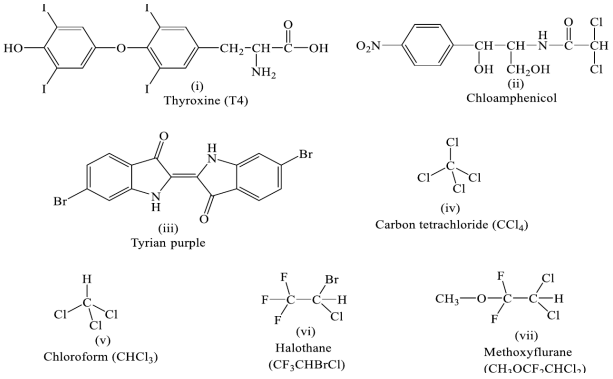
A. F, Cl, Br and I

B. Cl, Br, S and P

C. F, Cl, O and N

D. Only Cl and Br

Q2 Which compounds are organic solvents that may cause liver damage?



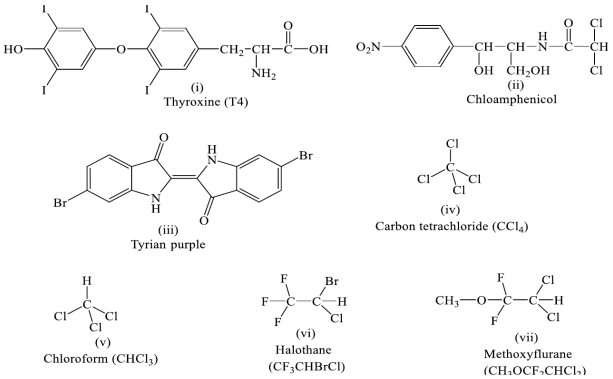
A. Halothane and methoxyflurane

B. Tyrian purple

C. Thyroxine and chloramphenicol

D. Carbon tetrachloride (CCl_4) and chloroform ($CHCl_3$)

Q3 Which alkyl halides are safe and useful as anesthetics?



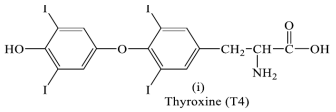
A. Halothane ($CF_3CHBrCl$) and methoxyflurane ($CH_3OCF_2CHCl_2$)

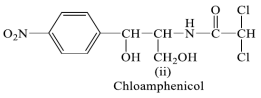
B. CCl_4 and $CHCl_3$

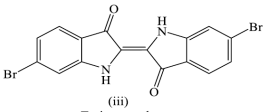
C. Thyroxine

D. Chloramphenicol

Q4 Match: a naturally occurring dye / a thyroid hormone component / an antibiotic from a soil bacterium.


(i)
Thyroxine (T4)

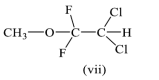

(ii)
Chloamphenicol


(iii)
Tyrian purple

(iv)
Carbon tetrachloride (CCl_4)

(v)
Chloroform ($CHCl_3$)

(vi)
Halothane
($CF_3CHBrCl$)


(vii)
Methoxyflurane
($CH_3OCF_2CHCl_2$)

A. Chloramphenicol / Tyrian purple / thyroxine

B. Halothane / thyroxine / CCl_4

C. Thyroxine / chloramphenicol / Tyrian purple

D. Tyrian purple / thyroxine (T4) / chloramphenicol

Q5 In which halide is the halogen bonded to an sp^2 carbon?

A. Allylic halide (CH_2=CH–CH_2–Cl)

B. Methyl halide (CH_3–Cl)

C. Benzylic halide (C_6H_5–CH_2–Cl)

D. Aryl halide (C_6H_5–Cl) and vinylic halide (CH_2=CH–Cl)

Q6 The halogen atom in an allylic halide is attached to an:

A. sp carbon

B. sp^3 carbon

C. sp^2 carbon

D. None of the above

Q7 A benzylic halide is:

A. X on a carbon attached to a benzene ring, e.g. $C_6H_5CH_2Cl$

B. X on a CH_3 group only

C. X on a C=C carbon

D. X directly on the benzene ring, e.g. C_6H_5Cl

Q8 How are alkyl halides classified as 1°, 2° and 3°?

A. By the number of halogen atoms

B. By the number of carbon atoms bonded to the carbon that carries the halogen (1, 2 or 3)

C. By the length of the chain

D. By the type of halogen

Q9 $(CH_3)_3C$–Br is a:

A. 2° alkyl halide

B. 1° alkyl halide

C. Methyl halide

D. 3° alkyl halide

12.2 Physical Properties

Q10 From the table, which compound has the highest and which the lowest boiling point?

R–X	Cl	Br	I
CH_3–X	−24 (gas)	5 (gas)	43 (liquid)
CH_3CH_2–X	12.5 (gas)	38 (liquid)	72 (liquid)
CH_2X_2	40 (liquid)	99 (liquid)	180 (liquid)
CHX_3	61 (liquid)	151 (liquid)	sublimes (solid)
CX_4	77 (liquid)	189 (liquid)	sublimes (solid)
CH_2=CH–X	−14 (gas)	14 (gas)	56 (gas)
Phenyl–X	131 (liquid)	156 (liquid)	188 (liquid)

b.p. in °C (physical state)

A. Highest: CH_3Cl; lowest: CBr_4

B. Highest: CBr_4 (189 °C) — and CHI_3/CI_4 are solids; lowest: CH_3Cl (−24 °C)

C. Highest: CH_2=CHCl; lowest: phenyl iodide

D. All have the same boiling point

Q11 For the same alkyl group, the boiling point order is:

A. R–I > R–Br > R–Cl

B. They are all equal

C. R–Br > R–I > R–Cl

D. R–Cl > R–Br > R–I

Q12 Are organohalogen compounds water-soluble?

A. Only the iodides are soluble

B. Yes — they are ionic

C. Yes — they form strong hydrogen bonds with water

D. No — they cannot form hydrogen bonds with water, so they are insoluble (but dissolve in organic solvents)

12.3 Nomenclature of Alkyl Halides

Q13 In IUPAC names, halogens are named as:

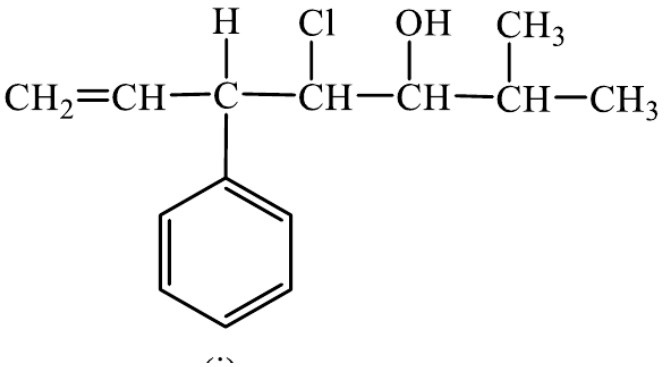
A. Suffixes: -fluoride, -chloride

B. They are not named

C. Suffixes: -ol

D. Prefixes: fluoro-, chloro-, bromo-, iodo-

Q14 Name this compound:



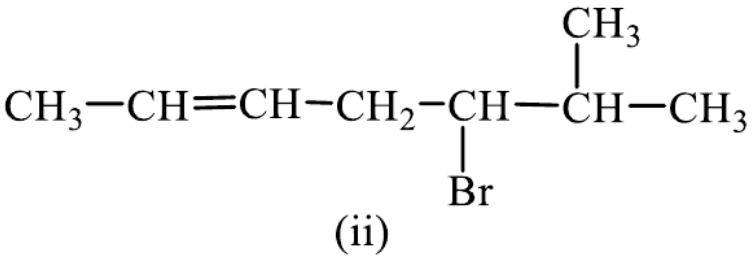
A. 5-Chloro-2-methyl-6-hepten-3-ol

B. 4-Chloro-2-methyl-5-phenylhept-6-en-3-ol

C. 4-Chloro-6-methyl-3-phenylhept-1-en-5-ol

D. 4-Chloro-2-methyl-5-phenylhept-6-en-3-one

Q15 Name this compound:



(ii)

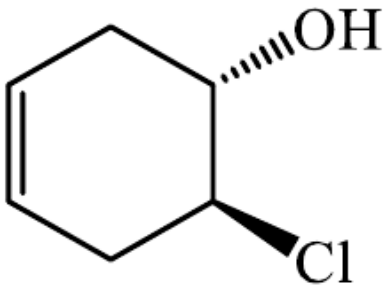
A. 2-Methyl-3-bromohept-5-ene

B. 5-Bromo-6-methylhept-2-ene

C. 5-Bromo-6-methylhept-3-ene

D. 3-Bromo-2-methylhept-5-ene

Q16 Name this compound:



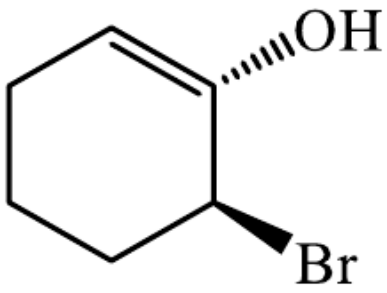
A. (1R,6S)-1-Chlorocyclohex-3-en-6-ol

B. (1R,2R)-2-Chlorocyclohex-4-en-1-ol

C. (1S,6R)-6-Chlorocyclohex-3-en-1-ol

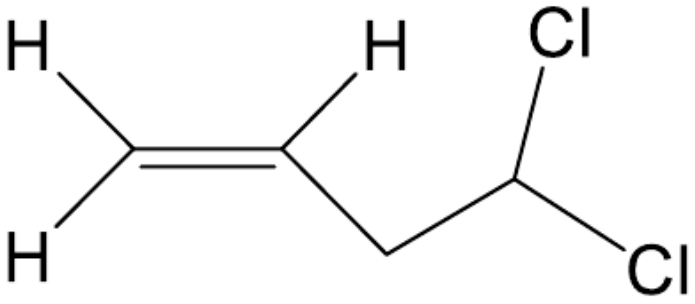
D. (1S,6S)-6-Chlorocyclohex-3-en-1-ol

Q17 Name this compound:



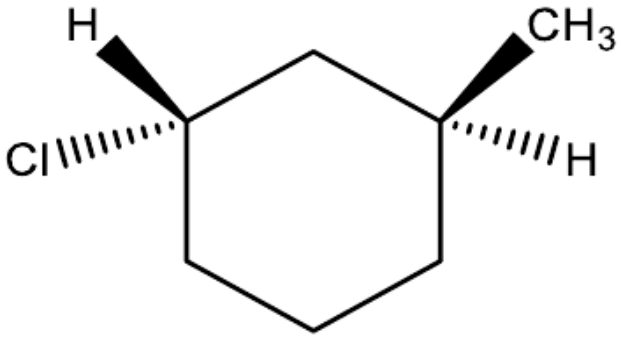
A. (S)-2-Bromocyclohex-6-en-1-ol
B. (R)-6-Bromocyclohex-1-en-1-ol
C. (S)-6-Bromocyclohex-1-en-1-ol
D. (R)-1-Bromocyclohex-2-en-3-ol

Q18 The correct IUPAC name for this compound is:



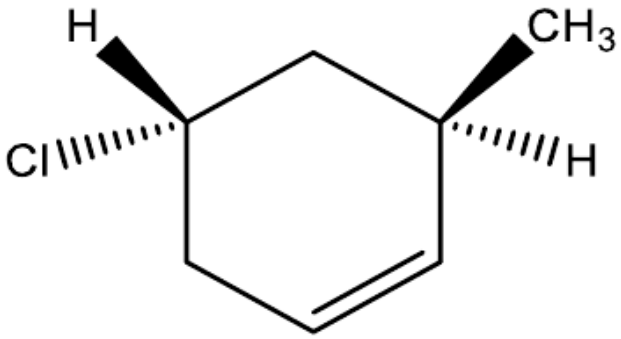
A. 1,1-Dichlorobut-1-ene
B. 1,1-Dichlorobut-3-ene
C. 1,1-Dichlorobutene
D. 4,4-Dichloro-1-butene

Q19 The correct IUPAC name for this compound is:



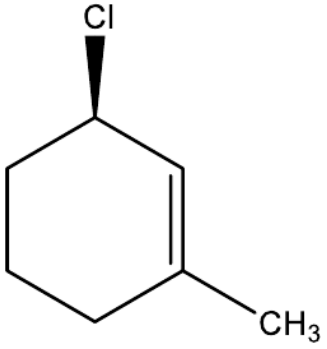
A. (1S,3S)-1-Chloro-3-methylcyclohexane
B. (1S,3R)-1-Chloro-3-methylcyclohexane
C. (1R,3S)-1-Chloro-3-methylcyclohexane
D. (1R,3R)-1-Chloro-3-methylcyclohexane

Q20 The correct IUPAC name for this compound is:



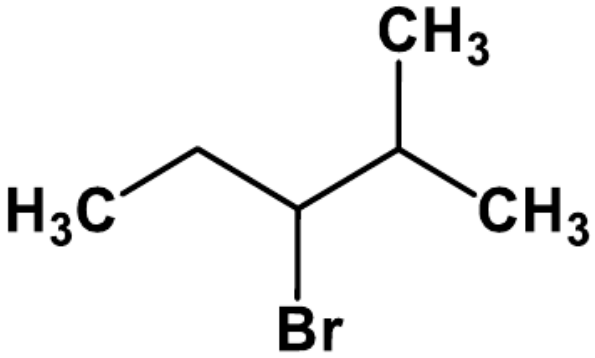
A. (4R,6S)-4-Chloro-6-methylcyclohex-1-ene
B. (4S,6R)-4-Chloro-6-methylcyclohex-1-ene
C. (3R,5S)-5-Chloro-3-methylcyclohex-1-ene
D. (3S,5R)-5-Chloro-3-methylcyclohex-1-ene

Q21 What is the IUPAC name of this compound?



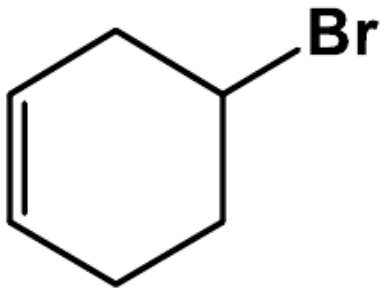
A. (R)-3-Chloro-1-methylcyclohexene
B. (R)-1-Chloro-3-methyl-2-cyclohexene
C. (S)-3-Chloro-1-methylcyclohexene
D. (S)-1-Chloro-3-methyl-2-cyclohexene

Q22 Name this compound:



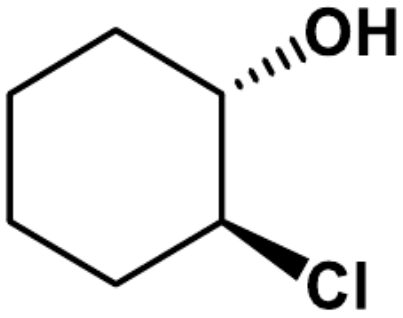
A. 3-Bromo-2-methylpentane
B. 2-Methyl-3-bromopentane
C. 3-Bromo-2-ethylbutane
D. 3-Bromo-4-methylpentane

Q23 Name this compound:



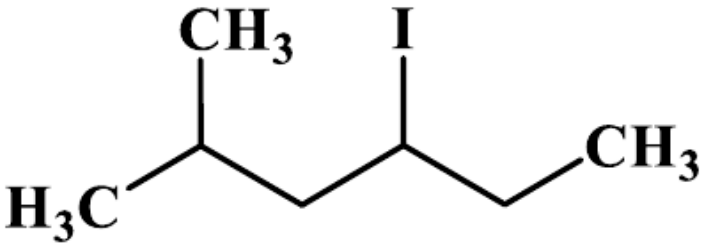
A. 5-Bromocyclohexene
B. 3-Bromocyclohexene
C. 1-Bromocyclohex-3-ene
D. 4-Bromocyclohexene

Q24 Name this compound:



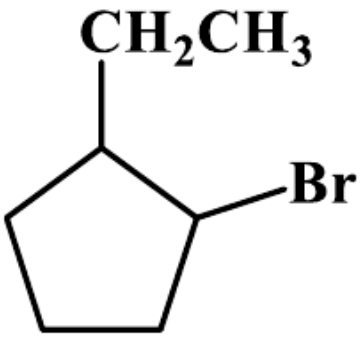
A. (1S,2S)-2-Chlorocyclohexan-1-ol
B. (1R,2S)-1-Chlorocyclohexan-2-ol
C. (1S,2R)-2-Chlorocyclohexan-1-ol
D. (1R,2R)-2-Chlorocyclohexan-1-ol

Q25 Name this compound:



A. 3-Iodo-5-methylhexane
B. 4-Iodo-2-ethylpentane
C. 4-Iodo-2-methylhexane
D. 2-Methyl-4-iodohexane

Q26 Name this compound:



A. 1-Bromo-2-ethylcyclopentane

B. 1-Bromo-2-ethylcyclohexane

C. 2-Bromo-1-ethylcyclopentane

D. 1-Ethyl-2-bromocyclopentane

12.4 Preparation of Alkyl Halides

Q27 Which reactions of alkenes and alcohols prepare alkyl halides?

A. Oxidation of alcohols with $KMnO_4$

B. Hydrogenation of alkenes

C. Addition of HX or X_2 to alkenes, and substitution of the OH of alcohols (HX, $SOCl_2$, PBr_3)

D. Ozonolysis of alkenes

Q28 $CH_3CH{=}CH_2$ + HBr gives (major product):

A. $CH_3CH_2CH_3$

B. $CH_3CH_2CH_2Br$ (1-bromopropane)

C. $CH_3CHBrCH_3$ (2-bromopropane) — Markovnikov

D. $CH_3CHBrCH_2Br$

Q29 $CH_2{=}CH_2$ + Br_2 gives:

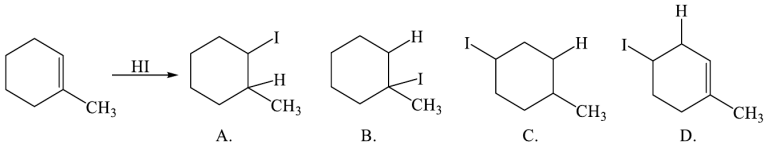
A. CH_3CH_2Br

B. $BrCH_2CH_2Br$ (1,2-dibromoethane, a vicinal dihalide)

C. No reaction

D. $CH_2{=}CHBr$

Q30 Predict the product of this reaction:



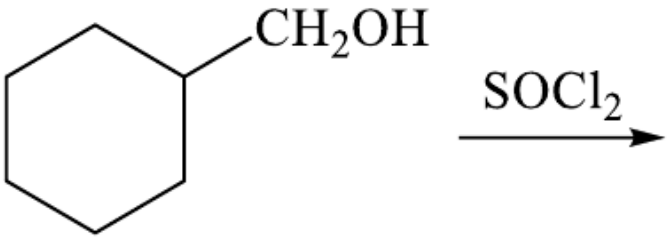
A. 2-Iodo-1-methylcyclohexane (structure A)

B. 4-Iodo-1-methylcyclohexene (structure D)

C. 1-Iodo-1-methylcyclohexane (structure B)

D. 1-Iodo-4-methylcyclohexane (structure C)

Q31 Predict the product of this reaction:



A. 2-Chloro-2-methylcyclohexane

B. Methylcyclohexene

C. (Chloromethyl)cyclohexane

D. 3-Chloro-1-methylcyclohexene

12.5 Reactions: Substitution & Elimination

Q32 Which two types of reactions do alkyl halides undergo?

A. Addition and polymerization

B. Nucleophilic substitution and elimination

C. Hydration and ozonolysis

D. Oxidation and reduction

Q33 In an elimination reaction of an alkyl halide, the product is:

A. An alcohol

B. A longer alkyl halide

C. An alkene (H and X are removed from neighbouring carbons)

D. An ether

Q34 What is a nucleophile?

A. An electron-rich species (anion or neutral molecule with a lone pair) that donates an electron pair to an electron-poor carbon

B. A positively charged ion such as H^+

C. A free radical

D. An electron-poor species that accepts electrons

Q35 Which are oxygen nucleophiles?

A. OH^-, RO^- (alkoxide), H_2O, ROH, $RCOO^-$ (carboxylate)

B. NH_3, RNH_2

C. CN^-, $RC{\equiv}C^-$

D. HS^-, RS^-

Q36 Which are nitrogen nucleophiles?

A. CN^-, $RC{\equiv}C^-$

B. OH^-, RO^-

C. I^-, Br^-, Cl^-

D. NH_3, RNH_2, R_2NH, R_3N, N_3^- (azide)

Q37 Which are sulphur nucleophiles?

A. OH^-, H_2O

B. HS^-, RS^- (thiolate), H_2S, RSH

C. CN^-

D. NH_3, N_3^-

Q38 Which are carbon nucleophiles?

A. NH_3 and RNH_2

B. Cl^- and Br^-

C. OH^- and RO^-

D. CN^- (cyanide) and $RC{\equiv}C^-$ (acetylide)

Q39 Which are halogen nucleophiles?

A. HCl gas only

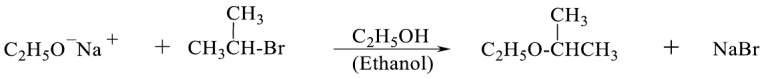
B. X^+ ions

C. I^-, Br^-, Cl^- (halide ions)

D. F_2, Cl_2

12.6 Nucleophilic Substitution Practice

Q40 In this reaction, identify the alkyl halide, the nucleophile and the solvent:



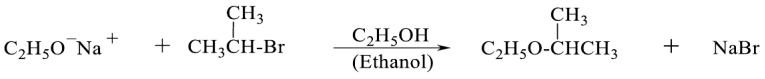
A. 3° alkyl halide; nucleophile = Br^-; solvent = ethanol

B. 2° alkyl halide; nucleophile = ethanol; solvent = NaBr

C. 2° alkyl halide (2-bromopropane); nucleophile = ethoxide ($C_2H_5O^-$); solvent = ethanol

D. 1° alkyl halide; nucleophile = Na^+; solvent = water

Q41 In the same reaction, identify the leaving group, the product and the by-product:



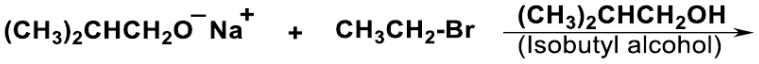
A. Leaving group = OH^-; product = alcohol; by-product = NaBr

B. Leaving group = Br^-; product = $C_2H_5OCH(CH_3)_2$ (2-ethoxypropane, an ether); by-product = NaBr

C. Leaving group = $C_2H_5O^-$; product = NaBr; by-product = ether

D. Leaving group = Na^+; product = propene; by-product = ethanol

Q42 Predict the product:



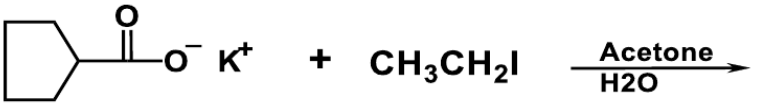
A. $CH_2{=}CH_2$

B. $(CH_3)_2CHCH_2{-}O{-}CH_2CH_3$ (ethyl isobutyl ether) + NaBr

C. $(CH_3)_2CHCH_2OH$ + CH_3CH_2Br

D. $(CH_3)_2CHCH_2{-}CH_2CH_3$

Q43 Predict the product:



A. Ethyl cyclobutanecarboxylate (an ester) + KI

B. Cyclobutanecarboxylic acid + ethane

C. Ethyl cyclobutyl ketone

D. Cyclobutylmethanol

Q44 Predict the product:

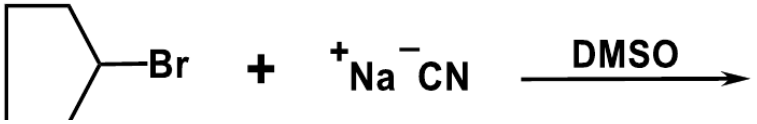
A. Cyclobutene

B. Cyclobutanethiol (cyclobutyl–SH) + KBr

C. Cyclobutanol

D. Dicyclobutyl sulfide only

Q45 Predict the product:



A. Cyclobutene

B. Cyclobutanol

C. Cyclobutylamine

D. Cyclobutanecarbonitrile (cyclobutyl–C≡N) + NaBr

Q46 Predict the product:

A. $CH_3CH_2C{\equiv}CCH_3$ (2-pentyne) + NaBr

B. $CH_3CH_2C{\equiv}CH + CH_3Na$

C. $CH_3CH_2C{\equiv}C–Br$

D. $CH_3CH_2CH{=}CHCH_3$

Q47 Predict the two products (??):

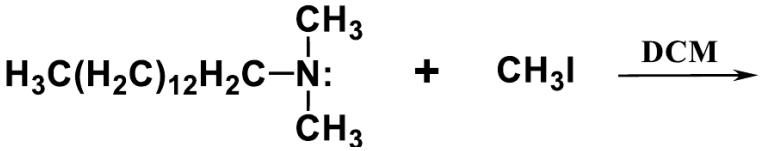
A. $(CH_3)_2N^+H–CH_2CH_3\ I^-$ (ammonium salt), then with base → $(CH_3)_2N–CH_2CH_3$ (a 3° amine)

B. $CH_3NH_2 + CH_3CH_2I$

C. No reaction

D. $CH_3–NH–CH_3 + CH_3CH_2I$

Q48 Predict the product:



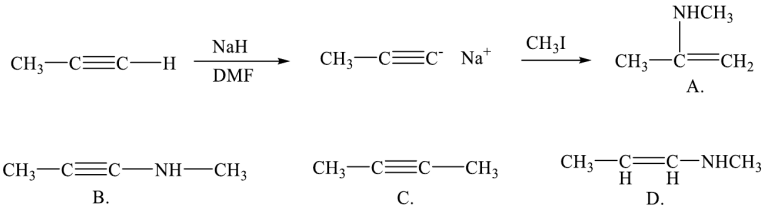
A. $CH_3(CH_2)_{13}I + (CH_3)_3N$

B. $CH_3(CH_2)_{13}N^+(CH_3)_3\ I^-$ — a quaternary ammonium salt

C. $CH_3(CH_2)_{13}N(CH_3)_2 + CH_3I$ (no reaction)

D. An alkene

Q49 The major product of this reaction will be:



A. $CH_3C(NHCH_3)=CH_2$ (structure A)

B. $CH_3CH=CH–NHCH_3$ (structure D)

C. $CH_3C{\equiv}CCH_3$, 2-butyne (structure C)

D. $CH_3C{\equiv}C–NHCH_3$ (structure B)

Q50 Which alkyl halide + nucleophile give $CH_3OCH_2CH_2CH_2CH_3$?

A. $CH_3Br + CH_3CH_2CH_2CH_3$

B. $CH_3CH_2CH_2CH_2OH + CH_3OH$

C. $CH_3CH_2CH_2CH_2Br + NaOH$

D. $CH_3CH_2CH_2CH_2Br + CH_3O^-Na^+$

Q51 Which alkyl halide + nucleophile give $HC{\equiv}CCH_2CH_2CH_3$?

A. $CH_3CH_2CH_2Br + NaOH$

B. $CH_3CH_2CH_2Br + HC{\equiv}C^-Na^+$

C. $CH_3CH_2CH_2OH + HC{\equiv}CH$

D. $HC{\equiv}CBr + CH_3CH_2CH_3$

Q52 Which alkyl halide + nucleophile give $CH_3CH_2CH_2NH_2$?

A. $CH_3CH_2CH_2Br + NaOH$

B. $CH_3CH_2CH_3 + NH_3$

C. $CH_3CH_2CH_2Br + NH_3$ (excess)

D. $CH_3CH_2CH_2OH + HCl$

Q53 Which alkyl halide + nucleophile give $CH_3CH_2CH_2SCH_2CH_2CH_3$?

A. $CH_3CH_2CH_2OH + H_2S$

B. $CH_3CH_2CH_2Br + CH_3CH_2CH_2S^-Na^+$

C. $CH_3CH_2CH_2Br$ + NaSH only

D. $CH_3CH_2CH_3 + S$

12.7 Safety of Organohalogen Compounds

Q54 Are all organohalogen compounds safe for oral use?

A. Yes, all are safe

B. Yes, because they are insoluble in water

C. No — e.g. CCl_4 and $CHCl_3$ are toxic and can cause liver damage

D. No — only the fluorides are toxic

Answers · Lecture 12

12.1 Types, Uses & Classification

Q1 **A. F, Cl, Br and I**

The four halogens: **fluorine (F), chlorine (Cl), bromine (Br), iodine (I)**. General formula R–X (X = F, Cl, Br, I).

Q2 **D. Carbon tetrachloride (CCl_4) and chloroform ($CHCl_3$)**

CCl_4 (iv) and $CHCl_3$ (v) are good organic solvents, but they are toxic to the liver.

Q3 **A. Halothane ($CF_3CHBrCl$) and methoxyflurane ($CH_3OCF_2CHCl_2$)**

Halothane (vi) and methoxyflurane (vii) are inhalation anesthetics.

Q4 **D. Tyrian purple / thyroxine (T4) / chloramphenicol**

Tyrian purple (iii) = natural dye (contains Br) · **thyroxine T4** (i) = thyroid hormone (contains I) · **chloramphenicol** (ii) = antibiotic from a soil bacterium (Streptomyces; contains Cl).

Q5 **D. Aryl halide (C_6H_5–Cl) and vinylic halide (CH_2=CH–Cl)**

Methyl CH_3X, alkyl (e.g. CH_3CH_2Br), allylic CH_2=$CHCH_2X$ and benzylic $C_6H_5CH_2X$ → X on an **sp^3** carbon. Aryl C_6H_5X and vinylic CH_2=CHX → X on an **sp^2** carbon.

Q6 **B. sp^3 carbon**

Allylic = the carbon **next to** a C=C (CH_2=CH–CH_2–X). That carbon is sp^3.

Q7 **A. X on a carbon attached to a benzene ring, e.g. $C_6H_5CH_2Cl$**

Benzylic = C_6H_5–CH_2–X (sp^3). Aryl = C_6H_5–X (sp^2).

Q8 **B. By the number of carbon atoms bonded to the carbon that carries the halogen (1, 2 or 3)**

1°: RCH_2–X (one C attached) · 2°: R_2CH–X (two C) · 3°: R_3C–X (three C).

Q9 **D. 3° alkyl halide**

The C–Br carbon is bonded to three CH_3 groups → tertiary.

12.2 Physical Properties

Q10 **B. Highest: CBr_4 (189 °C) — and CHI_3/CI_4 are solids; lowest: CH_3Cl (−24 °C)**

Boiling (and melting) point increases with **molecular weight**: bigger halogen (Cl < Br < I) and more halogens (CH_3X < CH_2X_2 < CHX_3 < CX_4). CHI_3 and CI_4 are even solids. CH_3Cl is the smallest → gas, lowest bp.

Q11 **A. R–I > R–Br > R–Cl**

Example CH_3CH_2–X: Cl 12.5 °C < Br 38 °C < I 72 °C. Larger atom → more polarizable → stronger van der Waals forces.

Q12 **D. No — they cannot form hydrogen bonds with water, so they are insoluble (but dissolve in organic solvents)**

Even though the C–X bond is polar, alkyl halides have no O–H or N–H, so they cannot H-bond with water → insoluble in water, soluble in non-polar organic solvents.

12.3 Nomenclature of Alkyl Halides

Q13 **D. Prefixes: fluoro-, chloro-, bromo-, iodo-**

Halogens are always prefixes; they get numbers like alkyl groups and are listed in alphabetical order. Suffix groups (–OH, C=C …) decide the numbering.

Q14 **B. 4-Chloro-2-methyl-5-phenylhept-6-en-3-ol**

Longest chain with OH and C=C = 7 C (**hept**). OH is the suffix (**-ol**) → number from the right so OH = C3. C=C at C6 (**hept-6-en**). Substituents: Cl on C4, CH_3 on C2, phenyl on C5 → alphabetical: **4-chloro-2-methyl-5-phenylhept-6-en-3-ol**.

Q15 **B. 5-Bromo-6-methylhept-2-ene**

Chain = 7 C. The C=C (suffix -ene) must get the lowest number → number from the left: C=C at C2. Br on C5, CH_3 on C6 → **5-bromo-6-methylhept-2-ene**.

Q16 **D. (1S,6S)-6-Chlorocyclohex-3-en-1-ol**

OH carbon = C1. Number towards the C=C so it gets C3–C4 (not 4–5); then Cl is on C6 → 6-chlorocyclohex-3-en-1-ol. OH (hashed) and Cl (wedge) are trans. C1: OH > C6 > C2 > H, H points to the viewer → **S**. C6: Cl > C1 > C5 > H, H away → **S**.

Q17 **C. (S)-6-Bromocyclohex-1-en-1-ol**

OH is on a C=C carbon: OH carbon = C1, its double-bond partner = C2 (cyclohex-1-en-1-ol). Br is on the other neighbour, C6. At C6: Br > C1 (=C–O) > C5 > H; Br is wedged (H away) → **S**.

Q18 **D. 4,4-Dichloro-1-butene**

CH_2=CH–CH_2–$CHCl_2$. The C=C (suffix -ene) gets the lowest number → C1, so the Cl atoms are on C4 → **4,4-dichloro-1-butene (4,4-dichlorobut-1-ene)**.

Q19 **A. (1S,3S)-1-Chloro-3-methylcyclohexane**

No suffix group, so Cl gets C1 by alphabetical order (chloro before methyl). C1: Cl > C2 (towards C3) > C6 > H; H is wedged (toward you) → reverse → **S**. C3: C2 (towards C–Cl) > C4 > CH_3 > H; H hashed (away) → counter-clockwise → **S**.

Q20 **C. (3R,5S)-5-Chloro-3-methylcyclohex-1-ene**

C=C = C1–C2, numbered so the substituents get {3,5} (not {4,6}) → 5-chloro-3-methylcyclohex-1-ene. C3: C2 (=CH) > C4 > CH_3 > H, H away → **R**. C5: Cl > C4 > C6 > H, H toward you → **S**.

Q21 **A. (R)-3-Chloro-1-methylcyclohexene**

The C=C carbons are C1 and C2 (C1 has the CH_3), Cl on C3 → 3-chloro-1-methylcyclohexene. At C3: Cl > C2 (=CH) > C4 > H; Cl is wedged (H away) → clockwise → **R**.

Q22 **A. 3-Bromo-2-methylpentane**

Longest chain = 5 C. Numbering from the right gives {2,3} (from the left {3,4}) → **3-bromo-2-methylpentane**.

Q23 **D. 4-Bromocyclohexene**

The C=C must be C1–C2; going around the shorter way puts Br on C4 → **4-bromocyclohexene**.

Q24 **A. (1S,2S)-2-Chlorocyclohexan-1-ol**

OH (suffix -ol) = C1, Cl on C2, trans. C1: OH > C2 > C6 > H, H toward you → **S**. C2: Cl > C1 > C3 > H, H away → **S**.

Q25 **C. 4-Iodo-2-methylhexane**

6-carbon chain. From the left: {2,4}; from the right: {3,5} → use {2,4}: **4-iodo-2-methylhexane** (iodo before methyl alphabetically).

Q26 **A. 1-Bromo-2-ethylcyclopentane**

Both substituents get {1,2}; when there is a tie, the first in the alphabet (bromo) gets C1 → **1-bromo-2-ethylcyclopentane**.

12.4 Preparation of Alkyl Halides

Q27 **C. Addition of HX or X_2 to alkenes, and substitution of the OH of alcohols (HX, $SOCl_2$, PBr_3)**

(i) Alkene + HX → alkyl halide (Markovnikov); alkene + X_2 → vicinal dihalide. (ii) R–OH + HX → R–X + H_2O; R–OH + $SOCl_2$ → R–Cl + SO_2 + HCl; R–OH + PBr_3 → R–Br.

Q28 **C. $CH_3CHBrCH_3$ (2-bromopropane) — Markovnikov**

H goes to the carbon with more H atoms; Br goes to the more substituted carbon (more stable carbocation).

Q29 **B. $BrCH_2CH_2Br$ (1,2-dibromoethane, a vicinal dihalide)**

X_2 adds one X to each carbon of the C=C (anti addition).

Q30 **C. 1-Iodo-1-methylcyclohexane (structure B)**

HI adds by Markovnikov's rule: H goes to the =CH carbon, I goes to the carbon carrying the CH_3 (3° carbocation) → I and CH_3 on the same carbon.

Q31 **C. (Chloromethyl)cyclohexane**

$SOCl_2$ replaces the OH of an alcohol with Cl: $C_6H_{11}CH_2OH$ → $C_6H_{11}CH_2Cl$ + SO_2 + HCl.

12.5 Reactions: Substitution & Elimination

Q32 **B. Nucleophilic substitution and elimination**

Substitution: Nu^- + R–X → R–Nu + X^- (e.g. $CH_3Br + OH^- \rightarrow CH_3OH + Br^-$). **Elimination (dehydrohalogenation):** a base removes H and X → alkene (e.g. $(CH_3)_3CBr + OH^- \rightarrow (CH_3)_2C{=}CH_2 + H_2O + Br^-$).

Q33 **C. An alkene (H and X are removed from neighbouring carbons)**

Base + H–C–C–X → C=C + H–Base + X^- (dehydrohalogenation).

Q34 **A. An electron-rich species (anion or neutral molecule with a lone pair) that donates an electron pair to an electron-poor carbon**

Nucleophile = "nucleus-loving"; a Lewis base. It attacks the δ+ carbon of the C–X bond and forms a new bond.

Q35 **A. OH^-, RO^- (alkoxide), H_2O, ROH, $RCOO^-$ (carboxylate)**

The nucleophilic atom is O.

Q36 **D. NH_3, RNH_2, R_2NH, R_3N, N_3^- (azide)**

Ammonia and amines attack with the N lone pair.

Q37 **B. HS^-, RS^- (thiolate), H_2S, RSH**

Sulphur nucleophiles are very good nucleophiles (large, polarizable).

Q38 **D. CN^- (cyanide) and $RC{\equiv}C^-$ (acetylide)**

Carbon nucleophiles make a new C–C bond: R–X + CN^- → R–CN; R–X + R'C≡C^- → R–C≡CR'.

Q39 **C. I^-, Br^-, Cl^- (halide ions)**

Halide ions can replace another halogen, e.g. R–Cl + I^- → R–I + Cl^-.

12.6 Nucleophilic Substitution Practice

Q40 **C. 2° alkyl halide (2-bromopropane); nucleophile = ethoxide ($C_2H_5O^-$); solvent = ethanol**

$CH_3CHBrCH_3$ = 2-bromopropane (Br on a carbon bonded to 2 carbons → 2°). The nucleophile is $C_2H_5O^-$ (Na^+ is only the counter-ion). Solvent: C_2H_5OH.

Q41 **B. Leaving group = Br^-; product = $C_2H_5OCH(CH_3)_2$ (2-ethoxypropane, an ether); by-product = NaBr**

Br^- leaves, ethoxide bonds to the carbon → ethyl isopropyl ether; Na^+ + Br^- = NaBr.

Q42 **B. $(CH_3)_2CHCH_2$–O–CH_2CH_3 (ethyl isobutyl ether) + NaBr**

Alkoxide (O nucleophile) + 1° alkyl halide → ether (Williamson ether synthesis).

Q43 **A. Ethyl cyclobutanecarboxylate (an ester) + KI**

The carboxylate ($RCOO^-$) is an O nucleophile; it replaces I^- → RCOO–CH_2CH_3 (ester).

Q44 **B. Cyclobutanethiol (cyclobutyl–SH) + KBr**

HS^- (S nucleophile) replaces Br^- → thiol R–SH.

Q45 **D. Cyclobutanecarbonitrile (cyclobutyl–C≡N) + NaBr**

CN^- (C nucleophile) replaces Br^- → nitrile R–CN (one more carbon).

Q46 **A. $CH_3CH_2C{\equiv}CCH_3$ (2-pentyne) + NaBr**

Acetylide (C nucleophile) + CH_3Br → the CH_3 joins the alkyne carbon → internal alkyne.

Q47 **A. $(CH_3)_2N^+H$–CH_2CH_3 I^- (ammonium salt), then with base → $(CH_3)_2N$–CH_2CH_3 (a 3° amine)**

The 2° amine (N nucleophile) attacks CH_3I → ammonium iodide salt. A base removes the H^+ from N → N,N-dimethylethylamine (3° amine).

Q48 **B. $CH_3(CH_2)_{13}N^+(CH_3)_3$ I^- — a quaternary ammonium salt**

A 3° amine + CH_3I → the N gets a fourth carbon and a + charge → quaternary ammonium iodide (no H on N, so it stays a salt).

Q49 **C. $CH_3C{\equiv}CCH_3$, 2-butyne (structure C)**

NaH removes the terminal alkyne H → acetylide $CH_3C{\equiv}C^-Na^+$ (C nucleophile). It attacks CH_3I → $CH_3C{\equiv}C$–CH_3.

Q50 **D. $CH_3CH_2CH_2CH_2Br$ + $CH_3O^-Na^+$**

Ether = alkoxide + alkyl halide. (CH_3Br + $CH_3CH_2CH_2CH_2O^-$ also works.)

Q51 **B. $CH_3CH_2CH_2Br$ + $HC{\equiv}C^-Na^+$**

Acetylide anion + 1° alkyl halide → new C–C bond.

Q52 **C. $CH_3CH_2CH_2Br$ + NH_3 (excess)**

NH_3 (N nucleophile) replaces Br → 1° amine (excess NH_3 prevents further alkylation).

Q53 **B. $CH_3CH_2CH_2Br$ + $CH_3CH_2CH_2S^-Na^+$**

Thiolate (RS^-) + alkyl halide → sulfide (thioether) R–S–R.

12.7 Safety of Organohalogen Compounds

Q54 **C. No — e.g. CCl_4 and $CHCl_3$ are toxic and can cause liver damage**

Some are useful drugs (chloramphenicol, halothane, thyroxine), but many are toxic: CCl_4 and $CHCl_3$ damage the liver (and some are carcinogenic).